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Government

FOR YOU

Health and Wellness



PACK
Practical Approach to Care Kit

Practical Approach to Care Kit

What is PACK Child?

The *Practical Approach to Care Kit (PACK) Child guide* is a comprehensive guide for the primary care of the child up to 13 years old. It uses simple algorithms to evaluate and treat the child with common symptoms and a standardised checklist format to care for the child with a long-term health condition. It supports the clinician to integrate the routine care of the child into every visit.

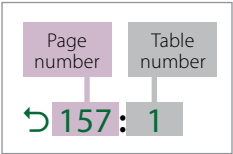
The PACK Child guide is not intended to replace the Integrated Management of Childhood Illnesses (IMCI). It aligns with the IMCI content but is arranged in a format that allows for expansion, including the management of the child over the age of 5, a greater number of symptom-based approaches as well as a new focus on long-term health conditions and the well child. This guide is also designed to comply with the Standard Treatment Guidelines (STG) and the Essential Medicines Lists (EML) as determined by the South African Essential Drugs Programme. This PACK Child 2026 edition has been tailored to local Western Cape policy and protocols including the Provincial Code List and Supplementary lists.

How to use PACK Child

- The PACK Child guide is designed to structure a clinical consultation with a child in primary care, providing preventive, curative and long-term care at the same visit. It is divided into three main sections:
- **Routine care section:** in the stable child, start the consultation on the routine care page relevant to the age of the child (< 2 months old or ≥ 2 months old). Use the standard 'Assess, Advise and Treat' framework to address the child's general health (feeding, development, immunisations, growth) and provide preventive treatment.
 - **Symptom section:** Use the Symptoms contents page to find the relevant symptom page in the guide. Decide if the child needs urgent attention (in the red box) and if not, follow the algorithm to either a management plan or to consider a long-term health condition.
 - **Long-term health condition section:** Use the Long-term health condition contents page to find that condition in the guide. Go to the colour-coded routine care pages to manage the child's long-term health condition using the 'Assess, Advise and Treat' framework.

Use these features to navigate PACK Child

- **Arrows** refer you to another page in the guide:
 - The **direct arrow** (➡) guides you to leave the page and continue on another page.
 - The **return arrow** (↩) guides you to consult another page but suggests you return and continue on the original page. Ideally, complete the original page and keep track of the other page/s you still need to consult, unless the other page is needed to continue the assessment. It can also direct you to the **Quick reference chart** on the back page for:
 - Normal ranges of respiratory rate, pulse rate and blood pressure
 - How to estimate weight according to age
 - How to calculate maintenance fluids
 - How to calculate endotracheal tube size
 - Level of consciousness assessments using the AVPU scale.
- ➡ **157:1** directs you to the **medication dosing tables** at the back of the guide (get there using the Medications Dosing Tables tab). The first number indicates the relevant page number to turn to and the second number indicates the number of the table to consult.
- The “**Assess**” tables are arranged in 3 tones to reflect those aspects of the history, examination and investigations to consider.



- **Medications are highlighted in orange, blue, green or pink:**
 - ♦ **Orange-highlighted** medications may be prescribed by a doctor or an authorised prescriber (clinical nurse practitioner/primary care nurse specialist or professional nurse) in accordance with his/her scope of practice within a specified field (like IMCI-trained nurse).
 - ➡ **Blue-highlighted** medications may be prescribed by a doctor or clinical nurse practitioner/primary care nurse specialist who is an authorised prescriber.
 - ■ **Green-highlighted** medications may be prescribed by a doctor only.
 - ● **Pink-highlighted** medications should be initiated by a doctor only, but may be continued by a clinical nurse practitioner/primary care nurse specialist who is an authorised prescriber.
 - Shape icons placed in front of colour-coded medications support users with colour vision deficiency.
- **How to interpret age and weight ranges:**
 - ‘10-12kg’ means ‘from and including 10kg, up to but not including 12kg’. The next range starts with 12kg.
 - ‘< 2 months old’ refers to a baby less than 2 months old and not including 2 months.
 - ‘≥ 2 months old’ indicates that the baby is 2 months old and older (this includes the whole month that the baby is 2 months old).
 - ‘< 5 years old’ refers to a child who is younger than 5 years, not including being 5 years old.
 - ‘≥ 5 years old’ refers to a child who is 5 years old and older, and starts on the day the child turns 5.
- Refer to the **Glossary** for abbreviations and units used in PACK Child.

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The PACK Adult guide is revised and improved every year, which includes feedback from end-users. Send us your feedback: kту@uct.ac.za.

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Overview of 2026 updates

The 2026 edition includes the following updates and additions:

General alignment with the Primary Care Standard Treatment Guidelines and Essential Medicine List, 2024 (8th edition) and updated chapters.

A **new page** on **Cholera** was added using the following guidelines and circulars:

- 2014 SA National Guidelines for Cholera Control
- 2023 SA National Circular on Cholera Outbreak
- Circular H17/2023 - Cholera alert, 2 February 2023: Strengthening preparedness in South Africa
- 2024 WHO Clinical tools for cholera treatment facilities

The **HIV section** has been revised to align with the following guidelines and circulars:

- National HIV Testing Services Policy 2025
- ARV DRUG DOSING CHART FOR CHILDREN ≥ 28 DAYS (2025). Compiled by National HIV & TB Health Care Worker Hotline and the Child and Adolescent Committee of SA HIV Clinicians Society in collaboration with the Department of Health
- Circular H108/2024 - Change in Resistance Testing Eligibility Timeframe for Patients on Second or Third Line Dolutegravir-Based Regimens
- Circular H135/2025 - Paediatric ARV Formulation Availability and Lopinavir/Ritonavir Review

The **TB section** has been revised. It aligns with the following guidelines and circulars:

- Management of tuberculosis in children and adolescents. A Clinical Guideline for the Diagnosis and Treatment of Drug-susceptible TB in Children and Adolescents in South Africa. September 2024
- Management of tuberculosis in children and adolescents (0 - <15 years old). A quick reference guide for frontline healthcare providers on the prevention (TPT), diagnosis and treatment of drug-susceptible TB (DS-TB) and rifampicin-resistant TB (RR-TB) in children and adolescents in South Africa.
- Circular H58/2022 - Rapid guidance for referral of children for collection of respiratory specimens for diagnosis of uncomplicated tuberculosis (TB)
- Circular H84/2023 - Systematic screening for tuberculosis (TB) using targeted universal TB testing (TUTT)
- Circular H08/2024 - Adoption of the updated NDOH guideline on TB preventive treatment (TPT) treatment of tuberculosis infection, 2023.
- Circular H68/2025 - Update on BCG vaccine administration in the Western Cape
- Circular H69/2025 - Revised National Department of Health (NDOH) guideline for management of drug-susceptible tuberculosis in children and adolescents < 16 years old

The sections related to **infectious diseases** other than HIV and TB have been revised and align with the following circulars:

- Circular H20/2024 - Changes in the expanded programme of vaccination and the maternal vaccination schedule
- Circular H60/2024 - COVID-19 endemic phase policy
- Circular H80/2024 - Alert and public health response to the confirmation of rabies in Cape fur seals
- Circular H135/2024 - Meningococcal disease management, prevention and control: update for healthcare workers
- Circular H150/2024 – Availability of Rabies vaccine and Rabies immunoglobulin
- Circular H13/2025 - Diphtheria alert and public health response to an increase of toxigenic respiratory cases in the Cape Town Metro District and Province
- Circular H125/2025 - Provincial Measles Outbreak Alert: Guidance for Epidemic Preparedness and Response

Additionally, the following pages or sections have been **updated**:

- The asthma pages align with circular H13/2024 - Changeover of inhaled corticosteroid
- Several symptoms pages have been revised to include the early SILUAN signs of childhood cancer
- The epilepsy pages align with circular H94/2023 - Valproate annual risk acknowledgment form and STG/EML chapter updates including indications for lamotrigine and levetiracetam

The **medication sections** align with:

- Circular H162/2024 - Supplementary list 166

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Glossary

A

AIDS	acquired immunodeficiency syndrome
ALP	alkaline phosphatase
ALT	alanine aminotransferase
ART	antiretroviral therapy

B

BCG	Bacillus Calmette-Guérin vaccine
BMI	body mass index
BP	blood pressure measured in millimeters of mercury [mmHg]

C

CD4	count of the lymphocytes with a CD4 surface marker
CHW	community health worker
CPR	cardiopulmonary resuscitation
Cr/Cl	creatinine clearance
CrAg	cryptococcal antigen
CLHIV	Child(ren) living with HIV

D

DR-TB	drug-resistant tuberculosis
DS-TB	drug-sensitive tuberculosis
DST	drug susceptibility testing
DTaP	diphtheria, tetanus, acellular pertussis vaccine

E

ECG	electrocardiogram
eGFR	estimated glomerular filtration rate
ELISA	enzyme-linked immunosorbent assay
ENT	ear, nose and throat specialist

F

FBC	full blood count
FBC+diff	full blood count and differential white cell count
FCS	Provincial Family Violence, Child Protection and Sexual Offences

G

GMFCS	Gross motor function classification system
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H

Hb	haemoglobin
HBsAb	hepatitis B surface antibody
HBsAg	hepatitis B surface antigen
HBV	hepatitis B virus
HFA	height-for-age
HiB	haemophilus influenza type b
HIV	human immunodeficiency virus
HPV	human papilloma virus

I

IM	intramuscular
IMCI	integrated management of childhood illness
IO	intraosseous
IPV	inactivated polio vaccine
IU	international units
IV	intravenous

K

KMC	kangaroo mother care
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L

L/HFA	length/height-for-age
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M

MAM	moderate acute malnutrition
MCV	mean corpuscular volume
MOU	maternity obstetric unit
MR	measles + rubella vaccine
MTB	<i>mycobacterium tuberculosis</i>
MU	million units
MUAC	mid-upper arm circumference

N

NaCl	sodium chloride
NGO	non-governmental organisation
NGT	nasogastric tube
NTP	nutritional therapeutic programme

O

ORS	oral rehydration solution
OPD	outpatients department

OPV	oral polio vaccine
-----	--------------------

P

PCAC	Provincial TB Clinical Advisory Committee
PCR	polymerase chain reaction
PCV	pneumococcal conjugate vaccine
PEP	post-exposure prophylaxis
PJP	<i>Pneumocystis jirovecii</i> pneumonia
PPE	papular pruritic eruption or personal protective equipment
Pulse rate	measured in beats per minute

R

Respiratory rate	measured in breaths per minute
RR-TB	Rifampicin resistant tuberculosis
RtHB	Road to Health Booklet
RUSF	Ready-to-use Supplementary food
RUTF	Ready-to-use Therapeutic Food
RV	rotavirus

S

SAM	severe acute malnutrition
SAPS	South African Police Service
Sats	oxygen saturation
SBP	systolic blood pressure

T

TB	tuberculosis
TB NAAT	tuberculosis nucleic acid amplification test
TBSA	total body surface area
Tdap	tetanus, reduced strength diphtheria and acellular pertussis vaccine
TPT	TB preventive treatment
TSB	total serum bilirubin
TSH	thyroid stimulating hormone
TST	tuberculin skin test

V

VL	viral load
----	------------

W

WFA	weight-for-age
WFL	weight-for-length
WFL/H	weight-for-length/height

Integrate routine care at every visit

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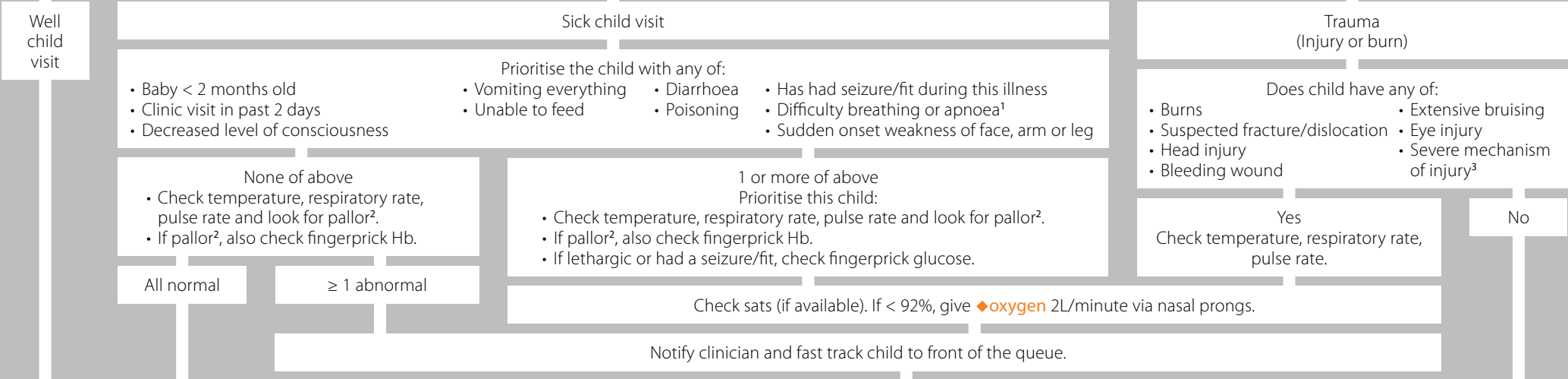
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Screen the child in the prep room

- Manage urgently if child not breathing or responding →24. If fitting →28. If baby newly born →23.
- Manage according to reason for clinic visit:



Measure and record growth: check weight (remove nappy) and if < 5 years old: mid-upper-arm circumference (MUAC) at every visit.
Measure length/height every 6 months. Measure head circumference at 14 weeks old and 12 months old.

How to measure length/height, head circumference and mid-upper arm circumference

Length/height

Remove shoes/hair ornaments that will interfere with measurements.



If < 2 years old, measure length:

- Lie baby on length board.
- Carer to hold head against headboard.
- Hold down legs and move foot board so that feet lie flat against it.



If ≥ 2 years old, measure height:

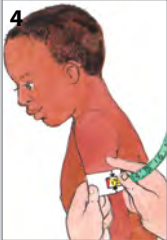
- Stand child against wall with head, shoulder blades, buttocks, calves and heels touching wall.
- Keep feet flat on ground.
- Get carer to hold knees and ankles.
- Hold child's chin to keep head straight.
- Pull down headboard⁴ to rest on head and measure in cm.

Head circumference



- Remove hair ornaments.
- Wrap tape measure around widest part of head:
 - Start from a point 2 fingers above eyebrow, past top of ears to around the widest part of back of head.
 - Measure in cm.

Mid-upper arm circumference (MUAC)



- Remove clothes to expose arm fully. Find tip of shoulder (acromion) and tip of elbow (olecranon) when elbow bent at right angle with palm up.
- Mark midpoint between these; then allow arm to hang straight down.
- Wrap tape around arm at midpoint and measure in cm.

¹Apnoea is episodes of no breathing > 10 seconds. ²Look for pale palms of the hands and conjunctival pallor: paleness of the lower inner eyelid. ³For example, a car or pedestrian accident, fall from height, stab wound, gunshot wound. ⁴If using headboard to measure height, ensure it is installed correctly (try measuring your own height). If using a tape measure, measure from the ground to top of head.

First assessment of the newborn

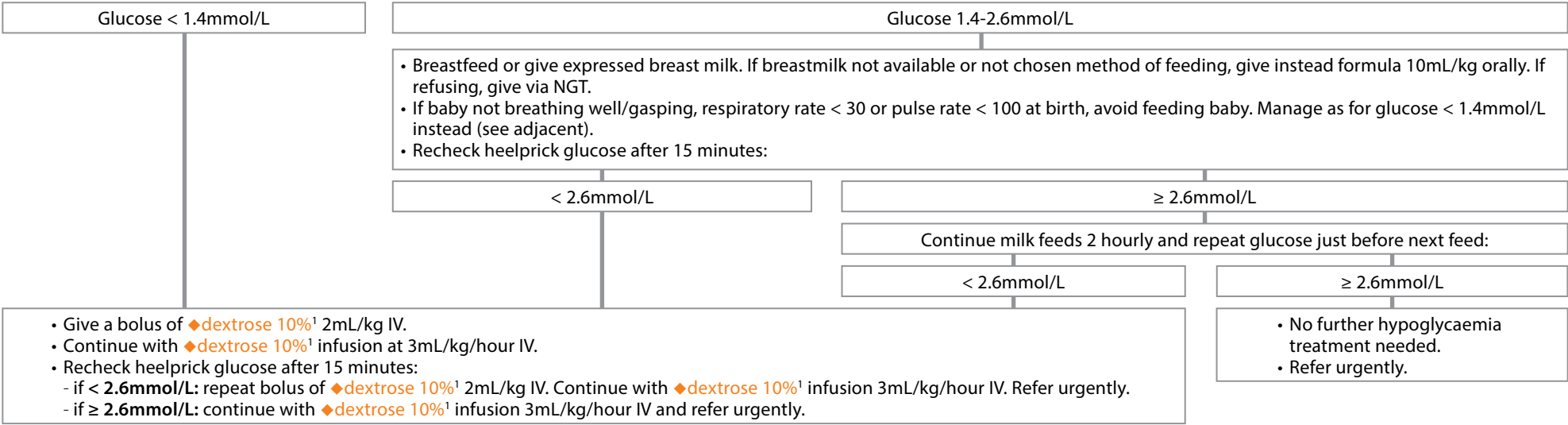
Assess the baby within 6 hours following birth.

Give urgent attention to the newborn with any of:

- < 2.5kg
 - Baby < 35 weeks gestation
 - Membranes ruptured > 18 hours before delivery/offensive liquor at birth
 - Difficulty breathing: blue lips/tongue, apnoea, respiratory rate > 60, grunting, nasal flaring, chest indrawing
 - Rapid swelling of head
 - Round, fluctuant swelling on one side of head, **cephalohaematoma** likely
 - Fontanelle full/bulging
- Severe bruising
 - Irregular jerky movements
 - Reduced movements/lethargy
 - Abnormal tone (floppy/stiff)
 - Abdominal distension
 - Jaundice →61
 - Pallor
- Imperforate anus
 - Cleft lip/palate
 - Ambiguous genitalia
 - Nostril/s not patent
 - Open area over spine
 - Head circumference > 39cm
 - Temperature < 36°C or > 37.5°C

Manage and refer urgently:

- If difficulty breathing, give ♦oxygen 1L/min via nasal prongs. If baby just born, help baby breathe →23. If nostril(s) not patent, discuss.
- Keep baby warm: place baby skin-to-skin with mother and cover with blanket or transport in incubator.
- Feed baby
 - If imperforate anus, delay feeding and give ►dextrose 10%¹ solution at 3mL/kg/hour IV.
 - If alert, encourage breastfeeding. If breastmilk not available or not chosen method of feeding, give instead formula/sugar water² 3mL/kg/hour orally.
 - If baby too sick to feed, give via NGT.
 - Feed at least 2 hourly until transfer.
- Check heelprick glucose. If < 2.6mmol/L, manage as **hypoglycaemia**:



Assess the newborn not needing urgent attention →10.

¹Neonatalyte. If dextrose 10% unavailable: in 20mL syringe, draw up 4mL of dextrose 50% and 16mL of water for injection. Syringe now contains 20mL of 10% dextrose. ²Dissolve 4 teaspoons of sugar (20g) into 200mL water.

Assess the newborn not needing urgent attention

Assess	Note (undress baby fully when examining but keep baby warm)
Mother/carer	- Give mother routine postnatal care ➔ PACK Adult. If mother/carer or you are worried about how mother/carer and family will cope with baby, refer to social worker/community health worker. - Look for increased psychosocial risk: parent/carer < 20 years old, family/relationship problems, violence at home, lack of partner/family support, financial difficulty, difficult life event in last year, foreigner. If any of these, give additional support, review more often if needed and link with support services/helpline ➔ 170.
Feeding	Assess suck/latch and give feeding advice ➔ 120 . If mother breastfeeding and has a BMI ¹ < 18.5 and/or mid-upper arm circumference (MUAC) < 23cm, refer to nutritional therapeutic programme (NTP).
TB risk	If mother/close contact on TB treatment ➔ 91 . Assess for TB in mother/carer ➔ PACK Adult .
Measurements	- Plot weight-for-age, weight-for-length and head circumference and record in RtHB. If head circumference < 32cm, discuss/refer. - If weight > 4.5kg or mother known with diabetes, check heelprick glucose hourly for 6 hours. If < 2.6mmol/L, manage as hypoglycaemia ➔ 9.
Head	- Check for swelling: if doughy swelling on head, that crosses suture lines, caput likely. Should resolve in under 5 days. - Check sutures: if sutures overlapping, review in 2-3 days. If still overlapping, refer.
Face and neck	If unusual appearance, abnormal shape of face or neck swelling/webbing, discuss with doctor.
Eyes	If white spot in pupil, white pupil, loss of red reflex with white light, discuss/refer same day. If thick yellow discharge in eyes ➔ 48 .
Nose	If nose blocked, give ♦ sodium chloride 0.9% 1 drop in each nostril.
Abdomen	If mass in abdomen, doctor to review. If soft collapsible mass around umbilicus or groin, hernia likely. If groin (inguinal) hernia, discuss with doctor/surgeon same day. If umbilical hernia, reassure, observe and refer to surgeon if still present when child ≥ 4 years old.
Chest/back	If abnormal, discuss.
Genitals	If displaced urethral opening (check tip of penis in male), refer. In male: if testes not felt in scrotum, review in 1 month. If still not felt, refer to surgical OPD.
Limbs	- If one arm rotated towards body or moving less than before, Erb's palsy likely, doctor to review. Follow up after 2 days: if problem persists, refer to orthopaedic OPD. - If extra digit with bone present, book surgical OPD appointment. If no bone present and on thin stalk of skin, doctor to tie off. - If foot/feet bent with sole/s facing inward, try to manipulate foot into normal position. If able to, refer to physiotherapist. If unable to, clubfoot likely, refer to orthopaedic specialist.
Hips for dislocation	- Hold thighs with thumb over inner thigh and first two fingers over outer thigh. Bend knees at right angles. Gently push downwards into the bed. If looseness or clunk felt, refer to orthopaedic specialist. - Next, move legs into frog-leg position and push hip upwards towards you from behind. If "click" heard/felt, refer to orthopaedic specialist.
Bilirubin	Check total serum bilirubin (TSB) 6 hours after birth if mother blood group O, rhesus negative or mother's blood group/rhesus unknown. Interpret TSB result ➔ 61 .
HIV risk	If HIV-exposed/abandoned baby, do a birth HIV PCR and assess for vertical transmission of HIV prevention (VTP) ➔ 92 .
Syphilis	If mother syphilis positive, assess baby ➔ 91 and treat mother ➔ PACK Adult .
Hepatitis B	If mother hepatitis B positive, assess and manage baby ➔ 91 .

Advise the carer of the newborn

- Encourage carer to hold, cuddle, talk/sing and make eye contact with baby. This helps with bonding and development. If carer finds this difficult, encourage carer to return more frequently.
- Carer to keep baby warm using skin-to-skin contact. If birthweight ≤ 2.5kg, advise both parents to practise Kangaroo Mother Care (KMC) until baby > 2.5kg.
- Advise carer to apply ♦**chlorhexidine gluconate 0.5% in 70% alcohol solution** or surgical spirits to the umbilical cord every 6 hours until it falls off (1-2 weeks). If area becomes red, to return to clinic.

Immunise newborn

- If well, give ♦**BCG** intradermally into right arm regardless of TB exposure or HIV exposure. Avoid BCG if CLHIV. If inborn errors of immunity or other immunodeficiency, discuss with specialist. If TB preventive treatment (TPT) or TB treatment started within 6 weeks of BCG, repeat BCG once treatment completed.
- Give ♦**OPV** orally.

Discharge newborn and plan review

- If newborn is well, urine passed and breastfeeding established, discharge after 6 hours. Issue RtHB and explain contents. Refer to community health worker for home visit and breastfeeding support.
- Review within 6 days, then at 6 weeks ➔ **11**. If preterm (< 37 weeks) or < 2.5kg, doctor also to review weekly until 2.5kg, then at 4 and 9 months old.
- Advise to return immediately if breastfeeding poorly, irritable/lethargic, fitting, vomiting everything, fever, cough with fast breathing, blood in stool or no stool within 48 hours of birth.

¹Body Mass Index = weight (kg) ÷ height (m) ÷ height (m).

Baby < 2 months old: routine care

Record problems and plot growth in notes and RtHB. Assess baby and mother within 6 days of delivery and at 6 weeks old. If never assessed →9.

Assess	When to assess	Note
Symptoms	Every visit	Manage symptoms first on symptom page. If cough/breathing problems → 55, if diarrhoea → 63, if vomiting → 62, if fits → 28, if lethargic → 45.
Urine/stool	1st visit (within 6 days)	If baby not passing urine or stool, discuss/refer.
↻ Feeding	Every visit	If breastfeeding or mixed feeding → 120, if formula feeding → 122.
Growth	Weight: every visit	- Assess weight-for-age at every visit and plot in RtHB. - If baby has lost > 10% of birth weight¹ in first week of life, or if baby has not regained birth weight at ≥ 10 days old, assess and manage poor growth → 125.
👤 Immunisation status	Every visit	Check if birth and 6 weeks immunisations are up to date → 13. If missed immunisations, catch up missed doses → 13. Report any adverse event following immunisation (AEFI) within 24 hours of presentation irrespective of causal relationship → 170.
Mother/carer	Every visit	Check mother received post-natal care → PACK Adult. Ask about HIV status, contraceptive needs and TB symptoms. If mother breastfeeding and BMI ² < 18.5 and/or mid-upper arm circumference (MUAC) < 23cm, refer to nutritional therapeutic programme (NTP).
Psychosocial risk	First visit	- If child support grant needed, advise to take child's birth certificate and carer's ID to SASSA³ to apply. If no birth certificate, refer to social worker. - Look for increased psychosocial risk: carer/parent < 20 years old, family/relationship problems, violence at home, lack of partner/family support, financial difficulty, difficult life event in last year, foreigner. If any of these, give additional support, review more often if needed and link with support services/helpline → 170. - Screen for depression/anxiety in carer: in the past month, has carer felt down/depressed/hopeless or felt little interest/pleasure in doing things? Are they often nervous, anxious or panicky, unable to stop worrying or thinking too much, or had thoughts of harming him/herself? If yes to any → PACK Adult. - If abuse or neglect suspected → 144.
🩺 HIV	If HIV-exposed ⁴ : every visit	If birth HIV PCR test not done, do now and follow up result → 95. If on post-exposure prophylaxis (PEP) → 92. If CLHIV → 97.
TB	Every visit	If mother/close contact on TB treatment → 91. Assess for TB in mother/carer → PACK Adult.
🩺 Basic examination	Every visit	Undress baby fully and check for obvious problems: pallor ⁵ → 46, skin problem (especially nappy area) → 75, injury → 33. If deformity, discuss/refer.
Skin	Every visit	If yellow skin/eyes → 61, pallor ⁵ → 46 or blue → 41. If rash/pustules → 75. If many/severe pustules, discuss and refer.
Eyes	Every visit	If white spot in pupil, white pupil, loss of red reflex with white light, discuss/refer same day. If pus and/or eyelid swelling → 48.
Mouth	Every visit	If white patches in mouth (inside of cheeks/lips and on tongue), oral thrush/candida likely → 51.
Umbilical cord	Every visit	If red stump, give ♦cephalexin → 159:13 or ♦flucloxacillin → 161:23. Apply ♦povidone iodine 5% cream or ♦povidone iodine 10% ointment to cord 6 hourly and expose to dry. Review in 1-2 days and if no better, refer; otherwise, continue treatment. If penicillin allergy or redness extends to skin and/or draining pus, discuss and refer.
💛 Hearing	Every visit	If carer has any concerns, book hearing test.

Advise the carer

- Encourage carer to keep baby warm using skin-to-skin contact. If birthweight ≤ 2.5kg, advise both parents to practise Kangaroo Mother Care (KMC) until baby > 2.5kg.
- Encourage parents to respond when baby cries and to hold, cuddle, talk/sing and make eye contact with baby. This helps with bonding and development. If mother finds this difficult, encourage her to return more frequently. The first 1000 days of a child's life are vital to his/her development.

Treat the baby

- Multivitamins (if birth weight < 2.5kg or baby < 37 weeks gestation): from 6 weeks old, give ♦multivitamin 0.6mL and ♦ferrous gluconate 2.5mL once daily until 6 months old. Review mother/carer and baby at 6 weeks old and 10 weeks old.
- Review more often if: premature, < 2.5kg, hospitalised for > 3 days after birth, neurological/congenital problem or suspected bonding problem (carer reluctant to hold or look at baby).

¹Birth weight (kg) ÷ 10 = 10% of birth weight: if weight loss in first week of life more than this, baby has poor growth. ²Body mass index: weight (kg) ÷ height (m) ÷ height (m). ³South African Social Security Agency. ⁴Mother HIV positive or abandoned/orphaned baby. ⁵Look for pale palms of the hands and paleness of the lower eyelid (conjunctival pallor).

Child ≥ 2 months old: routine care

Record problems and plot growth in notes and RtHB.

Assess	When to assess	Note
 Symptoms	If sick visit	Manage symptoms on symptom page.
 Feeding	Every visit	Ask carer if feeding problem. If yes, assess and manage further: if breastfeeding or mixed feeding ↗ 120 , if formula feeding ↗ 122 , if eating solids ↗ 123 .
 Growth	Every visit	Check chart ↗ 14 and interpret measurements ↗ 15 . If born premature, use corrected age ¹ until 2 years old.
 Development <i>Ask general screening questions at every visit. Then screen at specific ages.</i>	Every visit	- Ask: "Is child able to say and do what children of the same age can?" - If carer answers "No", manage problem: if vision problem ↗ 48, if communication problem ↗ 89, if not moving or sitting properly ↗ 90.
	14 weeks old	If unable to follow a close object with eyes ↗ 48 . If does not respond (stops sucking, blinks or turns) to sound ↗ 89 . If unable to lift head when held against shoulder ↗ 90 .
	6 months old	If unable to recognise familiar faces ↗ 48 . If squint, discuss. If does not turn to look for sound ↗ 89 . If unable to hold a toy in each hand ↗ 90 .
	9 months old	If unable to focus on a far object ↗ 48 . If does not turn when called ↗ 89 . If unable to sit and play without support ↗ 90 .
	12 months old	If unable to look closely at toys or pictures ↗ 48 . If not using ≥ 1 meaningful word ("dada, mama") ↗ 89 . If unable to stand with support ↗ 90 .
	18 months old	If not looking at or reaching for small objects or pictures ↗ 48 . If unable to point to 2 simple objects, uses < 3 words, does not obey simple commands ↗ 89 . If unable to walk unsupported or unable to feed using fingers ↗ 90 .
	3 years old	If unable to see small shapes clearly from 6 metres ↗ 48 . If unable to talk in simple 3-word sentences ↗ 89 . If unable to run or climb ↗ 90 .
 Routine treatment to protect from illness	Every visit	Check if immunisations, deworming and vitamin A are up to date in RtHB and what is due at this visit ↗ 13 . If missed doses, catch up ↗ 13 . Report any adverse event following immunisation (AEFI) within 24 hours of presentation irrespective of causal relationship ↗ 170 .
Mother/carer	Every visit	- Ask about general health, HIV status, contraceptive needs and TB symptoms ↗ PACK Adult. If breastfeeding and BMI² < 18.5 and/or MUAC³ < 23cm, refer to NTP⁴. - Screen for depression/anxiety in carer: in the past month, has carer felt down/depressed/hopeless or felt little interest/pleasure in doing things? Are they often nervous, anxious or panicky, unable to stop worrying or thinking too much, or had thoughts of harming him/herself? If yes to any ↗ PACK Adult.
Psychosocial risk	Every visit	- If any child < 18 years needs a child support grant, advise to take child's birth certificate and carer's ID to SASSA⁵ to apply. If no birth certificate, refer to social worker. - Look for increased psychosocial risk: carer/parent < 20 years old, family/relationship problems, violence at home, lack of partner/family support, financial difficulty, difficult life event in last year, foreigner. If any of these, give additional support, review more often if needed and link with support services/helpline ↗ 170. - If abuse or neglect suspected ↗ 144.
 HIV	Every visit	- If HIV status unknown, decide if HIV test is needed ↗ 95. If mother HIV negative and breastfeeding, check that mother tests for HIV every 3 months. - If HIV-exposed⁶, check when HIV test is due ↗ 95. Ensure baby is receiving post-exposure prophylaxis (PEP) if necessary ↗ 92. Ensure mother on ART ↗ PACK Adult. - If CLHIV, ensure child on ART and give routine HIV care ↗ 97.
TB	Every visit	If any of: significant TB exposure ⁷ , cough, especially if persistent, unexplained fever, drenching night sweats, poor weight gain, weight loss, failure to thrive, lethargy, decreased playfulness, visible neck mass, blood-stained sputum, headache, persistent vomiting (without diarrhoea), test for TB ↗ 108 .
Mental health	Every visit	If over past few months, has child been sad, withdrawn, irritable, worried, stressed, anxious or had emotional outbursts ↗ 139 or if problematic change in behaviour ↗ 136 .
School problems	If ≥ 6 years old: every visit	- Check if child at school. If not enrolled in school, refer to Western Cape Education Department ↗ 170. - If poor attendance, problematic change in behavior, bullying, learning problems, difficulty socialising at school ↗ 140.
 Basic examination	Every visit	If < 2 years, undress fully. Check for obvious problems: pallor (pale palms and paleness of lower inner eyelid) ↗ 46 , eye problem ↗ 48 , skin problem (especially nappy area) ↗ 75 , injury ↗ 33 , gum/teeth problem ↗ 54 . If deformity, discuss/refer.

Continue to advise and provide routine care treatment [→ 13](#).

¹Corrected age = actual age in months (or weeks) - number of months (or weeks) premature. E.g. to calculate corrected age of 9 month old, born premature at 32 weeks (8 weeks or 2 months premature): 9 months - 2 months = 7 months. ²Body mass index: weight (kg) ÷ height (m) ÷ height (m). ³Mid-upper arm circumference. ⁴Nutritional therapeutic programme. ⁵South African Social Security Agency. ⁶Mother HIV positive or abandoned/orphaned baby. ⁷Any child who in the last 12 months shared an enclosed space (school/crèche/home/similar) for ≥ 1 night or frequent/extended daytime periods with adult/adolescent with pulmonary TB ("index client"), during 3 months before index client started TB treatment.

Advise the child and carer and give health promotion messages

Advise about parenting:

- The first 1000 days (conception - 2 years old) are vital to a child's development.
- Stimulate development, respond when baby cries, talk to baby, read daily, tell stories, sing songs, play.
- Establish routines, provide discipline and actively listen to child. Avoid smoking in house or near child.

Educate about hygiene:

- Wash hands with soap and water, especially after using toilet/handling food/changing nappy.
- Wash fruit/vegetables. If no access to clean water, boil and cool water.

Discuss safety:

- Lock away toxic substances, safeguard fires/paraffin lamps/electrical sockets.
- Teach road safety, use seat belts/car seats.

Encourage a healthy lifestyle:

- Ensure a balanced diet. Limit sweets, chocolates, fizzy drinks, fast foods, chips and salt.
- Advise physical activity ≥ 1 hour/day (team sports/outside play).
- Limit screen time to < 1 hour/day. Advise no TV if < 2 years old. Monitor adult content.

Treat and immunise the child

- Multivitamins (if < 6 months old): if low birth weight (< 2.5kg) or born at < 37 weeks gestation, give ♦multivitamin 0.6mL once daily and ♦ferrous gluconate 2.5mL once daily until 6 months old.
- Vitamin A (from 6 months - 5 years old): give single dose ♦vitamin A ↗ 164:43 6 monthly.
- Deworm (from 12 months - 5 years old): give ♦mebendazole ↗ 162:31 6 monthly.
- Immunise: give immunisations (see table) even if born premature, unwell (delay only if temperature ≥ 38.5°C) or RthB missing¹. If missed immunisation, catch up missed immunisations in table below.
- Observe for 15 minutes after immunisation (or 30 minutes if known with severe allergies). Always keep fully equipped emergency tray at immunisation point. If collapse, check if responsive ↗ 24, call for help and place child on back with legs raised. If resolved and loss of consciousness < 1 minute, vasovagal episode likely, discuss with doctor. If any of: loss of consciousness > 1 minute/not resolved by raising legs, sudden onset of generalised itch/rash or face/tongue itch/swelling, sudden onset tight throat/hoarseness, difficulty breathing, wheeze, low blood pressure ↗ 175, dizziness, shock (≥ 2 of: cold hands/feet, weak/fast pulse ↗ 175, capillary refill time ≥ 3 seconds, decreased level of consciousness ↗ 174), anaphylaxis likely ↗37.
- If CLHIV or other long-term health condition, advise influenza vaccination during influenza vaccine campaign.

Age	Immunisation	Site
Birth	♦BCG ²	Intradermal right arm
	♦OPV 0	Oral
6 weeks	♦OPV 1	Oral
	♦RV 1	Oral
	Hexavalent: ♦DTaP-IPV-Hib-HBV 1	IM left thigh
	♦PCV 1	IM right thigh
10 weeks	Hexavalent: ♦DTaP-IPV-Hib-HBV 2	IM left thigh
14 weeks	♦RV 2	Oral
	Hexavalent: ♦DTaP-IPV-Hib-HBV 3	IM left thigh
	♦PCV 2	IM right thigh
6 months	♦Measles, Rubella (MR) ³ 1	Subcutaneous left thigh
9 months	♦PCV 3	IM right thigh
12 months	♦Measles, Rubella (MR) 2	Subcutaneous right arm
18 months	Hexavalent: ♦DTaP-IPV-Hib-HBV 4	IM left arm
6 years	♦Tdap	IM non-dominant arm
≥ 9 years ⁴	Tdap booster. If girl: HPV	
12 years	♦Tdap	IM non-dominant arm

Give routine immunisations:

Immunisation	Give first dose according to age:	Give next dose/s, if needed, after minimum interval:		
		Dose 2	Dose 3	Dose 4
♦BCG ²	If < 10 years, give now	-	-	-
	If ≥ 10 years, do not give	-	-	-
♦OPV	If < 6 months, give now	Give 4 weeks later	-	-
	If ≥ 6 months, do not give	-	-	-
♦DTaP-IPV-Hib-HBV ⁵	If < 5 years, give now	Give 4 weeks later	Give 4 weeks later	Give 12 months later ⁶
♦RV	If < 20 weeks, give now	Give 4 weeks later	-	-
	If 20-24 weeks, give now	-	-	-
	If > 24 weeks, do not give	-	-	-
♦PCV	If < 6 months, give now	Give 4 weeks later	Give at 9 months old	-
	If 6-9 months, give now	Give 4 weeks later	Give 8 weeks later	-
	If 9-24 months, give now	Give 4 weeks later	Give 8 weeks later	-
	If 2-6 years, give now	Only if long-term health condition: give 8 weeks later	-	-
♦Measles, Rubella (MR)	If < 11 months, give now ³	Give at 12 months old	-	-
	If ≥ 11 months, give now	Give 4 weeks later	-	-
♦Tdap	If ≥ 6 years, give now	Give at 12 years old	-	-

Catch up missed immunisation/s:
Refer to CHW. If concerns for poor parental care, refer to social worker.

Decide when child should return for next routine care visit ↗14.

¹Issue new RthB if lost. ²Give BCG regardless of TB/HIV exposure. If CLHIV, only give BCG if clinically well on ART and CD4 ≥ 25% (≤ 5 years) or ≥ 200 (> 5 years). If TPT/TB treatment started in first 6 weeks of life, repeat BCG after TPT/TB treatment completed: give BCG 24 hours after last TPT/DS-TB dose, or 5 days after last rifampentine dose. Avoid BCG if: CLHIV not on ART/unwell/CD4 < 25% (≤ 5 years) or < 200 (> 5 years). If inborn errors of immunity or other immunodeficiency, discuss with specialist. ³If < 9 months old, avoid giving measles-containing vaccine together with other immunisations. If needed, give measles-containing vaccine immediately and arrange visit to receive remaining immunisation 4 weeks later. ⁴HPV vaccine and Tdap booster given during integrated school health programme campaign. ⁵Avoid giving DTaP-IPV-Hib-HBV if ≥ 5 years old. ⁶Avoid giving DTaP-IPV-Hib-HBV dose 4 if < 18 months old.

Routine care chart

Check what routine care is needed at this visit and when to review next:

Age	Check growth and plot on growth charts						Test for HIV	Check TB risk	Check feeding	Immunise	Give vitamin A	Deworm	Check development	Arrange oral health visit ¹
	Weight	Length or height	Weight-for-length/height	BMI	MUAC	Head circumference								
3-6 days	x						x (only if HIV exposed and not yet done)	x	x					
1 week	x							x	x					
2 weeks	x							x	x					
6 weeks	x							x	x	x				
10 weeks	x						x (only if HIV exposed)	x	x	x				
14 weeks	x					x		x	x	x			x	
4 months	x							x	x					
5 months	x							x	x					
6 months	x	x	x		x		x (only if HIV exposed)	x	x	x	x		x	
7 months	x				x			x						
8 months	x				x			x						
9 months	x				x			x	x	x			x	
10 months	x				x			x						
11 months	x				x			x						
12 months	x	x	x		x	x		x	x	x	x	x	x	x
14 months	x				x			x						
15 months	x				x			x						
16 months	x				x			x						
18 months	x	x	x		x		x (even if not HIV exposed)	x	x	x	x	x	x	
20 months	x				x			x						
22 months	x				x			x						
24 months (2 years)	x	x	x		x			x			x	x		x
30 months (2 ½ years)	x	x	x		x			x			x	x		
36 months (3 years)	x	x	x		x			x			x	x	x	x
42 months (3 ½ years)	x	x	x		x			x			x	x		
48 months (4 years)	x	x	x		x			x			x	x		x
54 months (4 ½ years)	x	x	x		x			x			x	x		
60 months (5 years)	x	x		x	x			x			x	x	x	x
72 months (6 years)	x	x		x				x		x				x
9 years	x	x		x				x		x				
12 years	x	x		x				x		x				x

¹Refer to dentist, dental therapist or oral hygienist.

Assess and interpret growth

Use steps 1-5 to decide when to measure weight, height, mid-upper arm circumference (MUAC) and head circumference, then interpret and act on results. If ≥ 5 years, start at step 3.

Step

1

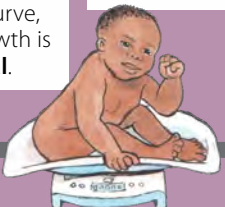
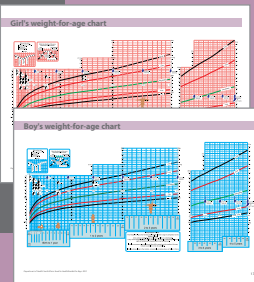
Weight-for-age

Plot weight-for-age at every visit:

Below -2 line
Child is likely **not growing well** →126.

Between -2 line and +2 line
Look at growth curve pattern: if growth curve flattening, falling or crossing z-score lines on 2 consecutive visits →126. If upwards growth curve, child's growth is **normal**.

On or above +2 line
Measure length/height (step 2) and then look at weight-for-length/height (step 3).



Step

2

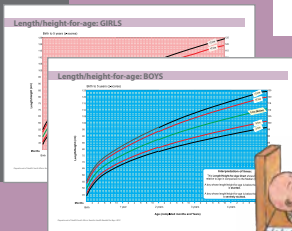
Length/height-for-age

If child < 2 years old, use length.
If child ≥ 2 years old, use height.

Plot length/height-for-age every 6 months:

Below -2 line
Child likely has **chronic malnutrition (stunting)** →126.

On or above -2 line
Child has a **normal** length/height. Continue to monitor length/height-for-age 6 monthly.



Step

3

Weight-for-length/height

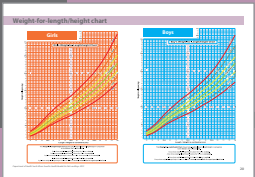
Plot weight-for-length/height, according to age, every 6 months:

Child < 5 years old
Plot weight-for-length/height:

Below -2 line
Child is **wasted** →126.

Between -2 line and +2 line
Weight-for-length/height **normal**

On or above +2 line
Child is **overweight** →129.

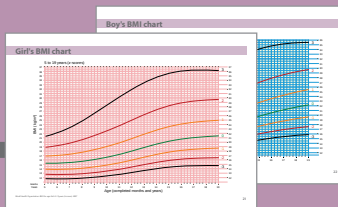


Child ≥ 5 years old
Calculate BMI¹ and plot on BMI-for-age chart → 21:

Below -2 line
Child is **wasted** →126.

Between -2 line and +1 line
BMI **normal**

On or above +1 line
Child is **overweight** →129.



Step

4

Mid-upper-arm circumference (MUAC)

Measure MUAC at every visit, from 6 months old until 5 years old.



< 12.5cm
Child has **acute malnutrition** →126.

≥ 12.5cm
MUAC normal.
Child is **growing well**.

Step

5

Head circumference

Measure head circumference at 14 weeks and 12 months old

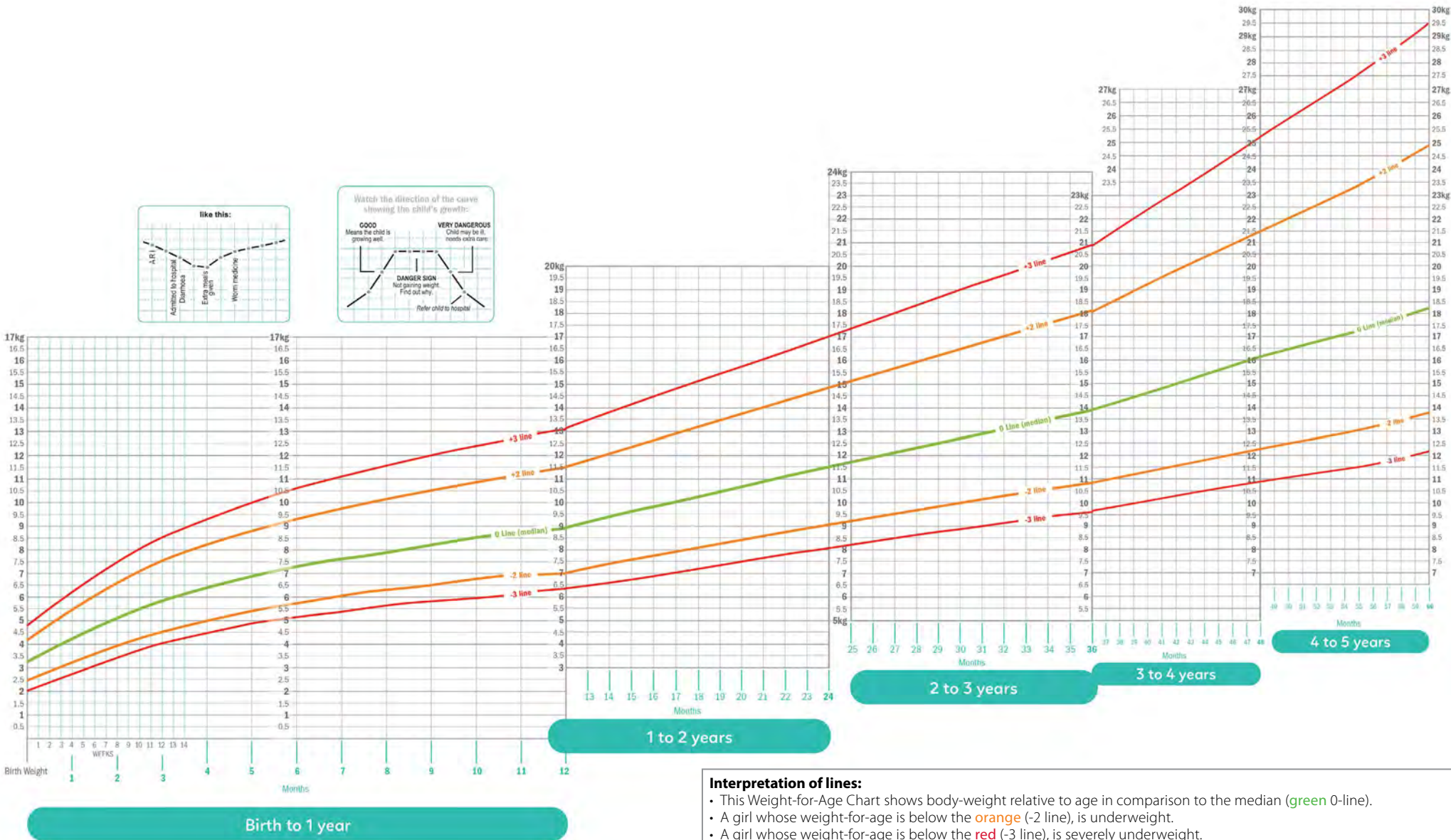


14 weeks old
Refer if < 38cm or > 43cm.

12 months old
Refer if < 43.5cm or > 48.5cm.

¹Body Mass Index: weight (kg) ÷ height (m) ÷ height (m).

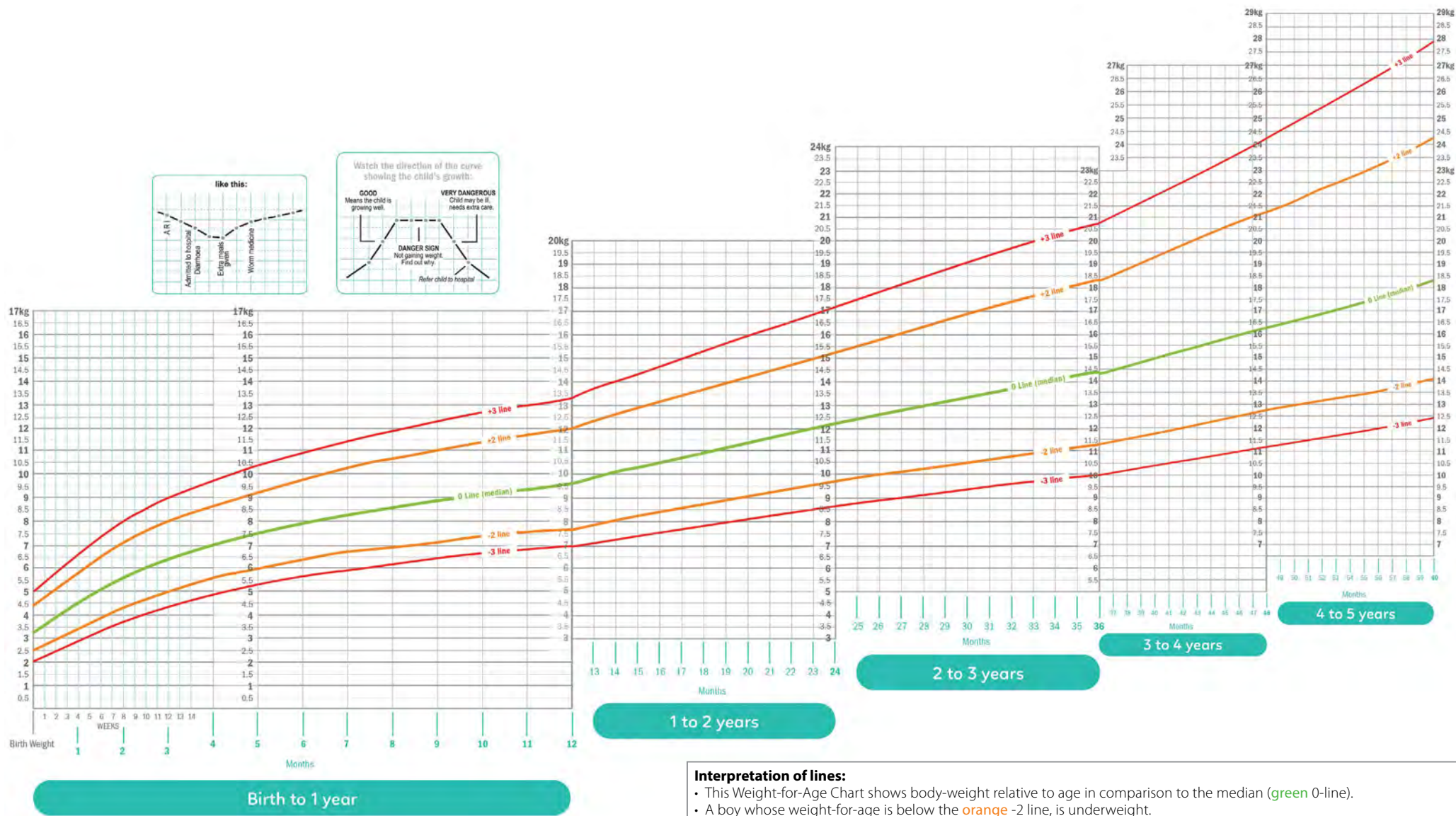
Girl's weight-for-age chart



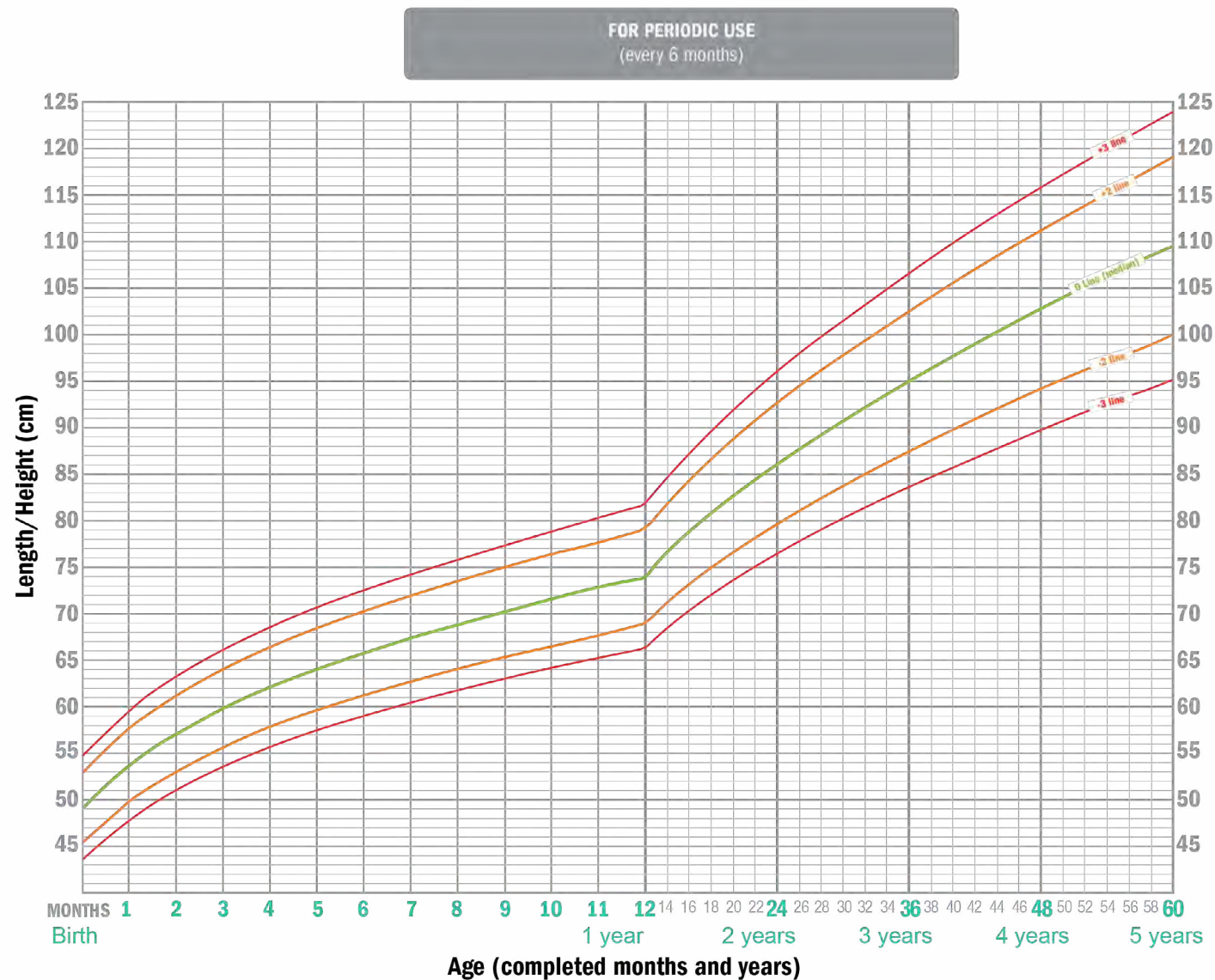
Interpretation of lines:

- This Weight-for-Age Chart shows body-weight relative to age in comparison to the median (green 0-line).
- A girl whose weight-for-age is below the orange (-2 line), is underweight.
- A girl whose weight-for-age is below the red (-3 line), is severely underweight.
- If her line crosses a z-score line and the shift is away from the median, this may indicate a problem or risk of a problem.
- If her line shifts away from her birth trend line, this may indicate a problem or a risk of a problem.

Boy's weight-for-age chart



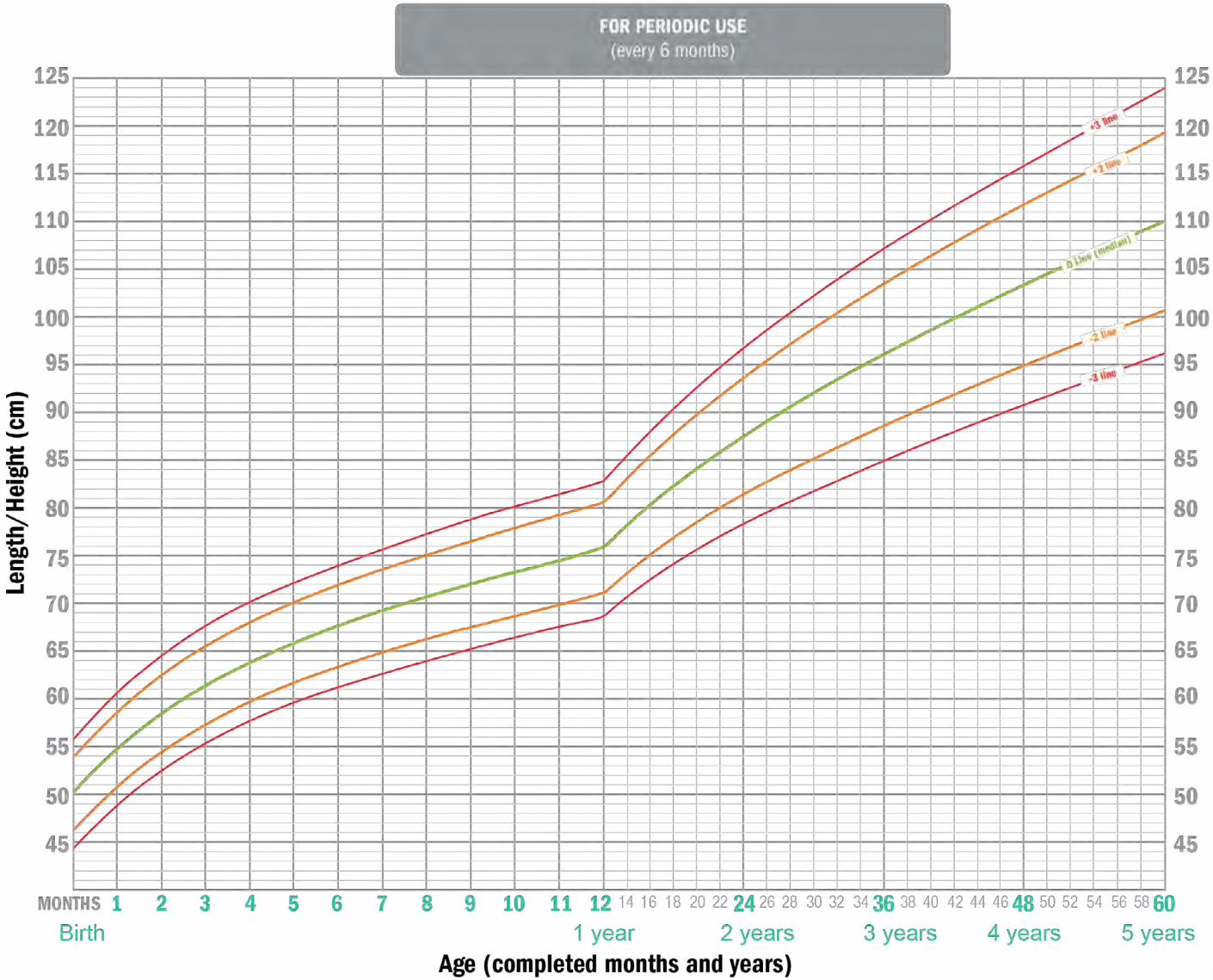
Girl's length/height-for-age chart



Interpretation of lines:

- This **length/height-for-age chart** shows height relative to age in comparison to the median **green** (0-line)
- A girl whose length/height-for-age is below the **orange** -2 line, is **stunted**
- A girl whose length/height-for-age is below the **red** -3 line, is **severely stunted**

Boy's length/height-for-age chart



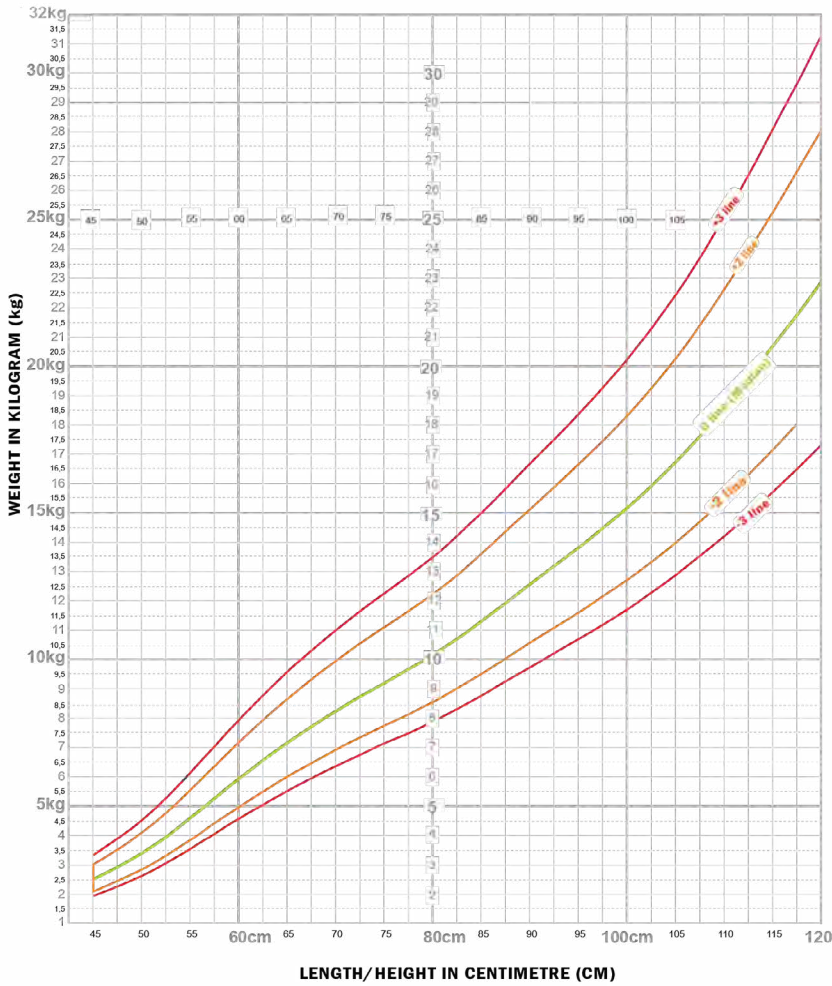
Interpretation of lines:

- This **length/height-for-age chart** shows height relative to age in comparison to the median **green** (0-line)
- A boy whose length/height-for-age is below the **orange** -2 line, is **stunted**
- A boy whose length/height-for-age is below the **red** -3 line, is **severely stunted**

Department of Health South Africa. Road to Health Booklet for Boys. 2018

Weight-for-length/height chart

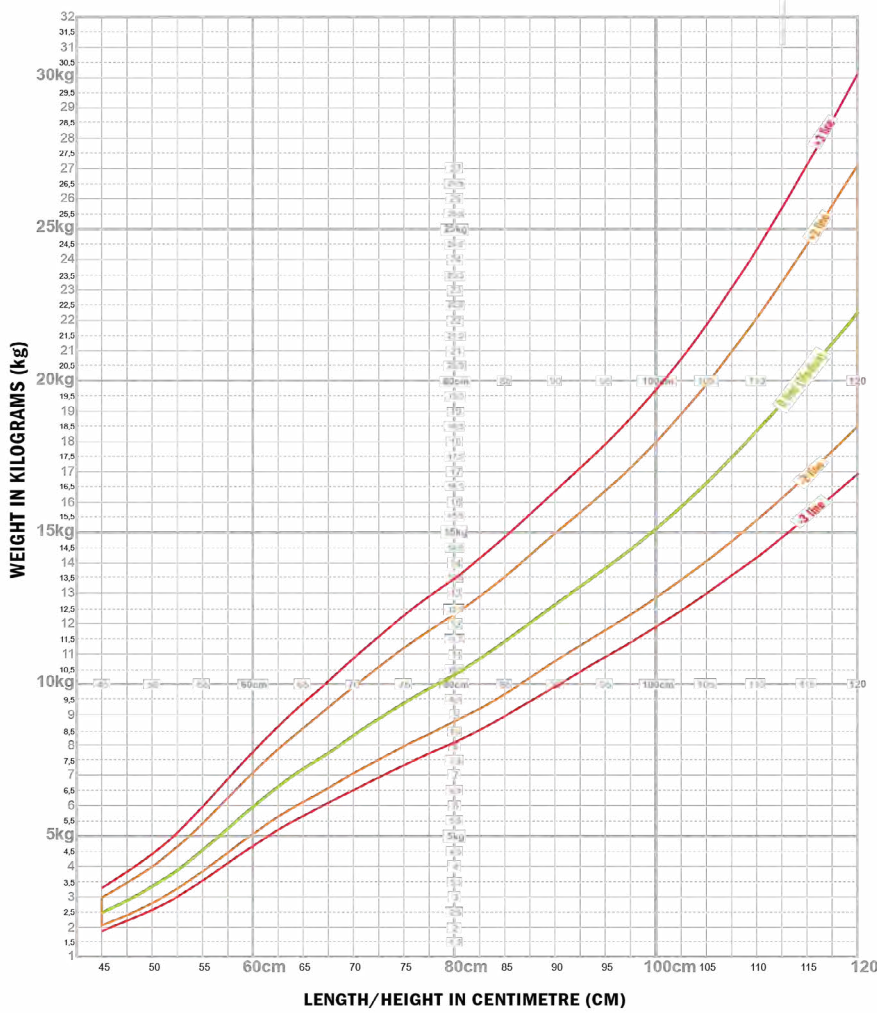
Girls



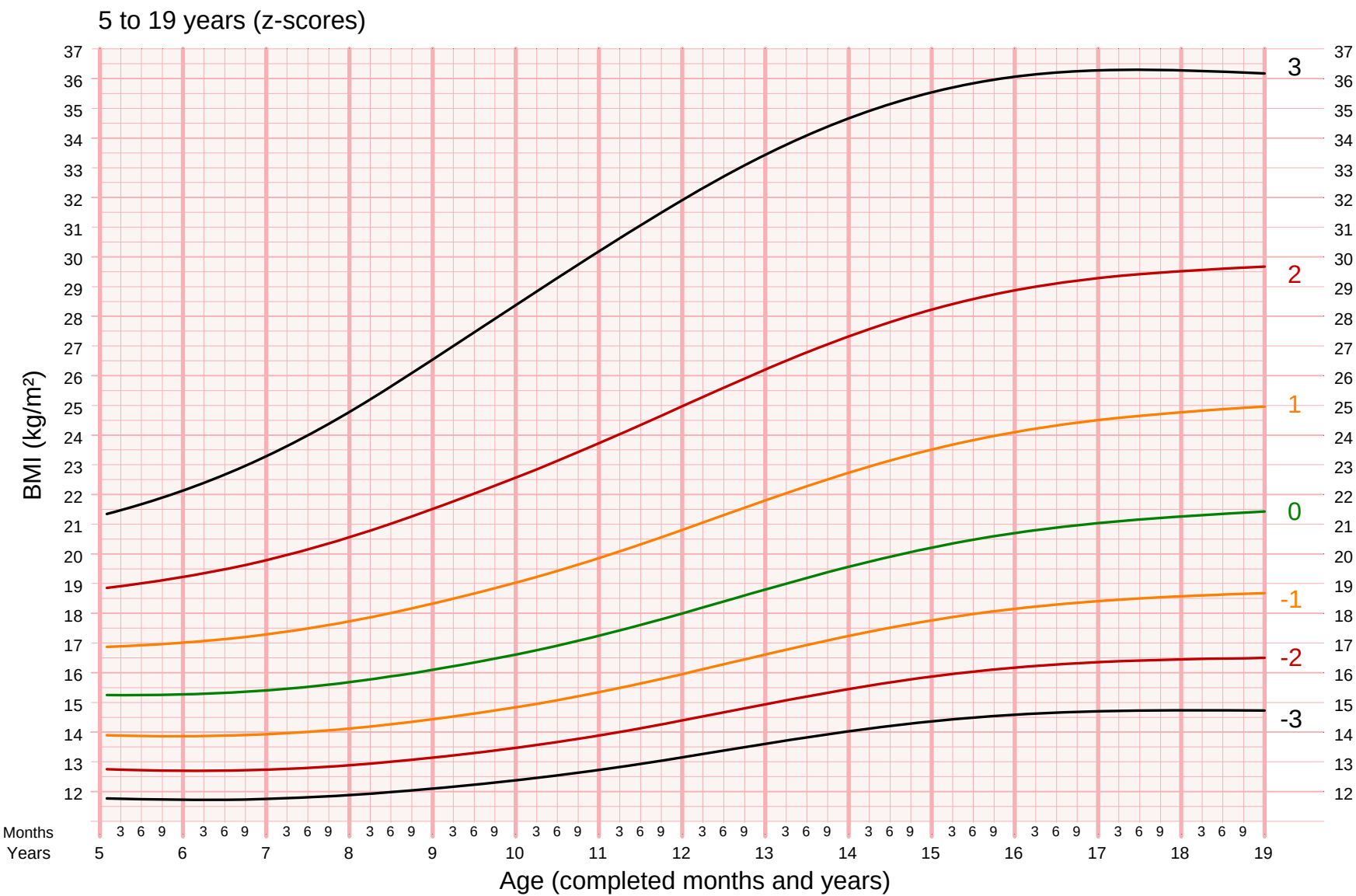
This **weight-for-height/length chart** shows body-weight relative to length/height in comparison to the median (the 0 z-score line).

- A child whose weight-for-length/height is above the +3 (red) line, is **obese**.
- A child whose weight-for-length/height is above the +2 (orange) line, is **overweight**.
- A child whose weight-for-length/height is below the -2 (orange) line, is **wasted**.
- A child whose weight-for-length/weight is below the -3 (red) line, is **severely wasted (SEVERE ACUTE MALNUTRITION)**. Refer for urgent specialised care.

Boys

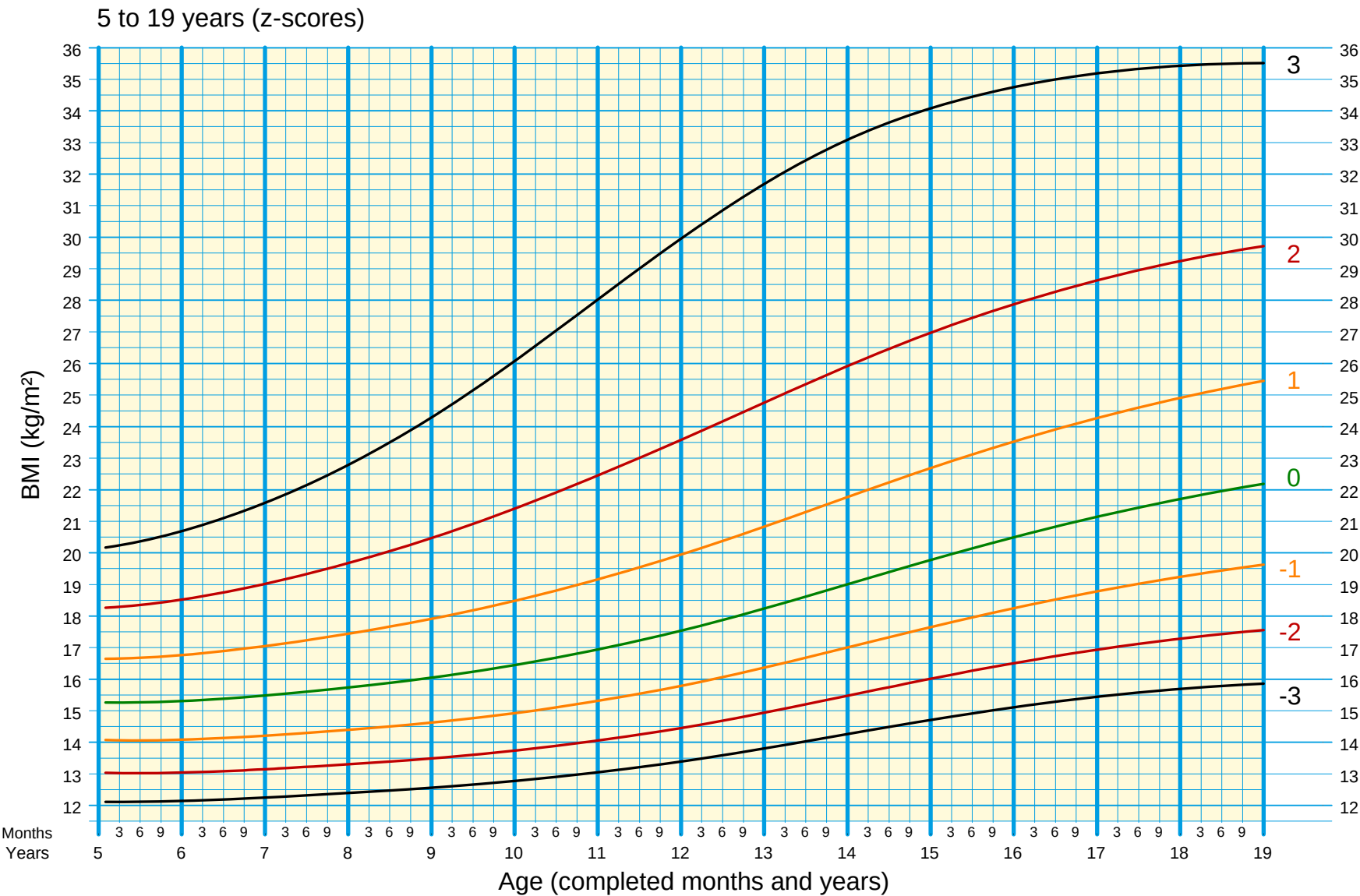


Girl's BMI chart



World Health Organization. BMI-for-age Girls 5-19 years (z-scores). 2007

Boy's BMI chart



World Health Organization. BMI-for-age Boys 5-19 years (z-scores). 2007

Help baby breathe at birth

Give urgent attention to the baby at birth

"The Golden Minute"

Do the following in the first 30 seconds:

- If meconium, avoid routine suctioning.
- Keep warm: if $\leq 1500\text{g}$ or < 30 weeks, avoid drying, wrap body (not head) in clear plastic wrap to prevent heat loss. If $> 1500\text{g}$ and ≥ 30 weeks, dry thoroughly and place baby on mother's chest skin-to-skin.
- Is baby crying?

Crying/breathing well

Not crying

- Clear airway, stimulate and keep warm.
- Then assess breathing:

Breathing well/crying

Check Apgar score and give routine care below

Determine the Apgar score at 1 and 5 minutes and record in RthB:

Score each criteria and add together to give 1 and 5 minute score.

	0	1	2
Crying?	No response	Grimace	Vigorous cry
Breathing?	Absent	Slow or irregular	Good cry
Colour?	Blue or pale	Pink but blue feet	Pink all over
Heart rate?	Absent	$< 100/\text{minute}$	$\geq 100/\text{minute}$
Is baby active?	Limp	Slight flexion	Active, moves

- Place baby on mother's chest skin-to-skin and cover. Keep warm and check breathing regularly. Clamp and cut the umbilical cord in 1-3 minutes. Start breastfeeding immediately.
- Give **vitamin K** 1mg IM and apply **chloramphenicol 1%** eye ointment into both eyes.
- Then fully assess baby in first few hours after birth $\rightarrow 9$.

Not breathing well/gasping/respiratory rate < 30

- Immediately give 1 breath every 2 seconds with bag-mask and room air (no oxygen). If preterm, connect bag mask to **oxygen** at no more than 2L/minute. Ensure chest rise with each breath.
- Clamp and cut umbilical cord.
- Assess breathing every 30-60 seconds:

Breathing well

Not breathing well/gasping/respiratory rate < 30

- Improve bag-mask ventilation: check neck slightly extended and that mask seal is adequate, suction if secretions and bag with mask more firmly.
- Check pulse rate:

Pulse rate < 100

Continue bag-mask ventilation (with oxygen as needed).

Pulse rate < 60

Start chest compressions; 3 compressions : 1 breath.

Assess breathing and heart rate every 30-60 seconds.

Pulse rate ≥ 100

Continue bag-mask ventilation until breathing well.

Pulse rate 60-100

Continue bag-mask ventilation with oxygen until referral.

Pulse rate < 60

Continue chest compressions and bag-mask ventilation.

Refer urgently to MOU/closest hospital.

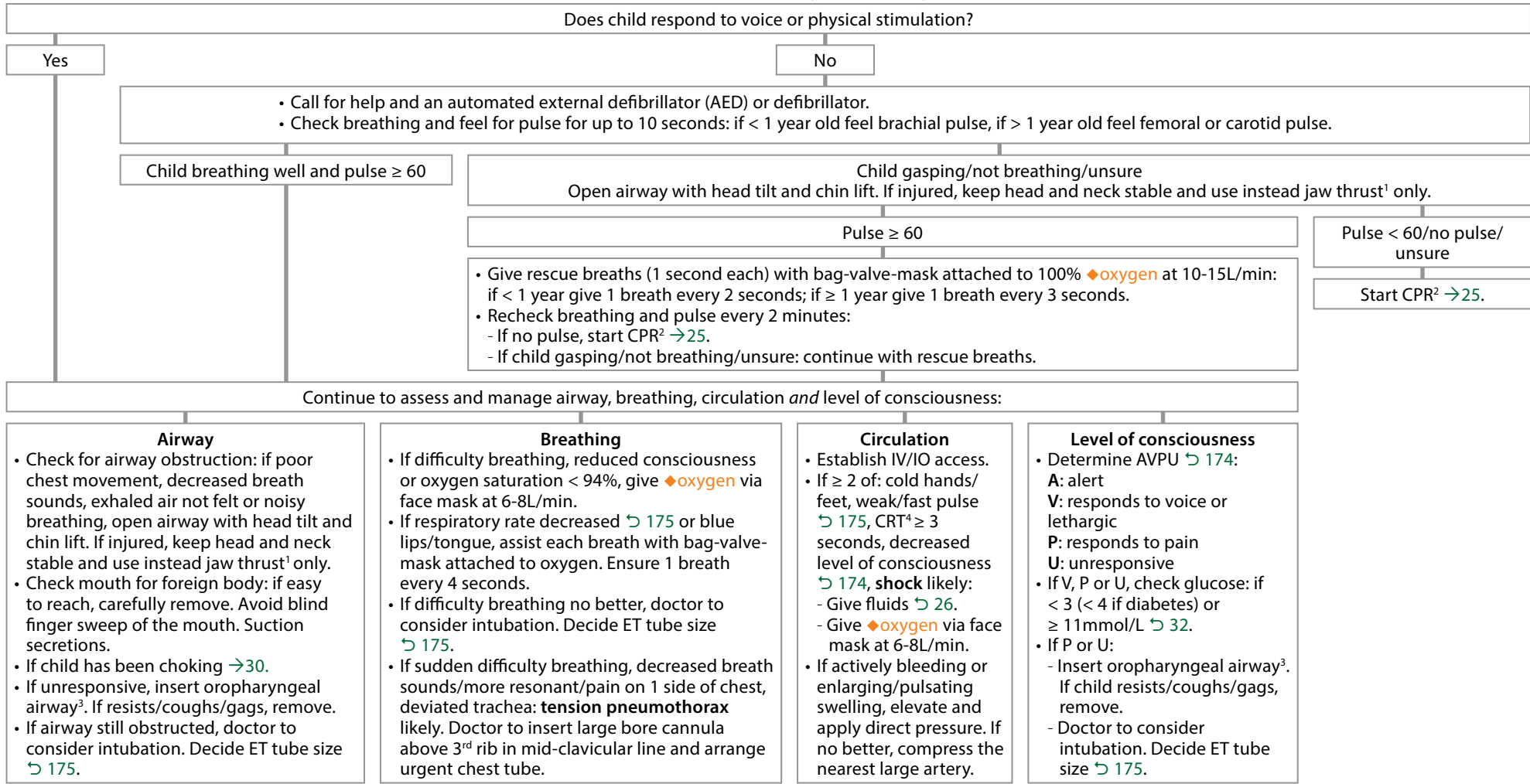
- Record Apgar score (see adjacent). Keep baby warm, do not overheat. Check glucose $\rightarrow 9$.
- Give **vitamin K** 1mg IM and apply **chloramphenicol 1%** eye ointment into both eyes.
- If doing chest compressions, doctor to establish IV/IO access and give **adrenaline** 0.1mL/kg (1:10 000 solution)¹ IV/IO. Repeat every 4 minutes until transfer.
- If transport delayed, only consider stopping resuscitation in consultation with a specialist if: no heart beat or breathing after 10 minutes; or no breathing after 20 minutes; or only gasping after 30 minutes.

- Record Apgar score and give routine care adjacent.
- Refer urgently to MOU/closest hospital if:
 - > 5 minutes to breathe well
 - Apgar score ≤ 7 at 5 minutes

¹To make adrenaline 1:10 000 solution, draw up 1 ampoule (1mg/mL, 1:1000) **adrenaline** and then 9mL **sodium chloride 0.9%** into a 10mL syringe.

The emergency child

Give urgent attention to the emergency child (if baby at birth →23)



Manage further and refer urgently:

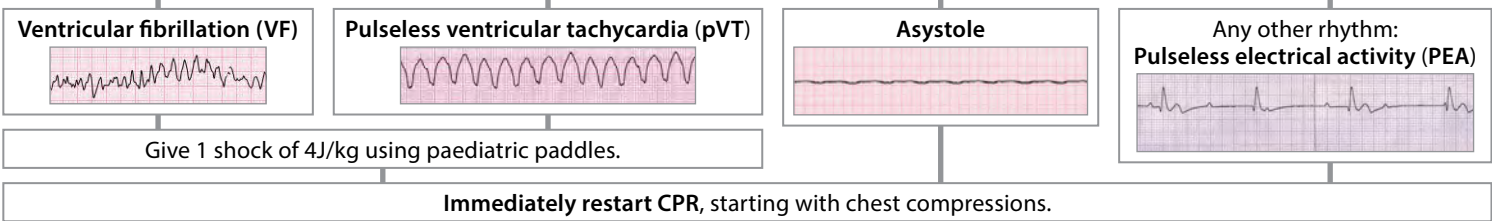
- While awaiting transport, continue to assess and manage airway, breathing, circulation and level of consciousness.
- If injured →33, if fitting →28, if just had fit →29, if decreased consciousness →31, if overdose/poisoning →36, if burns →38, if bites/stings →40, if fever →43, if rash →75, if anaphylaxis →37.
- If other symptom, manage as on symptom page.

¹Lift chin forward with fingers under bony tips of jaw. ²If known with life-limiting illness, follow advanced care plan and consider whether or not to proceed. ³For correct oropharyngeal size, place against cheek with one tip at corner of mouth and check other tip reaches front of earlobe. ⁴Capillary refill time (CRT): hold hand/foot higher than level of heart. Press soft pad of finger/toe until it turns pale, then release pressure and note time taken for colour to return.

Cardio-pulmonary resuscitation (CPR) of the child

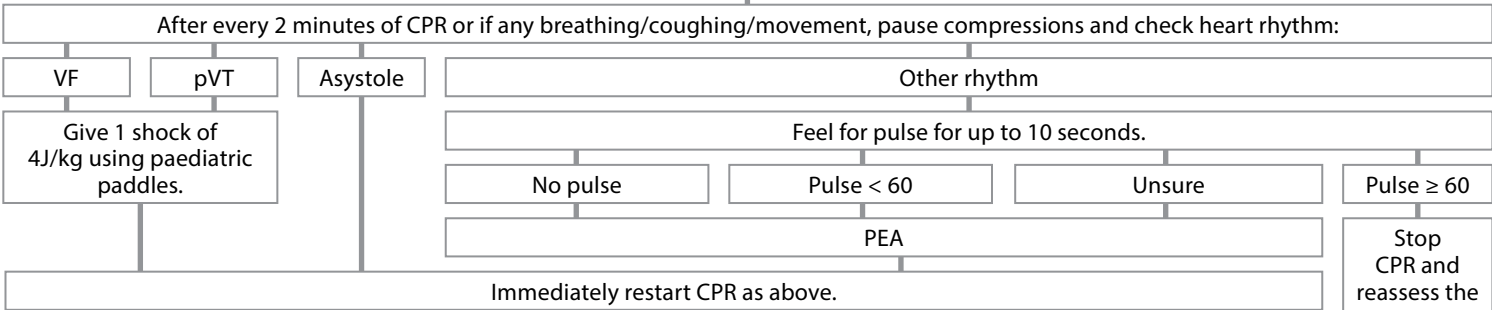
Start chest compressions in the unresponsive child with no pulse or pulse < 60:

- Record start time.
- Give continuous cycles of 15 chest compressions and 2 breaths with bag-valve-mask attached to 100% **oxygen** at of 10-15L/min. If you are alone, give cycles of 30 compressions and 2 breaths. If choking suspected, check mouth for foreign body after every cycle of compressions, before giving next 2 breaths.
- Attach monitor/defibrillator and pause compressions to check heart rhythm:



Dose IV/IO adrenaline (1:10 000 solution) ¹ according to weight. If weight unknown, use age.		
Weight	Volume	Age
2.5-7kg	0.5mL	0-6 months
7-11kg	1mL	6-18 months
11-17.5kg	1.5mL	18 months- 5 years
17.5-25kg	2mL	5-7 years
25-35kg	3mL	7-11 years
35-55kg	5mL	11-15 years

- While giving continuous cycles of CPR:
- Establish IV access. If unsuccessful after 1 minute, establish IO access.
 - Give **adrenaline** 0.01mg/kg IV/IO (see dose table), followed by 5mL **NaCl 0.9%**. Repeat every 4 minutes (every 2 cycles).
 - Doctor to consider intubation: decide ET tube size **175**. If intubated, give 1 breath every 6 seconds and continuous chest compressions.
 - Give **NaCl 0.9%** 10mL/kg IV/IO. Repeat if needed. If unsure, discuss with doctor.
 - Look for and manage possible cause:
 - If temperature < 35°C, **hypothermia** likely **31**. If glucose < 3mmol/L (< 4 if diabetes) **32**.
 - If overdose or poisoning **36**.
 - If decreased breath sounds/more resonant on 1 side of chest or deviated trachea, **tension pneumothorax** likely: doctor to insert large bore cannula above 3rd rib in mid-clavicular line.

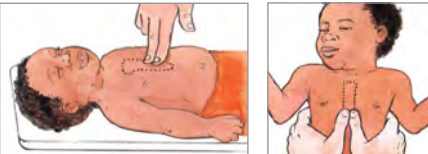


- Decide when to stop CPR:
- If no pulse after 30 minutes of continuous CPR:
 - If ongoing VF/pVT, temperature ≤ 35.5°C or overdose/poisoning, continue CPR and discuss/transfer urgently.
 - If none of above and fixed dilated pupils, consider stopping CPR in consultation with doctor. If CPR stopped pronounce dead and arrange bereavement counselling for family/carer.

Ensure correct chest compression technique

- Push hard (≥ 1/3 of depth of chest) and fast (100-120/minute).
- Compress over lower sternum (not on xiphisternum). Allow full chest recoil.
- Minimise interruptions in compressions.
- Rotate compressor every 2 minutes.

Baby < 1 year old
Use 2 fingers or 2 thumbs encircling chest.



Child ≥ 1 year old
Use heel of hand/s.



¹To make adrenaline 1:10 000 solution, draw up 1 ampoule (1mg/mL, 1:1000) **adrenaline** and then 9mL **NaCl 0.9%** into a 10mL syringe.

Assess and manage child's fluid needs

- First, weigh child or estimate weight¹ → 175.
- Then assess for signs of shock: are there ≥ 2 of: cold hands/feet, weak/fast pulse → 175, CRT² ≥ 3 seconds, decreased level of consciousness → 174?

Yes

No

Assess for dehydration
→ 27.

Shock likely

Manage and refer urgently:

- Give ♦oxygen 2L/min via nasal prongs.
- Check fingerprick glucose: if ≤ 28 days old and glucose < 2.6mmol/L → 9. If > 28 days old and glucose < 3mmol/L (< 4 if diabetes) → 32.
- Establish IV access: try 3 times for up to 90 seconds each. If unsuccessful, insert IO or external jugular line.
 - Give 1st bolus ♦NaCl 0.9% 10mL/kg IV/IO over 20 minutes.
 - If IV/IO access unsuccessful, discuss with doctor to give ♦ORS 10mL/kg over 20 minutes via NGT. If NGT not possible, give ♦ORS 10mL/kg over 20 minutes orally.
- Is the child injured?

Yes

- If actively bleeding or enlarging/pulsating swelling, apply direct pressure and call doctor. If still bleeding, apply tourniquet above injury if possible.
- Assess response to first fluid bolus after 20 minutes: feel hands, check pulse and CRT².

Hands still cold, pulse weak/not felt, CRT ≥ 3 seconds

Poor response

Give 2nd bolus ♦NaCl 0.9% 10mL/kg IV/IO. Consider giving ■O-negative emergency blood 10mL/kg IV, if available.

Hands warmer, CRT < 3 seconds, pulse stronger/slower

Good response
Avoid giving further IV/IO fluid.

- If poor response/hydration status not improving or unsure, discuss further fluid needs with referral centre.
- While awaiting transfer:
 - If shock not due to trauma or watery diarrhoea, give single dose ♦ceftriaxone⁶ 100mg/kg IV/IM → 159:12. Avoid injecting > 1g IM at one injection site.
 - Reassess fluid status hourly and keep warm: place child skin-to-skin with mother and cover with blanket.
 - Check fingerprick glucose every 15 minute: if ≤ 28 days old and glucose < 2.6mmol/L → 9. If > 28 days old and glucose < 3mmol/L (< 4 if diabetes) or ≥ 11mmol/L → 32.

No

After 20 minutes, assess response: feel hands, check pulse and CRT².

Hands still cold, pulse weak/not felt, CRT ≥ 3 seconds

Still shocked

- Give 2nd bolus ♦NaCl 0.9% 10mL/kg IV/IO.
- Discuss need for 3rd bolus with doctor.

Hands warmer, CRT < 3 seconds, pulse stronger/slower

No longer shocked

Are there any of: < 2 months old, ≥ 5 years old, SAM³ difficulty breathing, suspected meningitis⁴?

No

Give ♦NaCl 0.9% with dextrose 5% 10mL/kg/hour IV/IO⁵.

Yes

Give ♦NaCl 0.9% with dextrose 5% 4mL/kg/hour IV/IO⁵.

CRT - capillary refill time
IO - intra-osseous
IV - intravenous
NaCl - sodium chloride
NGT - nasogastric tube
ORS - oral rehydration solution
SAM - severe acute malnutrition

¹Use resuscitation tape (PAWPER or Broselow tape) to estimate weight if available. ²CRT: hold hand/foot higher than level of heart. Press soft pad of finger/toe until it turns pale, then release pressure and note time taken for colour to return. ³SAM: weight-for-length/height below -3 line or BMI below -3 line or MUAC < 11.5cm or any malnutrition with oedema. ⁴Temperature ≥ 38°C, headache, decreased level of consciousness → 174, neck stiffness: meningitis likely. ⁵Ideally use 200mL bag and give via buretrol/dial-a-flow device. ⁶Avoid giving Ringer's lactate together with IV/IM ceftriaxone. If severe penicillin allergy (previous anaphylaxis, urticaria or angioedema), discuss with doctor.

Assess for signs of dehydration: are there ≥ 2 of: lethargy, sunken eyes, slow skin pinch¹ ≥ 2 seconds, drinking poorly?

Yes

No

Severe dehydration likely

Manage and refer urgently

- Check fingerprick glucose: if ≤ 28 days old and glucose < 2.6 → 9. If > 28 days old and glucose < 3 (< 4 if diabetes) or ≥ 11mmol/L → 32.
- If breastfeeding and drinking well, continue feeds.
- If < 28 days old or SAM², give single dose ♦ceftriaxone³ 100mg/kg IV/IM → 159:12. Avoid injecting ≥ 1g IM at one site.

If > 2 of: restless/irritable, sunken eyes, slow skin pinch¹ < 2 seconds, thirsty/drinks eagerly

Yes

No

Some dehydration likely

- Check fingerprick glucose: if ≤ 28 days old and glucose < 2.6 → 9. If > 28 days old and glucose < 3 (< 4 if diabetes) or ≥ 11mmol/L → 32.
- If breastfeeding and drinking well, continue feeds.

Dehydration unlikely

- Manage symptoms on symptom page:
 - If diarrhoea → 63.
 - If vomiting → 62.
- Discharge child into reliable carer's care.
- Encourage caregiver to continue feeds, especially if breastfeeding.
- Show caregiver how to make ORS at home.
- Advise to give ♦ORS 10mL/kg after each loose stool:
 - If ≤ 2 years old, give 50-100mL
 - If > 2 years old, give 100-200mL.

Give fluids: Are there any of: < 2 months old, ≥ 5 years old, SAM², difficulty breathing, suspected meningitis⁴?

No

Yes

- Give ♦ORS 20mL/kg hourly for 4 hours. If child vomits, wait 10 minutes, then give ORS more slowly.
- If drinking poorly/not tolerating fluids, give ♦ORS 20mL/kg hourly for 4 hours via NGT.
- If NGT not possible, give ♦NaCl 0.9% 20mL/kg/hour IV/IO.

- Give ♦ORS 10mL/kg hourly for 4 hours. If child vomits, wait 10 minutes, then give ORS more slowly.
- If drinking poorly/not tolerating fluids, give ♦ORS 10mL/kg hourly for 4 hours via NGT.
- If NGT not possible, give ♦NaCl 0.9% 10mL/kg/hour IV/IO.

Assess hydration status after 1 hour.

Are there ≥ 2 of: cold hands/feet, weak/fast pulse → 175, CRT⁵ ≥ 3 seconds, decreased level of consciousness → 174?

Yes

No

Shock likely
→26

- Continue hourly fluids as above. If on oral ♦ORS and drinking poorly/not tolerating orally give via NGT instead. If NGT not possible, give ♦NaCl 0.9% IV/IO or ♦NaCl 0.9% with dextrose 5% if awaiting transfer.
- Encourage caregiver to report to healthcare worker if poor oral intake or concern about child's condition.
- Reassess hydration status after 3 hours. Is child drinking well with no vomiting and no signs of dehydration?

No

Yes

Ongoing dehydration: refer

While awaiting transfer:

- If poor response/hydration status not improving or unsure, discuss further fluid needs with referral centre.
- If receiving IV/IO fluids give ♦NaCl 0.9% with dextrose 5% instead.
- Keep warm: place child skin-to-skin with mother and cover with blanket.
- If initial hypoglycaemia, recheck fingerprick glucose every 15 minutes. If no initial hypoglycaemia, recheck every hour. If glucose < 3mmol/L → 32.

No dehydration

Child drinking well, *not* vomiting and has *no* signs of dehydration after 4 hours.

- Referral not needed.
- Treat as **dehydration unlikely** in adjacent column.

CRT - capillary refill time
IO - intra-osseous
IV - intravenous
NaCl - sodium chloride
NGT - nasogastric tube
ORS - oral rehydration solution
SAM - severe acute malnutrition

¹Pinch skin on abdomen between 2 fingers. Release. Skin usually snaps rapidly back to its normal position. A slow skin pinch takes longer. ²SAM: weight-for-length/height below -3 line or BMI below -3 line or MUAC < 11.5cm or any malnutrition with oedema. ³Avoid giving Ringer's lactate together with IV/IM ceftriaxone. If severe penicillin allergy (previous anaphylaxis, urticaria or angioedema), discuss with doctor. ⁴Temperature ≥ 38°C, headache, decreased level of consciousness → 174, neck stiffness: meningitis likely. ⁵CRT: hold hand/foot higher than level of heart. Press soft pad of finger/toe until it turns pale, then release pressure and note time taken for colour to return.

Seizures/fits

Give urgent attention to the child who is unconscious and fitting:

- Open airway: clear mouth, chin lift/jaw thrust, suction secretions. Place in recovery position¹. Do not place anything in mouth.
- Give ♦oxygen via face mask or nasal cannula to maintain oxygen saturation $\geq 95\%$.
- Check fingerprick glucose: If ≤ 28 days and < 2.6 ↗ 9. If > 28 days and < 3 (< 4 if diabetes) or unable to measure, manage as hypoglycaemia ↗ 32.
- Establish IV access. Do not delay treatment if first attempt unsuccessful.

If fit has lasted ≥ 5 minutes, give medication according to age to stop the fit:

- If < 28 days old: give ■phenobarbital² 20mg/kg IV over 10 minutes and refer urgently. If child still fitting, discuss with specialist.
- If ≥ 28 days old: give one of the following:
 - If IV access available: give ■midazolam² 0.25mg/kg IV over 1 minute ↗ 163:35 or ■diazepam² 0.25mg/kg IV ↗ 160:20 over 1 minute.
 - If IV access not available: give ■midazolam² 0.1mg/kg IM or 0.5mg/kg buccal³ (avoid buccal midazolam if < 6 months old) ↗ 162:34 or ■diazepam² 0.5mg/kg rectal⁴ ↗ 160:19.
- Monitor breathing: if respiratory rate < 20 , call doctor. If gasping or breathing stops, ventilate with bag-valve mask (1 breath every 3-5 seconds) ↗ 24.
- Expect a response within 5 minutes:

Child still fitting

Child stops fitting

Decide if child needs urgent referral and manage →29.

- If IV access available: give ■midazolam² 0.25mg/kg IV over 1 minute ↗ 163:35 or ■diazepam² 0.25mg/kg IV ↗ 160:20 over 1 minute.
- If IV access not available: give one of the following:
 - ■midazolam² 0.1mg/kg IM or 0.5mg/kg buccal³ (avoid buccal midazolam if < 6 months old) ↗ 162:34 or
 - If not already given; ■diazepam² 0.5mg/kg rectal⁴ ↗ 160:19.
- Monitor breathing: if respiratory rate < 20 , call doctor. If gasping or breathing stops, ventilate with bag-valve mask (1 breath every 3-5 seconds) ↗ 24.
- Expect a response within 5 minutes:

Child still fitting or repeated fits without regaining consciousness

Child stops fitting

Manage and refer urgently:

- If available, give single dose of ■phenobarbital² 20mg/kg IM ↗ 164:38 slow injection.
- If IM phenobarbital *not* available: give single dose of crushed ■phenobarbital² tablets, 20mg/kg via nasogastric tube ↗ 164:39.
- Monitor breathing: if respiratory rate < 20 , call doctor. If gasping or breathing stops, ventilate with bag-valve mask (1 breath every 3-5 seconds) ↗ 24.
- Expect a response within 5 minutes:

Child still fitting

Child stops fitting

- Discuss further management with referral hospital.
- Doctor to assess need for intubation ↗ 24.
- Refer urgently.

- Keep child in left lateral position with oxygen.
- Establish IV if not done already.
- Refer urgently.

¹Recovery position: turn onto left side, place left hand under cheek with neck slightly extended and bend the right leg to stabilise position. ²If no doctor available, nurse to get telephonic prescription. ³Buccal: use 5mL syringe to draw up correct dose, remove needle and give midazolam between the cheek and gum. ⁴Rectal: use 2mL syringe to draw up correct dose, remove needle, lubricate whole syringe barrel and insert into rectum, inject contents, remove syringe and hold buttocks together.

Approach to the child who is no longer fitting:

Confirm that child indeed had a fit: jerking movements, loss of consciousness, eyes open during fit, incontinence, post-fit drowsiness and confusion. If not, refer to specialist same week¹.

Give urgent attention to the child with recent fit and any of:

- Child < 2 years old
- Temperature $\geq 38^{\circ}\text{C}$ and < 18 months old or ≥ 5 years old
- Fit lasted ≥ 5 minutes
- Does not respond to voice > 1 hour after fit
- More than one fit in 24 hours
- ≥ 2 of: temperature $\geq 38^{\circ}\text{C}$, headache, decreased level of consciousness $\rightarrow$ 174, neck stiffness: meningitis likely
- Fit occurs only on one side of body
- Weakness of arm/leg/face, even if resolved
- Restless/irritable/lethargy/altered level of consciousness, sunken eyes, slow skin pinch² or drinking poorly/thirsty: dehydration likely
- Ingestion of medication/potentially harmful substance $\rightarrow$ 36
- Known with long term health condition
- Travel to malaria area in past 3 months: malaria likely
- CLHIV
- Head injury within past week
- Significant TB exposure³

Manage and refer urgently:

- Establish IV access.
- Check fingerprick glucose: If ≤ 28 days and < 2.6 $\rightarrow$ 9. If > 28 days and < 3 (< 4 if diabetes) $\rightarrow$ 32.
- If meningitis likely, give $\blacklozenge$ ceftriaxone⁴ 100mg/kg IV/IM $\rightarrow$ 159:12. Avoid injecting $\geq 1\text{g}$ IM at one injection site.
- If temperature $\geq 38^{\circ}\text{C}$, give $\blacklozenge$ paracetamol 15mg/kg orally $\rightarrow$ 163:37 or rectally if available.
- If dehydration, assess and manage child's fluid needs $\rightarrow$ 26.
- If in malaria area in past 3 months, malaria likely. Discuss urgently with referral centre.

Approach to the child with recent fit not needing urgent attention

- Check fingerprick glucose: If ≤ 28 days and < 2.6 $\rightarrow$ 9. If > 28 days and < 3 (< 4 if diabetes) $\rightarrow$ 32.
- Is temperature > 38.5°C?

Yes

If child ≥ 18 months - 5 years old and: fit is generalised, lasted < 15 minutes (typically 1-2 minutes) with no central nervous system infection.

Simple febrile seizure/fit likely

- Look for source of fever $\rightarrow$ 43. If none found, refer.
- Refer if: first episode and does not regain full consciousness immediately after seizure.
- Discharge if alert within 1 hour of fit.
- Reassure and advise carer:
 - Febrile seizures/fits common from 6 months - 5 years old.
 - There is a 30% chance that the child will fit with a fever in the next 2 years.
 - There is a very slight risk of epilepsy later but this will not effect intellect, academic performance or behaviour in future.
 - Give paracetamol for fever to relieve discomfort but explain that this will not prevent further fits.
- If > 3 febrile seizures/fits in 6 months, refer to paediatrician. Avoid starting antiseizure medication.

No

- If child known with epilepsy, give routine epilepsy care $\rightarrow$ 130.
- Is there history of birth trauma, head injury, meningitis, family history of epilepsy⁵?

Yes

Refer/discuss with paediatrician same day.

No

Has child had ≥ 2 fits in the last year on 2 different days?

Yes

No

- Doctor to review
- If not talking/understanding problems, refer.
 - If otherwise well, review in 3 months for further fits, new symptoms or delayed milestones.

Advise the carer on what to do if child fits at home

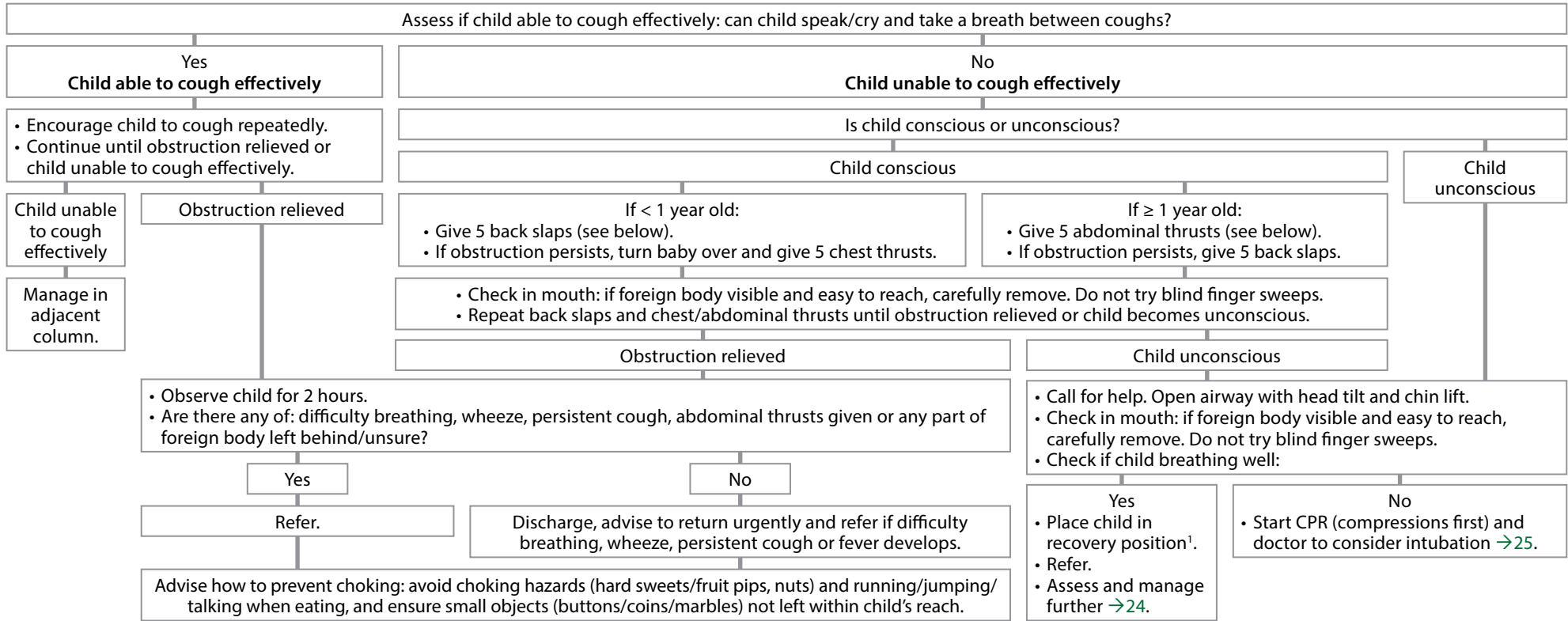
- Place child in safe place (on floor or bed) away from objects that may cause injury.
- Lie child in left lateral position. Do not place anything in mouth. Wipe away secretions.
- Time fit: get help if fit continues for more than 3 minutes or child does not wake up properly between fits.
- Encourage carer/s to have a plan ready if medical attention needed urgently: know where nearest clinic is, have reliable transport plan.



¹Encourage carer to take a video of event to show specialist. ²Skin pinch: pinch skin on abdomen between 2 fingers. Release. Skin normally snaps rapidly back to its normal position. A slow skin pinch takes longer. ³Significant TB exposure: any child who in the last 12 months shared an enclosed space (at school/cresche/home/similar), for ≥ 1 night or for frequent/extended daytime periods, with an adult/adolescent with pulmonary TB ("index patient"), during the 3 months before index client started TB treatment. ⁴Avoid giving Ringer's lactate together with IV/IM ceftriaxone. If severe penicillin allergy (previous anaphylaxis, urticaria or angioedema), discuss with doctor. ⁵Family history: a parent or sibling with childhood onset epilepsy.

Choking

Give urgent attention to the child who is choking or has been choking

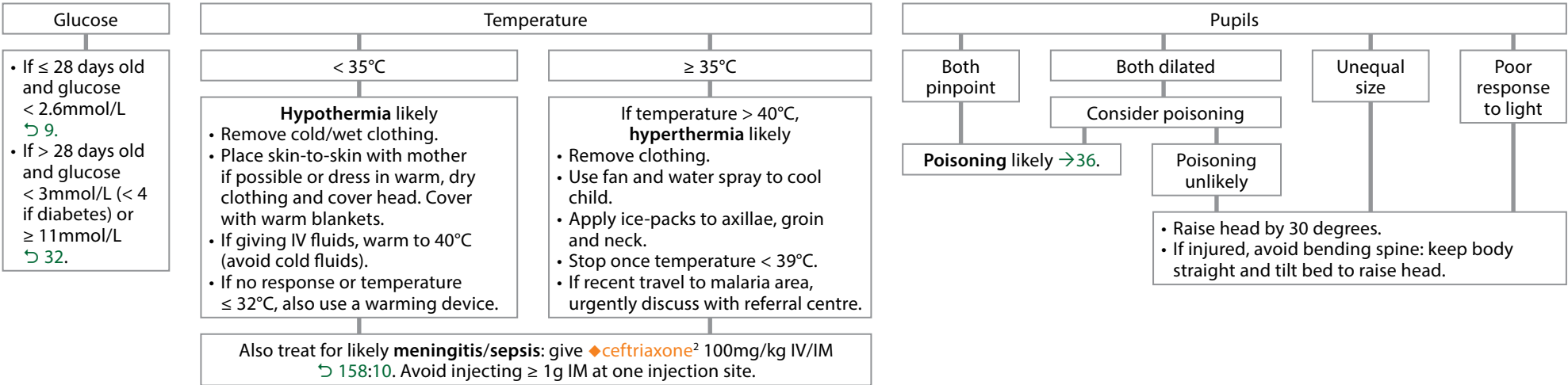


Decreased level of consciousness

- The child with a decreased level of consciousness is not alert. S/he may or may not respond to voice or pain.
- Assess level of consciousness ↗ 174.

Give urgent attention to the child with a decreased level of consciousness

- If not already done, assess and manage airway, breathing and circulation ↗ 24.
- Place child in recovery position: turn onto left side, place left hand under cheek with neck slightly extended and bend the right leg to stabilise position.
- Ask about possible causes and manage symptoms: trauma or injury ↗ 33, fitting ↗ 28, just had a fit ↗ 29, poisoning ↗ 36, burns ↗ 38.
- If sudden decreased consciousness and any of: generalised itch/rash, face/tongue itch/swelling, tight throat/hoarseness, difficulty breathing, wheeze, abdominal pain/nausea/vomiting/diarrhoea or exposure to possible allergen¹, **anaphylaxis** likely →37.
- Check fingerprick glucose, temperature and pupils:

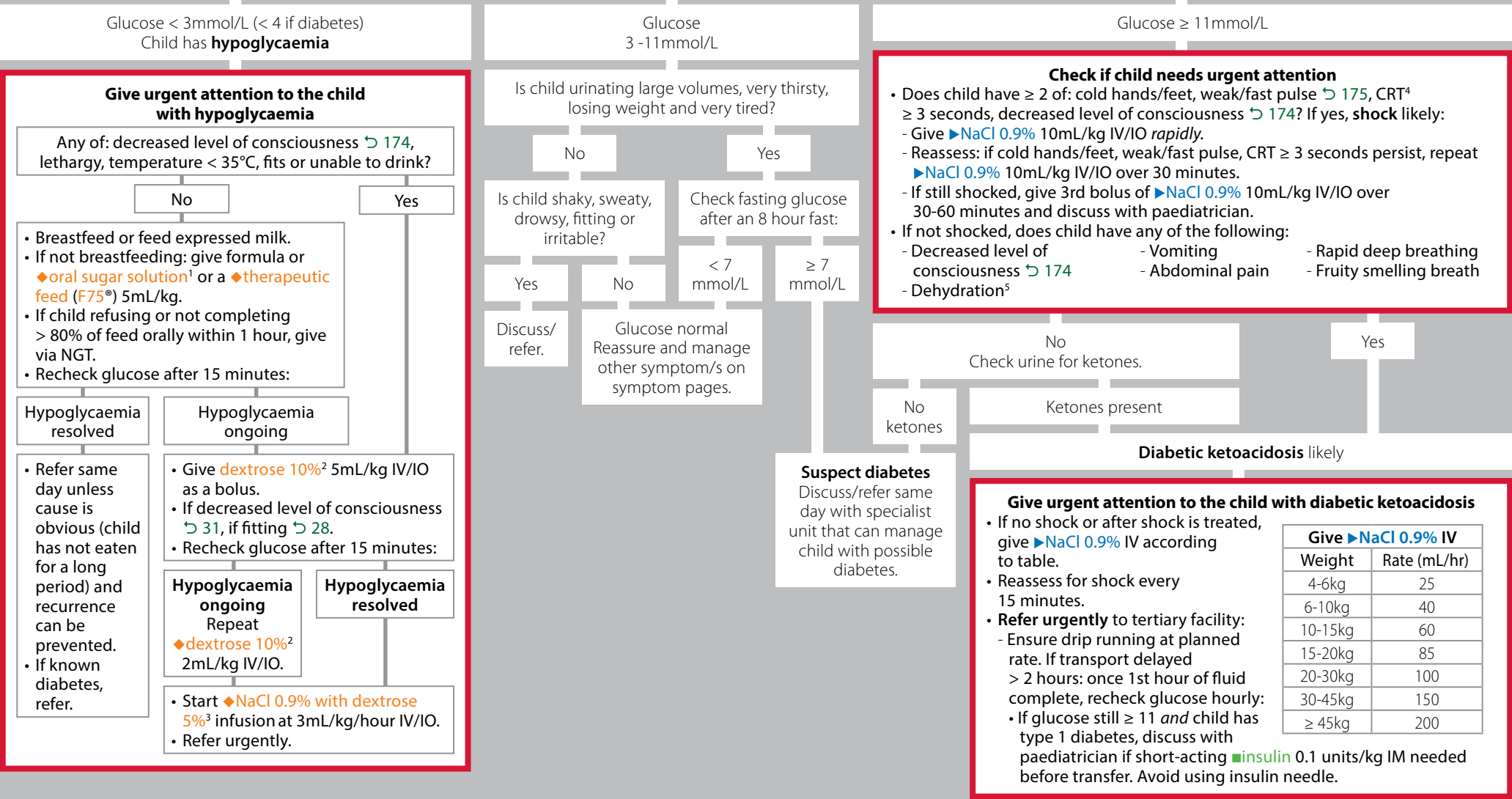


- **Refer urgently with advanced life support ambulance:**
 - If any of: history inconsistent with examination, delay in presentation, skull fracture, old and new scars on body, unusual/patterned wounds, burns, wounds around ano-genital region, consider child abuse ↗ 144.
 - If child aggressive or violent:
 - Ensure child and health workers safety.
 - Assess child with help of other staff. Use security personnel if needed.
 - If sedation needed, discuss with referral centre.
 - While awaiting transport:
 - Check pulse ↗ 175, respiratory rate ↗ 175, sats (if available), capillary refill time³ and AVPU/GCS ↗ 174 every 15 minutes.
 - If pulse/respiratory rate abnormal ↗ 175, sats drop ≤ 92%, capillary refill time ≥ 3 seconds, or AVPU/GCS worsens, reassess and manage airway, breathing and circulation ↗ 24.

¹Common allergens include medications, new food or an insect bite/sting within the last few hours. ²Avoid giving Ringer's lactate together with IV/IM ceftriaxone. If severe penicillin allergy (previous angioedema, anaphylaxis or urticaria), discuss with doctor. ³Capillary refill time: hold hand/foot higher than level of heart. Press soft pad of finger/toe until it turns pale, then release pressure and note time taken for colour to return.

Manage glucose

Interpret and manage random fingerprick glucose. If baby ≤ 28 days old →9.



¹Dissolve 3 teaspoons of sugar (15g) into 200mL water. ²If dextrose 10% unavailable: in 20mL syringe, draw up 4mL of **dextrose 50%** and 16mL of **water for injection** (syringe now contains 20mL of 10% dextrose). ³If 200mL bag not available: add 10mL **dextrose 50%** to each 100mL of **NaCl 0.9%**. ⁴Capillary refill time (CRT): hold hand/foot higher than level of heart. Press soft pad of finger/toe until it turns pale, then release pressure and note time taken for colour to return. ⁵Restless/irritable/lethargy/alter level of consciousness, sunken eyes, drinking poorly/thirsty or slow skin pinch (pinch skin on abdomen between 2 fingers. Release. Skin normally snaps rapidly back to normal position. A slow skin pinch takes longer).

The injured child

- **Airway:**
 - Decreased level of consciousness ↗ 174
 - **Breathing:**
 - Abnormal respiratory rate ↗ 175, noisy breathing/grunting/nasal flaring/chest indrawing
 - **Circulation:**
 - Bleeding despite direct pressure
 - Amputation
 - Distended abdomen

- **Disability:**
 - Weak/numb limb
 - Unexplained severe pain in a limb
 - **Type of injuries:**
 - Multiple injuries
 - Burns ↗ 38
 - Suspected fracture ↗ 34

- Poor perfusion below injury: cold, pale, numb, no pulse
 - Severe unexplained pain, muscle tightness, numbness in limb, **compartment syndrome** likely ↗ 34
 - Pulsatile/growing swelling
 - Stab/gunshot wound

- **Severe mechanism of Injury:**
 - Ejected from a vehicle
 - Fatality in same vehicle
 - Fall from more than twice child's height
- Lethargy
 - Fitting or recent seizures ↗ 28
 - History of loss of consciousness
 - Strange behaviour or memory loss since injury

- Suspected skull fracture
 - Vomiting ≥ 2 episodes
 - Blurry/double vision
 - Blood or clear fluid leaking from ear/nose

- Bruising around eyes or behind ears
 - Drug or alcohol intoxication
 - Severe headache

- Pupils unequal or respond poorly to light
 - Blood behind eardrum
 - Palpable swelling on head
- Manage and refer urgently:**

 - Assess and manage airway, breathing and circulation ↗ 24. Establish IV access and assess and manage fluid needs ↗ 26.
 - Manage pain: give ▶paracetamol 15mg/kg 6 hourly ↗ 163:37. If additional pain medication needed, discuss with doctor.
 - If actively bleeding or enlarging/pulsating swelling, apply direct pressure while calling doctor. If unsuccessful, apply tourniquet above injury.
 - If severe head injury, neck/spine tenderness, decreased level of consciousness or weak/numb limb, immobilise head with tape and sandbags/bags of IV fluid. Use spine board if child moving around.
 - If pupils unequal or respond poorly to light, keep body straight, raise head by 30 degrees (do not bend spine) and keep head in midline.
 - Identify all injuries: undress child fully and assess front and back using log-roll to turn. Then cover and keep warm.
 - While awaiting transport, monitor every 15 minutes: pulse, respiratory rate, sats (if available) and AVPU/GCS ↗ 31. If deteriorates, reassess and manage airway, breathing and circulation ↗ 24.

Approach to the injured child not needing urgent attention

- Wound**

 - Apply direct pressure to stop bleeding. If bite →40.
 - If open wound, give ♦tetanus toxoid 0.5mL IM if none in past 5 years (check RthB).
 - Remove foreign material, loose/dead skin. Irrigate with ♦sodium chloride 0.9% or if dirty, clean with ♦chlorhexidine 0.05% solution.
 - If sutures needed/wound > 4cm: suture and apply non-adherent dressing for 24 hours. Ensure correct ▶lidocaine without adrenaline dose given ↗ 162:30. Plan to remove sutures after 5 days (face), 4 days (neck), 10 days (leg) or 7 days (rest of body).
 - Do not suture if wound > 12 hours old (or > 24 hours old on head/neck):
 - Pack wound with saline-soaked gauze and
 - If dirty wound, impaired immunity¹ or wound infected, give: ■cephalexin² ↗ 159:13 or ■flucloxacillin² ↗ 161:23 for 5 days.
 - Give ♦paracetamol 15mg/kg 6 hourly as needed for up to 5 days ↗ 163:37.
 - Review in 2 days. If no infection, suture now if still needed.
 - Advise to return if skin red, warm, painful: **infection** likely.
 - Refer if infected, remaining foreign material, deep puncture/unable to close wound, cosmetic concerns or child needs sedation to suture.

Head injury

 - Advise carer to observe child carefully for 24 hours and limit activity for at least 48 hours.
 - Advise to return immediately if any of: blurred vision, vomiting, palpable swelling on head, headache despite paracetamol, difficult to wake, balance problem.
 - Give head injury form (if available).

Painful limb

 - Rest and elevate limb.
 - Apply firm, supportive bandage.
 - If limping for > 48 hours ↗ 72.
 - If knee/hip injury, refer to physiotherapist.
 - Give ♦paracetamol 15mg/kg 6 hourly as needed up to 5 days ↗ 163:37.
 - Review after 1 week: if no better, arrange x-ray and doctor review.

If any of: clear history of abuse, history inconsistent with exam, delayed presentation, skull fracture, old and new scars, burns, unusual or patterned wounds, grasp marks on arms/chest/face, bruises on trunk, different colour bruises, wounds around anus/genitals region, injuries to mouth, consider child abuse ↗ 144.

¹HIV not on ART, cancer, receiving chemotherapy, dialysis or long-term corticosteroids. ²If severe penicillin allergy (history of anaphylaxis, urticaria or angioedema), give instead ■azithromycin 10mg/kg daily for 3 days ↗ 158:10.

Fracture/s

Give urgent attention to the child with a fracture and any of:

- Poor perfusion (capillary refill¹ ≥ 3 seconds/no pulse, cold, pale, numb) below fracture
 - If ≥ 2 of: cold hands/feet, weak/fast pulse ↗ 175, CRT¹ ≥ 3 seconds, decreased level of consciousness ↗ 174: **shock** likely ↗ 26
 - Fracture of femur or pelvis
- Suspected spine fracture
 - Frequent fractures, even with minor injuries
 - If severe pain (more than expected)/numbness/tingling/muscle tightness, **compartment syndrome** likely
- Deformity
 - Open wound over fracture
 - > 1 fracture

Manage and refer urgently:

- If poor perfusion, deformity or weakness/numbness below fracture: give oral ■morphine according to age ↗ 163:36.
- If open fracture: remove foreign material, irrigate with ►sodium chloride 0.9%, cover with saline-soaked gauze. Give ►ceftriaxone² 80mg/kg IM³ ↗ 159:11.
- If **compartment syndrome** likely, remove any tight clothing/jewellery or dressings.
- Splint limb to immobilise joint above and below fracture. If pelvic fracture, tie sheet tightly around hips to immobilise.
- Manage pain: give ►paracetamol 15mg/kg 6 hourly ↗ 163:37 and if needed oral ■morphine according to age ↗ 163:36.
- Give ◆tetanus toxoid 0.5mL IM if not had in last 5 years (check RtHB).
- Keep nil per mouth.

Approach to the child with a fracture not needing urgent attention:

- Do x-ray and arrange doctor review.
- Is there a fracture seen on x-ray?

No
Manage as sprain/strain ↗35.

Yes
Is there displacement > 50% (two ends of fracture move and don't line up straight) or is there joint involvement?

- No
• Manage common fractures according to site ↗35. If uncertain discuss with orthopaedic specialist.
• If baby < 28 days old with humerus or clavicle fracture, manage below:



- If little/no movement in arm/shoulder, weakness/absent grasp: brachial plexus injury likely discuss/refer to orthopaedic specialist.
- If brachial plexus injury excluded, immobilise: lightly strap arm to chest (see picture above). Follow up in fracture clinic after 2 weeks.

¹Capillary refill: hold hand/foot higher than level of heart. Press pad of finger/toe until pale, then release and note time for colour to return. ²Avoid giving Ringer's lactate together with IV/IM ceftriaxone. If severe penicillin allergy (previous anaphylaxis, urticaria or angioedema), discuss with doctor. ³Avoid injecting ≥ 1g IM at one injection site

Assess and manage common fractures below:

Clavicle



Restrict shoulder movement with a sling for 3 weeks.



Elbow (supracondylar)
Look for displacement and black shadows around the joint.

Undisplaced (grade 1)



fat pad sign

Displaced fracture



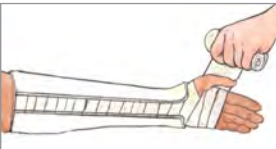
• Apply backslab from hand to below shoulder, elbow at 90°.
• If displaced fracture (fracture ends do not align), apply backslab in a comfortable position (slight flexion of elbow) from hand to above the elbow and refer same day.



Wrist (undisplaced radius)



Apply backslab from hand to below elbow, keeping fore arm in line with hand.



Ankle (fibula)



Apply backslab from foot to below knee, ankle at 90°.



Foot



Buddy strap finger/toe to the longer adjacent finger/toe. Check position of finger/toe, compared to other digits, is correct.



Finger or toe



Treat the child with a fracture:

- Give ►paracetamol 15mg/kg 6 hourly as needed up to 5 days ➔ 163:37. If no response, give ■ibuprofen 5-10mg/kg 8 hourly with food ➔ 161:28 up to 5 days. Avoid ibuprofen if asthma, heart failure or kidney disease.
- Elevate limb.
- Review next day: check pulses and sensation.
 - If problem, refer same day.
 - If wrist, ankle or foot fracture remove backslab and apply full Plaster of Paris in same position. Recheck pulses and sensation.
- Review again in 3 weeks. Remove POP/sling. If still pain or not moving limb well, refer to orthopaedic OPD.

If concerns about poor supervision at home, refer to social worker to arrange home visit. If any of : child < 1 year old, leg fracture in non-walking child, bilateral fractures, rib/skull fracture, multiple fractures in different stages of healing, any fracture out of keeping with history, delay in presentation to health facility: consider abuse ➔144.

How to apply a backslab

- Wrap limb in cotton bandage, to protect skin (cut hole for thumb if needed). Extend 3cm beyond area that backslab will be applied.
- Measure Plaster of Paris for backslab, slightly longer than needed. Fold open 10 layers for arm or 14 layers leg. Ensure wide enough to cover half limb circumference and long enough to reach past fracture site (include joint above/below if needed to immobilise fracture).
- Dip Plaster of Paris in room temperature water, squeeze out lightly and mould by rubbing smooth. Hold limb in position of function, for at least 5 minutes to form backslab. Reassure child that Plaster of Paris may heat up whilst setting.
- Secure with bandage.
- Check pulses and sensation. Advise to return if increased pain, numbness or discolouration in limb.

Poisoning

- The child has suspected poisoning if s/he has swallowed, inhaled or absorbed a potentially harmful substance like a medication, chemical/cleaning agent, pesticide, toxin, drug, gas, corrosive or plant.
- While assessing child contact Poison Information Helpline ↗ 170 or consult AfriTox Poisons Information database (if available) to help with management and to calculate toxic dose.

Give urgent attention to the child with suspected poisoning and any of:

- Attempted self-harm/suicide
 - Fitting ↗ 28
 - Decreased level of consciousness ↗ 174
 - Persistent vomiting
 - Excessive drooling/sweating
- Agitation, severe restlessness or hallucinations
 - Pupils dilated or pinpoint
 - Temperature ≥ 38°C
 - Difficulty breathing: abnormal respiratory rate ↗ 175, blue lips/tongue, sats < 94%, stridor, grunting, nasal flaring, chest indrawing
- If ≥ 2 of: cold hands/feet, weak/fast pulse ↗ 175, CRT¹ ≥ 3 seconds, decreased level of consciousness ↗ 174: **shock likely ↗ 26.**
 - Neck twisting, upward gaze, facial grimace, clenched jaw, speech difficulty: **dystonic reaction likely**

Manage and refer urgently:

- Assess and manage airway, breathing and circulation ↗ 24. If difficulty breathing, give ♦oxygen 2L/min via nasal prongs.
- If poison on skin: health worker to wear gloves, apron and mask. Remove clothes and wash skin with soap and water.
- If poison in eye/s: immediately flush under running water or ♦sodium chloride 0.9% for at least 20 minutes.
- Ask carers about time of exposure, amount and type of substance: examine tablets, packets, containers of suspected substance. Determine if toxic dose ingested:
 - If toxic dose swallowed within past hour *and* child fully conscious, give ▶activated charcoal 1g/kg with 50-100mL water ↗ 157:3. Avoid if child swallowed petrol/paraffin/iron/lithium/alcohol/corrosives (e.g. household detergents). Avoid inducing vomiting/doing gastric lavage unless specifically instructed by Poison Helpline due to aspiration risk.
- If pinpoint pupils, excessive drooling/sweating, coughing up or choking on secretions, vomiting/diarrhoea, slow pulse ↗ 175, **organophosphate poisoning likely**: give ▶atropine 0.05mg/kg IV ↗ 158:9. If no response, double the dose every 3-5 minutes until breathing and sats improve and heart rate normal ↗ 175 (pupils will take longer to improve). Monitor closely and give further doses if needed.
- If shallow breathing/decreased respiratory rate ↗ 175 and **opioid³ overdose** suspected: give 100% ♦oxygen via bag-valve mask at 10-15L/min and ■naloxone 0.1 mg/kg IV/IM (up to 2mg) immediately. Reassess every 2 minutes: if respiratory rate still decreased, give additional dose every 2 minutes up to a total dose of 10mg. Monitor closely and give further doses if needed.
- Monitor for recurrence of respiratory depression. If respiratory depression recurs discuss further management with specialist.
- If **dystonic reaction**: give ■biperiden 0.1mg/kg IM/slow IV (up to 2mg if < 7 years; 3mg if 7-10 years; 5mg if ≥ 10 years) or ■promethazine 0.5mg/kg IM (up to 12.5mg if < 10 years; 25mg if ≥ 10 years).
- Notify² if child with symptoms has been exposed to pesticides (like organophosphates, rat poison), mercury/lead.

Approach to a child with suspected poisoning not needing urgent attention

- Try to identify poison: obtain careful history and ask to see tablets, packets, containers of suspected agent used. Record time of exposure and how much child exposed to.
- Contact the Poison Information Helpline ↗ 170 or consult AfriTox poisons information database to determine if toxic dose has been swallowed, and manage according to symptoms:

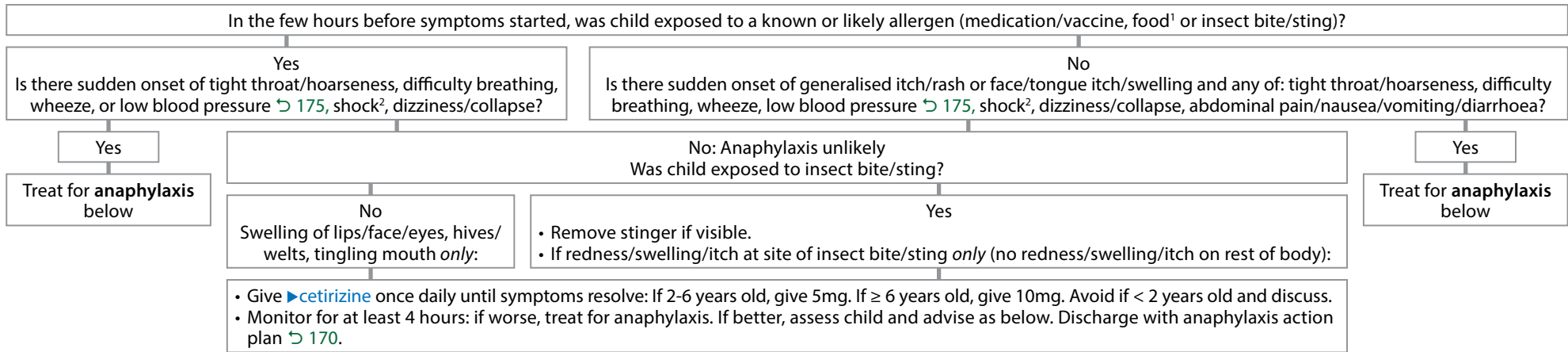
Toxic dose	Non-toxic dose		Unknown poison or dose
	Child has symptoms	Child has no symptoms	
• Refer urgently. • Discuss need for activated charcoal with Poison Helpline ↗ 170.		Discharge home.	• Refer if: - Swallowed any of: paracetamol, anti-epileptics, warfarin, tricyclic anti-depressants, sulphonylureas, iron, anti-hypertensives, anti-cholinergics, pesticides, chemical/cleaning agent. - Child swallowed paraffin and has increased respiratory rate ↗ 175. • If none of the above, observe for 4-6 hours: if no symptoms, discharge home. If symptoms persist/worsen, discuss/refer.
		• Advise carer to return if condition worsens. Prevent future poisoning, advise to lock away toxic substances. Share Poison Helpline with family ↗ 170. • If concerns about poor adult/parental supervision at home, refer to social worker to arrange home visit. • If concerns about stress, depression or substance use in carer ↗ PACK Adult.	

If child has had poisoning more than once, history not consistent with findings, or concern that carer intentionally exposed child to poison, manage as suspected child abuse →144.

¹Capillary refill time (CRT): hold hand/foot higher than level of heart. Press soft pad of finger/toe until it turns pale, then release pressure and note time taken for colour to return. ²Immediately report to provincial CDC telephonically and via email ↗ 170. Notify within 24 hours (7 days if lead poisoning) via NMC App or <https://www.nicd.ac.za/nmc-overview/notification-process>. Complete NMC case notification form and case investigation form and send to NMCsurveillanceReport@nicd.ac.za and provincialcdc@westerncape.gov.za. ³Examples of opioids include: codeine, tramadol, morphine, oxycodone, hydrocodone, fentanyl.

Anaphylaxis

Give urgent attention to the child with possible anaphylaxis



Manage anaphylaxis and refer urgently:

- Give immediately [♦adrenaline](#) 0.01mg/kg (1:1000) IM into mid-outer thigh (see table). Repeat every 5 minutes if no response. If child has prescribed auto-injector (like EpiPen® Junior 0.15mg or EpiPen® 0.3mg) available, use instead.
- Lie child flat, raise legs and give 100% face mask [♦oxygen](#) at 15L/min. Avoid lying child flat if difficulty breathing.
- If shock² likely, give [♦NaCl 0.9%](#) 10mL/kg IV over 5-10 minutes. Repeat until signs of shock resolved. Stop if breathing worsens.
- If difficulty breathing or wheeze, nebulise with 2mL [♦salbutamol 0.5% solution](#) and 2mL [♦ipratropium bromide 0.25mg/mL](#) or give [♦salbutamol](#) 400-600mcg (4-6 puffs) with metered dose inhaler (MDI) and large volume (500mL) spacer. Repeat every 5-10 minutes if no response. Assess and further manage airway if needed [↗ 24](#).
- If generalised itch/rash or face/tongue swelling, give [♦promethazine](#) IM/slow IV 0.25mg/kg: if 2-5 years old (12-17.5kg), give 2.5mg. If 5-7 years old (17.5-25kg), give 5mg. If 7-11 years old (25-35kg), give 7.5mg. If 11-15 years old (35-55kg), give 15mg [↗ 164:42](#). Avoid if < 2 years old or low BP [↗ 175](#).
- Give [■hydrocortisone](#) 5mg/kg IM/slow IV: if 1 month-2 years old (< 11kg), give 25mg. If 2-3 years old (11-14kg), give 50mg. If 3-5 years old (14-17.5kg), give 75mg. If > 5 years old (≥ 17.5kg), give 100mg [↗ 161:26](#).
- If delay in referral, collect blood in 2 yellow topped tubes (tryptase sampling) within 2 hours of symptom onset and send with child. If delay > 4 hours, store tubes on ice.
- If anaphylaxis due to medication/vaccine, report [↗ 156](#).

Adrenaline dosing

Weight	Injection 1mg/mL (1:1000)	Age (if weight unknown)
< 9kg	0.05mL	< 1 year
9-12kg	0.1mL	1-2 years
12-17.5kg	0.2mL	2-5 years
17.5-40kg	0.3mL	5-12 years
≥ 40kg	0.5mL	≥ 12 years

Assess the child with previous anaphylaxis

Assess	When to assess	Note
Trigger	At diagnosis	Ensure a specialist has reviewed the child with anaphylaxis to confirm trigger/s. Common triggers include medications, food ¹ and insect bites/stings.
Other allergy	At diagnosis	Check for other allergy ↗ 117 .

Advise the child with previous anaphylaxis and his/her carer

- Advise to avoid known trigger/s. If trigger is a medication/vaccine, inform health worker at every visit.
- Ensure child/carers has an anaphylaxis action plan available from Allergy foundation South Africa and arrange MedicAlert® bracelet [↗ 170](#). Ensure child/carers knows ambulance telephone number, where nearest hospital is and has reliable transport. If adrenaline prescribed, ensure child/carers knows when and how to use it [↗ 117](#).

¹Common foods causing anaphylaxis: peanuts, tree nuts, egg, milk and fish. ²If ≥ 2 of: cold hands/feet, weak/fast pulse [↗ 175](#), capillary refill time ≥ 3 seconds, decreased level of consciousness [↗ 174](#), shock likely.

Burns

Assess depth and area of burn and calculate percentage total body surface area (%TBSA) burnt ↗ 39. Accurate estimation of burn size is critical to ongoing fluid replacement and management.

Give urgent attention to the child with burn/s and any of:

- Child < 1 year old
- Likely inhalation burn (burns to face/neck, hoarse, stridor or black sputum)
- Burn of face, hand, foot, genitals, joints
- Circumferential¹ burn of chest/limbs
- Full-thickness burn
- Burn > 10% TBSA
- Burn > 5% TBSA in child 1-2 years old
- ≥ 2 of: cold hands/feet, weak/fast pulse ↗ 175, CRT² ≥ 3 seconds, decreased level of consciousness ↗ 174: **shock** likely ↗ 26
- Temperature ≥ 38°C
- Sudden skin swelling with redness, pain or warmth
- Electric/chemical burn
- Unable to drink/breastfeed

Manage and refer urgently to closest burns centre:

- Assess and manage airway, breathing and circulation ↗ 24.
- If burn > 10% TBSA, inhalational burn, drowsy/confused, give ♦oxygen 2L/minute via nasal prongs. Doctor to consider intubation.
- Remove non-sticking clothing. Cool burn with cold running tap water or wet towel for 20 minutes. Avoid hypothermia.
- Give ♦paracetamol 15mg/kg 6 hourly ↗ 163:37. If severe pain, give oral ■morphine according to age: if < 1 month give 0.05mg/kg 6 hourly, if 1-12 months give 0.1mg/kg 4-6 hourly, if ≥ 1 year give 0.2-0.4mg/kg 4-6 hourly ↗ 163:36. If oral morphine not available, give IV ■morphine 0.1mg/kg (up to 10mg) slowly over 3-5 minutes, 4 hourly as needed. If respiratory rate decreases ↗ 175 or sats ≤ 92%, give ♦oxygen 2L/minute via nasal prongs.
- Gently remove loose/dead skin, clean with ♦chlorhexidine 0.05%. Re-assess to confirm %TBSA burnt ↗ 39.
- Give IV fluids:
 - If **shock** likely, manage child's fluid needs ↗ 26.
 - If > 10% TBSA: give ►NaCl 0.9% IV 4mL x weight (kg) x %TBSA over first 24 hours. Give half this volume in first 8 hours from time of burn.
 - In addition, give maintenance fluids: ►ORS orally/via NGT or ►NaCl 0.9% with dextrose 5%³ IV, according to weight ↗ 175.
 - If other injuries, manage ↗ 33.
- Dress the burn:
 - If first aid burns dressing (e.g. Burnshield®) available: apply for 4 hours. If transfer delayed > 4 hours/or first aid burn dressing not available, use paraffin gauze dressing instead.
 - If full thickness or > 10% TBSA burn, apply paraffin gauze and cover with plastic wrap.
 - If hospital transfer delayed > 12 hours, apply paraffin gauze then cover with dry gauze and bandage.
- Keep child warm: cover with a clean dry sheet and blanket to prevent hypothermia. Monitor temperature.
- Give ♦tetanus toxoid 0.5mL IM if not had in last 5 years (check RthB).
- Reassess airway, breathing and circulation hourly ↗ 24.

Approach to the child with burn/s not needing urgent attention:

- if burn unrt area < 3 hours old, cool area with cold tap water for 20 minutes.
- Give ♦paracetamol 15mg/kg 6 hourly as needed for up to 5 days ↗ 163:37. Encourage oral fluid intake
- Remove loose/dead skin and gently clean burn with ♦chlorhexidine 0.05% or ♦sodium chloride 0.9%. Then cover with paraffin gauze dressing.
- Give ♦tetanus toxoid 0.5mL IM if not had in last 5 years (check RthB).
- If imprint burns, glove and stocking hot water burns, cigarette burns or burns not matching history, consider **child abuse** ↗ 144.
- Review wound/s and pain daily until burn healed:
 - Dress burn with paraffin gauze dressing. If pain/anxiety with dressing changes, give ♦paracetamol 15mg/kg ↗ 163:37 1 hour before changing dressing.
 - If signs of infection (redness/swelling), apply ►povidone iodine 5% cream daily.
 - If **severe infection** (extensive redness/swelling, foul-smell, pus or temperature ≥ 38°C), ongoing pain despite medication or burn not healing, refer

¹ Circumferential: extends right around the chest or limb and may affect breathing movement of chest or circulation of limb. ²Capillary refill time (CRT): hold hand/foot higher than level of heart. Press soft pad of finger/toe until it turns pale, then release pressure and note time taken for colour to return. ³If not available: add 10mL ►dextrose 50% to each 100mL of sodium chloride 0.9%.

Assess depth and area of burn

Step 1. Assess depth of burn:

Look to see if burn dry or moist, blisters and colour of skin.

Burn is dry and blanches. There may be minor blisters and/or redness. Painful.



© University of Cape Town

Superficial burn

Red, blistering burn that is moist and may weep. Painful.



© University of Cape Town

Superficial partial thickness burns

Burn may be moist or have waxy appearance. May be white/yellow slough or red, mottled. Less painful.



© University of Cape Town

Deep partial thickness burn

Burn is dry, charred whitish or brown or black. Painless and firm to touch.



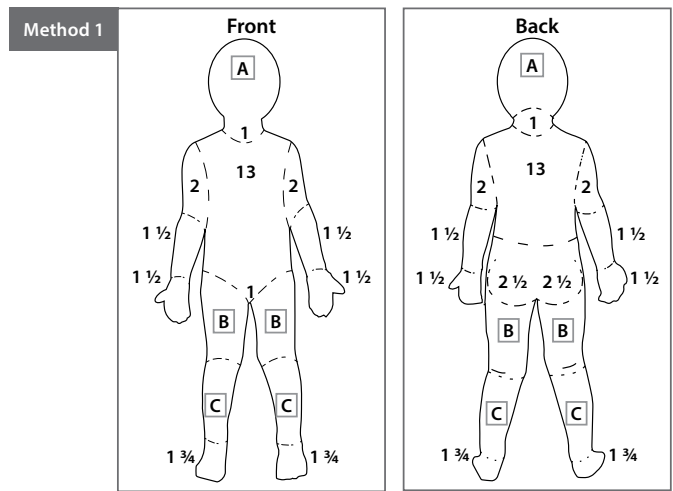
© University of Cape Town

Full thickness burn

Burns areas may vary in depth, with some areas being superficial and some areas deep.

Step 2. Calculate the body surface area of burn using one of the methods below:

- **Method 1:** Use the percentages shown in the figures and table to estimate the area the burn covers. If available use the Vula mobile application¹ burns referral section to assist with this calculation.
- **Method 2:** Use to calculate small burn wounds.



Lund and Browder chart

- Estimating percentage total body surface area in children affected by burns
- Do not include areas of simple erythema (redness)

Body Part	Age			
	0	1	5	10
A - 1/2 head	9 1/2	8 1/2	6 1/2	5 1/2
B - 1/2 of 1 thigh	2 3/4	3 1/4	4	4 1/2
C - 1/2 of 1 lower leg	2 1/2	2 1/2	2 3/4	3 1/2

Example A 2½-year old child has a superficial burn that covers half their face and neck (front), their full arm and their hand:

- The burn involves:
 - front of the head and neck, therefore: $8\frac{1}{2}\% + 1\% = 9\frac{1}{2}\%$
 - full arm $2\% + 2\% + 1\frac{1}{2}\% + 1\frac{1}{2}\% = 7\%$
 - full hand $1\frac{1}{2}\% + 1\frac{1}{2}\% = 3\%$
- Total body surface area affected: $9\frac{1}{2}\% + 7\% + 3\% = 19\frac{1}{2}\%$

Method 2

If burn wound is small: estimate using the area of child's open hand as a guide. The area of the palm of hand represents 1% TBSA. Do not include simple erythema (redness) in calculation.

If burn depth varies in different areas, add % superficial burns and % deep to get a total % burns. Continue to manage burn →38.

¹Vula mobile application: Mobile system used to improve referral system of patients to various hospitals and specialties.

Bites and stings

Give urgent attention to the child with bite/sting and any of:

- Snakebite¹ (even if bite not seen) or venom in eyes
 - Scorpion¹ sting
 - Deep or large wound needing sutures
 - Actively bleeding
 - Bite involves hand
- Weakness
 - Drooping eyelids
 - Difficulty swallowing/speaking
 - Double vision
- Spider¹ bite with drooling, restlessness, muscle cramps, erection
 - Sudden onset of generalised itch/rash or face/tongue itch/swelling and any of: tight throat/hoarseness, difficulty breathing, wheeze, shock², dizziness/collapse, abdominal pain/nausea/vomiting/diarrhoea or exposure to possible allergen³, **anaphylaxis** likely →37

Manage and refer urgently:

- If bleeding, apply direct pressure to stop bleeding. Assess and manage child's fluid needs → 26.
- Wash wound well with **chlorhexidine 0.05%** or **povidone iodine 10% solution** and irrigate under running water for 10-15 minutes then apply dressing. Do not suture.
- If snake/spider bite or scorpion sting: avoid applying tourniquet, squeezing/cutting the bite or sucking out venom. Contact Poisons Information Helpline → 170.
- If venom in eyes: instil 1 drop **oxybuprocaine 0.4%** eye drops then flush under running water or **sodium chloride 0.9%** for 20 minutes. Avoid giving oxybuprocaine to take home.
- If pain, give **paracetamol** 15mg/kg → 163:37. If very painful scorpion sting, doctor to inject **lidocaine 2%** 2mL around site.
- Give **tetanus toxoid** 0.5mL IM if not had in last 5 years (check RtHB).

Approach to the child with bite/sting not needing urgent attention

- Wash wound with soap under running water for 15 minutes (animal bites)/10 minutes (human bites) then clean with **chlorhexidine 0.05%** or **povidone iodine 10%** solution.
- Do not suture unless wound on head/face, < 24 hours old and uninfected. Dress (avoid compressive dressings) and check wound every 2 days.
- Give **tetanus toxoid** 0.5mL IM if none in past 5 years. If pain, give **paracetamol** 15mg/kg 6 hourly as needed → 163:37.
- Manage further according to type of bite or exposure:

Spider bite or insect bite/sting	Animal bite/exposure was from dog, cat, cow, goat, donkey, horse, sheep, mongoose, jackal, fox, seal, or any direct skin contact with bat			Human bite
- If bee sting, remove stinger immediately. - If severe pain, itch, redness or swelling, apply ice pack and give: ▶calamine lotion to apply to area. ▶chlorpheniramine 0,1mg/kg/dose 6-8 hourly up to 2 weeks ↗ 159:15. - If extensive local response to bite, papular urticaria likely ↗ 78. Advise to return if worse/no better after 2 days. If skin red, warm, painful, cellulitis likely →76.	No	Yes. Are there any of: strange behaviour in animal ⁴ , unprovoked attack, animal's immunisation status unknown/not immunised?		- Ensure immunisations up to date. Give catch up doses if needed ↗ 13. - If broken skin, assess for hepatitis B post-exposure prophylaxis ↗ 93. - Also give antibiotics:
	No	Yes		
	Are there any of: impaired immunity ⁵ , broken skin with blood; animal licked broken skin/eyes/mouth or any direct skin contact with bat?			
	No	Yes		
	- Discuss all cases with infectious disease specialist/ NICD helpline ↗ 170. - Inject rabies vaccine 1 ampoule IM (anterolateral thigh) on days 0, 3, 7 and between day 14-28 (4 doses). - Stop if animal tests negative for rabies/is still well after 10 days. - If bite involves hand, give antibiotics:	- Discuss all cases with infectious disease specialist/NICD helpline ↗ 170. - Inject rabies vaccine 1 ampoule IM (anterolateral thigh) on days 0, 3, 7 and between day 14-28 (4 doses). - If ≤ 7 days since first dose of rabies vaccine, inject rabies immunoglobulin 20 IU/kg (human derived) or 40 IU/kg (equine derived), in and around the bite⁶. - Give antibiotics:		
- Complete animal bite incident form and email to: provincialcdc@westerncape.gov.za and contact local state veterinarian. - Give amoxicillin/clavulanic acid 45mg/kg 12 hourly for 5 days ↗ 158:8. If severe penicillin allergy (history of anaphylaxis, urticaria or angioedema), give instead azithromycin 10mg/kg once daily for 3 days ↗ 158:10 plus metronidazole 7.5mg/kg 8 hourly for 5 days ↗ 162:33. - If wound infected, continue antibiotics for 10 days. If no better on antibiotics, refer.				

¹Obtain description of spider/snake/scorpion to help with specific management once referred. ²If ≥ 2 of: cold hands/feet, weak/fast pulse → 175, capillary refill time ≥ 3 seconds, decreased level of consciousness → 174, **shock** likely. ³Common allergens: medications, new food, insect bite/sting within last few hours. ⁴Strange behaviours: domestic animal with unusual aggression or wild animal that appeared weak/sick. ⁵HIV not on ART, cancer, receiving chemotherapy, dialysis or long-term corticosteroids. ⁶Inject as much rabies immunoglobulin as anatomical site of bite will allow. Discard any remaining immunoglobulin.

The blue child

Give urgent attention to the blue child with any of:

- Choking →30
 - Newborn →23
 - < 6 months old
 - Known heart or lung disease
 - If ≥ 2 of: cold hands/feet, weak/fast pulse → 175, CRT¹ ≥ 3 seconds, decreased level of consciousness → 174: **shock** likely → 26
- Difficulty breathing: abnormal respiratory rate → 175, noisy breathing, grunting, nasal flaring or chest indrawing → 55
 - Temperature < 35.5°C or ≥ 38°C
 - Blue lips/tongue and/or sats ≤ 94%
 - Has a life-limiting illness
 - Hb ≤ 7 g/dL

Manage and refer urgently:

- If not already done, manage airway, breathing and circulation → 24.
- Give oxygen 2L/min via nasal prongs. If difficulty breathing, assist each breath with bag valve mask attached to oxygen → 24.
- Doctor to check for decreased breath sounds/hyper-resonance on percussion/pain on 1 side/deviated trachea: **tension pneumothorax** likely: insert large bore cannula above 3rd rib in mid-clavicular line and arrange urgent chest tube. Assess need for intubation.
- Check fingerprick glucose: if ≤ 28 days old and glucose < 2.6mmol/L → 9. If > 28 days old and glucose < 3mmol/L (< 4 if diabetes) → 32.
- If temperature < 35.5°C or ≥ 38°C, give ♦ceftriaxone² 80mg/kg IM → 159:11. Avoid injecting ≥ 1g IM at one injection site.
- Keep child calm on carer's lap, if possible.
- If known heart disease, place in knee-chest position (knees bent in to chest). Discuss urgently with cardiology service.
- Warm child: place skin to skin with mother or dress warmly including head and feet, and cover with warm blanket.

Approach to the child with a history of turning blue

Does child have a persistently blocked nose, snore or stop breathing during sleep?

Yes

Obstructive sleep apnoea likely
Discuss/refer urgently to ENT specialist.

No

Does child usually cry, then hold breath until s/he turns blue, becomes limp, unconscious and perhaps fit?

Yes

Breath holding spells likely

- Check Hb: if Hb < 10g/dL in child < 5 years old or Hb < 11g/dL in child ≥ 5 years old, **anaemia** likely → 46.
- Reassure carer that breath holding spells are common in child 6 months - 4 years old. Prognosis is excellent and development is normal. No treatment is needed. During the spell, lay child on his/her side. Reassure carer that child will start to breathe on his/her own. Advise to avoid putting anything in mouth.
- If child fits at home, advise carer what to do → 29.
- Discuss/refer to paediatrician.

No

Discuss/refer same day.

¹Capillary refill time (CRT): hold hand/foot higher than level of heart. Press soft pad of finger/toe until it turns pale, then release pressure and note time taken for colour to return. ²Avoid giving Ringer's lactate together with IV/IM ceftriaxone. If severe penicillin allergy (previous anaphylaxis, urticaria or angioedema), discuss with doctor.

The inconsolable crying/irritable child

- Assess the child if s/he cries continuously and frequently for no obvious reason.
- If baby < 3 months old cries for short periods and is easily consoled, reassure.

Give urgent attention to the inconsolable/crying/irritable child with any of:

- Baby < 2 months and not feeding well
- Nasal flaring/chest in-drawing →55
- Tires/sweats during feeds
- If ≥ 2 of: cold hands/feet, weak/fast pulse → 175, CRT¹ ≥ 3 seconds, decreased level of consciousness → 174, shock likely → 26
- Restless/irritable/lethargy/altering level of consciousness, sunken eyes, slow skin pinch² or drinking poorly/thirsty, dehydration likely
- Likely meningitis:
 - If < 2 years old: bulging fontanelle (when not crying), lethargy or refusing food/drink
 - Any age, and ≥ 2 of: temperature ≥ 38°C, headache, decreased level of consciousness → 174, neck stiffness
- Decreased level of consciousness → 174
- Not moving properly →90
- Cramping abdominal pain and lethargy and/or jelly-like stool

Manage and refer urgently:

- Check fingerprick glucose → 32.
- Assess and manage child's fluid needs → 26.
- If meningitis likely, give ♦ceftioxone³ 100mg/kg IV/IM → 159:12 Avoid injecting ≥ 1g IM at one injection site.
- If temperature ≥ 38°C, give ♦paracetamol 15mg/kg → 163:37 or rectally if available.

Approach to the inconsolable/crying/irritable child not needing urgent attention

- Ask about other symptoms: if temperature ≥ 38°C or history of fever → 43, if abdominal pain/distension → 60, if vomiting → 62, if diarrhoea → 63, if burning urine → 69, if constipation with pain on passing stool, faecal impaction, involuntary leakage of stool, voluntarily withholding stool → 66, if recent injury → 33 or may have swallowed potentially harmful substance → 36.
- Assess growth → 15.

Then check for source of pain/discomfort

- Examine child from head to toe; check eyes, ears, mouth/throat, skin, groin, genitalia, anal and nappy area:

Check eyes

If foreign body or scratch → 48.

Check ears

If red, bulging eardrum or ear pain/discharge → 49.

Look in mouth and throat

If white patches/blisters/ulcers → 51 or red throat → 52.

Check skin

- Remove cause:
 - Thorns/splinters or something wrapped tightly around finger/toe/penis.
- If bruising/skin marks, consider child abuse → 144.
- If insect bites → 40.

Look for groin swelling

If bulge in groin on crying/cough/passing stool, inguinal hernia likely. Discuss with doctor/surgeon same day.

Check genitalia

If scrotal swelling, discuss/refer same day.

Check anus

Check for crack, lump/pile or red/raw skin → 67.

Check nappy area

If nappy rash → 85.

Check teeth

If < 3 years old, consider teething problem → 54.

- If no cause found, check urine. If blood, nitrites or leucocytes on urine dipstick →69.
- If child has a life-limiting illness, discuss/refer.

Screen for social risk/stressors

- Screen for depression in carer: in the past month, has carer felt down/depressed/hopeless or little interest/pleasure in doing things? If yes to either → PACK Adult.
- Ask carer if aware of any abuse of child. Ask child if anyone hurts/upsets him/her. If yes to either, consider child abuse → 144.
- If violence or substance abuse at home, involve social worker. If school problem → 140. If child is stressed, miserable or angry → 139.
- If newborn/breastfeeding, ask about maternal substance abuse. If found, refer baby to hospital same day and involve social worker.

- If < 4 months old and crying for ≥ 3 hours/day on ≥ 3 days a week, consider colic → 60. Do not leave a young child (< 2 years) to cry alone.
- If unable to find cause, discuss/refer.

¹Capillary refill time (CRT): hold hand/foot higher than level of heart. Press soft pad of finger/toe until it turns pale, then release pressure and note time taken for colour to return. ²Skin pinch: pinch skin on abdomen between 2 fingers. Release. Skin normally snaps rapidly back to its normal position. A slow skin pinch takes longer. ³Avoid giving Ringer's lactate together with IV/IM ceftriaxone. If severe penicillin allergy (previous anaphylaxis, urticaria or angioedema), discuss with doctor.

Fever

A child with a fever has a tympanic temperature of $\geq 38^{\circ}\text{C}$ (measured in the ear), or an axillary temperature of $\geq 37.5^{\circ}\text{C}$ (measured under the arm) now or in the past 3 days.

Give urgent attention to the child with a fever and any of:

- Baby < 2 months old
 - Just had a fit →29
 - Blue skin/lips →41
 - Unable to feed/drink
 - Decreased level of consciousness → 174
 - Fever > 2 weeks with no cause found
- Increased respiratory rate → 175 and/or difficulty breathing →55
 - ≥ 2 of: cold hands/feet, weak/fast pulse → 175, CRT¹ ≥ 3 seconds, decreased level of consciousness → 174: **shock** likely → 26
 - Headache/neck stiffness/bulging fontanelle
 - Purple/red rash that does not disappear with pressure
 - Severe abdominal pain or tender right lower abdomen, **appendicitis** likely
- Little or no urine →69
 - Jaundice
 - Travel to a malaria area in the past 3 months
 - ≥ 2 of: strange movements of limbs/face, lumps over joints/tendons, rash (round pink lesions with pale centre), joint pain/swelling, **rheumatic fever** likely
 - Previous rheumatic fever or known with rheumatic heart disease
- Manage and refer urgently:**
- Check fingerprick glucose → 32.
 - If headache, decreased level of consciousness → 174, neck stiffness, bulging fontanelle and/or purple/red rash, **meningitis** likely, give ♦**ceftriaxone**² 100mg/kg IV/IM³ → 159:12.
 - If baby < 2 months old or **appendicitis** likely, give ♦**ceftriaxone**² 80mg/kg IM³ → 159:11.
 - If in a malaria area in the past 3 months, consider malaria. Discuss/refer.
 - If able to feed/drink, give ♦**paracetamol** 15mg/kg → 163:37.

Approach to the child with a fever not needing urgent attention

Tick bite (red sore with dark centre) or tick present?

Yes

Tick bite fever likely

- May have body pains, headache, rash and lymphadenopathy.
- If present, grip tick close to skin using forceps and remove.
- Give treatment:
 - If < 8 years or < 45kg, give ■**azithromycin** once daily for 3 days → 158:10.
 - If ≥ 8 years and ≥ 45 kg, give ■**doxycycline** 100mg 12 hourly for at least 3 days after fever resolves and clinically better (up to 7 days).

No

Ask about other symptoms and manage on symptom page:

- Cough →55
 - Blocked/runny nose →50
 - Sore throat →52
- Ear pain →49
 - Nausea/vomiting →62
 - Diarrhoea →63
- Abdominal pain/swelling →60
 - Urinary symptoms →69
 - Painful/swollen joint →74
- Limping/difficulty moving limb →90
 - Lumps in neck, axilla or groin →47
 - Skin symptoms →75
- Face pain/swelling →53
 - Sore tooth →54
 - Eye swelling →48

- Check urine dipstick (get clean catch sample if possible): if blood, leucocytes or nitrites →69.
- If recently started abacavir, check for abacavir hypersensitivity reaction (AHR) → 104.
- If any of: significant TB exposure⁴, cough, unexplained fever, drenching night sweats, poor weight gain, weight loss, failure to thrive, lethargy, decreased playfulness, visible neck mass, blood-stained sputum, headache, persistent vomiting (without diarrhoea), test for TB → 108.
- Test for HIV if status unknown → 95.

If no obvious cause found, **acute viral infection** likely

- Advise on cough/sneeze hygiene, adequate fluid intake, to wash hands regularly, rest, and keep home from school until well.
- Advise that antibiotics are not needed.

- If fever causes discomfort, give ♦**paracetamol** 15mg/kg 6 hourly as needed up to 5 days → 163:37. Avoid tepid sponging or positioning child in front of a fan⁵. Dress child to suit weather condition.
- If fever persists ≥ 3 days and no obvious cause found or fever recurs, discuss/refer.

¹Capillary refill time (CRT): hold hand/foot higher than level of heart. Press soft pad of finger/toe until it turns pale, then release pressure and note time taken for colour to return. ²Avoid giving Ringer's lactate together with IV/IM ceftriaxone. If severe penicillin allergy (previous anaphylaxis, urticaria or angioedema), discuss with doctor. ³Avoid injecting $\geq 1\text{g}$ IM at one injection site. ⁴Any child who in the last 12 months shared an enclosed space (school/creche/home/similar) for ≥ 1 night or frequent/extended daytime periods with an adult/adolescent with pulmonary TB ("index client"), during the 3 months before index client started TB treatment. ⁵This will cause shivering and increase core temperature even further.

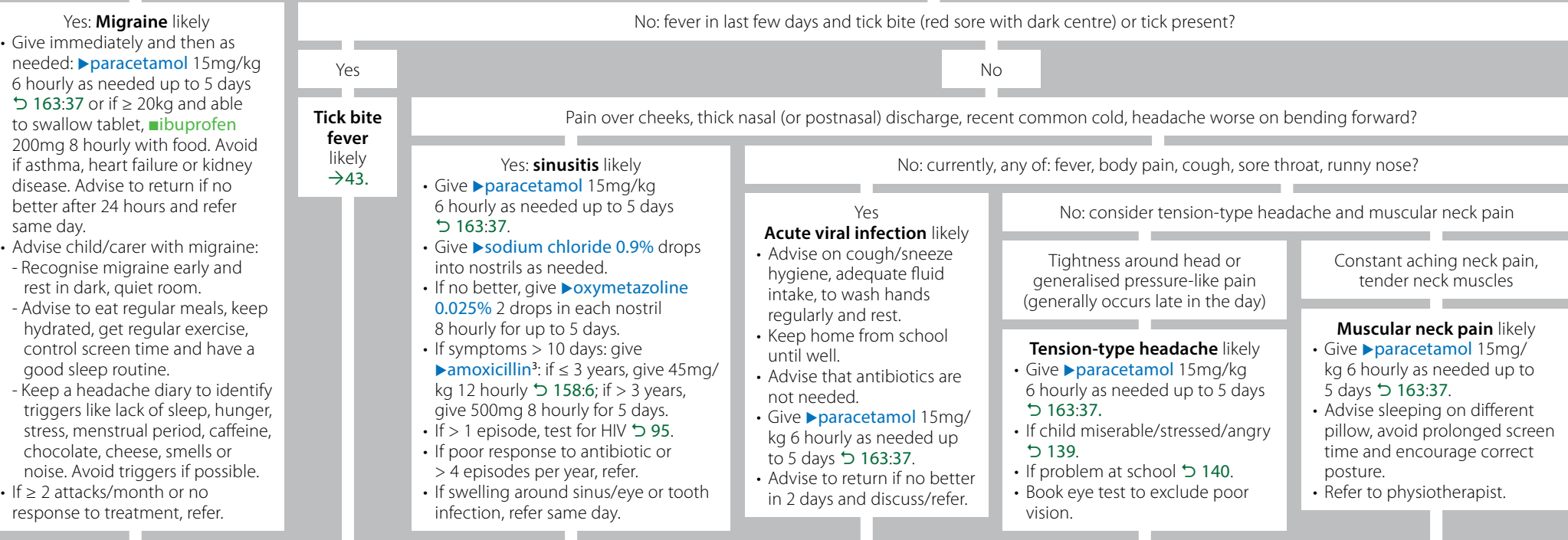
Headache

- Give urgent attention to the child with headache and any of:**

 - Sudden severe headache
 - Headache/vomiting on waking from sleep
 - Headache getting worse and more frequent or wakes child
 - Travel to malaria area in past 3 months: **malaria** likely
 - Persistent headache and TB symptoms or significant TB exposure¹ [↗ 108](#)
 - Temperature $\geq 38^{\circ}\text{C}$ and neck stiffness
 - Decreased level of consciousness [↗ 174](#)
 - Head tilted to one side (torticollis)
 - Weakness of arm or leg
 - > 2 weeks duration with no other symptoms
 - Vision problems (e.g. double vision)
 - Pupils different sizes
 - New squint or unable to move eyes as before
 - Head trauma in last week [↗ 33](#)
 - Abnormal walk or balance problem
 - Elevated BP [↗ 175](#)
- Manage and refer urgently:**
- If temperature $\geq 38^{\circ}\text{C}$, decreased level of consciousness [↗ 174](#), neck stiffness, **meningitis** likely: give **ceftriaxone**² 100mg/kg IV/IM [↗ 159:12](#). Avoid injecting $\geq 1\text{g}$ IM at one injection site.
 - Give **paracetamol** 15mg/kg [↗ 163:37](#).
 - If in a malaria area in past 3 months, consider malaria. Discuss/refer.

Approach to child with headache not needing urgent attention

Is headache throbbing, disabling and recurrent with nausea/vomiting or light/noise sensitivity, that resolves completely within 72 hours?



The tired or lethargic child

- If child is not moving or sitting properly (limb/s not working properly) →90.
- The lethargic child may be difficult to wake or excessively sleepy and wants to lie down and not move around for considerable periods.

Give urgent attention to the child with tiredness or lethargy and any of:

- Unable to drink/breastfeed
- Baby < 3 months old
- Vomiting everything
- Decreased level of consciousness → 174
- Increased respiratory rate → 175 or breathing problem →55
- Bone pain or continuous pain
- Easy bruising or bleeding
- Likely meningitis:
 - If < 2 years old: bulging fontanelle (when not crying), refusing food/drink
 - Any age, ≥ 2 of: temperature ≥ 38°C, headache, decreased level of consciousness → 174, neck stiffness
- Weight loss, poor weight gain/failure to thrive and pale palms/conjunctiva

Manage and refer urgently:

- Check fingerprick glucose → 32 and fingerprick Hb → 46.
- Assess and manage child's fluid needs → 26.
- If meningitis likely, give ♦ceftriaxone¹ 100mg/kg IV/IM → 159:12. Avoid injecting ≥ 1g IM at one injection site.

Approach to the child with tiredness or lethargy not needing urgent attention

- Assess and manage child's fluid needs → 26.
- Check child's temperature: if ≥ 38°C or fever in last 3 days → 43.
- Exclude heart problem: if < 1 year old and tires/sweats during feeds or if ≥ 1 year old and breathless on minimal exertion/exercise, discuss/refer same day.

If none of the above, ask about associated symptoms:

- If vomiting →62, diarrhoea →63, jaundice →61, headache →44, blocked/runny nose →50, sore throat →52, cough →55. Manage other symptoms on symptom pages.
- Exclude allergy: if persistent itchy blocked/runny nose and sneezing →50, if asthma →119.
- Check for lumps/swellings in neck, axilla or groin →47 and swollen joints →74.

Then exclude anaemia, diabetes, TB, HIV, growth problem and vision problem:

Test for anaemia

Do Hb: if Hb < 10g/dL in child < 5 years old or < 11g/dL in child ≥ 5 years old, **anaemia** likely → 46.

Screen for diabetes

Check fingerprick glucose → 32.

Screen for TB

→ 108.

Screen for HIV

If status unknown, test for HIV → 95.
If HIV, give routine care → 97.

Check for growth problems

Assess growth → 15.

Exclude vision problems

Assess visual milestones → 12.

- Check urine: if leucocytes, nitrites or blood on urine dipstick →69.
- If child has a life-limiting illness, also give palliative care →150.
- Assess child's mood: if withdrawn or change in mood, behaviour/feelings or not coping →139.

- Check if child sleeps enough daily:
 - baby < 12 months old should sleep ≥ 12 hours (including naps)
 - child 1 to 3 years old should sleep ≥ 11 hours (including naps)
 - child 3 to 6 years old should sleep ≥ 10 hours (including naps)
 - child 6 to 13 years old should sleep ≥ 9 hours
- If child not sleeping enough or on any medication that could be causing tiredness →88.

If no cause found, review in a week. If no better, refer.

¹Avoid giving Ringer's lactate together with IV/IM ceftriaxone. If severe penicillin allergy (previous anaphylaxis, urticaria or angioedema), discuss with doctor.

Pallor or anaemia

A child has pallor if s/he has pale palms and/or conjunctiva (lower inner eyelid). If possible, check fingerprick Hb. Child has **anaemia** if:

- Child < 5 years old and Hb < 10g/dL
- Child ≥ 5 years old and Hb < 11g/dL

Give urgent attention to the child with pallor and/or a low Hb and any of:

- Hb < 7g/dL
 - Baby < 6 months old
 - Jaundice
 - Swollen legs
 - Widespread/easy bruising or bleeding
- Increased respiratory rate ↗ 175
 - Increased pulse rate ↗ 175
 - Palpitations or chest pain
 - Bone or joint pain
- Lethargy or decreased level of consciousness ↗ 174
 - Purple/red rash that does not disappear with pressure
 - Recent travel to a malaria area, **malaria** likely
 - Weight loss, poor weight gain/failure to thrive

Manage and refer urgently:

- If increased respiratory rate, give ♦oxygen at 2L/minute via nasal prongs.

Approach to the child with pallor not needing urgent attention

Manage according to age of child:

Child < 5 years old

Iron deficiency anaemia likely

- If child > 12 months old, deworm with ♦mebendazole ↗ 162:31 or ♦albendazole (single dose) ↗ 157:5.
- Give 1-2mg/kg of elemental iron 8 hourly with food: give ♦ferrous gluconate ↗ 160:21 or ♦ferrous sulphate ↗ 160:22. Avoid giving iron if child is on Ready-to-use Therapeutic Food (RUTF) as this contains sufficient iron.
- Advise the carer:
 - Give foods rich in iron: liver, kidney, dark green leafy vegetables like spinach, egg yolk, beans, peas, lentils, fortified cereals.
 - Give food rich in vitamin C as this helps with iron absorption: oranges, naartjies, melons, tomatoes, broccoli, cauliflower, guavas, strawberries.
 - Avoid drinking tea/coffee with meals as this interferes with iron absorption.
 - Treatment with iron can make child's stools look black - no need to be concerned.
 - Iron can be extremely dangerous if child overdoses - keep out of reach of child.

Check Hb after 14 days

Hb lower

Hb same or higher

Refer.

Continue treatment as above. Repeat repeat Hb after 1 month

Hb same or lower, refer.

- Hb improving
- Continue treatment and review monthly.
 - Continue treatment for 3 months after Hb ≥ 10g/dL.

Child ≥ 5 years old

Take blood for full blood count (FBC) and manage further according to MCV¹ result:

MCV¹ low

Iron deficiency anaemia likely
Anaemia may be due to occult (hidden) blood loss: discuss/refer.

MCV¹ normal

Anaemia of chronic disorder likely

- Exclude TB ↗ 108 and HIV ↗ 95.
- If no cause found, refer/discuss same week.

MCV¹ high

Megaloblastic anaemia (folate and/or vitamin B12 deficiency) likely.
Refer/discuss same week.

¹MCV: Mean Corpuscular Volume. The MCV helps to decide the underlying cause of anaemia and can be found on FBC result sheet. Check if MCV high, low or normal compared to the reference range for age of child.

Lump/swelling in neck, axilla or groin

Give urgent attention to the child with a lump/swelling in neck, axilla or groin and any of:

- Firm painful swelling in groin area, vomiting or not passing stool, **incarcerated/strangulated inguinal hernia** likely
- Swelling in neck with wheeze or difficulty breathing
- Red overlying skin, painful to touch and soft (fluctuant) in middle of swelling, **abscess** likely

Refer urgently.

Approach to the child with lump/swelling in neck, axilla or groin not needing urgent attention

- If lump/swelling involves surface of the skin, manage as skin symptom →75.
- If lump/swelling beneath the skin, first exclude thyroid mass and hernia:
 - Neck lump compressible (cystic), in mid-line or moves when child swallows, **thyroid mass** likely: refer same week.
 - Lump in groin that bulges when child cries/coughs/passes stool, **inguinal hernia** likely: discuss with doctor/surgeon same day.
- If none of the above, a lump/swelling in neck, axilla or groin is likely an enlarged lymph node (lymphadenopathy). If unsure, refer.
- If lymphadenopathy likely, assess further according to child's age:

< 1 month old

- Test for HIV → 95. If CLHIV, give routine care → 97.
- Is lymphadenopathy localised (neck or axilla or groin) or generalised (≥ 2 areas)?

Localised Generalised

If axillary lymphadenopathy on BCG arm, **BCG lymphadenitis** likely

- Report adverse event → 156.
- Reassure that this is usually painless and resolves on its own. It may persist for up to 6 months.
- If < 1.5cm, review in one week.
- If ≥ 1.5cm or pus draining, refer.

Check mother's syphilis result, **congenital syphilis** likely if any of:

- Mother's RPR is unknown/unable to get result.
- Mother's RPR is positive and any of:
 - She is untreated
 - She was only partially treated (< 3 doses)
 - Treatment completed < 1 month before delivery
- Refer baby.
- Treat mother → PACK Adult.

≥ 1 month old

Is lymphadenopathy localised (neck or axilla or groin) or generalised (≥ 2 areas)?

Localised lymphadenopathy:

- If recent BCG vaccine and axillary lymphadenopathy on that side, **BCG lymphadenitis** likely (see column on the left).
- Is lymph node/s hot, red and painful?

Yes

- If sore throat with lymph nodes in neck, **tonsillitis** likely →52.
- If unilateral lymph nodes in neck, treat for **bacterial lymphadenitis**, give ▶cephalexin for 5 days → 159:13. If severe penicillin allergy (history of anaphylaxis, urticaria or angioedema), give instead ▶azithromycin once daily for 3 days → 158:10.
- Advise to return in 2 weeks if no better and refer.

No

- Look for sore throat → 52, skin infection or rash → 75, gums/teeth → 54 and ask about face symptoms → 53.
- If lymph node in groin and sexually active¹, treat for **bubo** →PACK Adult.

Local cause found

- Manage as on symptom page.
- Advise to return in 4 weeks if no better and refer.

No cause found

- If any of: significant TB exposure², cough, unexplained fever, drenching night sweats, poor weight gain, weight loss, failure to thrive, lethargy, decreased playfulness, visible neck mass, blood-stained sputum, headache, persistent vomiting (without diarrhoea), check for TB → 108.
- If lymph node suggestive of TB (≥ 2 cm, non-tender, asymmetrical, matted/firm/fluctuant, present for > 2 weeks, does not improve with antibiotics or discharging sinus), refer to doctor for fine needle aspiration (FNA).
- Test for HIV → 95. If CLHIV, give HIV routine care → 97.
- If HIV and TB excluded, discuss/refer same day if any of:
 - Generalised lymphadenopathy
 - Weight loss
 - Pallor³ → 46
 - Swollen abdomen
 - Lymph node growing quickly
 - Night sweats
 - Persistent tiredness
 - Shortness of breath

If none of the above, reassure. Advise to return if lymph node/s > 1 cm persists for > 2 weeks, new symptoms develop or lymph node grows: discuss/refer.

Generalised lymphadenopathy

If joint pain →74.

¹Report to DSD → 170. If any of: 1) sex not consensual, 2) < 12 years old, 3) ≥ 12 years old and partner not in peer group (partner must have < 2 year age gap), **sexual abuse** likely → 144. Otherwise advise reliable contraception and advise correct and consistent use of condoms → PACK Adult. ²Any child who in the last 12 months shared an enclosed space (school/creche/home/similar) for ≥ 1 night or frequent/extended daytime periods with an adult/adolescent with pulmonary TB ("index client"), during the 3 months before index client started TB treatment. ³Look for pale palms of the hands and conjunctival pallor: paleness of the lower inner eyelid.

Eye/vision symptoms

Give urgent attention to the child with eye/vision problems and any of:

- Baby < 1 month old with pus in eyes or swollen eyelids, conjunctivitis of the newborn likely
- Penetrating injury (with/without foreign body)
- Whole eyelid swollen/red or bulging eye, orbital/periorbital cellulitis likely
- Sudden onset redness and excessive eye discharge, with history of sexual contact/abuse → 144, gonococcal conjunctivitis likely
- Shingles involving eye or nose
- One painful red eye
- Corneal ulcer or hazy cornea
- Sudden onset/new squint
- Sudden loss/change in vision (blurred/reduced)
- Sudden drooping of eyelid
- Eyelid laceration
- Chemical burn
- Jaundice →61

Manage and refer urgently:

- If conjunctivitis of newborn likely, give ♦ceftriaxone¹ 50mg/kg (up to 2g) IM². Irrigate with ♦sodium chloride 0.9% hourly until referral. Give postnatal care to mother and her partner/s →PACK Adult.
- If gonococcal conjunctivitis likely: If < 12 years and < 45kg, give ▶ceftriaxone¹ 125mg IM. If > 12 years or ≥ 45kg, give ▶ceftriaxone¹ 250mg IM. If > 12 years also give ▶azithromycin 1g orally.
- If orbital/periorbital cellulitis likely and > 6 hours delay in referral expected, give ▶ceftriaxone¹ 80mg/kg IM → 159:11. Avoid injecting ≥ 1g IM at one injection site.
- If penetrating/metallic foreign body in eye, do not try to remove. Cover gently with protective shield and avoid lying flat.
- If chemical burn: wash eye/s continuously for at least 30 minutes with 1-2 litres of ♦sodium chloride 0.9% or sterile water.

Approach to child with eye/vision symptoms not needing urgent attention

- If white spot in pupil, white pupil, loss of red reflex with white light, cataract/retinoblastoma likely: discuss/refer same day. If longstanding squint, refer next eye OPD appointment.
- Look for discharge, red/swollen eyelids, foreign body and poor vision:

Both eyes are discharging/watery. Are eyes very itchy?		Red/swollen eyelids		Foreign body (FB)	Poor vision
Yes		No			
May have persistently blocked/runny nose.		Purulent discharge			
Allergic conjunctivitis likely		Bacterial conjunctivitis likely	Viral conjunctivitis likely		
• Advise to apply cold compresses ³ .		• Wipe eyes gently from inside out with clean cotton wool soaked in ♦sodium chloride 0.9% until pus clears.	• Apply cold compresses ³ .	• Wash with sterile water or ♦sodium chloride 0.9%.	• Refer to eye OPD within 1 month if: not meeting visual milestone, not responding to mother's face, wandering eye movements, pokes/prods own eye, staring at bright lights.
• If < 2 years, refer.		• Insert ♦chloramphenicol 1% ointment 6 hourly in each eye for 7 days.	• Give ▶paracetamol 15mg/kg 6 hourly as needed up to 5 days → 163:37.	• If FB not visible, use fluorescein stain and ultraviolet light.	• If ≥ 6 years and poor vision with Snellen E chart, refer to optometrist.
If 2-6 years: give ▶chlorpheniramine ⁴ 0.1mg/kg 6-8 hourly for 7 days → 159:15.		• May return to school after 2 days and no pus.	• If > 6 years old, give ▶oxymetazoline 0.025% 1 drop in each eye 6 hourly up to 7 days to reduce redness.	• Instil 1 drop ■oxybuprocaine 0.4% ⁵ eye drops, repeat at 90 second intervals if needed, up to 3 doses. Gently remove FB with cotton bud.	• If cannot see in dark, vitamin A deficiency likely, give single dose ▶vitamin A → 164:44 and refer. If eyes dry, also give lubricating eye drops.
If > 6 years: give ▶oxymetazoline 0.025% 1 drop 6 hourly up to 7 days.		• If no better in 2 days, refer to eye OPD.	• May return to school once discharge has cleared/after 1 week.	• Insert ■chloramphenicol 1% ointment 6 hourly for 5 days.	
• If no response after 7 days or if recurrent problem, change to ♦sodium cromoglycate 2% 1 drop 6 hourly for 1-3 months or long-term.		Advise to avoid rubbing eyes/sharing towels. Wash hands regularly.		• Refer same day if: removal unsuccessful, abnormal vision/eye movement, FB not visible, not opening eye after 24 hours.	
• If ongoing symptoms, add ▶cetirizine daily for 1-3 months or long-term: if 2-6 years give 5mg; if ≥ 6 years give 10mg.					
• If recurrent problem, also give routine allergy care → 117.					
• If brown discolouration of eyes, sensitivity to light or poor vision, refer urgently.					

¹Avoid giving Ringer's lactate together with IV/IM ceftriaxone. If severe penicillin allergy (previous anaphylaxis/urticaria/angioedema), discuss with doctor. ²IV not indicated. Avoid injecting ≥ 1g IM at one injection site. ³Wet clean facecloth with cold water for cold compresses and hot water (not boiling) for warm compresses, ring out facecloth, gently apply over eyes for 10 minutes. ⁴Chlorpheniramine is sedating. ⁵Strictly avoid giving oxybuprocaine to take home.

Ear symptoms/difficulty hearing

Ask about ear itch, pain, discharge from ear, foreign body, wax and difficulty hearing. Then look in ear.

Itchy ear

Ear canal red/swollen (pus may be present)



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Otitis externa likely

- Clean ear¹.
- Instil ►acetic acid 2% in alcohol after cleaning, 4 drops 6 hourly for 5 days.
- Give ►paracetamol 15mg/kg 6 hourly as needed up to 5 days ► 163:37.
- If severe pain, firm red swelling in canal, or temperature ≥ 38°C, give ►cephalexin² ► 159:13 or ►flucloxacillin² ► 161:23 for 5 days.
- Refer if:
 - No better after 5 days
 - Blisters on ear, herpes zoster likely

Painful ear

- Ear canal not red/swollen.
- Able to view eardrum?

No

Pain > 2 days or pain waking at night?

No

Has temperature been ≥ 38°C in last > 2 days?

No

Give ►paracetamol 15mg/kg 6 hourly as needed up to 5 days ► 163:37.

Advise to return in 2 days if no better.

Yes

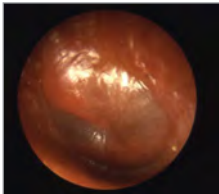
Acute otitis media likely

- Give ►paracetamol 15mg/kg 6 hourly as needed up to 5 days ► 163:37.
- If child has not had antibiotics in past 30 days, give ►amoxicillin² 45mg/kg 12 hourly for 5 days ► 158:6.
- If discharge, clean ear¹ and avoid getting ear wet.
- If > 1 episode, test for HIV ► 95.
- Advise to return if no better in 5 days: extend ►amoxicillin² for another 5 days and review after this.
- If child had amoxicillin in past 30 days or poor response to 10 days amoxicillin, give ►amoxicillin/clavulanic acid 45mg/kg 12 hourly for 5 days ► 158:8.
- If no response to treatment or >5 episodes per year, refer.
- Refer same day if:
 - Painful swelling behind ear, mastoiditis likely
 - Neck stiffness
 - Baby ≤ 1 month old
- If treated above but communication problem present ► 89.

Yes

If normal looking ear drum, referred pain likely, check for face, mouth, gum or tooth problems.

If red bulging eardrum:



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Discharge from ear³

Discharge for ≤ 2 weeks

Symptoms ≥ 2 weeks, hole in eardrum



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Chronic suppurative otitis media likely

- Explain that ear can only heal if dry.
- Clean ear¹.
- Avoid getting ear wet.
- If poor response to treatment, test for HIV ► 95 and TB ► 108.
- Refer if:
 - No better after 4 weeks
 - Bloody discharge for > 2 weeks
 - Large hole in drum
 - Difficulty hearing
- Refer same day if:
 - Neck stiffness
 - Painful swelling behind ear
 - Yellow/white deposit on eardrum, cholesteatoma likely.

Foreign body

Syringe ears⁴ with warm water.

Avoid syringing and refer instead if:

- Uncooperative
- Grommets
- Chronic suppurative otitis media
- Hole in eardrum
- Battery/food in ear
- Recent trauma to head or ear
- Neck stiffness

If unsuccessful after 3 attempts/ causes pain or if foreign body remains in ear, stop and refer to/ call doctor.

If no better, arrange for hearing test.

Wax

Difficulty hearing

- If on drug resistant TB medication, discuss with TB doctor.
- If normal looking ear: arrange for hearing test (same week, if concerns about complete deafness).
- If wax/foreign body, itchy/ painful ear, discharge from the ear, see adjacent.
- If fluid behind the eardrum, otitis media with effusion likely:

Otitis media with effusion likely

- Keep ear dry.
- Advise carer that usually resolves on own.
- If communication problem ► 89.
- Refer if still concerns about hearing after 3 months or if child clumsy/poor balance.



⁴How to syringe an ear: fill a 50-200mL syringe with warm water. Ask child/carer to hold container under ear to catch water. Pull ear upwards and backwards to straighten ear canal. Place tip of syringe at opening (no further than 8mm into canal) and spray water upwards into canal. Check after syringing to see if wax cleared.

¹Cleaning the ear (dry mopping): roll a piece of clean soft tissue into a wick. Insert wick into ear with twisting action. Remove and replace with clean dry wick. Repeat until wick is dry when removed. Never leave wick or other object inside ear. ²If severe penicillin allergy (history of anaphylaxis, urticaria or angioedema), give instead azithromycin 10mg/kg once daily for 3 days ► 158:10. ³If grommets (small tubes in eardrum) and purulent discharge persists > 2 weeks, discuss/refer.

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PALLIATIVE CARE

MEDICATIONS

Nose symptoms

Give urgent attention to the child with nose symptoms and any of:

- Newborn unable to breathe through nose
- Unable to feed because of blocked nose
- Head trauma with watery discharge from nose

Refer urgently.

Approach to the child with nose symptoms not needing urgent attention

Blocked/running nose or persistent snoring
Ask about duration and associated symptoms:

Sore throat or fever
• If rash →75.

Any of: temperature ≥ 38°C, chills, nausea, sore muscles?

No

Common cold likely

- Advise on cough/sneeze hygiene.
- If pain or fever causing discomfort, give ♦paracetamol 15mg/kg 6 hourly as needed up to 5 days → 163:37.
- Instil ♦sodium chloride 0.9% drops into nostrils as needed.
- Advise that antibiotics are not necessary.
- Advise to return if:
 - Symptoms persist > 7 days.
 - Initially better but fever returns and:
 - Productive cough →55.
 - Ear pain →49.
 - Pain over cheeks, sinusitis likely.
- If CLHIV or other long-term health condition, advise influenza vaccination during influenza vaccine campaign.
- If child known with asthma → 119.

Yes

Acute viral infection likely (like influenza or COVID-19)
Keep child home from school/ creche until well.

Headache worse on bending forward, pain over cheeks

Sinusitis likely

- Give ►paracetamol 15mg/kg 6 hourly as needed up to 5 days → 163:37.
- Give ►sodium chloride 0.9% drops into nostrils as needed.
- If no better, give ►oxymetazoline 0.025% 2 drops in each nostril 8 hourly for up to 5 days.
- If symptoms > 10 days, give ►amoxicillin¹: if ≤ 3 years, give 45mg/kg 12 hourly → 158:6; if > 3 years, give 500mg 8 hourly for 5 days.
- If > 1 episode, test for HIV → 95.
- Refer if:
 - Poor response to antibiotic
 - > 4 episodes per year
 - Neck stiffness
 - Temperature ≥ 38°C for > 2 days
- Refer same day if swelling around sinus/eye or tooth infection.

Recurrent sneezing or itchy/blocked nose/rubbing nose most days for > 4 weeks

Allergic rhinitis likely

- If < 4 years old, refer.
- If 4-6 years old: give ►fluticasone² nasal spray 50µg 12 hourly in each nostril. If symptoms > 1 month, give ►cetirizine 5mg once daily.
- If ≥ 6 years old, give ►fluticasone² nasal spray 100µg 12 hourly in each nostril. If symptoms > 1 month, give ►cetirizine 10mg once daily.
- For symptom relief:
 - Give ►chlorpheniramine 0.1mg/kg 6-8 hourly for up to 2 weeks → 159:15.
 - If blocked nose at night affecting sleep, give ►oxymetazoline 0.025% 2 drops in each nostril at night for up to 5 nights.
- Give routine care to child with allergy → 117.
- Review 3 monthly.

Only one nostril affected

Foreign body likely

- Examine nostrils and ears using good light and nasal speculum. Wrap child in sheet if need to restrain.
- Gently remove object with crocodile forceps. If unable, block clear nostril and ask child to blow out through his/her nose.
- If unsuccessful or object not visible, refer.

Persistent snoring and poor sleep

Obstructive sleep apnoea likely

- If child with allergy →117.
- If overweight, plot growth and give routine care →129.
- Refer if enlarged tonsils, episodes of no breathing for 10 seconds/ chokes/gasps while sleeping.

Bleeding nose

- Check for foreign body.
- If able, firmly pinch nostrils together for 5-10 minutes with child sitting and leaning forward.
- If still bleeding, insert ►bismuth iodoforn paraffin paste (BIPP) soaked ribbon gauze into nostril/s.
- If bleeding stops, advise to return next day to remove BIPP gauze.
- If still bleeding or unable to do above procedures, refer urgently.
- If recurrent bleeds:
 - Advise to avoid picking nose.
 - If rubbing nose, consider allergic rhinitis.
 - If no better, refer.

¹If severe penicillin allergy (history of anaphylaxis, urticaria or angioedema), give instead ►azithromycin once daily for 3 days → 158:10. ²Aim nozzle upwards and outwards (aim for the eye). Avoid aiming for back of throat or sniffing vigorously.

Mouth symptoms

Give urgent attention to the child with mouth symptoms and any of:

- Unable to open mouth
- Sudden onset of generalised itch/rash or face/tongue itch/swelling and any of: tight throat/hoarseness, difficulty breathing, wheeze, shock¹, dizziness/collapse, abdominal pain/nausea/vomiting/diarrhoea or exposure to possible allergen², anaphylaxis likely →37

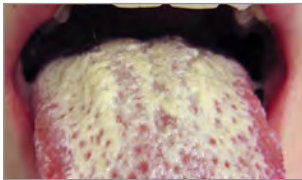
Manage and refer urgently:

- Check fingerprick glucose. If ≤ 28 days old and glucose < 2.6mmol/L → 9. If > 28 days old and glucose < 3mmol/L (< 4 if diabetes) or ≥ 11mmol/L → 32.

Assess the child with mouth symptoms not needing urgent attention

If problem with gums/teeth or if child < 3 years and drooling, consider teething → 54. Wear a mask while examining the mouth. Check for white patches, redness, cracks, blisters, or ulcers.

White patches on cheeks, gums, tongue, palate



James Heilman, MD via Wikimedia Commons (CC BY-SA 4.0)

Oral thrush/candida likely

- If fever, generalised red rash and white spots in mouth/on inside of cheeks, measles likely → 80.
- Give ◆nystatin suspension 1mL 6 hourly after meal/feed for 7 days. Keep inside mouth for as long as possible.
 - If baby, advise carer to apply inside mouth with clean finger. If breastfeeding and nipple painful, apply ►clotrimazole cream to nipples after feed.
 - Continue both for 2 days after cure.
- Give ►paracetamol 15mg/kg 6 hourly as needed up to 5 days → 163:37.
- If child on inhaled corticosteroids, use spacer and rinse mouth with water after use → 57.
- If CLHIV, give routine care → 97. If status unknown, test for HIV → 95.
- If recurrent candida in child with a life-limiting illness, also give palliative care → 150.

If difficulty/painful swallowing, refusing feeds, drooling or hoarse cry, oesophageal candida likely. Discuss/refer.

Red, cracked corners of mouth

Angular cheilitis/stomatitis likely

- Apply ◆zinc and castor oil ointment 8 hourly.
- If also oral candida, see adjacent column and apply ►clotrimazole cream 12 hourly for 2 weeks.
- If crusts/blisters around mouth, impetigo likely → 83.
- If very itchy, contact dermatitis likely. Identify and remove irritant.
- If on inhaled corticosteroids, advise to rinse mouth after use.
- If no better or uncertain of cause:
 - Assess growth → 15.
 - Test for HIV → 95.
 - Check fingerprick glucose → 32.
 - If still uncertain, refer.

Groups of painful blisters on lips/mouth
Are blisters also on palms and soles?

Yes



MidgleyDJ via Wikimedia Commons (CC BY-SA 4.0)

Hand, foot and mouth disease likely → 76.

No



Herpes simplex likely

- Apply ►petroleum jelly to blisters on outside of mouth to prevent spread.
- Give ►paracetamol 15mg/kg 6 hourly as needed up to 5 days → 163:37.
- If ≥ 6 years old, apply thin layer of ►tetracaine 1% gel to blisters 6 hourly.
- If CLHIV or extensive herpes (and < 72 hours from onset), give ►aciclovir 8 hourly for 7 days → 157:2.
- Refer if extensive/recurrent or no better after 7 days.
- If status unknown, test for HIV → 95.

Painful ulcer/s with central white patch



Litalm33 via Wikimedia Commons (CC BY-SA 4.0)

Aphthous ulcer/s likely

- Give ►paracetamol 15mg/kg 6 hourly as needed up to 5 days → 163:37.
- Rinse with salt water³ for 1 minute 12 hourly.
- If ≥ 6 years old, apply ►tetracaine 1% gel on ulcers 6 hourly until healed.
- If recurrent, test for HIV → 95.
- Refer if large (> 1cm) or not healed within 10 days.
- If recurrent ulcers in child with a life-limiting illness, also give palliative care → 150.

Advise the child/carers to avoid spicy, hot, sticky, dry or acidic food and to eat soft, moist food. Keep mouth and teeth clean by brushing and rinsing regularly.

¹If ≥ 2 of: cold hands/feet, weak/fast pulse → 175, capillary refill time (CRT) ≥ 3 seconds, decreased level of consciousness → 174, shock likely. ²Common allergens include medications, new food or an insect bite/sting within the last few hours. ³Mix 1/2 teaspoon of salt in 1/2 cup of lukewarm water.

Throat symptoms

Give urgent attention to the child with throat symptoms and any of:

- Unable to open mouth
- Unable to swallow at all (recent onset drooling)
- Red swelling blocking airway
- Sudden onset of generalised itch/rash or face/tongue itch/swelling and any of: tight throat/hoarseness, difficulty breathing, wheeze, shock¹, dizziness/collapse, abdominal pain/nausea/vomiting/diarrhoea or exposure to possible allergen², **anaphylaxis likely** →37

Manage and refer urgently:

- Check fingerprick glucose. If ≤ 28 days old and glucose < 2.6mmol/L → 9. If > 28 days old and glucose < 3mmol/L (< 4 if diabetes) or ≥ 11mmol/L → 32.

Assess the child with throat symptoms not needing urgent attention

Wear a mask while examining the throat. Check for redness, grey membrane, bleeding and white patches.

Is there sore throat and a thick grey layer (pseudomembrane) in throat/nose that bleeds easily if touched, with or without mild fever?

No

Yes

Are there swollen tonsils and any of: pus/white patches on tonsils or no cough or no runny nose?

No

Yes

Red throat and/or swollen tonsils, tender neck lymph nodes, cough, runny nose

Viral tonsillopharyngitis likely

- Give ♦paracetamol 15mg/kg 6 hourly as needed up to 5 days → 163:37.
- Advise salt water gargle (mix ½ teaspoon salt in ½ cup lukewarm water) twice a day.
- Explain that antibiotics are not necessary.

Bacterial tonsillopharyngitis likely

- Child may also have a generalised red rash, red spots in back of throat or a red tongue with bumps (strawberry tongue).
- Give ♦paracetamol 15mg/kg 6 hourly as needed for up to 5 days → 163:37.
- Give **one** of:
 - Single dose ♦benzathine benzylpenicillin^{3,4}: if < 30kg, give 600 000 units IM; if ≥ 30kg, give 1.2 million units IM or
 - ♦amoxicillin³ 50mg/kg once daily for 10 days → 158:7 or
 - ♦phenoxymethylpenicillin³ 12 hourly for 10 days: if 18 months-11 years, give 250mg; if ≥ 11 years, give 500mg.



Doc James via Wikimedia Commons (CC BY-SA 4.0)

- If red rash develops only after starting antibiotic, **glandular fever** likely.
 - Stop antibiotic. Reassure rash will resolve spontaneously.
 - Child may return to school when better but can only resume sporting activities > 3 weeks from onset of illness.
- If no better or worse after 5 days, review: assess and manage fluids → 26, continue antibiotics and review again in 5 days. If still no better, refer.
- If ≥ 6 episodes per year or persistent snoring, refer to ENT.



Dileepunnikri via Wikimedia Commons (CC BY-SA 4.0)

Consider diphtheria

- Isolate if possible, but ensure child can be safely monitored.
- Notify⁵.
- Refer.

Advise to return immediately if any of: painful or swollen joint/s, strange movements of limbs or face, lumps over joints/tendons or rash (round lesions with pale centre) to exclude **rheumatic fever**. Discuss/refer.

¹If ≥ 2 of: cold hands/feet, weak/fast pulse → 175, capillary refill time ≥ 3 seconds, decreased level of consciousness → 174, **shock** likely. ²Medications, new food, insect bite/sting within the last few hours. ³If severe penicillin allergy (history of anaphylaxis, urticaria or angioedema), give instead ♦azithromycin 10mg/kg once daily for 3 days → 158:10. ⁴Dissolve benzathine benzylpenicillin 1.2 million units in 3.2mL ♦lidocaine 1% without adrenaline. ⁵Immediately report to provincial CDC telephonically and via email → 170. Notify within 24 hours via NMC App or www.nicd.ac.za/nmc-overview/notification-process. Complete NMC case notification form and case investigation form and send to NMCsurveillanceReport@nicd.ac.za and provincialcdc@westerncape.gov.za.

Face symptoms

If eye symptoms →48, mouth symptoms →51, nose symptoms →50, teeth symptoms →54.

Give urgent attention to the child with face swelling and any of:

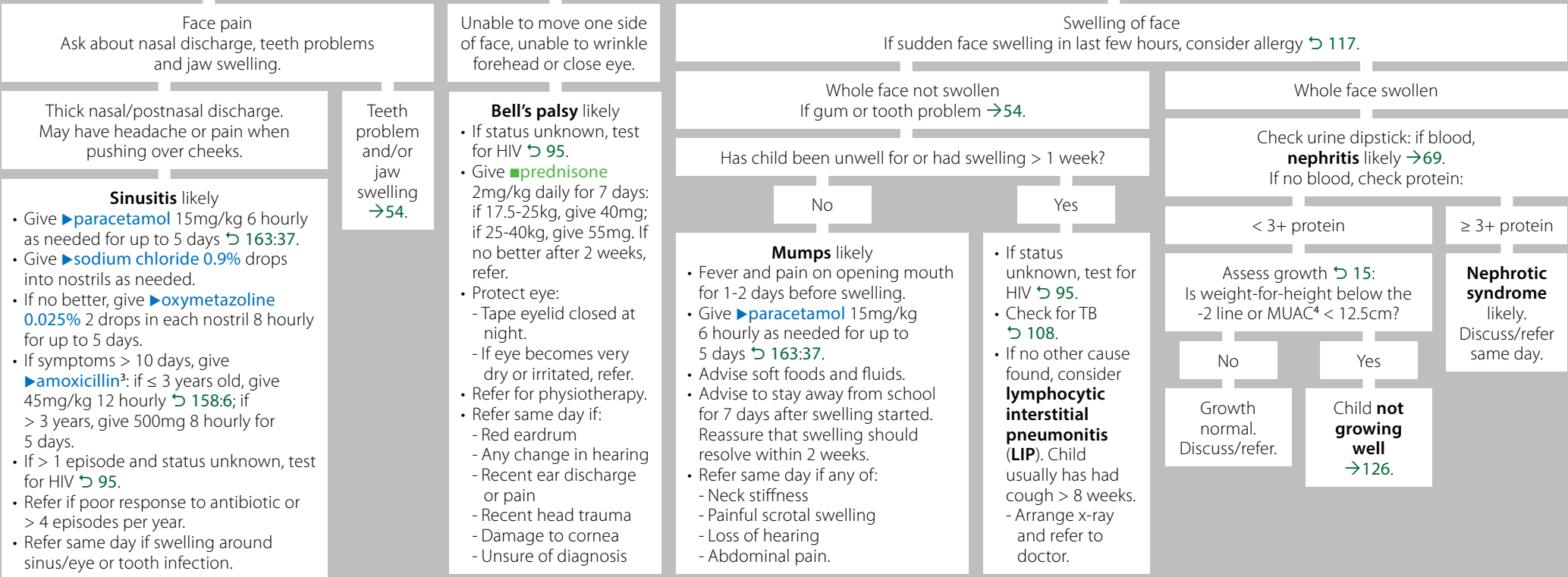
- Temperature ≥ 38°C, **cellulitis** likely
- Dark red/brown urine or little/no urine →69
- Sudden onset of generalised itch/rash or face/tongue itch/swelling and any of: tight throat/hoarseness, difficulty breathing, wheeze, shock¹, dizziness/collapse, abdominal pain/nausea/vomiting/diarrhoea or exposure to possible allergen², **anaphylaxis** likely →37

Manage and refer urgently:

- If **cellulitis** likely, give ▶**cephalexin**³ ↪ 159:13 or ▶**flucloxacillin**³ ↪ 161:23.

Approach to the child with face symptoms not needing urgent attention

- If rash on face →75.
- If unusual facial features or syndrome suspected, check milestones ↪ 12.
- Manage according to face symptom/s:



Gum/teeth symptoms

Give urgent attention to the child with gum/teeth symptoms and any of:

- Temperature $\geq 38^{\circ}\text{C}$ and red or painful cheek
- Unable to eat or drink

Refer urgently.

Approach to the child with gum and teeth symptoms not needing urgent attention

- If no teeth by 3 years old, refer to dentist.
- If grinding teeth, reassure carer that it should resolve by 10 years old. If child stressed, unhappy or angry → 139.
- Manage according to symptoms and findings:
 - Ask about drooling/biting on hard objects, bleeding gums, tooth pain or sensitivity, or face/jaw swelling
 - Lift the lip to look at teeth and gums:

Child < 3 years old, drooling or biting on hard objects

Teething likely

- If child unwell, look for cause, e.g. fever → 43, diarrhoea → 63.
- Reassure carer that the teething process is normal.
- Advise carer to massage gum gently and encourage biting on objects (eg. teething ring).
- Cooled objects to bite on can help ease symptoms.
- Advise against using local numbing preparations on gums.

Gums bleeding *and/or* red

Gum problem likely

- Advise to brush and floss teeth twice daily.
- Rinse mouth with salt water¹ for 1 minute twice daily.
- Give ►chlorhexidine 0.2% mouthwash twice daily for 5 days:
 - If ≥ 7 years old, swirl in mouth but do not swallow.
 - If < 7 years old, apply with gauze to area.
 - Avoid repeated use as can damage teeth.
 - Advise to wait 30 minutes before eating/drinking.
- If pain, give ►paracetamol 15mg/kg 6 hourly as needed up to 5 days → 163:37.
- Look for dental caries.
- Refer to dentist if:
 - No better after 5 days
 - Foul-smelling breath
 - Temperature $\geq 38^{\circ}\text{C}$
 - Loss of supporting bone and gum around tooth
 - HIV.

Brown/black staining of teeth at gumline, holes, pits or missing teeth



© University of the Western Cape

Dental caries likely

- Child might complain of tooth pain with hot/cold/sweet foods.
- Refer to dentist.
- If known with a heart valve problem, give antibiotic prophylaxis before extraction → 133.
- Check growth → 15 and Hb. If Hb < 10g/dL in child < 5 years old or Hb < 11g/dL in child ≥ 5 years old, anaemia likely → 46.
- Refer to dietitian and give dietary advice → 124.

Pus in mouth *and/or* swelling next to tooth or on face/jaw



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Dental abscess likely

- Give ►paracetamol 15mg/kg 6 hourly as needed up to 5 days → 163:37.
- If painful facial swelling and redness, give ►amoxicillin² 10-20mg/kg 8 hourly for 5 days: if 11-25kg, give 250mg. If ≥ 25 kg, give 500mg. Also give ►metronidazole 7.5mg/kg 8 hourly for 5 days → 162:33.
- Advise to return if temperature $\geq 38^{\circ}\text{C}$ or red/painful cheek over tooth/gum no better after 2 days and refer.
- Refer to dentist for tooth extraction.

¹Mix ½ teaspoon of salt in ½ cup lukewarm water. ²If severe penicillin allergy (history of anaphylaxis, urticaria or angioedema), give instead ►azithromycin 10mg/kg once daily for 3 days → 158:10.

Cough and/or breathing problems

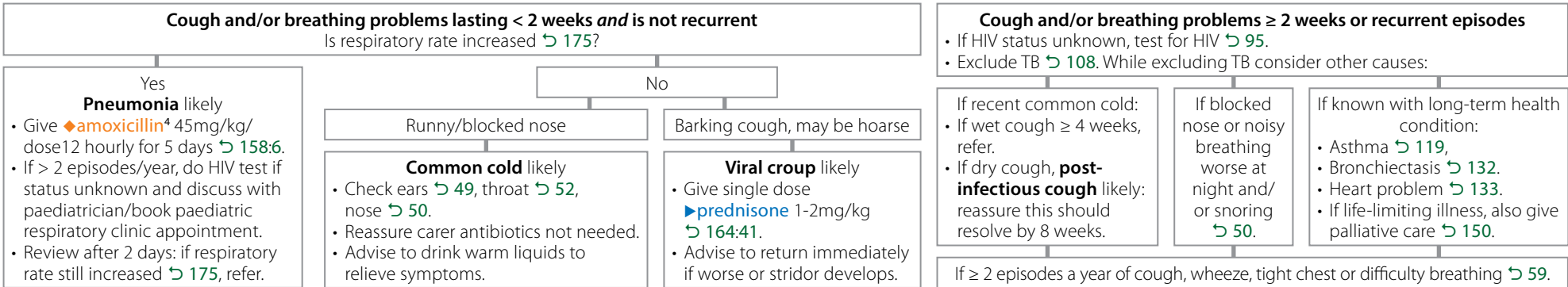
The child with breathing problems may have noisy breathing, wheeze, grunting, snoring or stridor (noisy, high-pitched breathing). If child not breathing → 24.

Give urgent attention to the child with cough/difficulty breathing and any of:

- Baby < 2 months old
 - History of apnoea (episodes of no breathing > 10 seconds)
 - Unable to drink/feed
 - Tires/sweats during feeds
 - Lower chest indrawing
- Nasal flaring
 - Grunting
 - Stridor
 - Blue lips/tongue
 - Sats ≤ 92%
- Restless or irritable
 - Lethargy or decreased level of consciousness → 174
 - Sudden onset of generalised itch/rash or face/tongue itch/swelling and any of: tight throat/hoarseness, difficulty breathing, wheeze, shock¹, dizziness/collapse, abdominal pain/nausea/vomiting/diarrhoea or exposure to possible allergen², **anaphylaxis** likely → 37.
- Manage and refer urgently:**
- Give ♦oxygen 2L/minute via nasal prongs. If < 1 year old with blocked nose, instil ♦sodium chloride 0.9% solution 1 drop into each nostril and suction the nose.
 - If wheeze → 56 or if known heart problem → 133.
 - Check fingerprick glucose. If ≤ 28 days old and glucose < 2.6mmol/L → 9. If > 28 days old and glucose < 3mmol/L (< 4 if diabetes) or ≥ 11mmol/L → 32.
 - Give ♦ceftriaxone³ 80mg/kg IM → 159:11. Avoid injecting ≥ 1g IM at one injection site. If < 1 year old and CLHIV/HIV status unknown, also give single dose ♦co-trimoxazole → 160:18.
 - If stridor, encourage carer to keep child calm:
 - Give ♦prednisone 1-2mg/kg → 164:41. Give 2mL ♦adrenaline (1:1000) in 2mL ♦sodium chloride 0.9% via nebuliser with ♦oxygen at 8L/minute every 15 minutes until stridor disappears.

Approach to the child with cough and/or breathing problems not needing urgent attention:

Ask about duration and number of episodes:



Wheeze/tight chest

If sudden wheeze/tight chest and any of: exposure to possible allergen¹ or sudden generalised itch/rash, face/tongue itch/swelling, shock², dizziness/collapse, abdominal pain/nausea/vomiting/diarrhoea, **anaphylaxis** likely →37. If history of recent choking, refer same day. If known heart problem: if gradual onset wheeze/tight chest or swelling of both legs, **heart failure** likely →133.

Give urgent attention to the child who is wheezing or currently has a tight chest with accessory muscle use³:

Does child have one or more of:
• Oxygen saturation < 92% (or blue lips/tongue) • Silent chest (tight chest with no wheeze) • Confused/drowsy/agitated • Unable to feed/speak • PEFR⁴ (Peak expiratory flow rate) < 33% → 58

Yes: **Life threatening**
Give ♦oxygen to keep saturation 93-95% while preparing nebuliser.

Refer urgently. While awaiting transport:
• Nebulise 2mL ♦salbutamol 0.5% solution and 2mL ♦ipratropium bromide 0.25mg/mL with ♦oxygen at 8L/min. Repeat every 20 minutes if needed, up to 4 doses in 2 hours.
• Give single dose ▶hydrocortisone 4mg/kg IM/slow IV immediately → 161:27.
• If child deteriorates, discuss urgently with doctor for possible IV magnesium sulphate or IV salbutamol and intubation.

No. Does child have one or more of:
• Increased respiratory rate → 175

Yes: **Severe**
Give ♦oxygen to keep saturation 93-95% while preparing nebuliser.

Refer urgently. While awaiting transport:
• Nebulise 1mL ♦salbutamol 0.5% solution⁴ and 3mL ♦sodium chloride 0.9% with ♦oxygen at 8L/min or give inhaled ♦salbutamol 400-1000mcg (4-10 puffs).
• If asthma likely, give single dose ♦prednisone 1-2mg/kg → 164:41. If unable to take orally, give single dose ▶hydrocortisone 4mg/kg IM/slow IV → 161:27.
Assess response:

Worsening
Treat as **life threatening** wheeze in adjacent column.

No response:
If < 2 years old, **severe bronchiolitis** likely: continue ♦oxygen while awaiting transfer.
If no response/unable to maintain good response and ≥ 2 years old:
• Give, 2mL ♦ipratropium bromide 0.25mg/mL with 0.5-1mL ♦salbutamol 0.5% solution and ♦sodium chloride 0.9% made up to 4mL. Repeat every 20 minutes if needed, up to 4 doses in 2 hours or
• Give inhaled ♦salbutamol 400-1000mcg (4-10 puffs) and ♦ipratropium bromide 80-160mcg (2-4 puffs) every 20 minutes as needed, for up to 3 hours.

Good response
Continue nebulised or inhaled ♦salbutamol as needed every 20 minutes.

No. Does child have one or more of:
• Difficulty to feed/speak • PEFR⁴ (Peak expiratory flow rate) < 60% → 58

No: **Mild or moderate**
• Give ♦salbutamol: if ≥ 5 years give 400-1000mcg (4-10 puffs); if < 5 years give 200-600mcg (2-6 puffs) with metered dose inhaler (MDI) and large volume (500mL) spacer. If < 3 years old, use spacer with mask or
• Nebulise ♦salbutamol 0.5% solution: if < 3 years give 0.5mL; if ≥ 3 years give 1mL in ♦sodium chloride 0.9% made up to 4mL, with ♦oxygen at 8L/min.
• If asthma likely → 59, also give single dose ♦prednisone 1-2mg/kg → 164:41.
Assess response after 20 minutes:

Worsening (≥ 1 of: increased respiratory rate → 175, sats < 92%, chest indrawing, < 20% improvement in PEFR → 58, difficulty to feed/speak.
Treat as **severe** wheeze in adjacent column.

If no response and < 2 years old, **bronchiolitis** likely →57.
• If no response/unable to maintain good response:
Repeat ♦salbutamol every 20 minutes for a total of 1 hour. Assess response every 20 minutes, for 1 hour:

• If symptoms recur/unable to maintain good response: discuss/refer.
• If able to maintain good response, discharge home: give ♦salbutamol 200mcg (2 puffs) 6 hourly for 5 days. Continue ▶prednisone 1-2mg/kg → 164:41 daily for 7 days. Avoid antibiotics unless temperature ≥ 38°C with either thick yellow/green sputum or increased respiratory rate → 175.; give ▶amoxicillin⁵ 45mg/kg 12 hourly for 5 days → 158:6. Check if appropriate device for age and correct inhaler technique → 57. Advise to identify and avoid triggers⁶. Advise to return if symptoms worsen or child needs > 10 puffs in 24 hours.
• If not known asthma with recurrent episode of wheeze, consider **asthma** →59. If known asthma, give routine care →119.
• Review in 5 days. Stress importance of follow up to prevent further episodes.

Good response (all of): normal respiratory rate → 175, no chest indrawing, > 20% improvement in PEFR, easily talking or drinking.
Monitor for at least 1 hour

Symptoms recur
Repeat ♦salbutamol every 20 minutes for a total of 1 hour. Assess response every 20 minutes, for 1 hour:

• If symptoms recur/unable to maintain good response: discuss/refer.
• If able to maintain good response, discharge home: give ♦salbutamol 200mcg (2 puffs) 6 hourly for 5 days. Continue ▶prednisone 1-2mg/kg → 164:41 daily for 7 days. Avoid antibiotics unless temperature ≥ 38°C with either thick yellow/green sputum or increased respiratory rate → 175.; give ▶amoxicillin⁵ 45mg/kg 12 hourly for 5 days → 158:6. Check if appropriate device for age and correct inhaler technique → 57. Advise to identify and avoid triggers⁶. Advise to return if symptoms worsen or child needs > 10 puffs in 24 hours.
• If not known asthma with recurrent episode of wheeze, consider **asthma** →59. If known asthma, give routine care →119.
• Review in 5 days. Stress importance of follow up to prevent further episodes.

Symptoms do not recur

¹Common allergens include medications, new food or insect bite/sting within the last few hours. ²If ≥ 2 of: cold hands/feet, weak/fast pulse → 175, capillary refill time (CRT) ≥ 3 seconds, decreased level of consciousness → 174, shock likely. ³Accessory muscle use is any of: indrawing inbetween and below the ribs (intercostal/subcostal recession), indrawing above the ribs at the base of the throat (tracheal tug), use of neck muscles, head bobbing. ⁴Percentage of predicted value. ⁵If penicillin allergy (history of anaphylaxis, urticaria or angioedema), give instead ▶azithromycin daily for 3 days → 158:10. ⁶Irritant triggers: cigarette smoke and chemicals. Allergic triggers like animals, dust, mould, pollen, grass to be avoided if confirmed to be an allergen.

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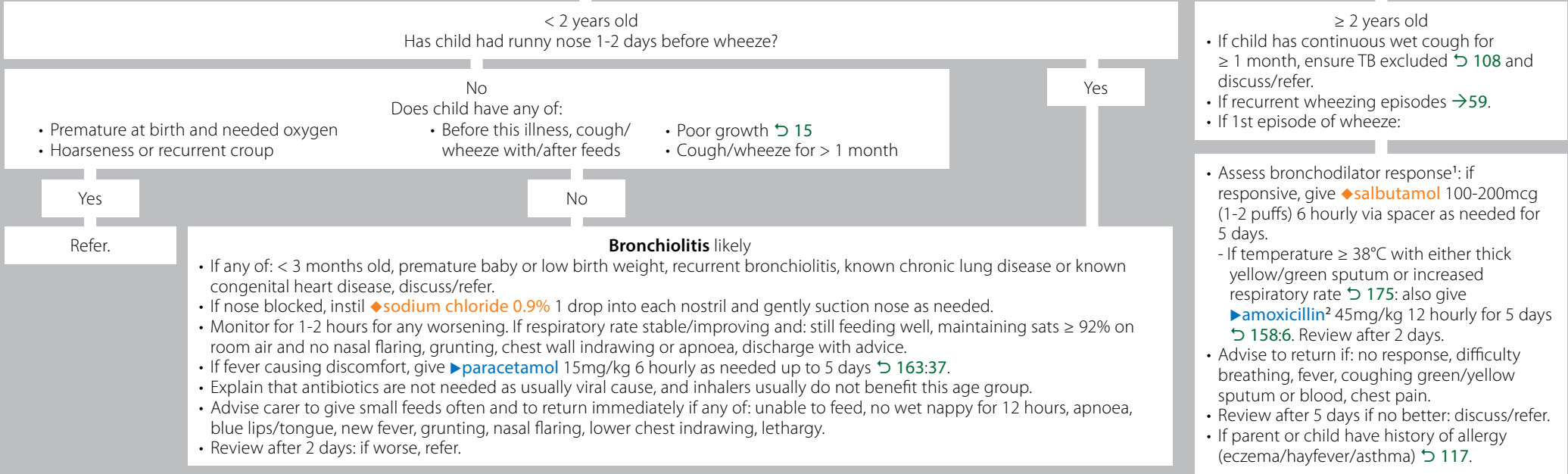
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Approach to the child with wheeze not needing urgent attention

- Confirm wheeze: ask carer to describe wheeze, if noisy breathing/snoring/gurgling noises/stridor without wheeze →55.
- If history of choking or wheeze only on one side of chest, refer same day.
- If longstanding persistent wheeze present most of the time, refer.
- If intermittent wheeze or wheeze for short duration (hours or days), manage further according to age:



How to use an inhaler with a spacer (with or without a mask)

- Spacers improve delivery of inhaled medications to lungs. Every child with asthma should have and use a spacer. If ≥ 3 years old, use mouthpiece with spacer. If < 3 years old, use face mask with spacer.
- Prime spacer initially with 10 puffs of medication. When medication is finished, replace only the inhaler.
- Clean spacer monthly: remove inhaler and wash spacer with soapy water. Do not rinse with water. Allow to drip dry (no need to re-prime).
- Demonstrate inhaler technique 2-3 times until child and/or carer understand.
- Then ask child and/or carer to show you how to use it.



1

- Remove cap from inhaler and spacer.
- Shake inhaler for 5 seconds and insert into spacer.



2a



2b

- If > 3 years old, child to put mouthpiece into mouth and close lips around it.
- Hold baby on lap and firmly hold mask over baby's nose and mouth.



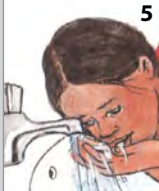
3

Press pump down once and allow 6 deep breaths before continuing³.



4

Remove inhaler and spacer and wait for 30 seconds before repeating. Repeat for each puff prescribed.



5

Rinse mouth after using inhaled corticosteroid (e.g. budesonide).

¹Give ►salbutamol via spacer 400mcg (4 puffs) and assess response after 15 minutes: if wheeze improves, child is responsive. If no better, child is bronchodilator unresponsive. ²If severe penicillin allergy (history of anaphylaxis, urticaria or angioedema), give instead ►azithromycin 10mg/kg once daily for 3 days → 158:10. ³Advise the older child to breathe out as far as possible, press pump down once and breathe in slowly but deeply for 1 full breath. Hold breath for 10 seconds, then breathe out through nose.

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How to measure and interpret peak expiratory flow rate (PEFR)

If child > 5 years old, measure and interpret PEFR to: 1) help diagnose asthma and 2) assess treatment response during acute asthma exacerbation.

How to measure peak expiratory flow rate (PEFR)



Move marker to bottom of numbered scale.



- Stand up and take a full, deep breath.
- Hold breath, place mouthpiece between teeth and tongue below mouthpiece.
- Form a seal with lips.



Breathe out as hard and as fast as possible (keeping fingers clear of scale).



- Read the result.
- Move marker back to bottom and repeat twice. Use the highest of the 3 results.

How to assess response to inhaled beta-agonist

Ideally, arrange lung function tests using spirometry to accurately diagnose and assess asthma. If spirometry unavailable, use PEFR as below.

Calculate % PEFR response to inhaled beta-agonist to:

1) Help diagnose asthma

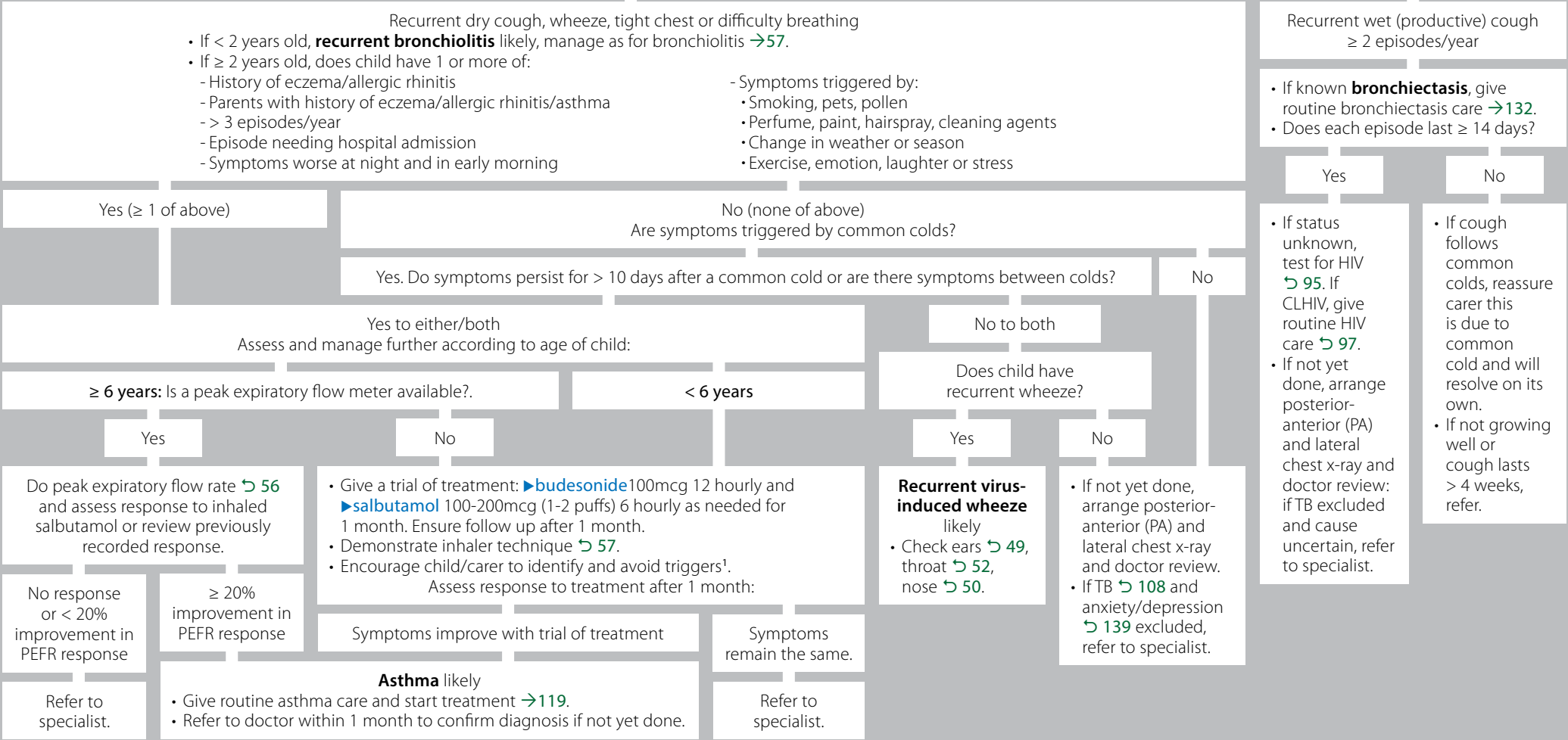
2) Assess response to treatment during acute asthma exacerbation

- Measure 'initial PEFR'. Use the highest reading of 3 results.
- Give inhaled [salbutamol](#) 400mcg (4 puffs via a spacer) and wait for 15 minutes.
- Repeat PEFR - this is the 'repeat PEFR'.
- Calculate % PEFR response = $\frac{\text{repeat PEFR} - \text{initial PEFR}}{\text{Initial PEFR}} \times 100$
- If % PEFR response is $\geq 20\%$, **asthma** likely or good response to treatment if acute asthma exacerbation [↪ 119](#).
- If % PEFR response is $< 20\%$, manage further [↪ 56](#).

Recurrent respiratory symptoms

The child with recurrent respiratory symptoms has repeated episodes of cough, wheeze, tight chest or difficulty breathing.

Approach to the child with recurrent respiratory symptoms:
Exclude TB → 108. Ask about nature of cough (wet or dry) and difficulty breathing:



Abdominal symptoms

Give urgent attention to the child with an abdominal problem and any of:

- Guarding and rigidity during peri-umbilical percussion, **peritonitis** likely
 - Tender in right lower abdomen and vomiting, **appendicitis** likely
 - Cramping pain and jelly-like stool, **intussusception** likely
- Tender, elevated testes
 - Painful groin/umbilical swelling
 - Rash and joint pain
- Vomiting, deep sighing respiration, fatigue, **diabetic ketoacidosis** likely: check blood glucose [↗ 32](#)
 - No stool/wind for 24 hours and vomiting
- Bile-stained vomiting

Manage and refer urgently:

- Assess and manage child's fluid needs [↗ 26](#). Keep nil by mouth. Give maintenance fluid IV according to reference guide [↗ 175](#).
- If baby < 1 month old or **peritonitis** or **appendicitis** likely, give [▶ ceftriaxone¹](#) 80mg/kg IM [↗ 159:11](#). Avoid injecting ≥ 1g IM at one injection site.
- If pain, give [▶ paracetamol](#) 15mg/kg [↗ 163:37](#). If pain and **peritonitis** or **appendicitis** likely, give oral [■ morphine](#) [↗ 163:36](#).

Approach to the child with an abdominal problem not needing urgent attention

- If recent injury/trauma [→ 33](#), if jaundice [→ 61](#). If temperature ≥ 38°C or history of fever [→ 43](#), check throat: if white patches on tonsils [→ 52](#).
- Check urine: if burning urine or nitrites/leucocytes/blood on dipstick [→ 69](#).
- If any of: significant TB exposure², cough, unexplained fever, drenching night sweats, poor weight gain, weight loss, failure to thrive, lethargy, decreased playfulness, visible neck mass, blood-stained sputum, headache, persistent vomiting (without diarrhoea), test for TB [↗ 108](#).
- Is there abdominal swelling or a mass?

Yes

No

Mass or localised swelling

- If mass felt in abdomen, refer to surgeon same week.
- If bulge on crying/cough/passing stool:
 - If in groin area, **inguinal hernia** likely, discuss with doctor/surgeon same day.
 - If in umbilical area, **umbilical hernia** likely. Reassure. Refer to surgeon if still present when child ≥ 4 years old.

Generalised swelling

- Exclude TB [↗ 108](#).
- Do urine dipstick:
 - ≥ 3+ protein, **nephrotic syndrome** likely, discuss/refer same day.
- Assess growth [↗ 15](#).
 - If growth problem [↗ 126](#).
 - If growth normal, discuss/refer.

- If stools infrequent and any of: pain, impaction, involuntary leakage or voluntary withholding, **constipation** likely [↗ 66](#).
- If child > 12 months old, deworm 6 monthly with [◆ mebendazole](#) [↗ 162:31](#). Educate on personal hygiene and avoid uncooked meat.
- If no constipation or worms, manage abdominal problem according to age of child:

≤ 4 months

4 months - 10 years old
Discuss/refer.

≥ 10 years old

- If crying for ≥ 3 hours/day on ≥ 3 days a week, doctor to assess baby:
- Exclude reflux [↗ 62](#).
 - Assess growth [↗ 15](#). If baby growing well and appears well, **colic** likely:
 - Give feeding tips:
 - Avoid over/under feeding.
 - Burp adequately after each feed.
 - Breastfeed if possible.
 - Feed in semi-upright position.
 - Assess and advise the carer:
 - Discourage smoking in household.
 - Promote a calm, quiet home environment.
 - Screen for carer depression [↗ PACK Adult](#).
 - Reassure colic should resolve by 4 months old.

- If girl and pain around time of period, dysmenorrhoea likely.
 - Give [▶ ibuprofen³](#) 200mg 8 hourly with food for 3 days.
 - Reassure that it is common, encourage to carry on with everyday activities.
- If girl and sexually active⁴:
 - If lower abdominal pain and/or vaginal discharge, **pelvic infection** likely, discuss with doctor.
 - If lower abdominal pain with amenorrhoea or vaginal bleeding 6-8 weeks after last period, **ectopic pregnancy** likely, refer same day.
 - If abuse suspected [↗ 144](#).
- Is there pain with bloating and diarrhoea?

Yes
Functional abdominal pain likely, refer to specialist.

No
Daily short-lasting pain not related to stool pattern/meal times, screen for anxiety [↗ 139](#).

If cause unclear or not resolved, refer.

¹Avoid giving Ringer's lactate together with IV/IM ceftriaxone. If severe penicillin allergy (previous anaphylaxis, urticaria, angioedema), discuss with doctor. ²Any child who in the last 12 months shared an enclosed space (school/creche/home/similar), for ≥ 1 night or frequent/extended daytime periods, with an adult/adolescent with pulmonary TB ("index patient"), during the 3 months before index client started TB treatment. ³Avoid ibuprofen if asthma, heart failure or kidney disease. ⁴Report to DSD [↗ 170](#). If any of: 1) sex not consensual 2) < 12 years old 3) ≥ 12 years old and partner not in peer group (partner must have < 2 year age gap) **sexual abuse** likely [↗ 144](#). Otherwise advise reliable contraception [↗ PACK Adult](#). Check that s/he knows how to use condoms.

Jaundice

Give urgent attention to the child with jaundice and any of:

- Newborn within first 24 hours of birth
 - Baby ≤ 10 days old with ≥ 10% weight loss
- Confused or decreased level of consciousness ↗ 174
 - Hb < 8g/dL
- Easy bruising or bleeding
 - Temperature ≥ 38°C

Manage and refer urgently:

- Check fingerprick glucose. If ≤ 28 days old and glucose < 2.6mmol/L ↗ 9. If > 28 days old and glucose < 3mmol/L (< 4 if diabetes) or ≥ 11mmol/L ↗ 32.

Approach to the child with jaundice not needing urgent attention

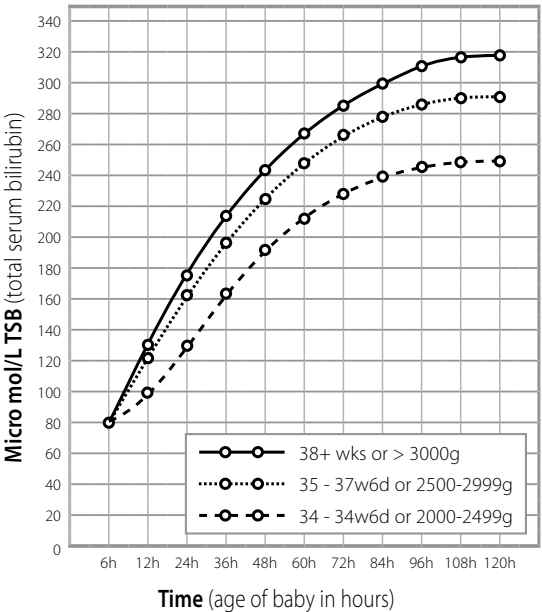
Is child older than 14 days?

No

Yes

Phototherapy

If gestational age is accurate, rather use gestational age (weeks) instead of body weight.



Adapted from: Horn AR. Phototherapy and exchange transfusion for neonatal hyperbilirubinaemia. SAMJ. 2006; Vol. 96, No.9, figure 4, Phototherapy guidelines for all gestational ages.

- Assess growth ↗ 15 and feeding ↗ 120.
- Look for enlarged organs.
- Refer to hospital same day if:
 - Enlarged liver or spleen felt.
 - Not feeding well.
- Do heelprick total serum bilirubin (TSB) and check graph:
 - If TSB above phototherapy line, refer to hospital same day.
 - If TSB on phototherapy line, refer to MOU or if no MOU, closest hospital for phototherapy.
 - If TSB under phototherapy line, repeat TSB level as follows:
 - If 1-20μmol/L below line: repeat TSB in 6 hours or refer for phototherapy.
 - If 21-50μmol/L below line: repeat TSB in 12-24hours.
 - If > 50μmol/L below line: repeat TSB until level is falling or jaundice improving.

Is child on any medication?

No. Manage according to age of child:

Yes

≤ 1 year old

> 1 year old

Discuss/refer same day.

Any of: tiredness, loss of appetite, vomiting, nausea, abdominal pain, pale stools, dark urine or bilirubin on urine dipstick?

- Stop medication.
- Is there vomiting, abdominal pain or nausea?

Yes

No

No

Yes

Hepatitis A infection likely

Discuss/refer same day.

Refer same day.

- Check blood hepatitis A IgM to confirm.
- Manage as hepatitis A while waiting for result.
- Advise child and carer to:
 - Wash hands after toilet and before eating/preparing food to prevent spread.
 - Keep child home from school until jaundice cleared.
- Review blood results in 3 days: if IgM negative, **hepatitis A** unlikely, discuss/refer same day. If IgM positive, notify¹ and advise to:
 - Avoid fatty foods and drink lots of fluid.
 - Return if persistent vomiting, starts bleeding/bruising easily, behaving strangely or drowsiness and refer urgently.
- Review in 2 weeks: if jaundice not resolved, refer.

¹Notify within 7 days via NMC App or www.nicd.ac.za/nmc-overview/notification-process. Complete notifiable medical conditions (NMC) case notification form and case investigation form and send to NMCsurveillanceReport@nicd.ac.za and provincialcdc@gov.za.

Vomiting or refluxing

First assess and manage child's fluid needs ↗ 26. If child has vomiting with watery diarrhoea and has been in cholera outbreak area in past 5 days, cholera likely ↗ 64.

Then give urgent attention to the child with vomiting and any of:

- < 2 months old and vomiting (not refluxing)
- Vomits fresh or digested blood
- Unable to drink
- Continuous vomiting/vomiting everything
- Severe abdominal pain
- Headache
- No stool or wind for last 24 hours
- Several episodes of early morning vomiting
- Decreased level of consciousness ↗ 174
- Neck stiffness/bulging fontanelle
- Guarding and rigidity during peri-umbilical percussion, **peritonitis** likely
- Tender in right lower abdomen, **appendicitis** likely
- Painful groin swelling that will not reduce
- Distended abdomen
- Abdominal mass
- Jelly-like stool
- Bile-stained vomiting

Manage and refer urgently:

- Check fingerprick glucose. If ≤ 28 days old and glucose < 2.6mmol/L ↗ 9. If > 28 days old and glucose < 3mmol/L (< 4 if diabetes) or ≥ 11mmol/L ↗ 32.
- Give maintenance fluid IV according to reference guide ↗ 175
- Keep nil by mouth.
- If ≥ 2 of: temperature ≥ 38°C, headache, decreased level of consciousness ↗ 174, neck stiffness/bulging fontanelle, **meningitis** likely: give ◆**ceftriaxone**¹ 100mg/kg IV/IM² ↗ 159:12.
- If baby < 2 months old and unable to feed: give ◆**ceftriaxone**¹ 80mg/kg IM² ↗ 159:11 and keep warm.
- If **appendicitis** or **peritonitis** likely: give ►**ceftriaxone**¹ 80mg/kg IM² ↗ 159:11. If pain, give oral ■**morphine** ↗ 163:36.

Approach to the child with vomiting not needing urgent attention

- A child is **vomiting** if there is tightening of stomach muscles and forceful expulsion of liquid/food.
- A child is **refluxing** if milk dribbles out mouth after a feed.
- If vomiting *and* refluxing, refer.

Vomiting

- Look for underlying cause:
 - If child bumped head in last few days ↗ 33.
 - Check throat ↗ 52 and ears ↗ 49, if jaundice ↗ 61.
 - Ask about urinary symptoms and check urine dipstick: if burning urine or leucocytes/nitrites/blood on dipstick ↗ 69. If glucose on dipstick, check fingerprick glucose ↗ 32. If bilirubin on dipstick, **hepatitis A infection** likely ↗ 61.
- Manage further according to duration of vomiting:

< 24 hours

- **Gastroenteritis** likely, especially if child has diarrhoea.
- Reassure this is likely due to viral infection or food poisoning and will resolve on its own.
- If not dehydrated, advise fluids at home: if breastfeeding, offer frequently. Offer older child ORS/fluids frequently.
- Review in 5 days or if initially dehydrated, review after 2 days.
- Advise to return immediately if unable to drink well or vomits everything, lethargic, sunken eyes, severe abdominal pain or vomits blood.

≥ 24 hours

- If sexually active³ girl, exclude pregnancy.
- If on ART, advise taking it with food. If vomiting persists, discuss with doctor.
- If older child induces vomiting after eating, refer.
- If child has a life-limiting illness, also give palliative care ↗ 150.
- Refer same day if vomits > 3 days.
- Refer if not growing well or none of the above.

Refluxing

- Check growth ↗ 15 and feeding ↗ 120.
- If baby growing and feeding well, reassure carer that reflux causes no harm. It may worsen between 4-6 months old.
- If smoking in house, alert to risks and encourage smoker to quit ↗ **PACK Adult**.

Refer if:

- Stridor
- Irritable/refuses feeds
- Not growing well and no cause found
- Episodes of no breathing > 20 seconds
- Recurrent wheeze/cough/chest infections
- If child known with cerebral palsy → 146
- Overweight for age, or obese
- Suspected allergy to cow's milk
- Still refluxing at 18 months old.

¹Avoid giving Ringer's lactate together with IV/IM ceftriaxone. If severe penicillin allergy (previous anaphylaxis, urticaria or angioedema), discuss with doctor. ²Avoid injecting ≥ 1g IM at one injection site. ³Report to DSD ↗ 170. If any of: 1) sex not consensual 2) < 12 years old 3) ≥ 12 years old and partner not in peer group (partner must have < 2 year age gap), **sexual abuse** likely ↗ 144.

Diarrhoea

If child has watery diarrhoea (with or without vomiting) and has been in cholera outbreak area in past 5 days, **cholera** likely ↗ 64.

Give urgent attention to the child with diarrhoea and any of:

- Decreased level of consciousness ↗ 174
- Seizures ↗ 28
- Blood in stool in child < 1 year old
- Unable to drink
- Persistent vomiting
- Persistent temperature ≥ 38°C
- Shock¹/severe dehydration²
- Guarding and rigidity during peri-umbilical percussion, **peritonitis** likely
- Swelling of legs/wasting, **severe acute malnutrition (SAM)** likely ↗ 126
- Persistent diarrhoea with dehydration despite treatment for 3 days

Manage and refer urgently:

- Check fingerprick glucose. If ≤ 28 days old and glucose < 2.6mmol/L ↗ 9. If > 28 days and glucose < 3 (< 4 if diabetes) or ≥ 11mmol/L ↗ 32. Also assess and manage child's fluids needs ↗ 26.
- If child < 1 year with blood in stool, or severe dehydration in baby < 28 days old/any child with SAM, give ▶ **ceftriaxone**² 100mg/kg IV/IM ↗ 159:12. Avoid injecting ≥ 1g IM at one injection site.
- Give ♦ **vitamin A**: if < 6 months old: 50 000IU; if 6-12 months old: 100 000IU; if ≥ 1 year old: 200 000IU.
- Give ♦ **zinc**: give 10mg (break scored 20mg tablet in half and give half a tablet).

Approach to the child with diarrhoea not needing urgent attention

- Confirm child has diarrhoea: watery stools and/or > 3 stools/day (changed from normal pattern). If not already done, assess and manage child's fluid needs ↗ 26.
- Ask about duration of diarrhoea:

Diarrhoea for < 7 days.
Is there blood in stool?

No

- If current cholera outbreak ↗ 64.
- If on antibiotic, it may be causing diarrhoea. Refer to doctor to stop/change.
- If not on antibiotic, **viral gastroenteritis** likely. Reassure this should resolve within 3 days.

Yes

Dysentery likely
Give ♦ **ciprofloxacin**
15mg/kg 12 hourly for 3 days
↗ 160:16.

Diarrhoea for ≥ 7 days

- If HIV unknown, test for HIV ↗ 95. If CLHIV, give routine care ↗ 97. Lopinavir/ritonavir cause diarrhoea, should resolve within 6 weeks.
- Check ears ↗ 49 and check urine ↗ 69.
- If abdominal pain/cramps, mucous in stools, poor growth or weight loss/low energy, check for TB ↗ 108 and discuss/refer.
- If child known with a life-limiting illness, also give palliative care ↗ 150.

- Assess growth (weight-for length/height) ↗ 15 and if child in nappies, check for nappy rash ↗ 85.
- If repeated episodes of diarrhoea and no access to clean water, refer to health promotion officer/social worker.
- Give treatment:
 - Give ♦ **oral rehydration solution** (dissolve ½ teaspoon salt + 8 teaspoons sugar in 1 litre cooled boiled water) 10mL/kg after each loose stool. Give extra fluids if child asks and continue until diarrhoea stops.
 - Give ♦ **zinc** 10mg daily (break scored 20mg tablet in half and give half a tablet), for 14 days.
 - If diarrhoea > 7 days, give additional dose of ♦ **vitamin A** ↗ 164:44.
- Advise the carer to practise good hygiene: wash hands before handling food, after using toilet/changing nappy, wash soiled clothing/bedding and dispose of dirty nappies.
- Discuss feeding: feed frequently and for longer if breastfeeding. If > 6 months of age, give soup, rice water, yoghurt or amasi. If diarrhoea > 7 days, give small, frequent meals 6 times/day. Give an extra meal daily for a week after diarrhoea has resolved.

- If initially dehydrated or treated for dysentery, review next day, otherwise arrange review in 5 days.
- If diarrhoea ≥ 7 days and well on review, review again in 14 days and assess growth (weight-for length/height) ↗ 15.

Advise to return immediately if unable to drink well, sunken eyes, lethargy or stools become bloody.

¹If ≥ 2 of: cold hands/feet, weak/fast pulse ↗ 175, capillary refill time (CRT) ≥ 3 seconds, decreased level of consciousness ↗ 174, **shock** likely. ²Any of: lethargy, sunken eyes, slow skin pinch ≥ 2 seconds, drinking poorly. ²Avoid giving Ringer's lactate together with IV/IM ceftriaxone. If severe penicillin allergy (previous anaphylaxis, urticaria or angioedema), discuss with doctor.

Cholera

- If child has watery diarrhoea (with or without vomiting) and has been in cholera outbreak area in past 5 days, **cholera** likely. Discuss all suspected cases of cholera with doctor.
- Isolate if possible, but ensure child can be safely monitored. Health worker to wear gloves and apron. Disinfect surfaces contaminated with secretions with 70% alcohol or chlorine-based disinfectant.
- Weigh child or estimate weight ➔ 175. Check fingerprick glucose. If ≤ 3 give **dextrose 10%** 5mL/kg IV or via NGT as a bolus. Recheck fingerprick glucose after 15 minutes. If still ≤ 3 , repeat dextrose.
- Record each episode of diarrhoea and vomiting and use this to calculate ongoing losses when giving fluid replacement below.

Give urgent attention to the child with suspected cholera and any of:

- Lethargy/decreased level of consciousness ➔ 174
- Sunken eyes
- Slow skin pinch¹ ≥ 2 seconds
- Drinking poorly/not able to drink
- Less than one month of age

Manage as severe dehydration and refer urgently:

Calculate ongoing losses and add this to maintenance fluids:

Age	Amount of IV fluids/ORS to be given after each episode of diarrhoea or vomiting
< 2 years	50–100mL
2–10 years	100–200mL
≥ 10 years	10–20mL/kg (as much as wanted)

Able to insert IV line?

Yes. Assess for severe acute malnutrition (SAM)² and give fluids as indicated below:

Give fluid bolus with **Ringers lactate** or **sodium chloride 0.9%**. If SAM, give 10mL/kg IV over 10 mins. If no SAM, give 20mL/kg as a rapid bolus

Assess response: check radial pulse

Radial pulse weak/undetected:

Give 2nd fluid bolus with **Ringers lactate** or **sodium chloride 0.9%**. If SAM, give 10mL/kg IV over 10 mins. If no SAM, give 20mL/kg rapidly

Assess response: check radial pulse

Radial pulse remains weak/undetected

Give 3rd fluid bolus with **Ringers lactate** or **sodium chloride 0.9%**. If SAM, give 10mL/kg IV over 10 mins. If no SAM, give 20mL/kg rapidly

Radial pulse stronger

Give maintenance fluids with **Ringers lactate** or **sodium chloride 0.9%** for the next 5 hours.

If SAM, give 10mL/kg/hour IV. If no SAM, give 20mL/kg/hour IV

- Also calculate ongoing losses and increase rate of IV fluids as needed (see table above).
- As soon as child is awake and able to drink, continue IV maintenance fluids, and switch ongoing losses to **oral rehydration solution (ORS)**.

- Give single dose **ciprofloxacin** 20mg/kg ➔ 160:17. If able to swallow, also give **zinc** 10mg (break scored 20mg tablet in half and give half a tablet).
- Send stool specimen for MCS and cholera. Ensure correct contact and address details appear on lab request form.
 - If laboratory processing delay > 2 hours: place specimen in Cary-Blair transport medium, if available. Place in fridge or cooler box with ice before transport. Avoid freezing.
- Notify³ immediately as suspected cholera.

While awaiting transfer:

- Reassess hydration status hourly. Keep warm (place child skin-to-skin with parent, cover with blanket). Check fingerprick glucose: if initial glucose ≤ 3 , recheck every 15 mins; if glucose > 3, recheck hourly.
- If poor response/hydration status not improving or unsure, discuss further fluid needs with referral centre.

No

Insert NGT

- Give **oral rehydration solution (ORS)** 20mL/kg/hour via NGT for 6 hours.
- Also calculate ongoing losses (see table above).
- Reassess every hour.
- If child vomits, decrease **ORS** to 10mL/kg/hr.
- If abdomen distended, stop fluids and discuss.

¹Pinch skin on abdomen between 2 fingers. Release. Skin usually snaps rapidly back to normal position. A slow skin pinch takes longer. ²SAM: weight-for-length/height below -3 line or BMI below -3 line or MUAC < 11.5cm or any malnutrition with oedema. ³Immediately report to provincial CDC telephonically and via email ➔ 170. Notify within 24 hours via the NMC App or www.nicd.ac.za/nmc-overview/notification-process. Complete notifiable medical conditions (NMC) case notification form and case investigation form and send to NMCsurveillanceReport@nicd.ac.za and provincialcdc@gov.za.

Approach to the child with likely cholera not needing urgent attention:

Are there any of: restless/irritable, sunken eyes, thirsty/drinks eagerly, slow skin pinch¹ < 2 seconds, increased pulse rate ➔ 175.

Yes

Some dehydration likely

- If drinking well, continue feeds, especially if breastfeeding.
- Use table below to give **ORS** 20mL/kg/hour for the next 4 hours. Give more if child wants more.

Age	< 4 months	4–12 months	1–2 years	2–5 years	≥ 5 years
Weight (kg)	< 5	5–8	8–11	11–16	≥ 16
ORS (mL) per hour	50–100	100–150	150–200	200–300	300–550

- Show caregiver how to give ORS with a cup and spoon. Give frequent small sips. If child vomits, wait 10 minutes, then continue more slowly.
- Also give additional **ORS** for each episode of diarrhoea/vomiting:

Age	Amount of ORS to give after each episode of diarrhoea or vomiting
< 2 years	50–100mL
2–10 years	100–200mL
≥ 10 years	As much as tolerated

Reassess hourly for the next 4 hours.

Is there good response to fluids and is hydration status improving?

No

Manage urgently as **severe dehydration** ➔64.

Yes

Does child have reliable transport to return if worse?

No

- Continue assessing and managing dehydration hourly, discuss.
- If some dehydration, long-term health condition (HIV, diabetes), or no response to fluid treatment, give antibiotic:
 - Give single dose **ciprofloxacin** 20mg/kg ➔ 160:17.

Yes

Discharge child into reliable carer's care.

- Advise to return if worse, unable to drink, bloody diarrhoea, fever, drowsy.
- Advise to continue ORS³ after each stool.
- Advise to isolate the child if possible.
- Advise handwashing with soap and water, before preparing food/after going to toilet. Use only safe/disinfected water for preparing food/drinks/ice.

- Give **zinc** 10mg (break scored 20mg tablet in half and give half a tablet) daily for 14 days.
- Send stool specimen for MCS and cholera. Ensure correct contact and address details appear on lab request form. If laboratory processing delay > 2 hours: place specimen in Cary-Blair transport medium, if available. Place in fridge or cooler box with ice before transport. Avoid freezing.
- Notify⁴ immediately as suspected cholera.

No

Dehydration unlikely

- If breastfeeding, advise to continue.
- Give **oral rehydration solution (ORS)** for each episode of diarrhoea/vomiting:

Age	Amount of ORS to give after each episode of diarrhoea or vomiting
< 2 years	50–100mL
2–10 years	100–200mL
≥ 10 years	As much as tolerated

- Show caregiver how to give ORS with a cup and spoon. Give frequent small sips. If child vomits, wait 10 minutes, then continue more slowly.

Reassess hourly for the next 4 hours.

Is child tolerating fluids and no ongoing diarrhoea/vomiting

Yes

No

Not tolerating fluids or ongoing diarrhoea/vomiting

Reassess hydration status ➔ 64.

¹Pinch skin on abdomen between 2 fingers. Release. Skin usually snaps rapidly back to its normal position. A slow skin pinch takes longer. ³Give at least 4 sachets of ORS at discharge. Advise home ORS solution: add 8 teaspoons sugar and ½ teaspoon salt to 1L cooled boiled water. ⁴Immediately report to provincial CDC telephonically and via email ➔ 170. Notify within 24 hours via NMC App or www.nicd.ac.za/nmc-overview/notification-process. Complete notifiable medical conditions (NMC) case notification form and case investigation form and send to NMCsurveillanceReport@nicd.ac.za and provincialcdc@gov.za.

Constipation

The child is constipated if s/he has infrequent stools and any of: pain on passing stool, impaction, involuntary leakage of stool or voluntary withholding.

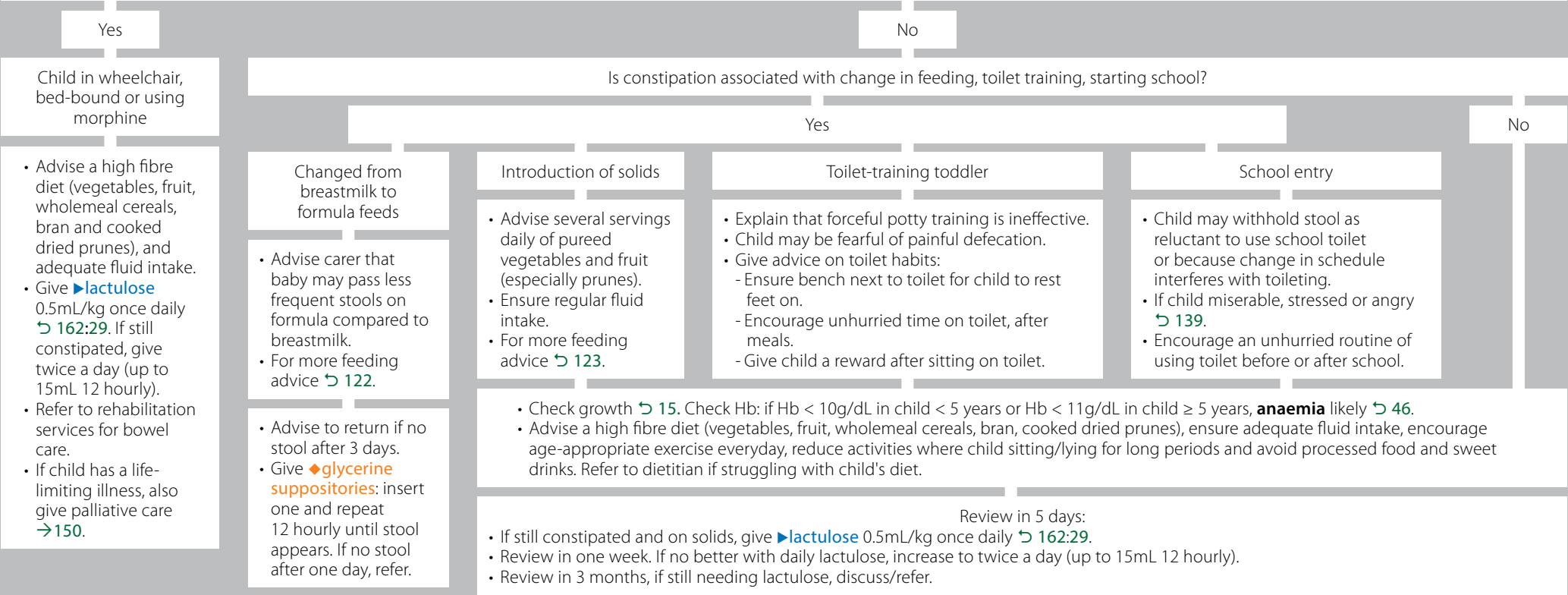
Give urgent attention to the child with constipation and any of:

- Distended abdomen and vomiting
- Mass in abdomen (not faecal mass)
- Newborn and no stool in first 2 days
- Temperature $\geq 38^{\circ}\text{C}$
- Bruising and lacerations around anus, consider child abuse [↗ 144](#)

Refer same day.

Approach to the child with constipation not needing urgent attention

- If constipation from birth or early infancy, refer to specialist same week.
- Inspect anus: if fissure (crack) [↗ 67](#). If rectal bleeding and no crack seen, refer.
- Refer to specialist same week if any of: anus does not tighten when child coughs/cries, abnormal spine, new urinary incontinence, involuntary leakage of retained stool, weight loss/poor weight gain, or delayed milestones (check milestones [↗ 12](#)).
- Does child have reduced mobility (is child bed bound, or using a wheelchair) or is child using constipating medications (like morphine)?



Anal symptoms

- Very painful lump on anus
- Unable to pass stool because of anal symptoms

Manage and refer same day: If pain, give ♦paracetamol 15mg/kg ↪ 163:37.

Give urgent attention to the child with anal symptoms and any of:

- Jelly-like stool
- If bruising or laceration around anus or history indicates possible inappropriate sexual conduct, consider child abuse ↪ 144

Assess the child with anal symptoms not needing urgent attention

- If bloody diarrhoea ↪63. If nappy rash ↪85. If burning urine, check urine dipstick: if leucocytes/nitrites ↪69. Review RtHB: check deworming up to date and that child is attending clinic visits.
- Manage according to symptom/s:

Itch/irritation Worms visible or history of worms visible near anus? No Yes Localised skin irritation likely • Advise to: - Avoid wiping anus excessively. - Avoid tight-fitting underwear. - Keep area dry. - Avoid soaps, detergents, perfumes. • Give ♦zinc and castor oil ointment to apply to area twice a day. • Review in 2 weeks, if no better, refer. Manage as below.	Pain or bleeding Fissure (crack) seen • Give ▶lactulose 0.5mL/kg once a day ↪ 162:29 until crack healed. • If constipation ↪ 66. • Refer if severe pain, > 1 episode or no response to lactulose after one week. No fissure (crack) seen, refer. • Give advice for soft stools: high fibre diet (vegetables, fruit, wholemeal cereals, bran and cooked dried prunes), adequate fluids , encourage age-appropriate exercise everyday, reduce activities where child sitting/lying inactive (like watching TV). • Give advice on toilet habits: - Ensure bench next to toilet for child to rest feet on. - Encourage unhurried time on toilet, after meals and give reward for sitting on toilet. - Address child's anxiety. They may be fearful of painful defecation. - Avoid straining when passing stool.	Perianal warts Manage as for genital warts ↪ 68.	Red/raw skin around anus • Apply ♦zinc and castor oil ointment after each nappy change/bowel action. • Do not wipe anus excessively. • Expose area to air as much as possible. • Keep area dry. • If ongoing diarrhoea ↪ 63.
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Worms

Carer may report seeing worm/s when child coughed, sneezed, vomited or in stool.

- If worm flat and white with worm segments (blunt ended), **tapeworm** likely. If child > 12 months old, give ▶albendazole once daily for 3 days ↪ 157:5.
- If worm is thin and white, with anal itch worse at night, **pinworm** likely. If child > 12 months, give single dose ▶mebendazole 100mg (and repeat dose after 2 weeks) or give ▶albendazole ↪ 157:5.
- If no worm seen or worm seen but not tapeworm/pinworm: if child > 12 months old, give ▶mebendazole ↪ 162:31 or ▶albendazole ↪ 157:5. Treat all house members at same time.

- Assess and interpret growth ↪ 15.
- If pale conjunctiva/palms of hands, check Hb: if Hb < 10g/dL in child < 5 years or Hb < 11g/dL in child ≥ 5 years, **anaemia** likely ↪ 46.
- Advise to avoid uncooked meat and use good hygiene: wash hands with soap and water before handling food and after using toilet/changing nappy, wash soiled clothing and bedding and dispose of faeces properly. Keep fingernails short and keep toilet seat clean.
- Refer if: abdominal tenderness/pain, abdominal mass felt, vomiting, itchy red line under the skin that may move (larva migrans).
- Review in 4 weeks and repeat treatment if still infected.



Tapeworm



©dermnetnz.org/topics/cutaneous-larva-migrans

Genital symptoms

Before examining the child with a genital symptom, ensure that a chaperone is present.

Give urgent attention to the child with genital symptoms and any of:

- Sudden testicular pain, **torsion of testicle** likely
- Painful swelling of scrotum
- Foreskin retracted, swollen and unable to return to normal position, **paraphimosis** likely
- Rape/sexual assault: if wound or soft tissue injury needing urgent attention ➔ 33

Management:

- If **paraphimosis** likely, try to replace foreskin: apply **lidocaine 2% gel** to glans, wrap in gauze and apply increasing pressure for 10-15 minutes until foreskin can be replaced over glans. If unsuccessful or glans blue/black; refer urgently.
- If rape/sexual assault ➔ 144.

Approach to child with genital symptoms not needing urgent attention

- If skin rash ➔ 75, if itching or red/raw skin around anal region ➔ 67, if burning/smelly urine ➔ 69, if lymph node in groin ➔ 47.
- If testicle/testes not present in scrotum, review in one month. If still not felt, refer to surgical OPD.
- If girl with fused labia: reassure carer. If fused from birth, unable to see urethral opening, still fused after > 6 years old, recurrent urinary tract infections or if unsure, refer to gynaecologist.
- Manage according to genital symptom:

Sore/red/itchy vulva/vagina	Vaginal discharge		Urethral discharge	Warts	Ulcers	Scrotal/testicular problem	Penis problem
Check urine dipstick: • If leucocytes, nitrites or blood ➔ 69. • If glucose, screen for diabetes ➔ 32.	Discharge white/thick/cheesy	Discharge not smelly and clear/white in girl ≥ 8 years old	Discharge copious/smelly/green/yellow			• If firm lump, refer same week. • If scrotum soft, painless and fluid-filled, hydrocoele likely. If unsure or becomes painful, refer. Otherwise refer to surgeon when child > 1 year old.	• If glans red and swollen, balanitis likely: discuss/refer same day. • If white patches on glans, candida infection likely: apply clotrimazole 1% ointment at bed time for up to 14 nights. Refer if no response. • If foreskin has become non-retractile or if puberty and always been non-retractile, phimosis likely. Book surgical OPD appointment within next week.
Vulvovaginitis likely • Ask carer/child if there is a foreign object in vagina and examine vulva/vagina. • Advise to wipe from front to back, avoid tights/leotards/leggings/wet swim-wear and bubble baths/perfumed soaps. • Advise good hygiene: wash hands with soap and water before handling food and after using toilet/changing nappy and keep fingernails short. • Apply clotrimazole 1% ointment at bed time until resolved. • Treat for worms: if child > 12 months old, give mebendazole ➔ 162:31 or albendazole ➔ 157:5. • If no better after 2 weeks or recurs, refer. • If recurrent, screen for diabetes ➔ 32 and refer. • Ask carer if aware of any abuse of child. Ask child if anyone hurts him/her. If yes, consider child abuse ➔ 144. • If none of the above, refer.		Physiological discharge likely. Reassure this is normal.	• If sexually active¹: - Manage genital symptom(s) ➔ PACK Adult. - Advise reliable contraception ➔ PACK Adult. • If not sexually active or unsure, refer same week. • If CLHIV, give routine care ➔ 97. • If HIV status unknown, test for HIV ➔ 95. • Ask carer if aware of any abuse of child. Ask child if anyone hurts him/her. If yes, consider child abuse ➔ 144. • If none of the above, refer.				

¹Report to DSD ➔ 170. If any of: 1) sex not consensual, 2) < 12 years old or 3) ≥ 12 years old with a partner not in peer group (partner must have < 2 year age gap), **sexual abuse** likely ➔ 144.

Urinary symptoms

The child with urinary symptoms may have pain on passing urine, urinating very often/large volumes, urgency, new incontinence, bed-wetting, bloody/brown urine, unable to pass urine or foul-smelling urine.

Give urgent attention to the child with urinary symptoms and any of:

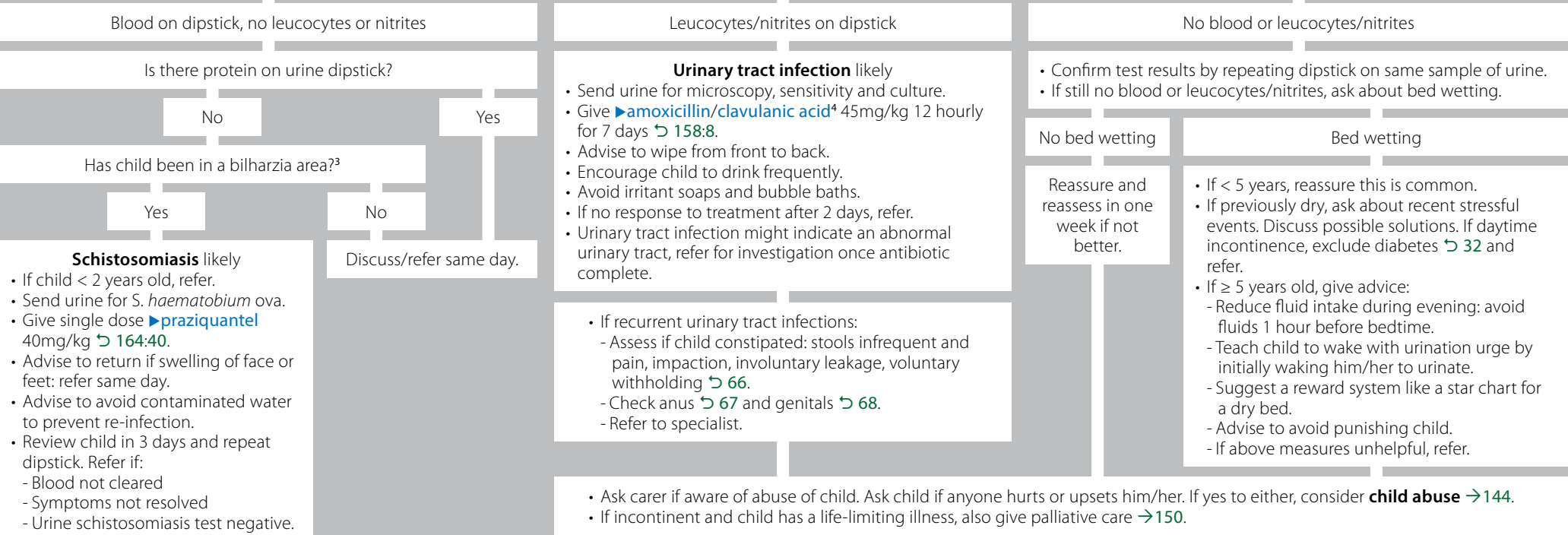
- ≤ 3 months old
- Passing little amounts or unable to pass urine
- > 3 months old with failure to thrive/abdominal pain/diarrhoea/ongoing fever
- Temperature ≥ 38°C/rigors/flank pain, **pyelonephritis** likely
- Swelling of face/feet and either blood in urine or passing little amounts of urine, **nephritis** likely

Manage and refer urgently:

- If **pyelonephritis** likely, give ▶**ceftriaxone**¹ 80mg/kg IM ↗ 159:11. Avoid injecting ≥ 1g IM at one injection site.
- If **nephritis** likely and signs of fluid overload (increased pulse/respiratory rate ↗ 175 or puffy eyes), give ♦**oxygen** 2L/minute via nasal prongs and give ■**furosemide** 1mg/kg IV over 5 minutes (avoid IV fluids) ↗ 161:25. Then check BP: if < 6 years and sBP > 120 or dBP > 90, or if > 6 years and sBP > 130 or dBP > 95, give ■**nifedipine** 0.25mg/kg (up to 10mg) squirted into mouth².
- If pain, give ♦**paracetamol** 15mg/kg ↗ 163:37.

Approach to the child with urinary symptoms not needing urgent attention

- Check urine dipstick. If child too young to urinate into specimen jar, clean perineum and apply urine bag:
 - If glucose/ketones in urine, check fingerprick glucose ↗ 32.
 - Look for blood, leucocytes and nitrites on dipstick:



¹Avoid giving Ringer's lactate together with IV/IM ceftriaxone. If severe penicillin allergy (previous anaphylaxis, urticaria, angioedema), discuss with doctor. ²Withdraw contents of 5mg nifedipine capsule with 1mL syringe: if 10–25kg: give 2.5mg; if 25–50kg: give 5mg; if > 50 kg: give 10mg. ³Bilharzia areas: Limpopo, Mpumalanga, KwaZulu-Natal, isolated areas in Eastern Cape. ⁴If severe penicillin allergy (history of anaphylaxis, urticaria, angioedema), give instead ■**ciprofloxacin** 10mg/kg/dose 12 hourly for 7 days.

Back pain

Give urgent attention to the child with back pain and any of:

- Child < 4 years old
 - Significant TB exposure¹ ➔ 106 or current/previous TB
 - Recent trauma or injury to back/spine ➔70
 - Severe pain causing night waking/affecting daily activities
 - Pain down leg
 - Urinary or bowel incontinence
- Pins and needles in limb/s
 - Temperature ≥ 38°C and/or weight loss
 - Weakness in limb/s
 - Sudden painful curve of spine
 - New onset limp

Manage and refer urgently:

- If open area on spine, cover with sterile saline-soaked gauze dressing.
- If difficulty breathing, give ♦oxygen 2L/min via nasal prongs.
- If pain, give ♦paracetamol 15mg/kg ➔ 163:37.

Approach to child with back pain not needing urgent attention

- Look at spine: if curved spine or deformity, refer to orthopaedic specialist.
- If asymmetric walk, refer to physiotherapist.
- Check for possible causes: carrying heavy weights, recent increase in physical activity, too little exercise, overweight:

If carrying heavy weights or recent increase in physical activity:

- If carrying heavy school bag (especially on one shoulder), advise to reduce weight of bag and carry on both shoulders.
- If recent increase in activity, advise to avoid exercise for 1-4 weeks/until no pain.

If < 1 hour/day of brisk exercise:

- Encourage child to be more active:
 - Go outside and play.
 - Join a team sport.
 - Take stairs instead of elevators or lifts, walk instead of taking transport.
- Limit screen time to less than 2 hours/day.

If overweight:

- If < 5 years old, plot weight and height ➔ 15. If weight-for-length/height is on or above +2 line ➔ 129.
- If ≥ 5 years old, plot body mass index (BMI) ➔ 15. If BMI on or above +1 line ➔ 129.

- Give ►paracetamol 15mg/kg 6 hourly as needed for up to 5 days ➔ 163:37 whilst addressing contributing causes.
- Refer same week if:
 - Early morning stiffness or pain lasting > 15 minutes.
 - Back pain persists longer than 4 weeks or worsens despite above measures.
 - History of cancer.

¹Significant TB exposure: any child who in the last 12 months shared an enclosed space (school/creche/home/similar) for ≥ 1 night or frequent/extended daytime periods with an adult/adolescent with pulmonary TB ("index client"), during the 3 months before index client started TB treatment.

Arm or hand symptoms

Give urgent attention to the child with arm or hand symptoms and any of:

- Temperature $\geq 38^{\circ}\text{C}$
- Injury in past 48 hours with deformity or swelling →33
- Sudden weakness or unable to move arm

Manage and refer urgently:

- Keep nil per mouth. If pain, give ♦paracetamol 15mg/kg ↪ 163:37.

Approach to the child with arm or hand symptom not needing urgent attention

If baby < 2 months old →10.

- If palms pale, check Hb: if Hb < 10g/dL in child < 5 years old or Hb < 11g/dL in child ≥ 5 years old, **anaemia** likely ↪ 46.
- Was there an injury?

Yes
Manage according to site of injury

Sudden pull of arm or forcefully lifted with straight arms and now refusing to move arm.

Pulled elbow likely, doctor to manage using steps below.
If swelling or deformity, **fracture** likely →34



Hold elbow bent at 90° with child's palm facing downward and apply gentle pressure over radial head.



Next, firmly turn wrist so that child's palm is facing upwards, keeping pressure on radial head.



Then fully bend elbow up to shoulder. You may hear or feel a small click.

If not wanting to move arm 10 minutes after manoeuvre: refer.

Injury or blow to finger

- Give ►paracetamol 15mg/kg 6 hourly as needed for up to 5 days ↪ 163:37.
- Arrange x-ray and doctor review.
- Buddy strap finger to the longer adjacent finger. Ensure finger rotation aligns:



- If fracture ends align by < 50% or fracture involves joint, buddy strap and refer same day.
- If fracture ends align by $\geq 50\%$ and does not involve joint, buddy strap and review after 1 week.

Fall on outstretched hand

Exclude fracture
→34.

No
Arm/hand swollen with no history of injury

Is there early morning stiffness lasting > 15 minutes?

Yes

Juvenile idiopathic arthritis likely.
Refer to specialist same week.

No

Refer to orthopaedic OPD same week.

- Refer/discuss with specialist if unsure of diagnosis or problem no better.
- If child abuse or neglect suspected →144.

Leg symptoms/limp/walking problems

If joint symptoms →74. If foot symptoms →73. If not moving or sitting properly e.g. abnormal tone, posture or weakness →90.

Give urgent attention to the child with leg symptoms and any of:

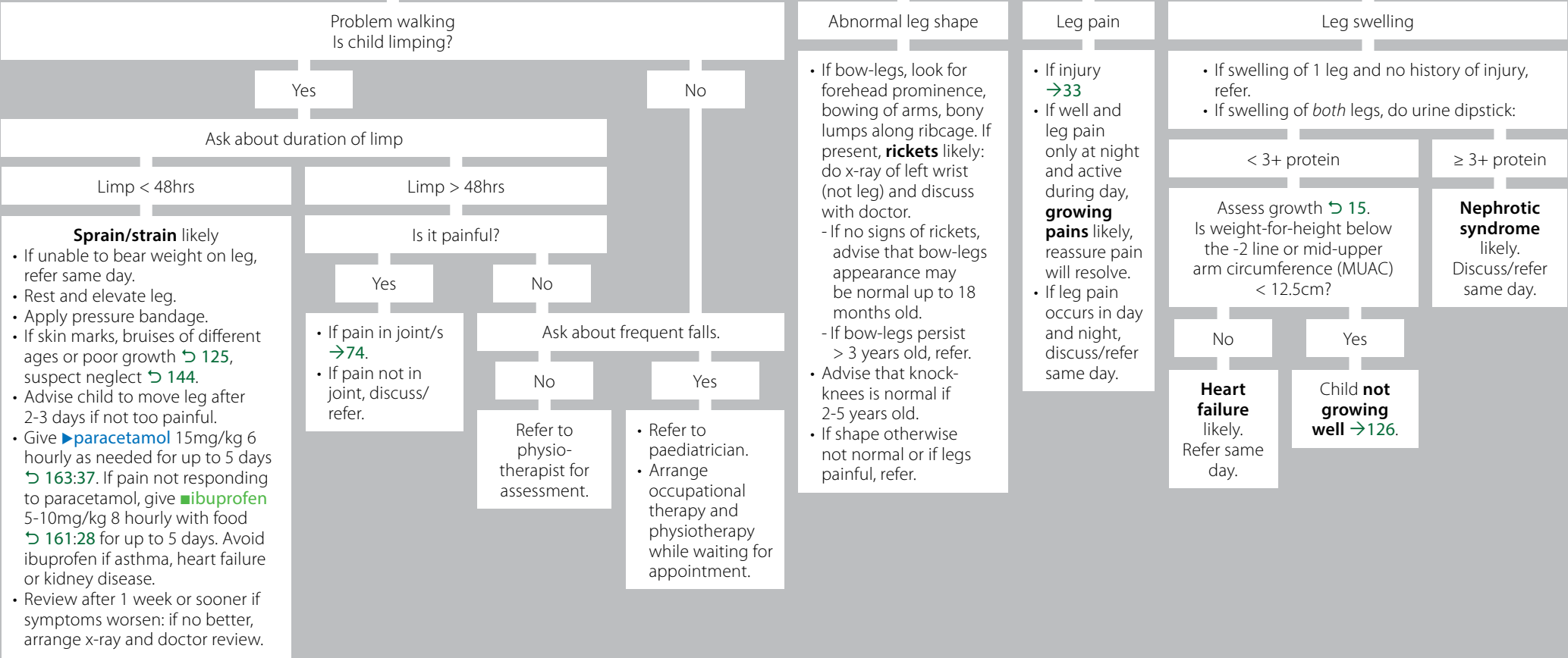
- Sudden refusal to sit, stand or walk
- Sudden onset weakness in leg/s
- Leg pain and fever ≥ 38°C
- Limping and weight loss/lethargy
- Unable to bear weight after leg injury →33

Refer urgently.

- If pain, give ♦paracetamol 15mg/kg ↪ 163:37.

Approach to the child with leg symptoms not needing urgent attention

Examine leg for abnormal shape, swelling, pain and assess walking.



Foot symptoms

- If joint symptoms →74.
- If leg symptoms →72.

Give urgent attention to the child with foot symptoms and any of:

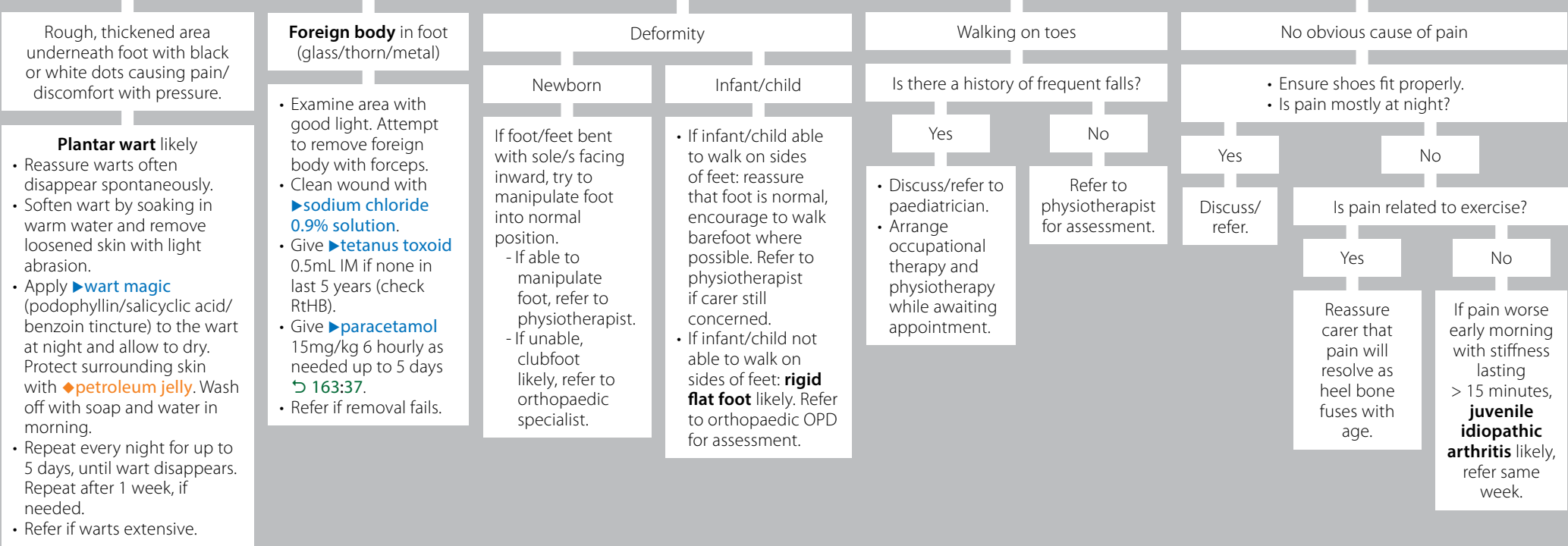
- Unable to/refuses to bear weight following injury →33
- Foot pain and temperature ≥ 38°C

Refer urgently.

- If pain, give ♦paracetamol 15mg/kg ↪ 163:37.

Approach to child with foot symptoms not needing urgent attention

- If injury →33.
- If nail symptoms →87.
- Examine foot for rough areas underneath foot with black or white dot/s, foreign body, deformity and assess walking.



Joint symptoms

- Check that problem is indeed in joint/s. If child old enough to understand, ask him/her to do the following: walk on heels then tip-toes. Make fist and open hand. Press palms together and back to back with elbows lifted (should be able to flex/extend to 90°). Put hands behind head. Touch shoulder with ear. Reach up and "touch the sky" (with straight arms) and "look at ceiling". Open mouth wide enough to fit 3 fingers. Bend forward and touch toes. Squat then stand up. If unable to do any of these actions comfortably, **joint problem** likely.
- If pain not over joint: if back problem →70, if arm problem →71, if leg problem →72, if foot problem →73.

Give urgent attention to the child with a joint problem and any of:

- Trauma in past 48 hours with severe pain: arrange x-ray and doctor review
- Refusing to weight-bear
- Pain occurs night and day
- Known haemophilia
- Sudden painful back curvature
- Temperature ≥ 38°C
- Joint swelling
- ≥ 2 of: strange movements of limbs/face, lumps over joints/tendons, breathlessness on exertion, rash (round pink lesions with pale centre), joint pain/swelling, **rheumatic fever** likely
- Previous rheumatic fever or known with rheumatic heart disease

Manage and refer urgently:

- If ≤ 2 months old and temperature ≥ 38°C, give ►**ceftriaxone**^{1,2} 80mg/kg IM ↗ 159:11. Avoid injecting ≥ 1g IM at one injection site.
- If **rheumatic fever likely**, give ►**benzathine benzylpenicillin**^{2,3} IM according to weight: if < 30kg, give 600 000 units. If ≥ 30kg, give 1.2 million units. Notify⁴.
- If pain, give ►**paracetamol** 15mg/kg ↗ 163:37. If not responding to paracetamol, give ►**ibuprofen** 5-10mg/kg with food ↗ 161:28. Avoid ibuprofen if asthma, heart failure or kidney disease.

Approach to the child with a joint symptom not needing urgent attention

- If recent injury →33.
- Screen for child abuse: ask carer if aware of any abuse of child, ask if anyone hurts or upsets him/her. If yes to either →144.
- Has child had a runny/blocked nose, sore throat or cough in last 2 weeks?

Yes

Viral reactive arthritis likely

- Reassure carer that it will resolve.
- Give ►**paracetamol** 15mg/kg 6 hourly as needed for up to 5 days ↗ 163:37. If pain not responding to paracetamol, give ►**ibuprofen** 5-10mg/kg 8 hourly with food ↗ 161:28 for up to 5 days. Avoid ibuprofen if asthma, heart failure or kidney disease.
- Follow up in 2 weeks.
- Refer if:
 - Joint pain persists on follow up.
 - Temperature ≥ 38°C.
 - Any signs of rheumatic fever appear (as above).

No

Does child have any of: fever, weight loss, night sweats, weakness, fatigue, rash, bruising?

No. Manage according to site of joint problem:

Pain in hip, knee or thigh

Pain in joint other than hip, knee or thigh

Yes

Discuss/refer same day.

Is there pain on hip movement?

Yes

No

- If available, do x-ray of pelvis (anterior/posterior) and frog lateral of both hips.
- Advise not to bear weight on painful side.
- Discuss/refer with doctor/orthopaedic specialist same week.

Is there early morning stiffness lasting > 15 minutes?

No

How long has joint been painful for?

Yes

< 6 weeks

≥ 6 weeks

- Reassure should resolve spontaneously.
- Refer if pain persists for ≥ 6 weeks or if temperature ≥ 38°C.

Refer same week.

¹Avoid giving Ringer's lactate together with IV/IM ceftriaxone. ²If severe penicillin allergy (previous anaphylaxis, urticaria or angioedema), discuss with doctor. ³Dissolve benzathine benzylpenicillin 1.2 million units in 3.2mL ►**lidocaine 1% without adrenaline**. ⁴Immediately report to provincial CDC telephonically and via email ↗ 170. Notify within 24 hours via the NMC App or www.nicd.ac.za/nmc-overview/notification-process. Complete notifiable medical conditions (NMC) case notification form and case investigation form and send to NMCsurveillanceReport@nicd.ac.za and provincialcdc@gov.za.

Skin symptoms

Start on this page for the child with skin symptoms.

Give urgent attention to the child with skin symptom/s and any of:

- Sudden onset of generalised itch/rash or face/tongue itch/swelling and any of: tight throat/hoarseness, difficulty breathing, wheeze, shock¹, dizziness/collapse, abdominal pain/nausea/vomiting/diarrhoea or exposure to possible allergen², **anaphylaxis** likely →37
- Purple/red rash that does not disappear when pressure is applied with any of: neck stiffness, drowsy/confused, temperature ≥ 38°C, headache, nausea/vomiting: **meningococcal disease** likely
- Baby < 1 month old with red, swollen skin around umbilical area and/or draining pus
- Weeping skin lesions in child with severe wasting (weight-for-length/height below -3 line), BMI³ below -3 line if ≥ 5 years, or MUAC⁴ < 11.5cm if < 5 years: **severe acute malnutrition with complications** likely →126
- Diffuse rash appearing within 3 months of starting a new medication and any of the following, **serious drug reaction** likely:
 - Temperature ≥ 38°C
 - Shock¹
 - Jaundice
 - Swollen lymph nodes
 - Abdominal pain/vomiting/diarrhoea
 - Rash involving mouth, eyes or genitals
 - Blisters, peeling or raw areas
 - Difficulty breathing → 55



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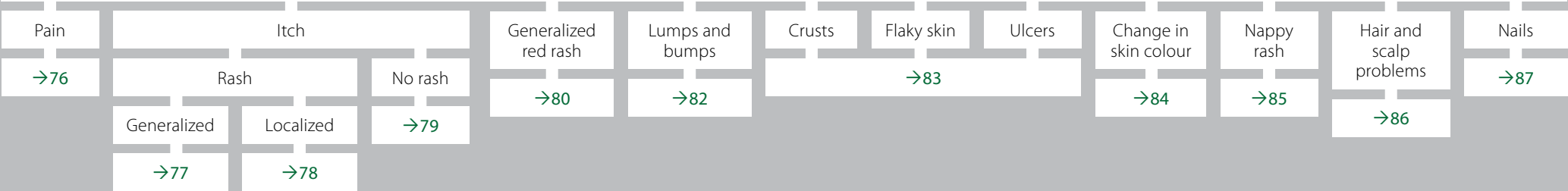


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Manage and refer urgently:

- Assess and manage child's fluid needs → 26. Keep child warm.
- If **meningococcal disease** likely: give **ceftriaxone**⁵ 100mg/kg IV/IM → 159:12. Avoid injecting ≥ 1g IM at one injection site.
 - Notify⁶ as suspected meningococcal disease. Isolate if possible, but ensure child can be safely monitored. Use respiratory droplet precautions.
 - Prevent disease in close contacts (household members, school/childcare contacts or anyone directly exposed to child's oral secretions: kissing, mouth-to-mouth resuscitation, intubation):
 - If < 6 years, give **ceftriaxone**⁵ 125mg IM.
 - If 6-12 years, give single dose **ciprofloxacin** 250mg.
 - If > 12 years, give single dose **ciprofloxacin** 500mg. If pregnant, give instead single dose **ceftriaxone**⁵ 250mg IM.
- If **serious drug reaction** likely: stop all medication. If peeling or raw skin, wrap in clean dry sheet.
- If baby < 1 month old with red, swollen skin around umbilical area and/or draining pus: check heelprick glucose. If < 2.6mmol/L, manage as **hypoglycaemia** → 9. If not jaundiced, give **ceftriaxone**⁵ 80mg/kg IM → 159:11.

Approach to the child with skin symptom/s not needing urgent attention



If status unknown, test for HIV → 95, especially if rash is extensive, recurrent or difficult to treat. If CLHIV, give routine HIV care → 97.

¹If ≥ 2 of: cold hands/feet, weak/fast pulse → 175, capillary refill time ≥ 3 seconds, decreased level of consciousness → 174, **shock** likely. ²Common allergens: medications, new food or insect bite/sting within the last few hours. ³Body mass index. ⁴Mid-upper arm circumference. ⁵Avoid giving Ringer's lactate together with IV/IM ceftriaxone. If severe penicillin allergy (previous anaphylaxis, urticaria, angioedema), discuss with doctor. ⁶Immediately report to provincial CDC telephonically and via email → 170. Notify within 24 hours via NMC App or <https://www.nicd.ac.za/nmc-overview/notification-process>. Complete NMC Case Notification Form and Case Investigation Form and send to NMCsurveillanceReport@nicd.ac.za and provincialcdc@westerncape.gov.za.

Painful skin

Check for crops of blisters in a band on one side of the body or inside mouth, on palms of hand and soles of feet. Also check for areas of redness, swelling and warmth:

Painful blisters in band along one side of body



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Crops of blisters inside mouth and on palms and soles



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Firm, red lump which softens in the centre to discharge pus



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Sudden onset swelling of an area of skin with redness, pain, local warmth



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Shingles (herpes zoster) likely

- Discuss and refer same day.
- If status unknown, test for HIV → 95.
- If CLHIV, give routine care → 97.
- Give ♦paracetamol 15mg/kg → 163:37.

Hand, foot and mouth disease likely

- May also have temperature $\geq 38^{\circ}\text{C}$, sore throat, fatigue, loss of appetite.
- Give ►paracetamol 15mg/kg 6 hourly as needed for up to 5 days → 163:37.
- Reassure that this is a viral illness that usually resolves on its own within 10 days and that antibiotics are not needed.
- Advise that the virus can spread without symptoms or after symptoms resolved.
- Advise child/carer to:
 - Rest and hydrate
 - Wash hands regularly and avoid popping blisters due to risk of infection.
- If infected (surrounding skin red, warm, swollen, painful): give ►cephalexin¹ → 159:13 or ►flucloxacillin¹ → 161:23.
- Advise to keep child home if:
 - Fever, unwell, extensive drooling from mouth lesions or many open blisters.
- Refer same day if any of: vomiting, neck stiffness, lethargy, balance problem, weakness of limbs, difficulty breathing or chest pain.

Boil/abscess likely

Discuss and refer urgently if any of:

- Systemic signs of infection: temperature $\geq 38^{\circ}\text{C}$, chills, increased heart rate → 175, shock² → 26, confusion/decreased level of consciousness → 174, difficulty breathing → 55, increased respiratory rate → 175
- Extensive surrounding cellulitis
- Baby < 1 month old

If none of the above:

- If area fluctuant³, arrange for incision and drainage by doctor.
- Give ►paracetamol 15mg/kg 6 hourly as needed up to 5 days → 163:37.
- If mild cellulitis, swollen painful lymph nodes in area, CLHIV or diabetes, give ►cephalexin¹ → 159:13 or ►flucloxacillin¹ → 161:23.
- Advise to wash with soap and water, keep nails short and avoid sharing clothing/towels.
- If recurrent boils/abscesses, test for HIV → 95, check glucose → 32, and discuss/refer.
- Refer if: involves face/genitals/hands, or no better within 48 hours.

Bacterial skin infection (cellulitis/erysipelas) likely

Discuss and refer urgently if any of:

- Severe swelling/pain/loss of function
- Involves arms, legs, hand, face or scalp
- Blisters or grey/black skin
- Extensive
- Systemic signs of infection: temperature $\geq 38^{\circ}\text{C}$, chills, increased heart rate → 175, shock² → 26, confusion/decreased level of consciousness → 174, difficulty breathing → 55, increased respiratory rate → 175
- Baby < 1 month old with red, swollen skin around umbilical area and/or draining pus → 75

If none of the above:

- Elevate area if possible.
- Give ►paracetamol 15mg/kg 6 hourly as needed for up to 5 days → 163:37.
- Give ►cephalexin¹ → 159:13 or ►flucloxacillin¹ → 161:23.
- Refer if no improvement within 48 hours.

¹If severe penicillin allergy (history of anaphylaxis, urticaria or angioedema), give instead ►azithromycin 10mg/kg once daily for 3 days → 158:10. ²If ≥ 2 of: cold hands/feet, weak/fast pulse → 175, capillary refill time ≥ 3 seconds, decreased level of consciousness → 174, shock likely. ³Fluid filled mass, that can be compressed/squeezed between two fingers.

Generalised itchy rash

If itchy, scaly skin that is usually red in the baby and dry in the older child, usually affects inner (flexural) surfaces of elbows and knees, cheeks, scalp and neck, consider **eczema** →118.

- Raised red bumps that become fluid-filled blisters (vesicles) which then break and leak and finally form crusts and scabs.
- Typically starts on chest, back, and face, then spreads over entire body.
- Usually fever and tiredness before onset of rash.



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Chickenpox likely

- Discuss and refer same day if any of: < 1 month old, difficulty breathing, severely ill, CLHIV or any signs of meningitis (≥ 2 of: temperature ≥ 38°C, headache, decreased level of consciousness, neck stiffness).
- Apply **calamine lotion** and give **paracetamol** 15mg/kg 6 hourly as needed up to 5 days →163:37.
- If very itchy, give **chlorpheniramine**¹ 0.1mg/kg 6-8 hourly as needed for up to 2 weeks →159:15.
- If rash extensive or child ≥ 10 years and rash started < 48 hours ago, give **aciclovir** 20mg/kg 6 hourly for 7 days →157:1.
- If surrounding skin red, warm, swollen, painful, give **cephalexin**² →159:13 or **flucloxacillin**² →161:23.
- If recurrent, test for HIV →95.
- Advise child/carer:
 - To ensure adequate hydration.
 - Chickenpox is highly contagious.
 - Avoid school, people with impaired immunity and pregnant women until all lesions have crusted.
- Refer if no better after 10 days.

Small, red, itchy, firm bumps (papules) that become dark (hyperpigmented) bumps on limbs, trunk or face.

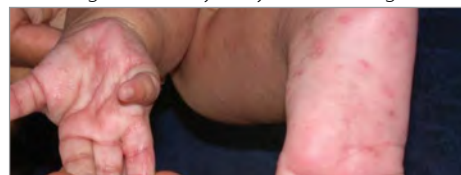


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Papular pruritic eruption (PPE) likely

- If CLHIV, give routine HIV care →97. If HIV unknown, test for HIV →95.
- Exclude scabies (see adjacent). PPE does not involve web-spaces between fingers and toes.
- Apply **hydrocortisone 1% cream** twice a day for 7 days (apply very thin layer to face).
- Give **chlorpheniramine**¹ 0.1mg/kg 6-8 hourly as needed for up to 2 weeks →159:15. If itch persists, switch to **cetirizine** daily until itch controlled/ up to 2 weeks: if 2-6 years, give 5mg; if ≥ 6 years, give 10mg.
- If poor response, refer.
- If red itchy crops of bumps that blister, and heal with darkening of skin, may have scratch marks, **insect bites** likely →78.
- Advise child/carer:
 - Explain that PPE may be long-standing and skin often remains hyperpigmented.
 - May temporarily worsen on starting ART.
 - Reduce exposure to insect bites.

Small red bumps (burrows) in web spaces of fingers and toes and in skin folds of axillae (armpits), waist or genitals. Very itchy, worse at night.



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Scabies likely

- If < 2 months: apply **sulphur ointment 5%** once daily for 3-5 days. If unsuccessful, discuss with doctor.
- If 2 months-6 years: apply **permethrin 5%** to whole body from neck to feet, at night. Wash off after 8-12 hours. If permethrin unavailable, use diluted³ **benzyl benzoate lotion 25%**.
- If ≥ 6 years old: apply **benzyl benzoate lotion 25%** to whole body from neck to feet. Wash off after 24 hours with soap and water. If severe, repeat once within 5 days. If no improvement, apply **permethrin 5%** as above.
- Treat itch: give **chlorpheniramine**¹ 0.1mg/kg at night or 6-8 hourly as needed for up to 2 weeks →159:15.
- Treat all household members at same time.
- Advise child/carer:
 - Put on clean, washed clothes after treatment.
 - Wash linen and clothes in hot water and expose bedding to direct sunlight.
 - Keep fingernails short and clean.
 - Permethrin and benzyl benzoate are toxic if swallowed. Avoid contact with eyes and broken skin. Stop if rash/swelling develops.
- If yellow crusts appear, **impetigo** likely →83.
- If no better after 1 week, repeat treatment.

Very itchy, red, raised wheals that appear suddenly and usually disappear within 24 hours.



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Urticaria likely

If sudden generalised itch/rash or face/tongue itch/swelling and any of: wheeze, difficulty breathing, dizziness/collapse, abdominal pain, vomiting or exposure to possible allergen⁴, **anaphylaxis** likely →37.

- If recently started new medication, check for drug reaction →80.
- If < 5 years old, ensure deworming up to date (check RtHB).
- Consider possible triggers: foods (milk/eggs/nuts/wheat/seafood), medications, insect bites/stings and latex.
- Apply **calamine lotion**.
- Give **chlorpheniramine**¹ 0.1mg/kg at night for up to 2 weeks →159:15. If recurrent or school going child ≥ 2 years, give instead **cetirizine** once daily until itch controlled/up to 2 weeks →159:14.
- If not better after 24 hours, refer to specialist.
- If repeated episodes, **allergy** likely →117.
- Advise to return immediately if any symptoms of anaphylaxis occur.

If no response to treatment, discuss/refer to specialist for review.

¹Chlorpheniramine is sedating. ²If severe penicillin allergy (history of anaphylaxis, urticaria or angioedema), give instead **azithromycin** 10mg/kg once daily for 3 days →158:10. ³Dilute benzyl benzoate according to age: if 2 months-1 year old: dilute **benzyl benzoate lotion 25%** 1:3 with water. If 1-6 years old: dilute **benzyl benzoate lotion 25%** 1:1 with water. ⁴Common allergens include medications, new food or insect bite/sting within the last few hours.

Localised itchy rash

- If rash on scalp →86.
- If very itchy, small red bumps (burrows) in web spaces of fingers and toes and in skin folds of axillae (armpits), waist or genitals, **scabies** likely →77.
- If patches of itchy, scaly skin that is usually red in the baby and dry in the older child, usually affects inner (flexural) surfaces of elbows and knees, cheeks, scalp and neck, **eczema** likely →118.
- Are there red itchy bumps that may have blistered or healed with darkening of skin?

Yes

Usually occurs in crops.



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Insect bites likely

- If extensive, test for HIV → 95.
- Reduce exposure to insects: treat pets, use bed nets, wash bedding, use insect repellents, clear away puddles of water around house.
- Apply ►calamine lotion as needed.
- If very itchy, give ►chlorpheniramine 0.1mg/kg 6-8 hourly as needed for up to 2 weeks → 159:15. Chlorpheniramine is sedating.
- If blisters or heals with darkened skin, ►papular urticaria likely. Apply ►hydrocortisone 1% cream daily for 5 days to new/inflamed lesions. If itch persists for > 2 weeks, stop chlorpheniramine and give instead ►cetirizine once daily until controlled/up to 2 weeks → 159:14.
- If yellow crusts, ►impetigo likely →83.
- Advise to return if no improvement.

No. Assess further according site of rash:

Rash involves head, face, trunk or limbs.
Does rash have ring-shaped lesion/s?

No

Well-defined, raised plaques covered with silvery scale. Often on knees, elbows, lower back, scalp. May have pitted nails.



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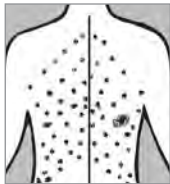
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Psoriasis likely

- Doctor to confirm diagnosis and refer to specialist, ideally within 2 weeks.
- If delay in specialist review and acute flare (worsening redness, dryness, itchiness, pain), apply ►betamethasone 0.1% ointment or ►fluocinolone acetonide 0.025% ointment twice a day for 1-2 weeks (avoid face and genitalia). Then apply ►hydrocortisone 1% ointment twice a day until flare subsides.
- Advise carer/child to: keep nails short, avoid scratching and to expose skin to sunlight before 11am for 30 minutes/day.

Yes

Oval scaly patches involving mostly the trunk (may be in pattern of Christmas tree on back). May have started with one large ring (herald patch) with fine scale in centre, usually on chest or back. Usually affects older children.



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Pityriasis rosea likely

- Reassure rash will resolve on its own within 2 months.
- If very itchy, give ►chlorpheniramine 0.1mg/kg at night for up to 2 weeks → 159:15 (chlorpheniramine is sedating). If not controlled/ more severe in day, switch to ►cetirizine once daily until itch controlled/up to 2 weeks: if 2-6 years old, give 5mg. If ≥ 6 years old, give 10mg.

Slow-growing lesion/s with raised ring of scale, clear in centre



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Tinea corporis (ringworm) likely

- Apply ►clotrimazole 1% cream three times daily. Continue for 2 weeks after rash has cleared.
- If yellow crusts, ►impetigo likely → 83.

Rash involves feet/toes

Cracks, peeling or scaly lesions between toes, or thickened scaly skin on soles, heels and sides of feet



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Tinea pedis (athlete's foot) likely

- Apply ►clotrimazole 1% cream twice a day for 4 weeks.
- Encourage open shoes or sandals.

- Advise to keep skin clean, to dry well and avoid sharing towels/clothes.
- If extensive/recurrent, test for HIV → 95 and diabetes → 32.
- If involves nails → 87. If on head or any hair loss → 86.
- If extensive/no better after 4 weeks, discuss/refer.

If diagnosis uncertain, discuss/refer.

Itch with no rash

- Confirm there is no rash, especially scabies, lice or insect bites.
 - If generalised itchy rash → 77.
 - If localised itchy rash → 78.
- If itch around anus only → 67.
- Is the skin very dry?

Yes

Dry skin (xerosis) likely



Source: Dr Nomphele Gantsho

- Advise to:
 - Avoid hot baths, wool/itchy fabrics and scratching as these may worsen itch.
 - Wash with ▶aqueous cream instead of soap.
 - Moisturise skin with ▶emulsifying ointment, twice a day.
 - Avoid scrubbing the skin and washing more than once a day. Gently pat skin dry.
 - Keep nails short.
- Apply ▶calamine lotion as needed.
- If no better with treatment, discuss/refer.

No

Did child start any new medication in the weeks before the itch started?

Yes

Medication side-effect likely

- Discuss with doctor whether to stop or change medication.
- Advise to:
 - Avoid hot baths, wool/itchy fabrics and scratching as these may worsen itch.
 - Wash with ▶aqueous cream instead of soap.
 - If skin dry, moisturise skin with ▶emulsifying ointment, twice a day.
 - Keep nails short.
- Apply ▶calamine lotion as needed.
- If itch causes difficulty sleeping, give ▶chlorpheniramine 0.1mg/kg at night for up to 2 weeks → 159:15. Chlorpheniramine is sedating.
- Advise to return if rash develops or itch persists.

No

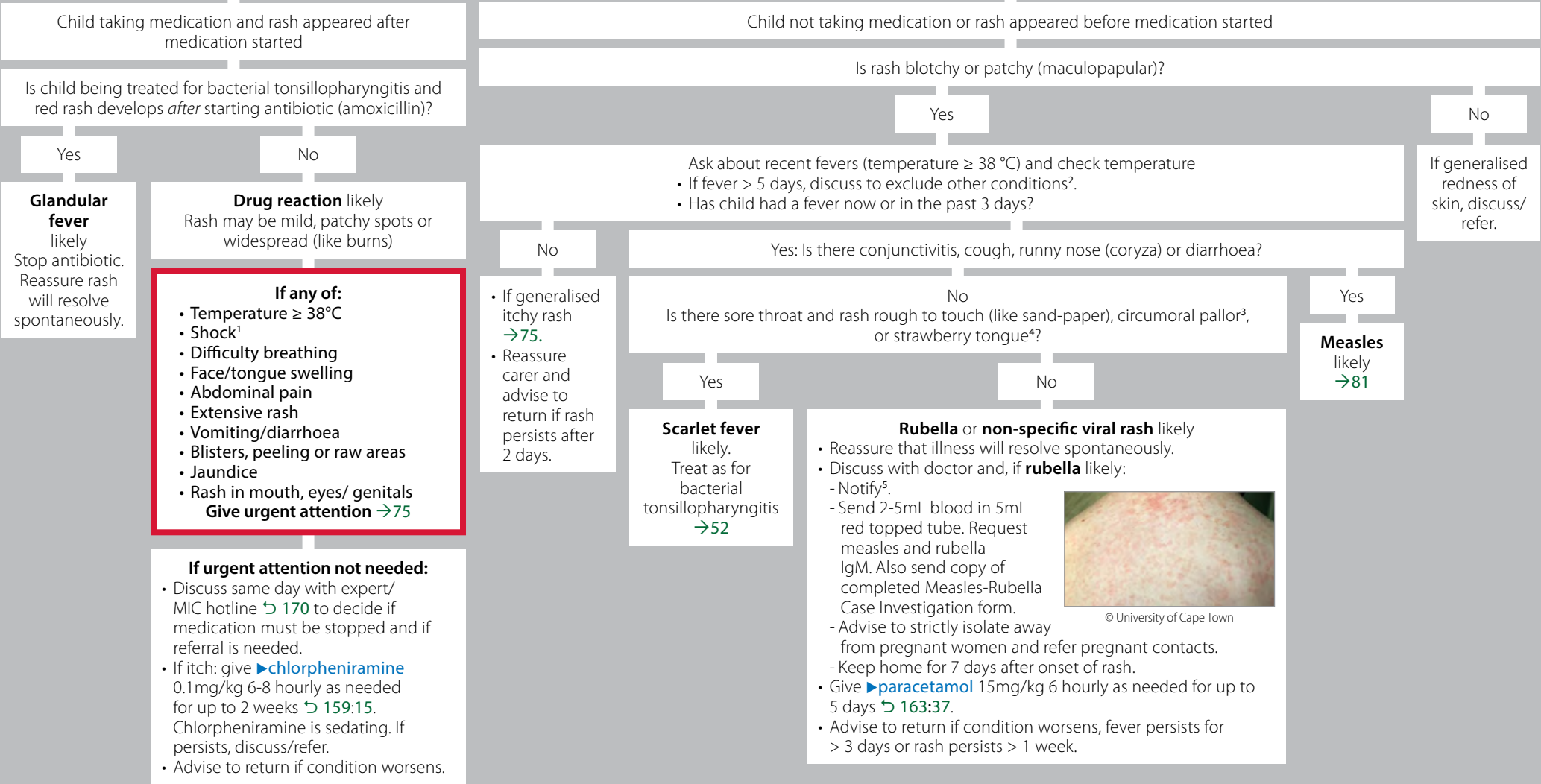
- If yellow skin/eyes, jaundice likely → 61.
- If pallor (pale conjunctiva/palms of hands), check Hb: if Hb < 10g/dL in child < 5 years or Hb < 11g/dL in child ≥ 5 years, anaemia likely → 46.
- If any of: dry skin, brittle hair, constipation, puffy face, intolerant to cold or thyroid enlargement, check TSH. If TSH abnormal, refer to doctor.
- Advise to:
 - Avoid hot baths, wool/itchy fabrics and scratching as these may worsen itch.
 - Wash with ▶aqueous cream instead of soap.
 - Moisturise skin with ▶emulsifying ointment, twice a day.
 - Keep nails short.
- Apply ▶calamine lotion as needed.
- If itch causes difficulty sleeping, give ▶chlorpheniramine 0.1mg/kg at night for up to 2 weeks → 159:15. Chlorpheniramine is sedating.
- If no better after 2 weeks, discuss/refer.

If known with a life-limiting illness, also give palliative care → 150.

If diagnosis uncertain, discuss/refer.

Generalised red rash

- Ask about: when and where rash started and spread, exposure to medications, other people who are ill, and associated symptoms. If joint symptoms →74.
- Look closely at type, pattern, and distribution of rash. Check eyes, inside of mouth (inside of cheeks, tongue, and palate), feel lymph nodes. Check for tick bite →43.
- Use responses to assess and manage further:



Approach to the child with suspected measles:

- Initial clinical features: runny nose (coryza), fever, diarrhoea, conjunctivitis (can be purulent), cough.
- After 2-3 days: small white spots in mouth appear (Koplik spots - see image below).
- 1-2 days later: skin rash appears as numerous, blanching flat red spots (some spots may have merged together). Usually starts behind ears and on neck, then on face and trunk, then limbs.



Image used with permission from VisualDx (www.visualdx.com)



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Koplik spots. CC BY-NC-ND 4.0
Image used with permission from DermNet (dermnetnz.org)

Isolate if possible, ensure child can be safely monitored. Use airborne precautions, including an N95 facemask.
Notify¹ all suspected measles cases.

Give urgent attention to the child with measles and any of:

- Child < 6 months old
 - Difficulty breathing or stidor ↗ 55
 - Headache, drowsy, neck stiffness, meningitis likely ↗ 43
 - Lethargy/altered level of consciousness, sunken eyes, slow skin pinch² ≥ 2 secs, dry mouth/mucous membranes or drinking poorly: dehydration likely
 - Swelling of legs/hands/feet
 - Malnourished³
 - Eye symptoms: purulent discharge, loss/change in vision (blurred/reduced), light sensitivity, hazy cornea, corneal ulcer, eye pain
 - Mouth ulcers/drooling
 - Impaired immunity⁴
 - HIV
 - TB
- Manage and refer urgently:**
- Assess and interpret glucose ↗ 32 and manage child's fluid needs ↗ 26.
 - If meningitis likely: give ♦ceftriaxone 100mg/kg IV/IM ↗ 159:12. Avoid injecting ≥ 1g IM at one injection site. Avoid giving Ringer's lactate together with ceftriaxone. If severe penicillin allergy (previous anaphylaxis, urticaria or angioedema), discuss with doctor.

Approach to the child with measles not needing urgent attention:

- Send 2-5mL blood in 5mL red topped tube. Request measles and rubella IgM. Mark laboratory form with: "NICD Measles Laboratory- attention CVI". Also send copy of completed Measles-Rubella Case Investigation form.
- Advise that measles is contagious and child needs to isolate for 4 days after start of rash. Reduce exposure to other children < 12 months old and pregnant women.
- If < 5 years, give additional ♦vitamin A ↗ 164:44.
- Give ▶paracetamol 15mg/kg 6 hourly as needed for up to 5 days ↗ 163:37.
- If diarrhoea ↗ 63. If painful ear or discharge from ear ↗ 49. If purulent conjunctivitis ↗ 48.
- Give healthy close contacts ≥ 6 months old measles-containing vaccination within 72 hours. If any of: contraindication to measles vaccine (congenital immunodeficiency, leukaemia, lymphoma, other malignancy of bone marrow/lymphatic system, on systemic immunosuppressive therapy (including corticosteroids), confirmed anaphylactic reaction to previous measles containing vaccine/neomycin/gelatine), pregnant, or living with HIV, discuss post-exposure prophylaxis with provincial CDC/NICD ↗ 170.
- Advise to return if not improving or worsening. Counsel parent/carer on signs/symptoms needing urgent attention (see red box).

¹Immediately report to provincial CDC telephonically and via email ↗ 170. Notify within 24 hours via NMC App or <https://www.nicd.ac.za/nmc-overview/notification-process>. Complete NMC Case Notification Form and Case Investigation Form and send to NMCsurveillanceReport@nicd.ac.za and provincialcdc@westerncape.gov.za. ²Pinch skin on abdomen between 2 fingers. Release. Skin normally snaps rapidly back to its normal position. A slow skin pinch takes longer. ³Any of: weight-for-age, length/height-for-age, weight-for-length/height or BMI-for age below -2 line or MUAC < 12.5cm or abnormal growth curve pattern on growth chart (flattening, falling or crossing z-score lines on 2 consecutive visits) or unintentional weight loss ≥ 5% of body weight (weight lost ÷ weight at last visit x 100 = % weight loss). ⁴Impaired immunity: cancer, child without spleen, autoimmune disease, inherited immune deficiency, child awaiting/received blood/organ transplant or receiving chemotherapy, dialysis, TNF-inhibitor therapy or long-term corticosteroids.

Lumps and bumps on skin

- If painful, red, warm lump, **boil/abscess** likely →76.
- If pimples/pustules on scalp, **folliculitis** likely →86.
- If red itchy crops of bumps that may have blistered or healed with darkening of skin, may have scratch marks, **insect bites** likely →78.

< 1 year old and bumpy rash

Tiny red bumps progressing to pustules, appears within 72 hours of birth



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Erythema toxicum likely

Reassure carer that rash is common and will resolve on its own within 1 week.

White bumps on nose and cheeks, appears before 1 month old



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Milia likely

Reassure carer that no treatment is needed - it will resolve on its own within a few months.

Red bumps and pustules on face only (usually cheeks), appears around 3 weeks old



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Baby acne likely

- Advise carer:
 - Wash skin with mild soap daily.
 - Avoid moisturisers/oils on affected areas.
- Reassure that it will resolve within 2 months with no scarring.
- If extensive, scarring or no better after 2 months, refer.

Small clear/red, itchy bumps on head/neck/trunk in hot, humid climate, heavily wrapped baby, or following fever



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Miliaria (heat rash) likely

- Advise carer:
 - Keep baby cool and avoid sweating.
 - Dress in loose cotton clothing.
 - Wash daily and gently rub skin with cloth.
- If fever → 43.
- For itch/discomfort, apply ►calamine lotion as needed.

Pink, red or blue lump/s, warm to the touch. Present from birth or in first few weeks



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Haemangioma likely

- Advise carer:
 - Grows rapidly over first year of life, then starts to shrink.
 - No treatment is needed.
 - It should resolve by age 9 but small red mark may remain.
- Refer if:
 - Lesion on lips, around eyes/nose/ear, over spine/front of neck.
 - Ulcerating or bleeding.
 - Infected (skin red, warm, painful).
 - Multiple or extensive.
 - Not starting to shrink after first year.

Skin coloured or pearly white bump/s with central dimple



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Molluscum contagiosum likely

- If extensive or large, test for HIV → 95.
- Reassure child/carers that bumps should resolve on their own or with ART.
- Refer for liquid nitrogen or treatment with podophyllin or topical retinoids if:
 - Extensive
 - On eyelid
 - Intolerable

Raised, rough lumps and bumps, often on hands, fingers, elbows or knees



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Warts likely

- If under foot, **plantar wart** likely →73.
- Reassure that warts often disappear on their own.
- If treatment desired:
 - Soften wart by soaking in warm water and scrub gently with clean nail file.
 - After drying well, protect surrounding skin with ►petroleum jelly.
 - Apply ►wart magic (podophyllin/salicylic acid/benzoin tincture) to wart at night and allow to dry. Wash off with soap and water in morning.
 - Repeat every night for up to 5 days, until wart disappears. Repeat after 1 week, if needed.
- If extensive or no better with treatment, test for HIV → 95.
- If extensive, refer.

If diagnosis uncertain, discuss/refer.

Crusts, flaky skin and ulcers

Ask how and when skin symptoms started and look at the colour, characteristics and distribution. Also check for additional clues like pus-filled blisters.

Pus-filled blisters which dry to form honey coloured crusts. Often around mouth/nose/buttocks. May have started as insect bite, scabies, injury, eczema or tinea capitis



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Impetigo likely

- Keep nails short. Wash and soak sores in soapy water to soften and remove crusts. Cover draining lesions with saline-soaked gauze dressing.
- Apply **povidone iodine 5% cream** or **povidone iodine 10% ointment** 8 hourly.
- If extensive, temperature $\geq 38^{\circ}\text{C}$, unwell or fatigue: give **cephalexin¹** $\rightarrow 159:13$ or **flucloxacillin¹** $\rightarrow 161:23$.
- If not completely resolved, repeat course of antibiotics.
- Look for cause: if insect bites $\rightarrow 40$, lice/nits $\rightarrow 86$, scabies $\rightarrow 78$, eczema $\rightarrow 118$, tinea capitis $\rightarrow 86$.
- Advise carer that impetigo is contagious:
 - Ensure regular hand-washing to prevent spread.
 - May return to school 1 day after starting antibiotic.
- If sores present for > 1 week, check urine dipstick.
- Refer if:
 - Cellulitis or abscess.
 - No better after 2 courses of antibiotics.
 - $\geq 1+$ blood/protein on urine dipstick, visible blood in urine, little/no urine, or limb/face/feet swelling $\rightarrow 69$.
- Advise child/carers to return immediately if blood in urine or limb/face/feet swelling and refer same day $\rightarrow 69$.

Flaky or greasy crusts with underlying red base. Often affects face, forehead, behind ears, eyebrows, eyelids and nose creases. May be itchy. Usually in infant < 1 year old



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Seborrheic dermatitis likely

- If on scalp, $\rightarrow 86$. If in nappy area $\rightarrow 85$.
- If extensive or > 1 year old, and HIV status unknown, test for HIV $\rightarrow 95$.
- Advise carer to:
 - Trim nails and avoid scratching.
 - Wash body with **aqueous cream** and avoid perfumed soap.
- Apply **hydrocortisone 1% cream** twice a day for 4 weeks or until improved. Then as maintenance, apply once or twice weekly as needed.
- If poor response or if severe, apply **betamethasone ointment 0.1%** or **fluocinolone acetonide 0.025% ointment** once a day for 7 days. Avoid face, neck and flexures (skin creases around joints).
- Refer if extensive and no response to treatment.

Well-defined, raised plaques covered with silvery scale. Often on knees, elbows, lower back, scalp. May have pitted nails

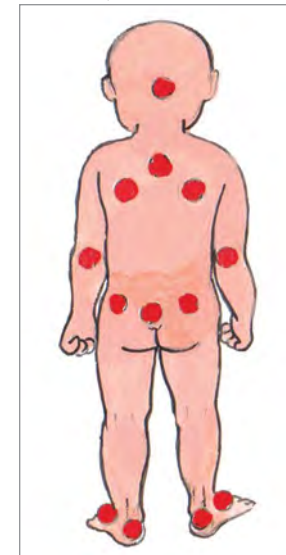


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Psoriasis likely

- Doctor to confirm diagnosis and refer to specialist, ideally within 2 weeks.
- If delay in specialist review and acute flare (worsening redness, dryness, itchiness, pain), apply **betamethasone 0.1% ointment** or **fluocinolone acetonide 0.025% ointment** twice a day for 1-2 weeks (avoid face and genitalia). Then apply **hydrocortisone 1% ointment** twice a day until flare subsides.
- Advise carer/child to:
 - Keep nails short and avoid scratching.
 - Expose skin to sunlight before 11am for 30 minutes/day.

Ulcer in common bed sore site, child usually in bed or wheelchair



Pressure ulcer likely

- Gently clean ulcer with **sodium chloride 0.9% solution** then apply **zinc and castor oil ointment**.
- Educate carer to prevent further bedsores:
 - Wash and dry skin daily.
 - Look daily for skin changes.
 - Keep linen/wheelchair seat dry.
 - Move (lift, do not drag) child every 1-2 hours if unable to shift own weight.
- If child known with a life-limiting illness, also give palliative care $\rightarrow 150$.
- Refer to hospital.

If skin lesions in child with severe wasting (weight-for-length/height below -3 line), BMI² below -3 line if ≥ 5 years, or MUAC³ $< 11.5\text{cm}$ if < 5 years, **severe acute malnutrition with complications** likely $\rightarrow 126$.

¹If severe penicillin allergy (history of anaphylaxis, urticaria or angioedema), give instead **azithromycin** 10mg/kg once daily for 3 days $\rightarrow 158:10$. ²Body mass index. ³Mid-upper arm circumference.

Altered skin colour

- If yellow skin, **jaundice** likely →61.
- Is area of skin red/pink, dark, light or absent in colour?

Red/pink areas present from birth

Is there a clear edge?

Sharply defined edge



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Port wine stain likely (congenital malformation)

- Advise child/carer:
 - Gently explain that this is likely permanent. It usually grows with child and may get darker and thicker in adulthood.
 - If causing distress, try covering with cosmetics.
- As child gets older, ask about school refusal and bullying/teasing → 140.
- Refer if:
 - Becomes warm, painful or raised
 - Around eye
 - Vision problems
 - Seizures/fits
 - Weakness of limb/s
 - Developmental delay
 - Causing significant distress

No clear edge



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Salmon patch/storkbite likely

- Reassure carer it will fade, usually within 2 years.
- No treatment needed.

Dark patches

Fine scale may be seen if scraped.



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Tinea versicolor likely

- Apply ►**selenium sulphide 2.5% shampoo**. Lather shampoo on affected areas.
 - Apply once weekly for 3 weeks: leave on overnight, then wash off in the morning.
- Advise that colour may take months to return to normal, but absence of scale indicates adequate treatment.
- Recurrence is common.

Light patches

Where is patch on body?

Trunk

Face



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Pityriasis alba likely

- Mild form of eczema.
- Reassure carer it will resolve in a few months.
- Apply ►**hydrocortisone 1% cream** twice a day until resolved.
- Advise child/carer:
 - Apply sunscreen daily.
 - Avoid sun exposure, wear hats/long-sleeved tops.

Patchy



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Vitiligo likely

- Apply ►**sun barrier cream** to sun-exposed areas, 15 minutes before going out into the sun.
- Advise child/carer:
 - If causing distress, try covering with cosmetics.
 - Skin colour may return, although usually not on hands, feet, lips and genitals.
- As child gets older, ask about school refusal and bullying/teasing → 140.
- Refer to specialist.

Absence of colour

Is absence of colour patchy or generalised?

Generalised

Present from birth. Involves skin, hair and eyes.



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Albinism likely

- Apply ►**sun barrier cream** to sun-exposed areas (face/neck/ears/chest/legs/arms/feet) *every day*, and 15 minutes before going into sun.
- Advise child/carer:
 - Avoid sun exposure as much as possible, wear hats, long-sleeved tops, long pants, and use an umbrella to shade from sun.
 - Wear sunglasses, to prevent eye damage.
 - Check skin regularly for new or changing skin lesions, especially in sun-exposed areas.
- Refer to specialist for eye care.
- Refer to dermatologist for regular skin checks or if any skin changes.
- Ask about school refusal/bullying/teasing → 140.

Refer if diagnosis uncertain.

Nappy rash

- Ask about: when and where rash started, type of nappies used, frequency of nappy changes, stools (if watery stools and/or > 3 stools/day ➔ 63), changes in diet, treatment tried already.
- Look at: the rest of the skin, check buttocks, area around anus and open groin folds to see if rash involves creases. Look for: scaling/satellite spots (small red flat dots near edge of rash).

Does rash involve inguinal creases (skin folds between the abdomen and thigh)?

No

Irritant nappy dermatitis likely
Most common rash. Red patches on surfaces that are in direct contact with the nappy.



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Explain that moist environment of nappy and prolonged contact and friction of soiled nappy irritates baby's sensitive skin. May also be caused by harsh soaps or detergents used to clean nappies.

- Give **zinc and castor oil ointment** to apply at every nappy change.
- If rash persists > 3 days: add **clotrimazole 1% cream** before covering with **zinc and castor oil ointment**, with each nappy change.

Yes

Does infant have a rash anywhere else on the body (face, scalp, and skin folds like armpits, neck, and behind ears)?

Yes
Seborrheic dermatitis likely

- Usually presents in the first month of life.
- Salmon-pink patches and scaling or flakiness.
- Often begins on scalp and face.
- More likely to develop irritant nappy dermatitis.



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No
Candida likely
Usually involves inguinal creases, with discrete/separate satellite pustules and spots, and scaling along the margins.



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Apply **clotrimazole 1% cream** and then **zinc and castor oil ointment** with each nappy change until resolved.

Advise the carer on good nappy practice:

- Change nappy frequently, as soon as soiled or at least every 2 hours, more often if diarrhoea or if newborn. Avoid waterproof pants. If able, use disposable nappies while rash healing.
- Use warm water and a soft cloth to clean area. Avoid soaps and bubble baths. Pat gently or air dry, avoid rubbing. Allow time without nappy on to expose skin to air and sunlight.
- Apply thick layer of zinc and castor oil ointment at every change and avoid removing it after each nappy change, rather apply another layer. Continue for at least 2 weeks after rash has resolved.

- For pain or discomfort, give **paracetamol** 15mg/kg 6 hourly as needed until rash improved or for up to 5 days ➔ 163:37.
- If new pustules, blisters, honey-crusted lesions develop, **bacterial infection** (like **impetigo**) likely: give **povidone iodine 5% cream** or **povidone iodine 10% ointment** to apply before zinc and castor oil ointment 8 hourly. If severe, also give **cephalexin**¹ ➔ 159:13 or **flucloxacillin**¹ ➔ 161:23 for 5 days.
- If rash worsens or no better after 3 days, discuss/refer.

If burns, bruising or unusual/patterned wounds in nappy area, or poor hygiene, consider abuse/neglect ➔ 144.

¹If severe penicillin allergy (history of anaphylaxis, urticaria or angioedema), give instead **azithromycin** 10mg/kg once daily for 3 days ➔ 158:10.

Hair and scalp symptoms

- If dark hair has turned reddish or hair has become sparse/brittle, assess growth → 15. If itchy, scaly, dry skin affecting scalp and inside of elbows, knees, cheeks or neck, **eczema** likely → 118.
- Ask about rash, itch and hair loss:

Rash with or without itch

Greasy scale over red/pink patches. May also occur between eyebrows, in nose folds, behind ears. Usually itchy.



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Seborrhoeic dermatitis likely

- If < 1 year old and only on scalp, **cradle cap** likely:
 - Apply baby shampoo to scalp, massage and rinse off, then gently brush scalp with fine-toothed comb or soft brush to remove loosened scale.
 - Repeat daily until resolved.
- If ≥ 1 year old:
 - Apply ► **selenium sulphide 2.5%** shampoo to scalp and massage.
 - Rinse off after 10 minutes.
 - Use once a week until resolved (usually 2-4 weeks) then every second week for maintenance.
- If involving face, forehead, behind ears, eyebrows, eyelids and nasal creases → 83.
- If extensive and HIV status unknown, test for HIV → 95.
- If no better after 1 month/extensive, refer.

Red pimples, pustules or nodules around hair follicles



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Folliculitis likely

- Wash scalp with ► **povidone iodine scrub** once a day until lesions resolved.
- Advise to wash hands regularly to prevent spread.
- If deep, extensive, recurrent or no response:
 - Give ► **cephalexin**¹ → 159:13 or ► **flucloxacillin**¹ → 161:23 for 5 days.
 - Test for HIV → 95.

Scaly patches, itchy scalp with patchy (usually circular) hair loss



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Tinea capitis likely

- Give ► **fluconazole** 6mg/kg once daily for 28 days → 161:24.
- Advise child/carer to wash combs and hairbrushes with bleach and avoid sharing them.
- Explain to carer that child can go to school once treatment started and there is no need to shave head or cut hair.

Itch without rash

Severe itch with lice or eggs in hair. May have small red bites on back of neck.

Lice likely

- If < 2 months old: comb wet hair with fine toothed comb after shampooing with normal shampoo.
- If ≥ 2 months old: apply ► **permethrin 5%** lotion to towel-dried hair:
 - Ensure whole scalp covered and hair saturated up to 1 hour.
 - Avoid broken skin/eyes.
 - Comb hair with fine toothed comb, rinse or wipe comb frequently.
 - Rinse off with normal shampoo after 1 hour.
 - Repeat every 5 days for 3 weeks.
- Wash comb/hair items, clothes and linen used in past 2 days in very hot water.
- Treat household contacts if infected or sharing a bed with child.
- Consider child abuse if lice on pubic, perianal areas or eyelashes/eyebrows → 144 and refer.

Fine, white flakes on hair and clothing

Dandruff likely

- Apply ► **selenium sulphide 2.5%** shampoo:
 - Massage into scalp.
 - Rinse off after 10 minutes.
 - Use once a week until resolved (usually 2-4 weeks) then every second week for maintenance.

Hair loss without rash/itch

Does child wear tightly-pulled ponytails/ braids/weaves, with hair loss along hairline/in area of braids/weave or does child pull at hair a lot and are hair follicles visible?

Yes

No

Discuss/
refer.



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Traction alopecia likely

- Explain cause and advise to avoid tight hairstyles.
- Reassure hair will usually grow again once cause removed.
- If pulling at hair, check:
 - If angry, withdrawn or change in mood/behaviour/feelings and not coping → 139.
 - If school problems → 140.
 - Ask child if anyone hurts/upsets him/her. If appropriate, ask carer if aware of any abuse of child or siblings. If yes to either → 144.
- If no better after 3 months, refer.

If diagnosis uncertain, discuss/refer.

¹If severe penicillin allergy (previous angioedema, anaphylaxis or urticaria), give instead ► **azithromycin** 10mg/kg once daily for 3 days → 158:10.

Nail symptoms

Look at nail/s and ask about trauma to nail:

Pain, redness and swelling of nail folds/edges.
Often with history of trauma, such as nail biting, cutting nails too short or pushing the cuticle/edges back.

Is there pus visible?

Yes



Source: BMJ Best Practice

Acute paronychia likely

- Advise to avoid trauma to nail.
- If pus collection and able to tolerate procedure, doctor to incise and drain.
- Advise soaks in warm salt water for 20 minutes twice a day.
- Give **paracetamol** 15mg/kg 6 hourly as needed for up to 5 days **163:37**.
- Apply **povidone iodine 5% cream** or **povidone iodine 10% ointment** after soaking.
- If severe pain/infection or temperature $\geq 38^{\circ}\text{C}$, also give **cephalexin**¹ **159:13** or **flucloxacillin**¹ **161:23** for 5 days.
- If no response, refer.

No



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Ingrown toenail likely

- If mild, pain not too severe and no signs of infection: trim nails in straight line to allow growth. Advise well fitting shoes.
- If severe redness and swelling/very painful and no pus: clean then elevate nail, separating nail from skin. Insert cotton swab to keep it separated for 2-12 weeks. Advise to apply dry dressings.
- If chronic/severe refer to surgical/orthopaedic service.

If signs of infection, manage as for acute paronychia in adjacent box.

White/yellow disfigured nails, nail/s lifting from nail bed



pepsyrock, Public domain, via Wikimedia Commons

Fungal infection likely

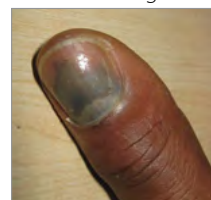
- Test for HIV **95** and check fingerprick glucose **32**.
- Psoriasis may cause nail thickening. Check for psoriasis on skin **83**.
- Advise that fungal nail infection is difficult to treat.
- Discuss/refer to dermatologist.

Blue/brown/black discolouration of nail

Has there been recent trauma to nail?

Yes

Blood and swelling under nail



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Haematoma likely

- Treat if injury < 2 days old, painful and able to tolerate procedure:
 - Clean nail with **povidone iodine 10% solution**.
 - Hold finger securely and gently twist a large bore needle into nail over centre of haematoma. Stop when blood drains through hole.
 - Cover with sterile gauze dressing.
 - Advise that nail may drain for up to 2 days.
- If no pain and no need to refer, reassure no treatment is needed.
- Advise to return if:
 - New pain occurs
 - Infection develops (red, swollen, warm, painful), **cellulitis** likely **76**.
- Refer if:
 - Finger fracture likely
 - Extensive damage to nail bed
 - Unable to perform/tolerate procedure
 - No improvement after treating infection

No

- Ask about onset and appearance:
 - If changing, refer.
 - If present for some time and unchanged, review in 6 months: if still unchanged, reassure
- **pigmented naevus** likely and no treatment needed. Advise to return and refer if starts to change.
- Psoriasis may discolour nails. If psoriasis on skin **83**.
- Review medication: chloroquine, fluconazole, ibuprofen, lamivudine, phenytoin and zidovudine can cause discolouration of nails. Discuss with doctor.

Transverse dents in nail/s (Beau's lines)



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- If fever **43**.
- Consider malnutrition: assess growth **15**.
- Check for paronychia in adjacent column.
- If above excluded, reassure dents are likely due to previous illness/injury and will grow out with nail.

If diagnosis uncertain, discuss/refer.

¹If severe penicillin allergy (previous angioedema, anaphylaxis or urticaria), give instead **azithromycin** 10mg/kg once daily for 3 days **158:10**.

Sleep problems

Baby/child has a **sleep problem** if:

- Baby < 12 months old is sleeping < 12 hours per day (including naps).
- Child 1 to 3 years old is sleeping < 11 hours per day (including naps).
- Child 3 to 6 years old is sleeping < 10 hours per day (including naps).
- Child 6 to 13 years old is sleeping < 9 hours per day.

Approach to the child with sleep problems

- **Check medication (review with doctor):**
 - Methylphenidate (used in ADHD) and salbutamol inhalers (used in asthma) may cause difficulty falling asleep.
 - If CLHIV on ART: if on efavirenz, check if ART can be switched → 97. If on dolutegravir, advise child and carer to take dolutegravir in the morning to avoid difficulty falling asleep or staying asleep.
 - Phenobarbital, valproate and carbamazepine may cause daytime sleepiness.
- **Ask about social risk/stressors:**
 - Ask who looks after child most of time and if carer aware of any abuse of child. Ask child if anyone hurts/upsets him/her. If yes to either, **child abuse** likely → 144.
 - Ask if different carers have different sleeping time rules.
 - If violence or drug/alcohol abuse at home, involve social worker.
 - Manage further according to type of sleep problem. Ask about difficulty falling asleep/staying asleep, bed-wetting/soiling, abnormal movements/behaviour or breathing problem/s:

Difficulty falling asleep or staying asleep

- If tight chest →56, persistent runny, itchy nose →50, itchy skin →79.
- If anal itch/irritation, exclude worms → 67.
- If miserable, stressed or angry →139.
- If behaviour problem →136.
- If child has a life-limiting illness, also give palliative care → 150.

Bed-wetting or soiling

- If previously dry, ask about recent stressful events. Discuss possible solutions.
- If bed wetting and ≥ 5 years old, →68. If bed soiling and ≥ 4 years old →63.
- If fits →28.

Abnormal movements or behaviour

- If teeth grinding →54.
- If child has fits →28.
- If wakes up suddenly screaming or confused and inconsolable and > 3 years old, **night terrors** likely.
 - Advise carer there is no need to wake child. Advise to stay with child until s/he is asleep peacefully.
 - Reassure night terrors are not dangerous and will resolve by age 12.

Breathing problem/s

- If cough/wheeze →55.
- If snoring →50.
- If episodes of no breathing > 10 seconds, **apnoea** likely, discuss/refer same day.

If none of the above, disrupted sleep may be due to bad habits. See below for advice.

Advise the carer of the child with sleep problems to develop sensible sleep habits: inform carer that correcting a poor sleeping cycle can take a few weeks.

Prepare the sleeping environment:

- Make sure space is safe, warm, quiet and not brightly lit. Check child has enough space.
- Remove television, electronic games and cell phones from bedroom.

Establish a good bedtime routine:

- Advise on a consistent bed time and wake up time. Ensure all carers know the sleeping rules.
- Sit quietly with child and read a bedtime story. An object of attachment, like a soft toy can help.
- In older child, allow time to unwind/relax before sleep. Avoid screen time in the hour before bed.
- If struggling with parenting or if child disobedient → 145.

Physical activity:

- Ensure child has > 1 hour of brisk exercise every day.

Food:

- Check child has adequate food and not going to bed hungry.
- Avoid drinking fluids in the 2 hours before bed. Avoid caffeine (coffee/tea, excluding rooibos) and sugar.

School environment:

- If school refusal/bullying/poor school grades → 140.
- If communication/learning problem → 89.

Advise the carer with a baby:

- Place baby on back to sleep (reduces risk of Sudden Infant Death Syndrome).
- Put baby to bed slightly awake, after a nappy change, food and comfort.
- Ensure night-time feed/s continue until 6 months old.

If sleep problem is causing significant distress, unable to find cause and no response to sensible sleep habits, discuss/refer.

Communication problem

Give urgent attention to the child with a communication problem if:
Suddenly unable to communicate as before. Refer urgently.

Approach to the child with a communication problem not needing urgent attention:
Ask carer if child appears to hear as other children do.

Child does not appear to hear as other children do.
Check ears for pain, discharge or eardrum problem.

Child appears to hear as other children do.

- Arrange hearing test.
- If hearing test abnormal, ensure follow up with audiology and speech services.
- If hearing test normal:

No ear pain, discharge or eardrum problem
Refer to audiologist for hearing screen.

Hearing screen normal

Child does not have hearing problem. Assess communication (talking) problem:

- If child ≥ 6 years old, ask if talking problem affecting school work (failing grades): refer to school-based support team or speech therapist, where available.
- Check if understanding is appropriate for age by asking the following in child's home language:

≥ 1 year old

- Points to common items (cup, shoe, bottle).
- Uses simple words ("mama, dada, ball")

2 years old

- Points to a few body parts.
- Does a one-step command ("fetch your bottle").
- Uses 2 word combinations ("come mommy").

3 years old

- Understands opposites ("go-stop", "big-small", "up-down").
- Follows 2 part command ("Pick up the book and give it to your mommy").
- Answers simple questions ("Who is your best friend?" and "What is your favourite food?")

4 years old

- Understands words like "cold", "hot", "hungry", "tired".
- Stranger can understand what child is saying.
- Says name, age, sex.

5 years old

- Speaks clearly.
- Answers questions about school.

Is child's understanding appropriate for age?

No, **understanding problem** likely.

- Assess milestones →12.
- Refer to paediatrician. While waiting for appointment, also assess:
 - If child uses language in an unusual way such as copies sounds, has difficulty initiating or sustaining age-appropriate conversation, makes unusual or repetitive sounds or appears not to respond to normal language cues, consider **autism spectrum disorder**, assess further → 137, and refer to paediatrician.
 - If adequate parental supervision. If harm or neglect suspected →144
 - If violence or substance abuse in home, refer to social worker.
 - If behavioural problems → 136 or if child miserable, stressed or angry → 139.
 - If carer struggling with parenting → 145.
 - Screen for depression in carer: in the past month, has carer: felt down/depressed/hopeless or little interest/pleasure in doing things? If yes to either → PACK Adult.
- Encourage carer to play and interact with child: provide contact/hold child regularly, sing/dance with child, encourage activity outside, kick/throw a ball, read books daily.

Yes, but child has **speaking problem**

- Assess mouth and throat: look for cleft palate or tongue-tie. If found, refer to ENT same week.
- If stuttering or other speaking problem, refer to speech therapist.

If red, bulging/abnormal looking eardrum or ear pain/discharge

→49

Hearing screen abnormal

- Ensure child has follow up with audiology and speech therapy.
- Help access support → 170.

Not moving or sitting properly

Give urgent attention to the child not moving/sitting properly and any of:

- Baby < 2 months old
 - Acute onset (unable to move/sit properly for < 72 hours)
 - Not talking or using hands as before
 - Decreased level of consciousness ↗ 174
- Injury ↗ 33
 - Recent loss of milestones
 - Headache > 1 week
- Painful movement
 - Temperature ≥ 38°C
 - Size of head increasing

Manage and refer urgently:

- If temperature ≥ 38°C, decreased level of consciousness or baby < 2 months old, infection likely. Give ♦ ceftriaxone¹ 100mg/kg IV/IM ↗ 159:12. Avoid injecting ≥ 1g IM at one injection site.

Approach to the child not moving/sitting properly not needing urgent attention:

- If problem only in leg/s ↗ 72. If problem only in arm/s ↗ 71.
- If painful or stiff joint, check if problem in joint ↗ 74.

Check the child’s motor milestones. If born premature, use corrected age² until 2 years old.

14 weeks old	6 months old	9 months old	15 months old	18 months old	3 years old	5-6 years old
Lifts head when held against shoulder.	Holds toy in each hand.	Sits and plays without support.	Stands on own.	Walks unsupported, uses fingers to feed.	Runs well, can climb.	Hops on one foot, draws a stick person.

Has child achieved milestone at the appropriate age?

Yes

No

Child achieved motor milestone at appropriate age. Now not moving or sitting properly.

Refer to paediatrician.

Check for HIV, TB, thyroid problem, anaemia, growth problems and correct/treat if abnormal:

If status unknown, test for HIV ↗ 95.
If HIV positive, give routine HIV care ↗ 97.

Exclude TB ↗ 108.

If ≤ 1 year old, check TSH.
If abnormal, refer to doctor.

Do Hb: if Hb < 10g/dL in child < 5 years old or Hb < 11g/dL in child ≥ 5 years old, anaemia likely ↗ 46.

Assess growth ↗ 15.

Then check for abnormal spine, head circumference ↗ 12, tone (child floppy or stiff) and posture.

Normal head circumference, tone, posture and spine

- Review motor milestones in 8 weeks:
 - If child has not achieved milestone, refer to paediatrician.
 - Arrange occupational therapy and physiotherapy appointments in meantime.
- Encourage carer to play and interact with child: provide contact/hold child regularly, sing/dance with child, encourage activity outside, kick/throw a ball, read books daily.
- Check if carer coping. If struggling, check carer's mental health ↗ PACK Adult.

Abnormal head circumference, tone, posture or spine

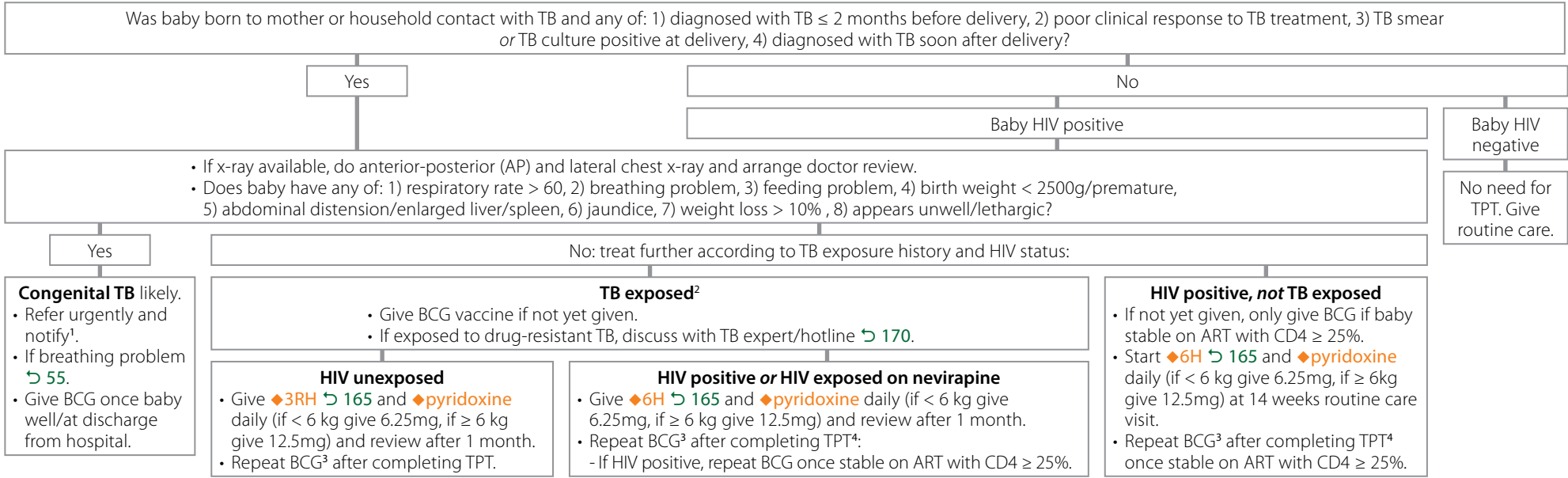
- Check visual, hearing and communication milestones ↗ 89.
- If abnormal behaviour ↗ 136.
- Refer same day if abnormal spine.
- Refer to next level of care.

¹Avoid giving Ringer's lactate together with IV/IM ceftriaxone. If severe penicillin allergy (previous anaphylaxis, urticaria or angioedema), discuss with doctor. ²Corrected age = actual age in months (or weeks) - number of months (or weeks) premature. To calculate corrected age of 9 month old baby born premature at 32 weeks (this is 8 weeks or 2 months premature): 9 months - 2 months = 7 months old.

Prevent communicable infections in the newborn

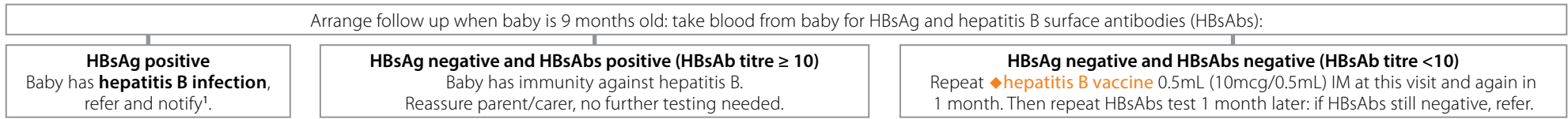
Assess and manage the newborn exposed to HIV, TB, hepatitis B or syphilis. If exposed to HIV, assess and manage further ➔ 92.

Assess the newborn exposed to TB, or the newborn who tests positive for HIV, for TB preventive treatment (TPT)



Manage the baby born to mother with hepatitis B infection

- Arrange delivery at facility that stocks immunoglobulin (HBIG) and the monovalent hepatitis B vaccine:
 - Give **hepatitis B immunoglobulin (HBIG)** 200IU IM and **hepatitis B vaccine** 0.5mL (10mcg/0.5mL) IM within 12 hours of delivery.
 - Continue routine hepatitis B immunisations at 6, 10, 14 weeks and 18 months.



Manage the baby born to mother with syphilis

- If rash (peeling rash, red/blue spots or bruising especially on soles and palms), jaundice, pale conjunctiva/palms, distended abdomen, swelling, birth weight < 2500g, runny nose, respiratory distress, hypoglycaemia, **congenital syphilis** likely. Refer urgently and notify¹ within 7 days.
- If no signs/symptoms and any of the following, **congenital syphilis** still likely. Give baby single dose **benzathine benzylpenicillin** 50 000 units/kg IM into outer thigh, notify¹ and discuss/refer:
 - 1) Mother received < 3 doses of benzathine benzylpenicillin
 - 2) Mother received antibiotic other than benzathine benzylpenicillin to treat syphilis
 - 3) Delay (> 14 days) between maternal doses of benzathine benzylpenicillin
 - 4) Baby delivered within 30 days of mother receiving last dose of benzathine benzylpenicillin

¹Complete notifiable medical conditions (NMC) case notification form and send to NMCsurveillanceReport@nicd.ac.za or notify electronically: <https://www.nicd.ac.za/nmc-overview/notification-process>. ²If mother with TB advise to adhere to TB treatment. Encourage to continue breastfeeding, wear a mask when holding baby and open windows in the home. ³Give BCG vaccination 24 hours after the last TPT dose. ⁴Continue TPT until infant is eligible to receive BCG.

Manage the HIV-exposed infant

Approach to the HIV-exposed baby

- Do mother's viral load at delivery and do HIV PCR test on baby as soon after birth as possible, within 48 hours. Place barcodes on discharge form and RtHB (with mother's consent).
- If abandoned/orphaned baby, do HIV PCR test and HIV rapid test on baby same day. Manage as higher risk formula feeding baby below, even if HIV rapid test negative¹.
- If mother newly diagnosed during breastfeeding do child's HIV test according to age ➔ 95. Manage as higher risk breastfeeding baby below.

Start post-exposure prophylaxis (PEP) as soon as possible, ideally within 1 hour of birth

- Give baby ♦zidovudine (AZT) 12 hourly and ♦nevirapine (NVP) once daily (see dosing tables below). Give supply for 6 weeks and advise carer to bring all medication to next visit.
- Advise to return for baby's HIV PCR and mother's viral load results in 3-6 days.

At 3-6 day postnatal visit, check results of baby's HIV PCR and mother's viral load and manage according to results:

If results not available, continue AZT and NVP and follow-up after 1 week. If no HIV PCR done, do at this visit and follow-up after 1 week.

Baby's HIV PCR negative

Baby's HIV PCR positive

Baby's HIV PCR indeterminate

Mother's VL < 50 at delivery

Mother's VL ≥ 50 or unknown at delivery

Low risk

Higher risk

- Manage mother's unsuppressed VL ➔ PACK Adult.
- If mother's VL ≥ 1000, discuss need for HIV resistance test for mother and baby with HIV expert/hotline ➔ 170.

Breastfeeding³

Formula feeding

- Stop AZT².
- Give ♦NVP daily for 6 weeks (see table).
- Repeat mother's VL 6 monthly if breastfeeding and support mother's adherence to ART ➔ PACK Adult.

- Give ♦AZT 12 hourly for 6 weeks (see dosing table below) and
- Give ♦NVP daily for at least 12 weeks (see dosing table below).
- Stop ♦NVP only once mother's VL < 50 or 4 weeks after final breastfeed.
- If mother on 3rd line ART for ≥ 3 months and VL ≥ 1000, alert to risks of breastfeeding, discuss changing to formula feeding and refer to nutritional therapeutic programme (NTP). Discuss with HIV expert/hotline ➔ 170.

Give ♦AZT (12 hourly) and ♦NVP (daily) for 6 weeks (see dosing table below).

- Send 2nd HIV PCR test and refer to doctor to change PEP to ART ➔ 97.
- Advise mother to breastfeed for at least 2 years.
- If formula feeding, consider feasibility of re-establishing breastfeeding.
- Check if baby needs TB preventive treatment (TPT) ➔ 91.

- Continue HIV PEP according to mother's delivery VL result (see adjacent).
- Do HIV PCR test and HIV viral load, review baby and check results within 3 days.

- Repeat baby's HIV test at 10 weeks, 6 months, 18 months, 6 weeks after final breastfeed/nevirapine stopped or any time if baby unwell ➔ 95.
- If breastfeeding mother's VL ≥ 50 at any time after NVP and/or AZT stopped or mother tests HIV positive for the first time during breastfeeding: repeat/do baby's HIV test at this visit, restart/start ♦NVP and ♦AZT (see above), and reassess once baby's HIV test result available.
- If baby started on Rifampicin-containing TB treatment while using NVP, discuss with HIV expert/hotline.

♦Nevirapine (NVP) syrup (10mg/mL)		
Age	Weight	Dose
Birth to 6 weeks	2-2.49kg ⁴	1mL (10mg) daily
	≥ 2.5kg	1.5mL (15mg) daily
6 weeks to 6 months		2mL (20mg) daily
6 to 9 months		3mL (30mg) daily
≥ 9 months		4mL (40mg) daily

♦Zidovudine (AZT) syrup (10mg/mL)		
Age	Weight	Dose
Birth to 6 weeks	2-2.49kg ⁴	1mL (10mg) 12 hourly
	≥ 2.5kg	1.5mL (15mg) 12 hourly
6 weeks to 6 months		6mL (60mg) 12 hourly
≥ 6 months		Dose 12 hourly according to weight ➔ 168.

¹An HIV rapid test shows whether baby was exposed to HIV, but cannot determine whether baby is infected with HIV. An HIV PCR test determines if baby is infected with HIV. ²Return unused AZT to pharmacy to be discarded. ³A breastfed baby has breastfed in the past 7 days or is mixed feeding. ⁴If weight < 2kg discuss with HIV expert/hotline ➔ 170.

Exposed to infectious fluid: post-exposure prophylaxis (PEP)

- Prevent HIV and hepatitis B with PEP following sexual exposure or accidental exposure to potentially infectious blood/bodily fluids. For prevention of vertical transmission of HIV and hepatitis B → 91.
- If exposure involved intact skin, or if exposed to vomit, stool, urine, saliva or sweat (non-blood stained), PEP is **not needed**. If unsure, discuss with HIV expert/hotline → 170.

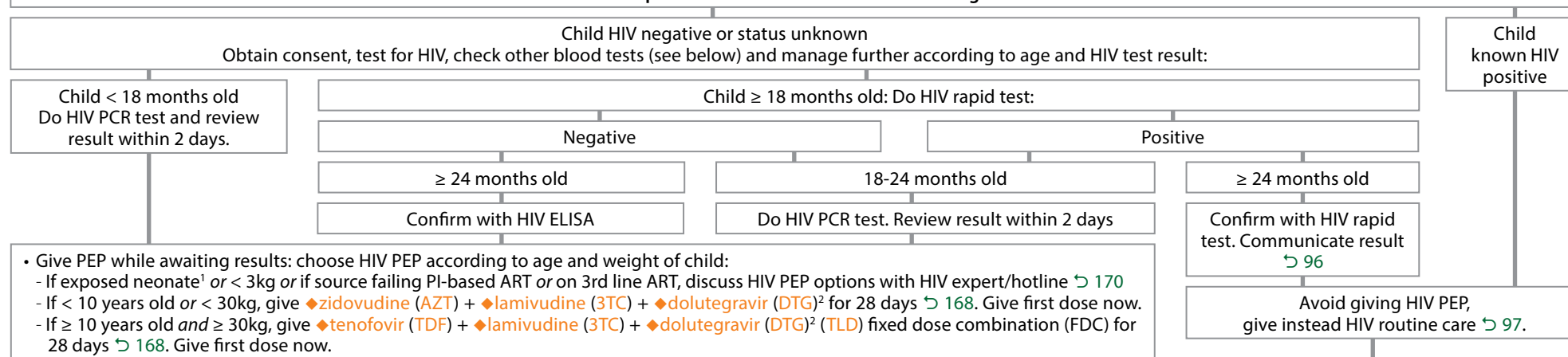
Give urgent attention to the child exposed to infectious fluids in the last 72 hours and any of:

- Exposure through mucosa or broken skin to any of: blood or blood-stained fluid, semen, vaginal secretions, breast milk from woman other than mother, wound secretions
- Sexual assault/rape (oral, vaginal or anal) → 144
- Needle-stick or sharps injury
- Received or gave a human bite that broke the skin

STEP 1: Give child immediate attention

- If broken skin, clean area immediately with soap and water. If human bite severe enough to cause bleeding only, prevent only hepatitis B if needed: move to Step 3 below.
- If splash to eyes, mouth or nose, immediately rinse mouth/nose or irrigate eye thoroughly with water.
- If child exposed to breast milk from woman other than mother, discuss immediately with HIV expert/hotline if milk needs to be aspirated via nasogastric tube (NGT) → 170.

STEP 2: Assess exposed child's need for HIV PEP according to HIV status



STEP 3: Assess exposed child's need for hepatitis B PEP and give first dose if needed

- Check RtHB: a fully vaccinated child received 4 doses of hepatitis B vaccine, usually at 6, 10, 14 weeks and 18 months.
- If not fully vaccinated/vaccination status unknown, give ♦**hepatitis B vaccine**: If < 11 years give 0.5mL (10mcg/0.5mL) IM; if ≥ 11 years, give 1mL (20mcg/mL) IM.
- Continue to assess child for hepatitis B PEP, even if fully vaccinated → 94.

STEP 4: Check other blood tests in exposed child and source client

- Exposed child:
 - **All:** hepatitis B surface antibodies (HBsAb)
 - **If HIV PEP:** if using TDF, do creatinine. If using AZT, do FBC + diff
 - **If sexual exposure:** syphilis
 - **If needle-stick/sharps injury or source hepatitis C positive/unknown:** hepatitis C antibody (HCV Ab)
- Source (if available and consents to test/s):
 - **If HIV unknown:** HIV ELISA
 - **If hepatitis B unknown:** hepatitis B surface antigen (HBsAg)
 - **If hepatitis C unknown:** hepatitis C antibody (HCV Ab)
 - **If sexual exposure:** syphilis

STEP 5. Review with blood results within 3 days → 94.

If sexual assault, ensure child transferred to Thuthuzela care centre/district hospital same day.

¹Baby < 28 days old. ²If child on rifampicin, adjust dose → 168. If < 20kg and DTG 10mg dispersible tablets unavailable, give instead ♦**lopinavir** + ♦**ritonavir (LPVr)** → 168.

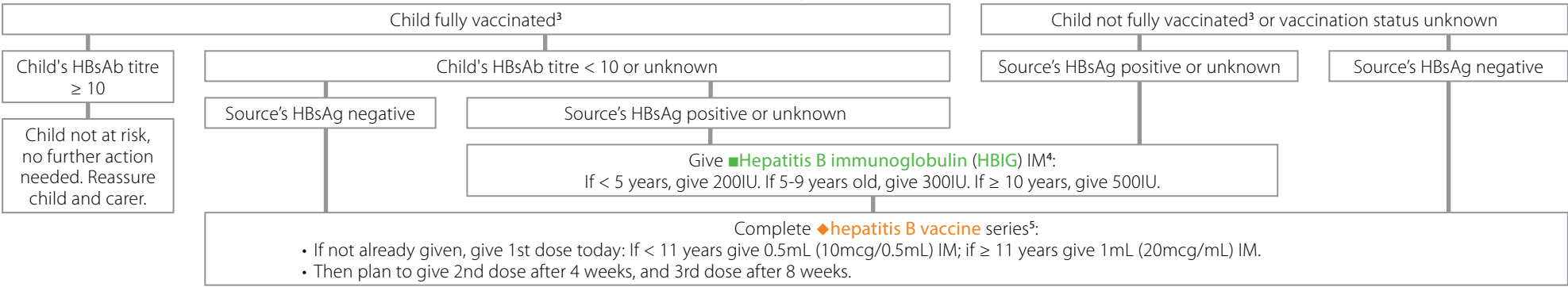
Review the child on post-exposure prophylaxis (PEP)

Review the child on PEP within 3 days, then at 2 weeks, 4 weeks, 6 weeks and 4 months after exposure. Check results of first visit within 3 days.

Assess	When to assess	Note
Adherence, side effects	3 days, 2 weeks	Ask about nausea, vomiting, diarrhoea and manage on symptom pages. Advise these should resolve within 2 weeks and to return immediately if side effects worsen or persist ≥ 2 weeks.
Mental health	Every visit	Assess emotional support and address anxieties around exposure. Link with counselor or support helpline ↗ 170.
Sexual health	Every visit	If sexually active ¹ , emphasise condom use, especially for 4 weeks after exposure. If sexual assault ↗ 144.
Source blood results (if done)	3 days	- If source HIV ELISA negative, discuss continuation of child's HIV PEP with HIV expert/hotline ↗ 170. - If source HIV ELISA, HBsAg, hepatitis C antibody, or syphilis positive, refer source for care. Use results to decide when to do tests in child (below).
HIV PCR	- Done at 1st visit 6 weeks, 4 months	- If HIV PCR positive: communicate result ↗ 96, confirm with 2nd HIV PCR, and give HIV routine care ↗ 97. Change HIV PEP to ART. - If HIV PCR negative: repeat after 6 weeks and 4 months. If < 24 months old use HIV PCR, if ≥ 24 months old, use HIV ELISA test instead.
HIV ELISA	- Done at 1st visit 6 weeks, 4 months	- If HIV ELISA positive: communicate result ↗ 96, confirm with 2nd HIV ELISA, and give HIV routine care ↗ 97. Change HIV PEP to ART. - If HIV ELISA negative: repeat after 6 weeks and 4 months. If ≥ 24 months old use HIV ELISA, if < 24 months old, use HIV PCR test instead.
FBC + diff	- Done at 1st visit - If on AZT: 2 weeks, 4 weeks	If Hb < 8g/dL or neutrophils < 1.0 x 10 ⁹ /L or platelets < 50 x 10 ⁹ /L, discuss/refer.
Creatinine and eGFR ²	- If done at 1st visit - If on TDF: 2 weeks	- If eGFR² < 80: avoid TDF, give instead ♦ zidovudine (AZT) ↗ 168. - If Hb < 8g/dL or neutrophils < 1.0 x 10⁹/L or platelets < 50 x 10⁹/L, avoid AZT and discuss/refer.
Hepatitis B surface antibodies (HBsAb)	Done at 1st visit	Use HBsAb titre result to decide if further hepatitis B PEP needed (see below).
Hepatitis C antibody (HCV Ab)	If done at 1st visit	If negative, do hepatitis C PCR at 6 weeks. If positive, refer/discuss.
Hepatitis C PCR	If needed: 6 weeks	Only check if child's HCV Ab negative and source's positive/unknown. If hepatitis C PCR positive, refer.
Hepatitis B surface antigen (HBsAg)	If needed: 4 months	Only check if source HBsAg positive/unknown. If positive, discuss/refer.
Syphilis	If needed: 4 months	Only check if sexual exposure or source syphilis result positive/unknown. If positive, discuss/refer.

Decide if further hepatitis B prophylaxis needed

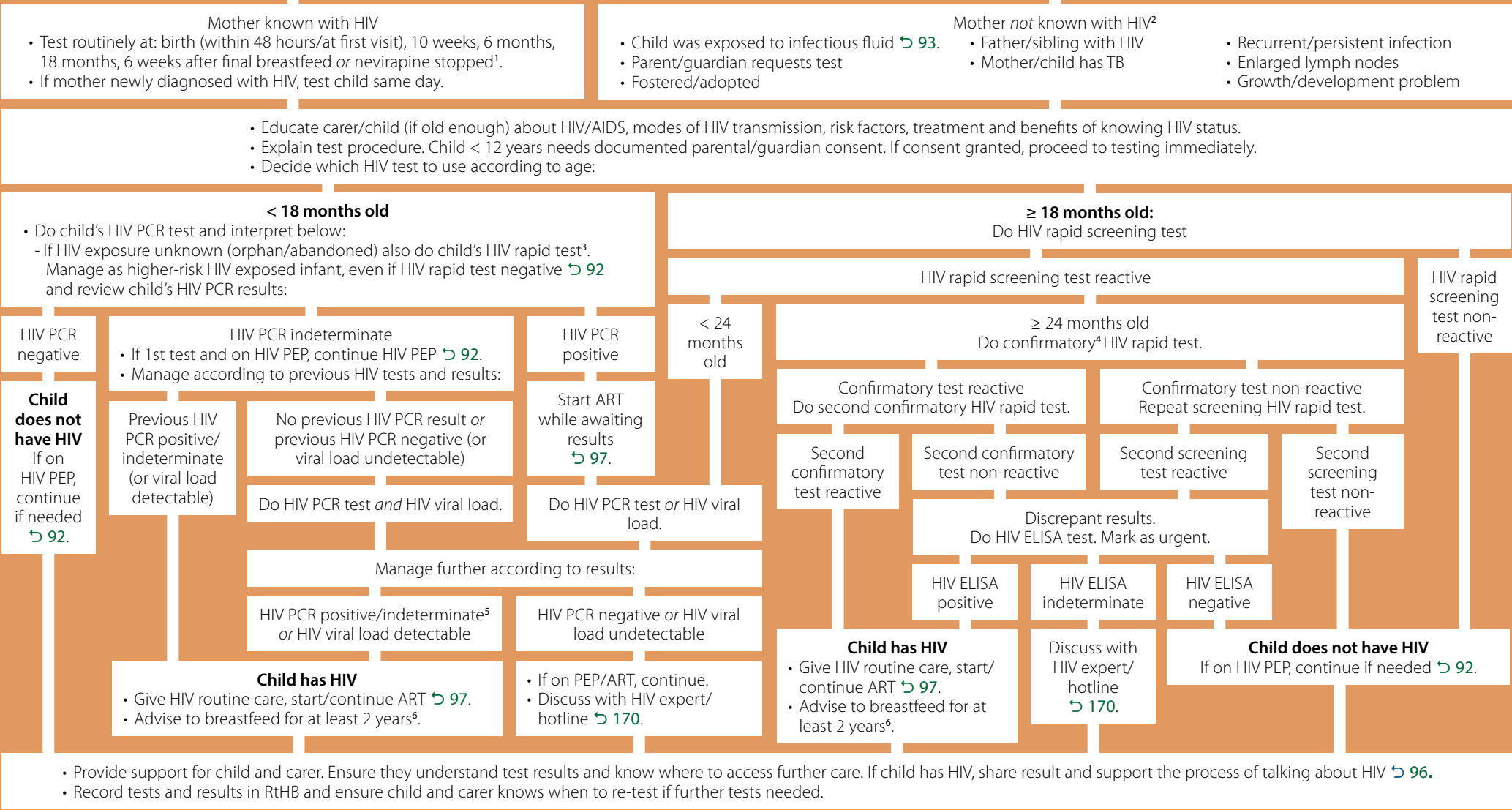
Use child's vaccination status³ and HBsAb result together with source's HBsAg results to decide if further prophylaxis needed:



¹If any of: 1) sex not consensual 2) < 12 years 3) ≥ 12 years old and partner not in peer group (partner must have < 2 year age gap), **sexual abuse** likely ↗ 144. Otherwise advise reliable contraception ↗ PACK Adult. Check that s/he knows how to use condoms. ²eGFR = [height (cm) x 40] ÷ creatinine (μmol/L). ³Check RtHB for vaccination status: fully vaccinated child received 3 doses of hepatitis B vaccine, usually at 6, 10, 14 weeks. If < 3 doses, child is not fully vaccinated. ⁴Use different injection sites for hepatitis B vaccine and HBIG. If HBIG unavailable, refer. ⁵If child not fully vaccinated and < 5 years old give catch-up hexavalent (DTaP-IPV-HiB-HBV) doses instead ↗ 13.

HIV: diagnosis

Regardless of HIV exposure, test child for HIV when child 18 months old. Decide when else to test for HIV and review child and result after 2 days if needed:



Talk to child and carer about HIV

Communicate child’s HIV positive test result

Plan to share result with child and/or carer in a quiet, private area according to child’s age:

< 12 years old	≥ 12 years old
Share result with primary carer ¹ first.	If HIV test done <i>without</i> carer's consent (child presented alone), offer child the opportunity to receive result with carer present.

Share result by following the steps below:

Step 1	Assess knowledge and expectation	- Assess what is already known and what is expected from this visit. Remind child/carers that HIV test was done. - Ask questions like: “What do you understand about what’s been happening so far?” or “Is there anything specific you were hoping to discuss today?”
Step 2	Prepare for news	Prepare child/carers for the news: “I have the result of your HIV test. Unfortunately, it is not what I hoped. I’m going to explain more to you now.”
Step 3	Give information	- Talk clearly, slowly and use simple language: “The result of your child’s HIV test came back positive. This means your child has HIV.” - Emphasise that HIV is treatable: “HIV is a serious infection with no cure but we are able to treat it with medicines everyday, that keep the immune system strong, so that your child can live a long, normal life.” - If HIV was vertically transmitted, reassure that it’s not the mother’s fault and that she is doing the best thing for her child by accessing HIV testing and treatment.
Step 4	Check understanding	Encourage questions and check understanding: “I know this is a lot to understand. Has anything I have said been unclear?” or “Do you have any questions?” or “Do you need more information?” Allow a moment of silence after each question to allow for processing and questions.
Step 5	Offer support and discuss next steps	- Reassure that ongoing support is available. Encourage child/carers to share this news with someone they trust and can rely on. - Give HIV routine care ↗ 97 and support child and carers through their journey with HIV (see below). - Refer to psychologist if needed and available.

Support child and carer through their journey with HIV

- Advise that disclosure is a gradual process. Emphasise the importance of fully disclosing a child’s HIV status to them, as appropriate for their age, and explain the risks of not doing this. Inadequate disclosure may result in poor adherence, poor school performance, emotional problems, transmission of HIV and distrust of carer and health workers.
- Assist carer with a plan of how to explain HIV to their child and share the level of understanding that a child should have according to their age/level of maturity²:

Child < 5 years old - Child should know how to take medications correctly and be praised for this. - If asking questions and seems ready, start to give basic HIV information (next column).	Child 5-10 years old (partial disclosure) - Child should know the basics of HIV: - How to live a healthy life - Basics of the immune system (white cells are fighter cells against germs) - The importance of taking daily treatment (to keep the fighter cells healthy so they can ensure the germs stay ‘sleeping’) - If asking questions and seems ready, give more information (next column).	 Child ≥ 10 years old (full disclosure) - Child should know: - That s/he has HIV - who to trust/share status with - What HIV is, how it is spread (what safe sex is) - Importance of medication adherence (to prevent illness/treatment resistance developing) - Emphasise that it is not child’s or parent’s fault and they can live a normal life.
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- Support the child taking long-term medication ↗ 152.

¹Primary carer: child’s parent or legal guardian. ²A child < 6 years old has sufficient maturity for age if s/he has no delayed milestones (see RtHB). A child ≥ 6 years old has reached sufficient maturity for age if s/he progresses through school grades at appropriate age and does not have a poorly controlled mental illness or severe neurological condition.

HIV: routine care

Assess the child with HIV

Assess	When to assess	Note	
Symptoms	Every visit	Ask if child has any problems, especially new skin rash, yellow eyes, vomiting, difficulty breathing: manage on symptom pages.	
TB	Every visit	- Check for TB at diagnosis, yearly and any time if TB symptoms¹ ➔ 108. Assess if eligible for TB preventive treatment (TPT) ➔ 106. - If starting/on rifampicin-based TB treatment and on dolutegravir or lopinavir/ritonavir, adjust ART doses ➔ 168.	
Adherence	Every visit	Ask about missed doses and look at pharmacy refill records. If poor adherence ² , give increased support ➔ 152.	
Side effects	Every visit	Ask about ART side effects ➔ 104.	
Medication interactions	Every visit	- Ask about other medications like TB or epilepsy treatment, contraceptives, calcium, iron, zinc, antacids, metformin. Adjust medications ➔ 105. - For any other medication interactions, check SAMF, EMGuidance app, use web-based drug interaction checker³ or discuss with doctor.	
Mental health	Every visit	If low mood, anger, stress or anxiety ➔ 139. If behaviour problems ➔ 136.	
Understanding of HIV	Every visit	Support the process of talking about HIV at appropriate age ➔ 96.	
Sexual health	If sexually active	When age appropriate (usually ≥ 12 year old), start to discuss safe sex. Educate that when or if child chooses to start having sex: - It needs to be consensual⁴ and within own peer group⁵. Consider child abuse if < 12 years old or partner is not within peer group⁵ ➔ 170. - Use protection against STIs. Demonstrate condoms. If genital symptoms ➔ 68. - Use reliable contraception (injectable, intra-uterine device, subdermal implant) <i>plus condoms</i> ➔ PACK Adult.	
Carer	Every visit	Ask about health (including mental health) of carer: if unwell, assess using PACK Adult . Ask about HIV status and if HIV positive, ensure parent on ART.	
Other conditions	Every visit	If other long-term health conditions, ensure these are adequately treated. If known with epilepsy, check seizure control and if medication adjustment needed ➔ 130.	
Routine care	Every visit	Integrate routine care into every visit: if < 2 months old ➔ 11; if ≥ 2 months old ➔ 12. Ensure immunisations are up to date. If girl ≥ 9 years, educate on importance of HPV vaccine (given during integrated school health programme campaign).	
Growth	Every visit	Check weight at every visit, height every 6 months, mid-upper arm circumference and head circumference if needed: interpret ➔ 15. Adjust ART doses as child grows ➔ 168.	
Development	Every visit	- If ≤ 5 years old, assess milestones ➔ 12. - If > 5 years old: if there is a delay in reading and writing or delay in self-care (such as dressing, bathing, brushing teeth), refer. If problems at school ➔ 140.	
WHO clinical stage	Every visit	Check weight, mouth, skin, previous and current problems to determine HIV stage (see table below). If stage worsens on ART, discuss/refer.	
Need for tests	Every visit	At diagnosis, decide which ART regimen the child needs before deciding which bloods to check ➔ 98. If on ART, check routine tests and interpret results ➔ 98.	
Stage 1	Stage 2	Stage 3	Stage 4
- No symptoms - Persistent generalised lymphadenopathy	- Recurrent/chronic sinusitis, otitis media, tonsillitis - Fungal nail infections - Chronic dermatitis/papular pruritic eruption (PPE) - Extensive warts/molluscum contagiosum - Herpes zoster (shingles) - Recurrent mouth ulcers - Angular cheilitis/stomatitis - Enlarged parotid glands/liver/spleen	- Pulmonary TB or TB lymphadenopathy - Severe recurrent bacterial pneumonia - Lymphoid interstitial pneumonitis (LIP) or bronchiectasis - Oral thrush if child > 6 weeks old - Oral hairy leukoplakia or acute necrotising ulcerative gingivitis/periodontitis - Unexplained conditions unresponsive to treatment: diarrhoea ≥ 14 days, fever ≥ 1 month, anaemia (Hb < 8g/dL), neutropaenia (neutrophils < 0.5x10⁹/L) or thrombocytopaenia (platelets < 50x10⁹/L) ≥ 1 month. - Moderate acute malnutrition (MAM) not responding to treatment	- Extrapulmonary TB (not TB lymphadenopathy) - Cryptococcal meningitis - Oesophageal thrush (pain on swallowing) ≥ 2 severe bacterial infections per year (not pneumonia) <i>Pneumocystis jirovecii</i> pneumonia (PJP) - Herpes simplex ulcers ≥ 1 month - Kaposi's sarcoma - HIV encephalopathy - Toxoplasmosis if child > 6 weeks old - Severe wasting, stunting or severe acute malnutrition (SAM)

Continue to assess the child with HIV → 98.

¹TB symptoms: current cough/fever, drenching night sweats, poor weight gain/failure to thrive, weight loss, decreased playfulness/fatigue, visible neck mass, blood-stained sputum, persistent headache/vomiting (without diarrhoea). ²Check script for pharmacy refills and notes for clinic appointment attendance. If available, do drug level on urine or blood specimen. Adherence is considered good if ≥ 1 of: 1) child has collected medication at least 80% of the time, 2) child has attended > 80% of scheduled clinic visits, 3) medications are detected in child's urine/blood. ³www.hiv-druginteractions.org/checker. ⁴Consensual: all partners are willing and agree to participate. ⁵Partner should also be 12-16 years old with < 2 year age gap.

Check tests according to table and review results below. Results do not have to be available to start child on ART same day. Record correct contact details in case of abnormal results to recall child/carer.

At diagnosis or if restarting ART	Starting/ changing ART	1 month on ART	3 months on ART	4 months on ART	6 months on ART	1 year on ART	6 monthly	Yearly	Also
• Blood: CD4, FBC + diff, syphilis (if sexually active¹) - If starting TDF: creatinine - If < 2 years old and mother failing DTG or PI: genotype testing • TB test: If able to produce sputum: TB NAAT. If unable to produce sputum <i>and</i> TB symptoms², do chest x-ray. • Urine: if ≥ 10 years old, urine dipstick and pregnancy test³	• Changing to TDF: creatinine • Changing from TDF: HBsAg • Starting AZT: FBC + diff	• TDF: creatinine • AZT: FBC + diff	• AZT: FBC + diff • ATVr/LPVr: random total cholesterol, triglycerides	• Viral load • TDF: creatinine	• AZT: FBC + diff	• Viral load • CD4 • TDF: creatinine	• CD4 until child stops co-trimoxazole	• Viral load • TB test: If able to produce sputum: TB NAAT. If unable to produce sputum, no further TB tests needed. • TDF: creatinine, urine dipstick	Any time if TB diagnosed: CD4, viral load.

ABC- abacavir; ALD- ABC + 3TC + DTG; AZT- zidovudine; DTG- dolutegravir; LPVr- lopinavir/ritonavir; TDF- tenofovir; TLD- TDF + 3TC + DTG; PI- protease inhibitor

CD4	• If CD4 < 200, check CrAg result and manage below. • Use CD4 to guide co-trimoxazole preventive therapy (CPT) ➔ 99. Stop monitoring CD4 once child stops CPT and viral load < 1000. If VL ≥ 1000, do CD4 6 monthly until VL < 1000. • If child returning to care > 90 days after missing ART appointment, check CD4.
Cryptococcal antigen (CrAg)	• Laboratory will automatically do this if CD4 ≤ 200. • If CrAg positive, refer. If symptomatic (severe/recurrent/persistent headache, neck stiffness, vomiting) or pregnant, refer urgently.
FBC + differential count)	• If Hb < 10 in child < 5 years old or Hb < 11 in child ≥ 5 years old, anaemia likely. Check FBC + diff and manage further ➔ 46. • If on AZT and Hb < 8g/dL or neutrophils < 1.0 x 10⁹/L or platelets < 50 x 10⁹/L, refer to doctor.
Syphilis	If syphilis positive ➔ PACK Adult .
Creatinine and eGFR ⁴	• If eGFR < 30, stop TDF and refer same day. • If eGFR 30-80, recall child/carer: - Refer to doctor to check BP, glucose, urine dipstick, send urine for protein/creatinine ratio, arrange kidney ultrasound and check if other medication doses need adjusting. - Treat according to HBsAg result: • If HBsAg positive: refer. • If HBsAg negative: stop TDF, use ■ABC instead. If on TLD: switch to ■ALD. If previous hypersensitivity to ABC, use ■AZT instead of ABC.
TB test	Interpret TB test results ➔ 108. Repeat TB test yearly, at same time as yearly viral load done.
Chest x-ray	Doctor to interpret chest x-ray ➔ 109.
Urine dipstick	• If proteinuria, check creatinine (eGFR) if not already done. Interpret result above. • If glucose in urine: check random fingerprick glucose ➔ 32.
Urine pregnancy test	• If pregnancy test positive, give antenatal care ➔ PACK Adult and if not on ART, start same day. • If pregnancy test negative and ≥ 12 years old, advise to use reliable contraception (intra-uterine device, subdermal implant, <i>plus</i> condoms) ➔ PACK Adult.
Hepatitis B surface antigen (HBsAg)	• If HBsAg positive, discuss/refer. • If HBsAg negative, ensure routine immunisations are up to date ➔ 13, check immune response and re-vaccinate against hepatitis B if needed ➔ 93.
Total cholesterol, triglycerides	• If triglycerides ≥ 10, discuss/refer same day. • If total cholesterol > 6 or triglycerides > 5, avoid LPVr. Check if child eligible to switch to DTG ➔ 102.
Viral load (VL)	• If VL < 50, continue routine VL monitoring. If not yet on DTG, switch ART ➔ 102. Check if eligible to collect medications from a repeat prescription collection point (RPC) ➔ 99. • If VL ≥ 50, manage unsuppressed viral load ➔ 103.

Advise and treat the child with HIV ➔99.

¹If any of: 1) sex not consensual 2) < 12 years 3) ≥ 12 years and partner not in peer group (partner must have < 2 year age gap), **sexual abuse** likely ➔ 144. ²Current cough/fever, drenching night sweats, poor weight gain/failure to thrive, weight loss, decreased playfulness/fatigue, visible neck mass, blood-stained sputum, persistent headache/vomiting (without diarrhoea). ³Only do pregnancy test if girl sexually active, missed period and not on contraception. ⁴eGFR = [height (cm) x 40] ÷ creatinine (μmol/L).

Advise the child with HIV and his/her carer

Support starting or restarting ART

- Identify a willing carer able to correctly administer/supervise ART when needed. If this is not possible and child needs ART, discuss/refer.
- Refer for fast-track initiation counselling (FTIC), explain need for lifelong treatment adherence and risks of resistance.
- If child starting abacavir, educate about the signs of abacavir hypersensitivity reaction (AHR) ➔ 104. Alert child/carers to special “patient alert card” in abacavir packaging.

Support carer to talk to child about HIV

- Support with the process of communicating HIV positive test result and talking about HIV at appropriate age ➔ 96.
- Encourage carer/s to disclose own status to child and encourage talking about HIV at home. Recommend that carer finds at least one other adult to help with ART.

Give increased adherence support if poor adherence¹ (or viral load ≥ 50):

- Support the child taking ART ➔ 152.
- Explore child/carers concerns about harms of ART and educate about side effects ➔ 104.
- Ensure carer on ART adherent to his/her own medication.
- If intentional neglect suspected ➔ 144.

Treat the child with HIV

- If *not* on ART, plan to start or restart ART, same day if possible. If starting ART, follow Steps 1-5 ➔ 100. If restarting ART ➔ 101.
- If on ART, check if child on best possible regimen for age and weight:
 - If not on dolutegravir, check if eligible to switch to dolutegravir ➔ 102.
 - Check if ART dose needs adjusting as child grows. Where possible, use most effective ART, fixed dose combination medications, once a day dosing and best tasting options.
 - If child on abacavir + lamivudine + dolutegravir or zidovudine + lamivudine + dolutegravir and is ≥ 10 years old and ≥ 30kg with no history of kidney disease, switch to tenofovir + lamivudine + dolutegravir.
 - If child not tolerating syrups, switch to tablets/capsules as soon as child able to swallow these. If child on lopinavir/ritonavir pellets/granules in capsules, avoid swallowing these whole ➔ 169.
 - If on dolutegravir/abacavir/lamivudine dispersible tablets or lopinavir/ritonavir pellets/granules in capsules, support carer to give ART correctly ➔ 169.
 - If on lopinavir/ritonavir or atazanavir/ritonavir tablets or capsules, child to avoid chewing, crushing or splitting these.

Give prophylaxis/preventive therapy to prevent against infections

Medication	Age	Note
■Co-trimoxazole ➔ 168	< 6 weeks	Only start when baby 6 weeks old.
	6 weeks - 12 months	Give regardless of CD4 or stage. Continue until at least 1 year old, then reassess according to CD4 and stage below.
	1 - 5 years	Give if CD4 ≤ 25% or stage 3 or 4. Stop once CD4 > 25%, unless previous <i>Pneumocystis jirovecii</i> pneumonia (PJP): then continue and reassess at age 5.
	≥ 5 years	Give if CD4 ≤ 200 or stage 3 or 4. Stop once CD4 > 200, regardless of clinical stage. If started for TB and CD4 > 200, stop once TB treatment completed.

- If never completed TB preventive treatment (TPT), assess eligibility for TPT ➔ 106.
- If cryptococcal antigen positive and lumbar puncture negative for cryptococcal meningitis, start or continue ■fluconazole in consultation with HIV expert/hotline ➔ 170.
- If ≥ 6 months old and CD4 > 100, advise influenza vaccination during influenza vaccine campaign.

Review the child with HIV

Child < 5 years old

- Review monthly until on ART for at least 6 months and stable². Then review 3 monthly.
- Where possible, align visit dates with immunisation or routine care visit.

Child ≥ 5 years old

- Review monthly until on ART for at least 4 months and stable². Then review 3 monthly.
- Consider enrolling child for repeat prescription collection (RPC)³:
 - Child is eligible to collect medications from a RPC if all of: 1) child on ART for ≥ 6 months, 2) no medication/dose changes in last 3 months, 3) VL < 50 in last 6 months, 4) all disclosure sessions attended by child/carers, 5) child/carers consents to RPC, 6) child stable², 7) child not pregnant.

¹Estimate adherence by checking script for pharmacy refills and notes for clinic appointment attendance. Adherence is ‘good’ if child tolerates medications (swallows medication, does not spit/vomit it out) and either: has come to collect medication at least 80% of the time; or has attended > 80% of scheduled clinic visits. ²A stable child on ART attends appointment within 28 days of given appointment date, is growing and developing well, does not have opportunistic infections (like TB) and has VL < 50 in the last 6 months. ³RPC points include ‘facility pick-up points’ (FAC-PUP), ‘external pick-up points’ (EX-PUP), clubs. Medications are pre-dispensed by Central Dispensing Unit (CDU) or Central Chronic Medicine Dispensing and Delivery (CCMDD).

Start ART

- ART experienced doctor to start ART if any of:**
- Weight < 3kg
 - Age < 4 weeks
 - Severe acute malnutrition (SAM)
 - TB

STEP 1. Plan to start ART same day according to age and weight.

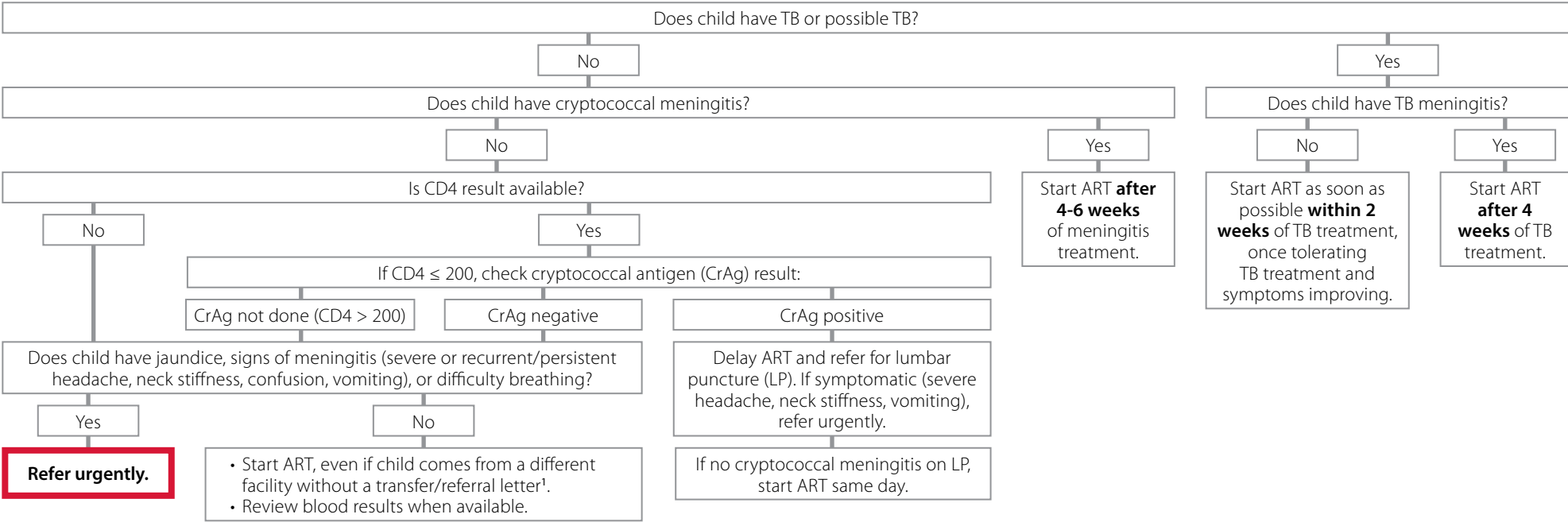
If child < 10 years old or < 30kg:
Plan to give **abacavir (ABC)** + **lamivudine (3TC)** + **dolutegravir (DTG)**, known as **ALD1**.
Check if any reason not to start this regimen (see below).

If child ≥ 10 years old and ≥ 30kg:
Plan to give **tenofovir (TDF)** + **lamivudine (3TC)** + **dolutegravir (DTG)**, known as **TLD1**.
Check if any reason not to start this regimen (see below).

STEP 2. Check for possible medication interactions and adjust ART or dosing if needed.
Check for possible medication interactions: ask specifically about TB, epilepsy, contraception, iron, zinc, antacids, metformin. If taking any of these [↗ 105](#).

STEP 3. Check tests according to chosen ART regimen if not yet done [↗ 98](#).

STEP 4. Check if there is a reason to delay starting ART



STEP 5. Dose ART correctly according to weight [↗ 168](#). Give practical tips for giving ART correctly if needed [↗ 169](#).

¹Avoid delaying treatment until transfer/referral letter available. Contact previous treatment facility to check treatment and clinical history, if possible.

Restart ART

Explore and address reasons for treatment interruption and try to resolve these [↪ 152](#).

STEP 1. Decide what ART regimen to restart

If previously on AZT + 3TC + DTG or third-line ART, discuss with experienced ART doctor, HIV expert/hotline [↪ 170](#).

Was child taking LPVr or ATVr for 2 or more years?

No

Yes

Previously on: LPVr or ATVr for < 2 years, or a DTG-, NVP- or EFV-based regimen, or unknown regimen.

Did child interrupt treatment for > 90 days?

Yes

No

Plan to restart DTG-based regimen same day based on age and weight:

< 10 years old or weight < 30kg

≥ 10 years old and ≥ 30kg

- Choose **ALD**: **abacavir** (ABC) + **lamivudine** (3TC) + **dolutegravir** (DTG).
- Record as **ALD1** if child never failed¹ a previous ART regimen and **ALD2** if child failed¹ a previous ART regimen.
- If previous hypersensitivity reaction to ABC, use instead **AZT** + **3TC** + **DTG**.

- Choose **TLD**: **tenofovir** (TDF) + **lamivudine** (3TC) + **dolutegravir** (DTG).
- Record as **TLD1** if child never failed¹ a previous ART regimen and **TLD2** if child failed¹ a previous ART regimen.
- If known kidney disease (eGFR² ≤ 80):
 - Use instead **ABC** + **3TC** + **DTG** (**ALD**). Record as **ALD1** if child never failed¹ a previous ART regimen and **ALD2** if child failed¹ a previous ART regimen.
 - If previous hypersensitivity reaction to ABC, use instead **AZT** + **3TC** + **DTG**.

First check eligibility for restarting ART and switching to DTG-based regimen same day [↪ 102](#).

- Manage any symptoms as on symptom page.
- Do VL as per routine schedule.

- Manage any symptoms as on symptom page.
- If no TB test done in past year, check for TB [↪ 108](#).
- Do CD4 at this visit, and repeat VL after 3 months.

STEP 2. Check for possible medication interactions and adjust ART or dosing if needed.

Ask about other medications child is taking: especially TB and epilepsy treatment, contraceptives and other common medications: calcium, iron, zinc, antacids, metformin [↪ 105](#).

STEP 3. Take bloods according to chosen ART regimen [↪ 98](#).

STEP 4. Check if there is a reason to delay ART [→100](#).

ABC - abacavir; **ATVr** - atazanavir/ritonavir; **AZT** - zidovudine; **DTG** - dolutegravir; **EFV** - efavirenz; **NVP** - nevirapine; **LPVr** - lopinavir/ritonavir; **TDF** - tenofovir; **3TC** - lamivudine

¹Child failed previous ART regimen if two consecutive VL ≥ 1000 while on regimen. ²eGFR = [height (cm) x 40] ÷ creatinine (µmol/L).

Switch ART

Aim to switch all children who are ≥ 3kg and ≥ 4 weeks old to dolutegravir (DTG). Resistance to DTG is rare and DTG provides rapid VL suppression and has minimal side effects.

Check if child eligible to switch to DTG-based regimen same day:

Does child's current or last known ART regimen include LPVr or ATVr?

No

- If child on a regimen *without* EFV or NVP, discuss with HIV expert/hotline → 170.
- If child on EFV- or NVP-based regimen:

Yes

Has child been on LPVr or ATVr for 2 or more years?

No

Yes

Manage further according to last two consecutive VL results, taken at least 2 years after starting LPVr or ATVr (if VL unknown, discuss):

Both VL results < 1000

Latest VL result ≥ 1000

Both VL results ≥ 1000

- Continue same regimen and assess and support adherence → 152.
- Repeat VL in 3 months:

VL < 1000

VL ≥ 1000

Switch to DTG-regimen today

- Check VL result in last 12 months:
 - If VL ≥ 50: continue to switch but assess and support adherence → 152.
 - If VL not done in last 12 months, do it at this visit (no need if switching from AZT + 3TC + NVP). No need to wait for results before switching.
- Switch to DTG according to child's age and weight:

< 10 years old or weight < 30kg

- Switch to ■ALD: ■abacavir (ABC) + ■lamivudine (3TC) + ■dolutegravir (DTG).
- Only if previous hypersensitivity reaction to ABC, use instead AZT + 3TC + DTG.

≥ 10 years old and ≥ 30kg

- Switch to ■TLD: ■tenofovir (TDF) + ■lamivudine (3TC) + ■dolutegravir (DTG).
- If known kidney disease (eGFR² ≤ 80), use instead ABC + 3TC + DTG (ALD).

- If VL done in last 12 months < 50, continue routine VL checks.
- If VL done in last 12 months ≥ 50, repeat VL in 3 months.

Virological failure confirmed

Estimate adherence¹ in the last 6-12 month. Has adherence been good?

No

Yes

Aim to improve adherence by improving medication tolerability

Is child on LPVr solution/tablets/pellets in capsules or ATVr or unknown

Yes

No

- Switch to ■ALD2 or ■TLD2 according to child's age and weight (see adjacent). If unsure, discuss with HIV expert/hotline → 170.
- If restarting ART, manage further → 101.
- Repeat VL in 3 months.

- Discuss need for resistance test and choice of new regimen with HIV expert, infectious disease specialist, third line ART committee or HIV hotline → 170.
- Avoid switching ART at this visit.

Check for possible medication interactions → 105, then dose ART correctly → 168 and give practical tips if needed → 169.

ABC - abacavir; ATVr - atazanavir/ritonavir; AZT - zidovudine; DTG - dolutegravir; EFV - efavirenz; NVP - nevirapine; LPVr - lopinavir/ritonavir; TDF - tenofovir; 3TC - lamivudine

¹Estimate adherence by checking script for pharmacy refills and notes for clinic appointment attendance. Adherence is 'good' if child tolerates medications (swallows medication, does not spit/vomit it out) and either: has come to collect medication at least 80% of the time; or has attended > 80% of scheduled clinic visits. Calculate adherence % for pharmacy refills: 'number of actual refills done during period assessed' ÷ 'number of months in period assessed'. Then x by 100. Calculate adherence % for clinic attendance: 'number of scheduled visits actually attended by child during period assessed' ÷ 'number of scheduled visits during period assessed'. Then x by 100. ²eGFR = [height (cm) x 40] ÷ creatinine (µmol/L).

Manage the child with an unsuppressed viral load (VL ≥ 50)

ALD - abacavir + lamivudine + dolutegravir; DTG - dolutegravir; TLD - tenofovir + lamivudine + dolutegravir

Assess and manage possible causes of unsuppressed viral load (VL ≥ 50):

- Check for underlying causes of unsuppressed VL, especially adherence issues → 152 and medication interactions → 105.
- If sexually active¹, emphasise condom use and contraception.

Repeat VL after 3 months:

VL < 50

VL ≥ 50

Continue routine VL monitoring → 97.

- If not on ALD or TLD, check if same-day switch to DTG is appropriate → 102. If on DTG-based regimen, continue below.
- Increase efforts to resolve adherence issues → 152 and address possible medication interactions → 105.

Estimate adherence in the last 6-12 months: check pharmacy refills and clinic appointment attendance
Does child tolerate medications (swallows and does not spit/vomit it out) and either collects medication > 80%² of time or attends > 80%³ of clinic visits?

No

Yes

Adherence poor

Adherence is good. Manage further according to duration on DTG

No resistance test needed.

Child has been on DTG ≥ 2 years

Child has been on DTG < 2 years

Has child had 2 or more consecutive VL results ≥ 1000 at least 2 years after starting ALD2/TLD2⁴

Repeat VL in 3 months

No

Yes

Has child had at least one VL ≥ 1000 with either CD4 ≤ 200 (≤ 25% if < 5 years old) or opportunistic infection⁵ or medication interaction suspected

VL ≥ 50,

- Assess adherence and possible drug interactions.
- Has child had 3 or more consecutive VL results ≥ 1000 and good adherence > 80% on 2 occasions after starting ALD2/TLD2⁴

VL < 50

No

Yes

Yes

No

- Continue to address adherence and possible medication interactions.
- If VL ≥ 1000, monitor CD4 6 monthly. If CD4 ≤ 200, restart co-trimoxazole → 99.
- Continue current ART and repeat VL after 6 months:
 - If VL < 50, continue HIV routine care → 97.
 - If VL ≥ 50, discuss with HIV expert/hotline → 170.

- Discuss resistance test and choice of new individualised ART regimen with HIV expert/hotline → 170.
- Monitor CD4 6 monthly. If CD4 ≤ 200 restart co-trimoxazole → 99.
- Repeat VL 3 months after starting new regimen:
 - If VL < 50, continue HIV routine care → 97.
 - If VL ≥ 50, discuss with HIV expert/hotline → 170.

Continue routine VL monitoring → 97.

¹Report to DSD → 170. If any of: 1) sex not consensual 2) < 12 years 3) ≥ 12 years and partner not in peer group (partner must have < 2 year age gap), **sexual abuse** likely → 144. ²Calculate adherence % for pharmacy refills: 'number of actual refills done during period assessed' ÷ 'number of months in period assessed'. Then x by 100. ³Calculate adherence % for clinic attendance: 'number of scheduled visits actually attended' by child during period assessed' ÷ 'number of scheduled visits during period assessed'. Then x by 100. ⁴ALD2: child was switched to ABC + 3TC + DTG after failing any other regimen. TLD2: child was switched to TDF + 3TC + DTG after failing any other regimen. If unsure of ART history, discuss. ⁵TB, oesophageal thrush, *pneumocystis jirovecii pneumonia* (PJP).

Manage ART side effects

- If recently started abacavir and ≥ 2 of: 1) fever, 2) rash, 3) fatigue or body pain, 4) nausea, vomiting, diarrhoea, abdominal pain, 5) sore throat, cough or difficulty breathing, **abacavir hypersensitivity reaction (AHR)** likely. Discuss/refer urgently.
- Look for and manage ART side effects:

Side effect	Possible cause	Management
Jaundice	LPVr, EFV, DTG, ATVr	• If on LPVr or EFV or DTG, refer urgently. • If on ATVr: - If no other symptoms (nausea, vomiting, diarrhoea, abdominal pain), advise that ATVr can cause jaundice without hepatitis. Consider switch to DTG ↗ 102. Check ALT and bilirubin. If raised, discuss with HIV expert/hotline ↗ 170. - If other symptoms (nausea, vomiting, diarrhoea, abdominal pain), discuss/refer.
Nausea, vomiting, diarrhoea, abdominal pain	ABC, LPVr, ATVr, AZT, EFV, TDF, DTG	• If recently started ABC, check if abacavir hypersensitivity reaction (AHR) likely (see above). If AHR unlikely: - Advise that symptoms should resolve within 2 weeks. Assess and manage child's fluid needs ↗ 26. - If persists ≥ 2 weeks, discuss with doctor. • If on LPVr, ATVr or EFV, consider switch to DTG ↗ 102. • If on AZT, discuss with doctor. • If on TDF and/or DTG: - Advise that it should resolve within 2 weeks. Assess and manage child's fluid needs ↗ 26. - If persists ≥ 2 weeks, discuss with doctor.
Fatigue/lethargy/tiredness	AZT	Look for pallor (pale palms of the hands and paleness of the lower inner eyelid) and check fingerprick Hb: • If pallor or Hb < 10 in child < 5 years old or Hb < 11 in child ≥ 5 years old, anaemia likely. Check FBC + diff and manage further ↗ 46. • If no pallor and Hb normal, reassure that tiredness should resolve. If persists ≥ 2 weeks, discuss with doctor.
Rash	ABC, EFV	• If recently started ABC, check if AHR likely (see above). If AHR unlikely, assess rash further ↗ 75. • If on EFV: switch to DTG ↗ 102 and assess rash further ↗ 75.
Psychosis (hallucinations, delusions, confused and disturbed thoughts)	EFV	Refer urgently.
Headache	DTG, AZT, EFV	• Doctor to assess further ↗ 44. • If no other cause found, give ■paracetamol 15mg/kg 6 hourly as needed for up to 5 days ↗ 163:37. • Advise that headache should resolve within 2 weeks. If persists ≥ 2 weeks, discuss
Difficulty sleeping	DTG, EFV	• If on DTG, advise to take in the morning. • If on EFV, switch to DTG ↗ 102.
Dizziness or low mood	EFV	Switch to DTG ↗ 102.
Lipoatrophy (fat loss in face, limbs and buttocks)	AZT	Refer to doctor.
Muscle weakness/cramps/stiffness/spasms		
Tingling or numbness in feet and hands		
Dyslipidaemia	LPVr, EFV	Check eligibility for switch to DTG ↗ 102. Discuss dyslipidaemia with doctor.
Gynaecomastia (breast enlargement)	EFV	Switch to DTG ↗ 102.

ABC - abacavir; ATVr - atazanavir/ritonavir; AZT - zidovudine; DTG - dolutegravir; EFV - efavirenz; LPVr - lopinavir/ritonavir

Manage ART medication interactions



Check for HIV medication interactions

- Ask child and carer about any over the counter or herbal/traditional medication use.
- If child ≥ 12 years using contraception, reassure there are no interactions with DTG. If on LPVr/ATVr or EFV, check eligibility to switch to DTG ➔ 102. Assess contraception interactions ➔ PACK Adult.
- For other medication/multivitamin/supplement interactions: check SAMF, EMGuidance app, Liverpool HIV iChart app, web-based drug interaction checker (see QR code) or discuss with HIV expert/hotline ➔ 170.
- Assess and manage common ART medication interactions below:

ART drug	Interacting medications	Management
Dolutegravir (DTG)	Rifampicin	Give additional DTG dose 12 hours after TLD or ALD dose, until 2 weeks after TB treatment completed ➔ 168.
	• Carbamazepine • Phenobarbital • Phenytoin	Avoid giving these anticonvulsants together with DTG. Discuss medication adjustments with paediatrician: • Discuss switch to lamotrigine, levetiracetam or valproate. Avoid valproate if older girl¹. • If unable to switch anticonvulsant, give additional DTG dose 12 hours after TLD or ALD dose while on anticonvulsants ➔ 168.
	Iron and/or calcium	• If taking iron only, take iron and DTG together with food. • If taking calcium only, take calcium and DTG together with food. • If taking iron <i>and</i> calcium, take DTG and calcium together with food, then take iron at least 4 hours later.
	Zinc	Take zinc at least 6 hours before or 2 hours after DTG.
	Magnesium/aluminium (antacids)	Take antacid at least 6 hours before or 2 hours after DTG.
	Metformin	Avoid giving more than 2g metformin per day.
Lopinavir/ritonavir (LPVr)	Rifampicin	• Avoid LPVr. Assess eligibility to switch to DTG ➔ 102. • If not eligible to switch, doctor to increase LPVr dose gradually: - If able to swallow tablets, double ■LPVr dose until 2 weeks after TB treatment completed ➔ 168. - If unable to swallow tablets, super-boost standard ■LPVr dose with ■ritonavir powder until 2 weeks after TB treatment completed ➔ 168. • If jaundice, refer urgently. • If nausea/vomiting/diarrhoea/abdominal pain, check ALT: - If ALT raised, discuss with HIV/TB expert/hotline ➔ 170. - If ALT normal, continue TB treatment and follow up more regularly. Recheck ALT according to weight: • If < 40kg and ongoing symptoms (nausea/vomiting/diarrhoea/abdominal pain), repeat ALT. If jaundice/ALT raised, refer. • If ≥ 40kg, double ■LPVr dose gradually and monitor ALT ➔ PACK Adult.
	• Carbamazepine • Phenobarbital • Phenytoin	• Avoid giving these anticonvulsants together with LPVr. Discuss medication adjustments with paediatrician: - Assess eligibility to switch to DTG ➔ 102. - Discuss switch to lamotrigine or valproate. Avoid valproate if older girl¹.
	Fluticasone/budesonide	Avoid giving LPVr together with fluticasone/budesonide. Discuss medication adjustments with paediatrician.
Atazanavir/ritonavir (ATVr)	Rifampicin	Avoid ATVr. Assess eligibility to switch to DTG ➔ 102. If not eligible, discuss with TB expert/hotline to switch rifampicin to rifabutin or switch ATVr to LPVr ➔ 170.
	Fluticasone/budesonide	Avoid giving ATVr and fluticasone/budesonide together. Discuss medication adjustments with paediatrician.
Efavirenz (EFV)	Bedaquiline	Avoid EFV. Switch to DTG ➔ 102.
	• Carbamazepine • Phenobarbital • Phenytoin	• Avoid giving these anticonvulsants together with EFV. Discuss medication adjustments with paediatrician: • Assess eligibility to switch to DTG ➔ 102. • Discuss switch to lamotrigine or valproate. Avoid valproate if older girl¹.
Zidovudine (AZT)	Linezolid	Avoid AZT. Discuss switch to ABC or TDF with doctor.

¹If valproate used, ensure risk acknowledgment form is signed by child/carer yearly (available from www.sahpra.org.za/wp-content/uploads/2022/08/GLF-CEM-PV-S01_v1-Valproate-Annual-Risk-Acknowledgement-Form.pdf). If older girl becomes pregnant, valproate can cause problems with development of spine, brain and other learning problems in baby. Ensure older girl on reliable contraception ➔ PACK Adult.

Assess and manage TB infection

‘TB infection’ is different from ‘TB disease’. TB infection refers to TB bacteria that has entered the body but is not yet making the body sick – often called latent TB, which means hidden/inactive.

Assess the need for TB preventive treatment (TPT)

- If TB symptoms: cough, fever, drenching night sweats, poor weight gain, weight loss, failure to thrive, lethargy, decreased playfulness, visible neck mass, blood-stained sputum, headache, persistent vomiting (without diarrhoea), avoid TPT and start TB work up instead →108.
- If HIV status unknown, test for HIV → 95.
- Does child have significant TB exposure: In the last 12 months, has child shared an enclosed space (at school/creche/home/similar) for ≥ 1 night or for frequent/extended daytime periods, with an adult/adolescent with pulmonary TB ("index client"), during the 3 months before index client started TB treatment?

Yes

No. If CLHIV/impaired immunity¹ and never completed TPT before?

Yes

No

- Check for TB disease → 108:
 - If unable to collect specimens for TB testing, arrange posterior-anterior (PA) and lateral chest x-ray, if available. Doctor to review chest x-ray within 2 days. If TB disease unlikely, continue below.
 - If chest x-ray unavailable or child/carer unable to travel for chest x-ray if not on site: if no TB symptoms and well, continue below.
 - Choose TPT regimen based on TB exposure history and, if available, index client's TB drug sensitivity results² and clinical response³ to TB treatment:
- TPT is not needed.
- Assess for TPT with every new TB exposure.
 - Continue routine care.

No known TB exposure	Susceptible to rifampicin and INH (or unknown) If index client has poor clinical response ³ to TB treatment, discuss with TB expert/hotline → 170.				Resistance to rifampicin or INH
Choose TPT regimen according to age/weight and HIV status:					
Newborn	Child < 25kg (excluding newborn)			Child ≥ 25kg	
If newborn exposed to TB, or newborn tests positive for HIV, assess and manage →91.	HIV exposed or HIV positive		HIV unexposed or HIV negative	HIV positive	HIV negative
	Child on ART	Child on PEP	Child no longer on PEP	Give ♦3HP. If ♦3HP unavailable, give instead ♦3RH or ♦6H.	
	Give ♦6H.	Give ♦3RH (preferred) or ♦6H.		- If newly diagnosed HIV and starting ALD/TLD: give ♦12H. - If child on ART (even ALD/TLD) with VL < 50 in last 6 months: give ♦3HP (preferred) or ♦12H⁴ or ♦6H⁵. - If child already on ART with VL ≥ 50 → 97 in last 6 months: give ♦12H⁴ or ♦6H⁵.	
- If severe peripheral neuropathy, active liver disease or previous adverse reaction to TPT, discuss TPT options with TB expert/hotline → 170. - Ensure each TB contact is screened for TB and TPT with every new TB exposure. Where possible, individuals in one household/group should get the same TPT regimen. - It is not necessary to do a tuberculin skin test (TST) to start TPT. - Treat the child needing TPT →107.					

6H – 6 months isoniazid; 12H – 12 months isoniazid; 3RH – 3 months rifampicin and isoniazid; 3HP – 3 months isoniazid and rifapentine; 4R – 4 months rifampicin

¹Impaired immunity: severe acute malnutrition (SAM), cancer, child without spleen, autoimmune disease, inherited immune deficiency, child awaiting/received blood/organ transplant or receiving chemotherapy, dialysis, TNF-inhibitor therapy or long-term corticosteroids. ²If treated at different clinic, contact treating clinic/laboratory to get index client's TB history and results. If child exposed to > 1 person with different drug sensitivity results, child needs TPT that will treat the most drug resistant TB child was exposed to. ³Poor clinical response to TB treatment: if smear/culture remains positive while on TB treatment, TB symptoms worsening/not resolving, missed appointments/poor adherence to TB treatment. ⁴If first course of TPT, give 12H. ⁵If child has received TPT before, give 6H.

Index client's TB shows resistance to rifampicin

Arrange posterior-anterior (PA) and lateral chest x-ray if not yet done. Doctor to review chest x-ray (and child's TB test results if done):

Chest x-ray *not* suggestive of TB

Doctor to decide on TPT regimen based on DST results of index client. Discuss with TB expert/hotline if unsure ↗ 170

Changes on chest x-ray suggestive of TB

Rifampicin mono-resistant TB (INH susceptible¹)

Resistance to INH or resistance unknown

Give ●6H

Discuss TPT options with TB expert/hotline ↗ 170 or provincial clinical advisory committee (PCAC)².

- Avoid giving TPT.
- **Diagnose RR-TB** and refer to doctor to discuss with PCAC/TB expert and start treatment.

Treat the child needing TPT: record all TB information in RthB

- If CLHIV and starting ART: start ART immediately and TPT within 2 weeks.
- Give routine care: If < 2 months old ↗ 11. If ≥ 2 months old ↗ 12. If child starts TPT in first 6 weeks of life, repeat BCG after completing TPT (see adjacent). Assess and interpret growth ↗ 15 monthly while on TPT and adjust dose according to weight ↗ 165.
- Give TPT according to chosen TPT regimen and weight ↗ 165.
- Also give ♦pyridoxine (not needed if on 4R):
 - If < 5 years old: if ≥ 6kg give 12.5mg (½ of 25mg tablet) daily. If < 6kg give 6.25mg (¼ of 25mg tablet) daily.
 - If ≥ 5 years old: give 25mg daily (1 tablet).
- Explain possible minor side effects like orange discolouration of secretions, and advise to return earlier if unwell.
- **Review monthly** while on TPT. Check weight and ask about TB symptoms and side effects at every visit:
 - If weight loss/not gaining weight or TB symptoms develop, test for TB ↗ 108 and discuss with TB expert/hotline ↗ 170.
 - If side effects develop ↗ 115.
- If index client's TB shows resistance to rifampicin, doctor to review child at month 2, 4, 6, 9, and 12, even if TPT completed.

Repeat BCG

- If child started TPT in first 6 weeks of life, repeat BCG after completing TPT.
 - If CLHIV, only give BCG if stable on ART *and* CD4 ≥ 25%³ (≤ 5 years) or ≥ 200³ (> 5 years).
 - Give BCG 24 hours after the last TPT dose, or if child received rifapentine give BCG 5 days after the last dose.
 - Avoid BCG if: CLHIV not on ART/unwell/ CD4 < 25%/CD4 < 200³ - continue TPT until the child is eligible for BCG. If inborn errors of immunity or other immunodeficiency, discuss with specialist.

At TPT initiation, decide patient category

- If never had TPT before or took TPT < 4 weeks, document as **new**.
- If completed TPT before or took TPT ≥ 4 weeks and stopped (due to adverse event, developed TB or was lost to follow up), document as **previously treated**.

Manage the child who interrupts TPT

Missed 1 dose

- If on 3RH, 4R, 6H, 12H: child to take missed dose as soon as child/carer remembers same day. If missed 1 day, take next dose as scheduled and continue daily dosing. Advise to avoid taking 2 doses on same day.
- If on 3HP: child to take missed dose as soon as child/carer remembers within 3 days. Advise to take next dose as scheduled or start new weekly schedule from day missed dose was taken.

Interrupted ≥ 1 month on 3RH, 3HP, 4R or interrupted ≥ 3 months on 6H or 12H

- Support the child taking long-term medication ↗ 152.
- Check if eligible for new course of TPT ↗ 106.
- Refer to psychologist and/or social worker if available/as needed. If intentional neglect suspected ↗ 144.

Interrupted < 1 month on 3RH, 3HP, 4R or interrupted < 3 months on 6H or 12H:

- Support the child taking long-term medication ↗ 152.
- If TB symptoms, avoid TPT and check for active TB ↗ 108.
- If no TB symptoms, continue TPT. Add missed doses at the end of treatment.

Interrupted TPT for a second time, regardless of duration of interruption

- Avoid restarting TPT.
- Reassess child for TPT if new significant TB exposure⁴.

Once TPT completed, decide on treatment outcome

- If completes full duration of TPT, document as **treatment completed**.
- If on 3HP, 3RH or 4R: If interrupts treatment for ≥ 4 weeks, document as **lost to follow-up**.
- If on 6H: If interrupts treatment for 2 consecutive months, document as **lost to follow-up**.
- If on 12H: if interrupts treatment for 3 consecutive months, document as **lost to follow-up**.
- If stops TPT due to serious adverse event or developed TB disease, document as **treatment stopped**.
- If died during TPT, document as **died**.

¹Check index client's INH phenotypic DST/culture-based result to confirm INH susceptibility. ²Provincial clinical advisory committee (PCAC): PCAC-WC@westerncape.gov.za. ³Do a CD4 count at TPT completion (do not wait for the annual count). ⁴Significant TB exposure: any child who in the last 12 months shared an enclosed space (school/creche/home/similar) for ≥ 1 night or frequent/extended daytime periods with an adult/adolescent with pulmonary TB ("index client"), during the 3 months before index client started TB treatment.

TB: diagnosis

Decide when to test for TB

- **If TB symptom/s:** cough, especially if persistent, unexplained fever, drenching night sweats, poor weight gain, weight loss, failure to thrive, lethargy, decreased playfulness, visible neck mass, blood-stained sputum, headache, persistent vomiting (without diarrhoea).
- **If no TB symptoms:** routinely test for TB if:
 - Significant TB exposure¹
 - CLHIV: at diagnosis, then yearly when viral load checked
 - Completed TB treatment in last 2 years: test for TB yearly for 2 years after completing TB treatment
 - Abnormal chest x-ray (CXR), even if no known significant TB exposure¹.

Give urgent attention to the child with possible TB and any of:

- Difficulty breathing, sats < 92%, stridor, chest indrawing, blue lips/tongue ↗ 41
- Coughs up ≥ 1 tablespoon of fresh blood
- Hb < 7 g/dL or very pale palms ↗ 46
- Severe acute malnutrition (SAM)² ↗ 126
- Lethargic/confused/decreased consciousness ↗ 174
- Fitting/seizures ↗ 28
- Neck stiffness, bulging fontanelle, restless or always irritable: **meningitis** likely
- Persistent vomiting/headache
- Pupils different sizes
- Change in mobility ↗ 90
- Swollen abdomen
- Unable to feed/drink/eat
- ≥ 2 of: cold hands/feet, weak/fast pulse ↗ 175, capillary refill time ≥ 3 seconds, decreased level of consciousness ↗ 174: **shock** likely ↗ 26
- Restless/irritable/lethargy/alterd level of consciousness, sunken eyes, slow skin pinch or drinking poorly/thirsty: **dehydration** likely ↗ 26

Manage and refer urgently: If able, send 1 specimen for TB NAAT. If CLHIV, also do rapid urine LAM test.
 • If meningitis likely, give ♦ceftriaxone³ 100mg/kg IV/IM ↗ 159:12. Avoid injecting ≥ 1g IM at one injection site.

Start the workup to diagnose TB in the child not needing urgent attention

- If HIV status unknown, test for HIV ↗ 95.
- If < 3 months old with TB symptoms, discuss same day with paediatrician.

Is the child < 2 years or ≥ 10 years, living with HIV, or being routinely checked for TB (with no TB symptoms)?

Yes

No: are TB symptoms present for > 2 weeks?

Yes

No: Manage symptoms on symptom page and follow up in 1-2 weeks⁴. Advise to return sooner if worsens. If no improvement, continue to assess for TB:

Is child able to produce sputum?

Yes

Send 1 specimen for TB NAAT ↗ 111.
If sputum sufficient, send 2 specimens.

No

- If routinely checking for TB (no TB symptoms) and no significant TB exposure¹, offer test at next visit and advise to return if TB symptoms occur.
- Use other method to collect respiratory specimen for TB testing ↗ 111, especially if any of the following:
 - CLHIV with TB symptoms
 - CLHIV with no TB symptoms *and* known significant TB exposure¹ *and* abnormal CXR
 - No TB symptoms *and* known significant TB exposure¹ to DR-TB *and* abnormal CXR
 - Severe CXR disease ↗ 110
 - TB diagnosis is uncertain
 - TB symptoms *and* known significant TB exposure¹ to DR-TB



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- If TB symptoms or significant TB exposure¹: arrange posterior-anterior (PA) and lateral CXR if available ↗ 109. If no TB symptoms and well, assess for TPT if needed, even if CXR not available ↗ 106.
- If CLHIV with TB symptoms *and* advanced HIV disease⁵, do rapid urine LAM test (see QR code above). If positive, **diagnose TB** and manage further with other results/information.
- If child < 5 years with TB symptoms and no known TB exposure: consider doing a tuberculin skin test (TST). See QR code above.
- Advise to return after 2 days and manage further according to results →110.

¹Significant TB exposure: any child who in the last 12 months shared an enclosed space (school/creche/home/similar) for ≥ 1 night or frequent/extended daytime periods with an adult/adolescent with pulmonary TB ("index client"), during the 3 months before index client started TB treatment. ²Any of: swelling of both feet, weight-for-length/height below -3 line, BMI-for-age below -3 line or MUAC < 11.5cm. ³Avoid giving Ringer's lactate together with IV/IM ceftriaxone. If severe penicillin allergy (previous anaphylaxis, urticaria or angioedema), discuss with doctor. ⁴If unable to easily return, continue TB workup same day. ⁵Child > 5 years with CD4 ≤ 200/stage 3 or 4 condition; or child ≤ 5 years *unless* on ART > 1 year and clinically stable on ART (well, CD4 > 25% and VL < 50).

Is chest x-ray (CXR) available?

Yes

Arrange PA and lateral CXR and doctor review. Compare to previous CXR if available.

No

Are diagnostic tests (TB-NAAT/culture/smear/ULAM) negative or unavailable?

Yes

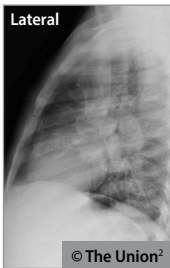
No

Manage according to result.

CXRs suggestive of TB¹



Hilar lymphadenopathy



Upper lobe cavitation



Pleural effusion



Lobar consolidation/collapse



Lower lobe cavitation



Miliary TB



Pericardial effusion

- CXR suggestive of TB
- If new changes on CXR, **diagnose TB on chest x-ray**. If unsure, discuss.
 - If miliary TB/pericardial effusion, refer same day.
 - If pleural effusion, consider referral. If unsure, discuss.

CXR not suggestive of TB *and* diagnostic tests negative.

Does the child have TB symptoms/signs?

Yes: is there significant TB exposure² or positive tuberculin skin test (TST)?

Yes

No

- If < 10 years old: Use the symptom scoring tool (see below).
- If ≥ 10 years old, discuss with TB expert.

No

TB disease unlikely

- If needed, continue to assess for TPT ➔ 106.
- If TPT not needed, continue routine care and advise to return if symptoms recur.

Treat for TB. Review TB NAAT, culture and DST results, if done ➔ 110.

Also look for extrapulmonary TB

Site	TB symptoms	Action
TB meningitis	If headache, vomiting (mostly without diarrhoea), unilateral weakness, behaviour changes, regression of milestones ³ or cranial nerve palsy ⁴ .	Manage and refer urgently as in red box ➔ 43.
TB lymphadenitis	Typical TB Lymph node(s): often in neck, ≥ 2cm, asymmetrical, non-tender, present > 2 weeks, does not improve with antibiotics ± discharging sinus.	Refer for/doctor to do fine needle aspiration (FNA).
Abdominal TB	Abdominal pain/swelling, vomiting, diarrhoea, constipation, jaundice ➔ 61.	Doctor review and abdominal ultrasound.
Spinal TB	Chronic back pain, lower limb weakness, unable to walk/limping, bulge on back.	Doctor review and spinal x-ray.
Joint/bone TB	One-sided effusion/chronic pain, usually in knee/hip. Swelling at end of long bones which limits movement.	Doctor review and xray
Ear TB	Discharge from ear ≥ 2 weeks.	Send aspirated fluid/pus swab for TB NAAT.

The symptom scoring tool:

Clinical feature	Score
Cough > 2 weeks	+5
Fever > 2 weeks	+10
Lethargy/reduced playfulness	+4
Weight loss	+5
Coughing up blood	+9
Night sweats	+6
Enlarged typical lymph nodes	+7
Increased pulse rate ➔ 175	+4
Increased respiratory rate ➔ 175	+2
Total score	?

Interpretation:
If score ≥ 11, consider TB.

However, this tool does not replace clinical judgment:

- TB treatment can be started if score < 11.
- Alternative diagnoses can be considered even if the score is ≥ 11.
- If unsure, discuss.

¹Other TB related chest x-ray abnormalities include: paratracheal lymphadenopathy, Ghon focus, airway narrowing, abnormal thoracic vertebra or paravertebral abscess. X-ray images: atlaschild.theunion.org. ²A significant TB exposure: any child who in the last 12 months shared an enclosed space (school/creche/home/similar) for ≥ 1 night or frequent/extended daytime periods with an adult/adolescent with pulmonary TB ("index client"), during the 3 months before index client started TB treatment. ³Loss of previously learned skills like sitting, crawling, talking or toilet training. ⁴Facial droop, hearing loss, new squint, speech problems.

If the CXR is suggestive of TB, determine if the child has features of non-severe or severe pulmonary TB - use this to determine if child eligible for treatment shortening ➔ 114.

Non-severe											
	Uncomplicated lymph node disease		Primary (Ghon) focus	Simple pleural effusion	Alveolar opacification: < 1 lobe	Other: Interstitial pneumonia Perihilar infiltrates					
Severe											
	Complicated lymph node disease Airway compression or deviation and/or hyperinflation or collapse				Consolidation involving ≥ 1 lobe	Primary (Ghon) focus with cavitation	Complicated pleural effusion¹	Alveolar opacification: involving a whole lobe or multiple lobes	Other: All cavitary disease Expansile pneumonia Miliary TB TB broncho-pneumonia		

Adapted from Palmer M, Seddon JA, Goussard P, Schaaf HS. Diagnostic CXR atlas for tuberculosis in children: A guide to chest X-ray interpretation, Second edition. Paris, France: International Union Against Tuberculosis and Lung Disease (The Union); 2022.

Interpret TB NAAT: If child < 13 years: samples sent for TB NAAT will also be processed for culture and DST (smear will not be done routinely).

MTB complex NOT detected	MTB trace detected	Test unsuccessful	MTB complex detected		
- If <i>no</i> TB symptoms: TB unlikely. If needed, continue to assess for TPT ➔ 106. - If TB symptoms: doctor to assess child clinically, arrange PA and lateral CXR if available. - Review TB culture and DST results.	- If treated for TB in last 2 years and <i>no</i> TB symptoms: TB unlikely. If needed, continue to assess for TPT ➔ 106. - If TB symptoms or <i>not</i> treated for TB in last 2 years: doctor to assess child clinically, arrange PA and lateral CXR if available. - Review TB culture and DST results.	- Collect new specimen for TB NAAT ➔ 111. - If unable to collect new specimen or TB symptoms: doctor to assess child clinically, arrange PA and lateral CXR if available.	RIF resistance NOT detected	RIF resistance detected	RIF unsuccessful
			If child has significant TB exposure² to an index client with drug-resistant TB: avoid starting TB treatment, discuss instead with TB expert/hotline ➔ 170. - If INH resistance detected, diagnose INH mono-resistant TB and refer to doctor to start treatment. - If <i>not</i> treated for TB in last 2 years, diagnose DS-TB and start TB treatment ➔ 112. - If treated for TB in last 2 years, doctor to assess child clinically, arrange PA and lateral CXR if available and review TB culture and DST results. If unsure, discuss.	- If <i>not</i> treated for RR-TB in last 2 years: diagnose RR-TB and refer to doctor to discuss with PCAC/TB expert and start treatment. - If treated for RR-TB in last 2 years, doctor to assess child clinically, arrange PA and lateral CXR if available and review culture and DST result. If unsure, discuss.	- If <i>not</i> treated for TB in last 2 years: diagnose TB, doctor to assess child clinically, arrange PA and lateral CXR, review culture and DST results and start TB treatment in consultation with TB expert/hotline ➔ 170. - If treated for TB in last 2 years: doctor to assess child clinically, arrange PA and lateral CXR and review culture and DST results.

If done, review culture and DST results. Follow up weekly until culture result confirmed:

Culture positive for MTB ³			Culture negative for MTB
No resistance to rifampicin and INH detected Diagnose DS-TB: start treatment if not yet started ➔ 112.	Resistance to <i>only</i> INH detected Diagnose INH mono-resistant TB: refer to doctor to start treatment, if not yet started.	Resistance to rifampicin detected Diagnose RR-TB: doctor to discuss with PCAC/TB expert and start treatment, if not yet started.	- If on TPT or TB treatment <i>and</i> well, continue. - If not on TPT/TB treatment <i>and</i> TB symptoms persist or unwell on TPT/TB treatment, refer.

¹Loculated effusion, empyema or pneumothorax. ²Significant TB exposure: any child who in the last 12 months shared an enclosed space (school/creche/home/similar) for ≥ 1 night or frequent/extended daytime periods with an adult/adolescent with pulmonary TB ("index client"), during the 3 months before index client started TB treatment. ³If culture positive for *mycobacterium other than tuberculosis* (MOTT) or *non-tuberculous mycobacterium* (NTB), discuss with TB expert/hotline ➔ 170.

How to collect a good specimen for TB testing

Aim to collect sputum in the early morning. This improves the chance of an accurate result. However, avoid missing the opportunity to collect a specimen anytime during a consultation.

If child able to produce sputum (usually if ≥ 5 years old), follow the steps below

- Explain that a good quality sputum specimen is important to make an accurate diagnosis of TB.
- Advise to avoid putting saliva or nasal secretions into specimen jar. Sputum is the secretion that comes from deep within the lungs and a forceful cough is needed to bring it up for collection.
- If observing sputum collection, the health worker/carer should use a mask (N95/FFP2), gloves and plastic apron in a well ventilated area/outside. Use a designated sputum collection area if available.
- Explain how to collect a good sputum specimen:

Step 1	Child to blow nose and rinse mouth with water to remove food, mouth wash or medication.	Step 2	• Child to breathe in and out deeply two times. • Have an open specimen jar ready. • Keep the jar sterile (clean), avoid touching inside it.	Step 3	• On the third breath, child to give a strong cough. Child to try to cough 2-10mL (1-2 teaspoons) sputum into the jar. • Child may need several coughs to get 2mL.	Step 4	Replace lid and screw tightly to prevent leaking. Wipe outside with alcohol swabs and allow to dry.	Step 5	Child/carer and health worker to wash hands after sputum collection.
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How to collect induced sputum and gastric washings in a child?



If child unable to produce sputum (usually if < 5 years old), use another method to collect specimen for TB testing

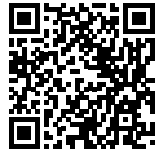
Respiratory specimen:

- **Induce sputum:** Follow same steps as above, only use nebuliser to help produce sputum after child fasted for a least 2 hours.
 - Monitor oxygen saturation and pulse rate; stop if sats drop < 92% or pulse ≥180 and discuss with doctor. Check for contraindications/precautions³ to procedure; if unsure discuss.
 - Pre-treat to prevent bronchospasm: nebulise 1ml ♦sodium chloride 0.9% and 1ml ♦salbutamol 0.5% solution with ♦oxygen at 5-8L/min or give ♦salbutamol 100mcg (1 puff) into spacer with mask, hold spacer in place and count for 10 seconds as child breathes in and out. Repeat once. Wait 5 minutes before starting nebulisation.
 - Nebulise with 2ml (if < 2 years) or 4ml (if ≥ 2 years) of ♦3% or ♦5% saline with ♦oxygen at 5-8L/min until empty or child coughs 2-3 times in quick succession.
 - If child able to follow instructions: ask child to remove mask, breathe in and out deeply 2-3 times, followed by a strong cough into specimen jar.
 - If child unable to follow instructions: measure correct length for catheter insertion: tip of child's nose to tragus of ear and suction secretions from nasopharynx using a soft-tipped suction catheter. Avoid suctioning for longer than 10 seconds.
 - Repeat until at least 1.5mL sputum collected. If no sputum collected after 10 minutes, repeat nebulisation once.
- If unable to induce sputum after nebulisation, do **gastric washings** after child fasted for at least 4 hours.
- A **stool sample** can also be sent for TB-NAAT (in child < 10 years), but should be sent together with another respiratory specimen if possible. Note: stool samples cannot be cultured.

Other specimen type:

- If lymph node(s) suggestive of TB (≥ 2 cm, non-tender, asymmetrical, matted, firm or fluctuant, present for > 2 weeks, does not improve with antibiotics or has discharging sinus), refer for/doctor to do **fine needle aspiration (FNA)**.

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Prepare specimens for transport to the laboratory

- Check specimens are adequate: if child unable to produce 2-5mL (1 teaspoon) but quality of sputum is still good¹, still send specimens to laboratory. If quality of specimen is poor, see below.
- Ensure lid is closed tightly and correctly, and that the specimen jar is correctly labelled. Place pre-printed barcode from lab request form horizontally on specimen jar. Wash hands after handling specimens.
- If room temperature is > 25°C or transport delayed for > 24 hours, store in refrigerator (2-8°C). Keep cool but do not freeze.
- Complete specimen request form: include how specimen was collected and ensure correct client contact details/cell phone number. Advise child/carer to return for results in 2 days ➔ 110.

If specimen inadequate

- If specimen is inadequate and/or of poor quality after repeated attempts, discard used jar in medical waste bin and give child/carer new labelled specimen jar. Instruct on how to collect sputum at home:
 - Collect sputum specimen early in the morning after waking up, before eating or taking any medications. Collect sputum specimen outside home. Follow the same steps tried above.
 - Once collected, protect sputum specimen sample from heat and light. Keep at room temperature and bring to the clinic as soon as possible.
- If specimen from home is adequate, prepare for transport to laboratory (above).

¹A good quality sputum specimen contains a thicker secretion than saliva, with or without pus, mucous or blood, from deep within the lungs. A poor quality specimen contains saliva (with or without blood). ²YouTube video provided through the courtesy of the Desmond Tutu TB Centre (DTTC). ³Avoid if unstable respiratory/cardiovascular status, recent surgery, child unable to protect own airway (e.g. decreased level of consciousness). If asthma/severely impaired lung function, perform under doctor supervision.

Drug-sensitive TB (DS-TB): routine care

Assess the child with DS-TB

Assess	When to assess	Note
Registration	At diagnosis	Ensure client record completed and captured in TB register. Record all TB information in RtHB.
Symptoms	Every visit	- Check if urgent attention needed ↗ 108. - Expect gradual improvement on TB treatment. If symptoms worsen or do not get better, refer to doctor. - If no significant reduction in size of lymph nodes after 4 months, discuss/refer.
Adherence	Every visit	Request child/carer brings all medication to each visit. Check adherence on TB card. Manage the child who interrupts TB treatment ↗ 116 .
Side effects	Every visit	Ask about TB treatment side effects ↗ 115 .
Medication interactions	Every visit	Ask about other medications like epilepsy treatment, contraception and ART/PEP. If unsure, discuss.
TB contacts	At diagnosis	- Identify index client if possible. Advise that all TB contacts¹ visit the clinic for TB screening/prevention. Ensure community health worker (CHW) does a home visit for TB screening and testing. Follow up results and drug susceptibility of index client. - Older children (typically ≥ 8 years old) are more likely to be infectious. All close contacts¹ should be assessed for TB/TPT eligibility.
Family planning	Every visit if sexually active ²	Advise on reliable contraception ↗ PACK Adult . Check that s/he knows how to use condoms.
Mental health	Every visit	- Screen for depression/anxiety in carer: in the past month, has carer felt down/depressed/hopeless or little interest/pleasure in doing things, or often nervous, anxious or panicky, unable to stop worrying or thinking too much or had thoughts of harming him/herself? If yes to any ↗ PACK Adult. - If over past few months, child has been miserable, stressed or angry ↗ 139 or if problematic change in behaviour ↗ 136.
Routine care	Every visit	Give routine care: If < 2 months old ↗ 11 . If ≥ 2 months old ↗ 12 .
Vision	If on ethambutol: every visit	Ask child/carer about change in vision. If any complaint of visual disturbance, stop ethambutol and refer to ophthalmologist. Discuss if unsure.
Weight/growth	Every visit	Assess and interpret growth ↗ 15 . Expect weight gain on treatment and adjust TB treatment dose ↗ 166 .
Chest x-ray	If needed	- If poor treatment response (ongoing symptoms/poor weight gain), arrange CXR. - If child has severe pulmonary TB, also do CXR at end of treatment to look for post-TB lung disease.
HIV	If > 3 months since last test	Test for HIV ↗ 95 . If CLHIV, give HIV routine care ↗ 97 . Manage the child with DS-TB and HIV ↗ 113 .
TB NAAT	If sent during diagnostic workup	Register according to laboratory result.
TB microscopy (smear) <i>Not routinely done at diagnosis in children, done in special circumstances.</i>	- If sent during diagnostic workup - If sent at week 7 and/or week 20	- If smear positive at diagnosis, child is not eligible for treatment shortening. Consider repeating smear at week 7 and 20 (no need to refer if unable to give sample). - If week 7 or week 20 smear positive, discuss.
TB culture and DST <i>Do in clinic if child can give sample. No need to refer if unable to give sample.</i>	If sent during diagnostic workup	- If culture positive at diagnosis, repeat: at 8 and 12 weeks if on shortened regimen or at 16 and 20 weeks if on 6 month regimen. - If MTB detected on culture, check DST result: - If sensitive to rifampicin and INH: if at diagnosis, continue treatment. If other/subsequent culture positive, discuss/refer. - If resistant to INH only, diagnose INH mono-resistant TB and refer to doctor to start treatment, if not yet started. - If resistant to rifampicin, diagnose rifampicin-resistant TB (RR-TB) and refer to doctor to discuss with PCAC/TB expert and start treatment. - If culture contaminated, repeat. - Use culture results to decide on treatment outcome ↗ 116.
Treatment outcome	At completion of TB treatment	Decide on treatment outcome ↗ 116 .

Advise and treat the child with DS-TB [↗ 113](#).

¹A TB contact refers to any person who shared an enclosed space (at school, socially, or in a household setting), for ≥ 1 night or for frequent/extended daytime periods, with an adult/adolescent with pulmonary TB ("index client"), during the 3-month period before the index client started their TB treatment. ²Report to DSD [↗ 170](#). If any of: 1) sex not consensual 2) < 12 years old 3) ≥ 12 and partner not in peer group (partner must have < 2 year age gap.), **sexual abuse** likely [↗ 144](#).

Advise the child with DS-TB and his/her carer

- Arrange TB counselling and refer for community adherence support. Explain how to administer medication and ensure carer understands. Use "how to" posters (see QR code).
- Educate about TB treatment side effects ↗ 115 and advise to return promptly should they occur.
- Educate about infection control: adequate ventilation/open windows (if area is not well ventilated, to wear a face mask), cough/sneeze into upper sleeve or elbow. Wash hands with soap regularly.
- Advise to stay home from school for the first 2 weeks of treatment.
- If older child uses tobacco or alcohol/drugs, alert to risks and support change ↗ PACK Adult.
- Give **enhanced adherence support** if poor adherence ↗ 152.
- Children with TB need more energy. Support caregivers to meet this need using home foods or supplements. Refer to dietician and social worker, if needed and available.

TB medication for children:
Easy dosing guide



Treat the child with DS-TB

- Assess the child for shortened TB treatment regimen at diagnosis and, if eligible, again at 1 month and 4 months ↗ 114.
- If difficulty administering medication, discuss with expert/hotline ↗ 170.
- If HIV exposed on nevirapine, discuss with expert/hotline ↗ 170.
- Extrapulmonary TB other than isolated cervical TB lymphadenitis will be diagnosed and treatment started at higher level of care, but may be down referred to continue TB treatment once stable.
- Does child have miliary TB or central nervous system TB (including TB meningitis) which has been down referred?

No

Treat 7 days a week: Give intensive phase **HRZ** + **E** for 8 weeks (2 months) ↗ 166.

After 8 weeks of treatment decide if regimen should change to continuation phase

Is child well (TB symptoms are improving and child is gaining weight)?

Yes

- Change to continuation phase: give **HR**.
- Decide on duration of continuation phase: is child eligible for shortened regimen ↗ 114?

Yes

Give 8 weeks (2 months) of continuation phase

No

- Decide on duration of continuation phase:
 - PTB or most EPTB: 16 weeks (4 months)
 - Bone/joint TB: 40 weeks (10 months) - may be extended as guided by a specialist.

No

Discuss with TB expert before changing

Yes

- Hospital letter will advise doctor on treatment and follow up needs.
- Treat with single phase for 6-9 months as guided by a TB expert. Give **HR**, **Z** and **Eto** ↗ 167 for 7 days a week for the full duration of treatment.

- Dose TB treatment according to weight and adjust as weight increases ↗ 166. Refer to doctor if not growing well.
- Give **pyridoxine** until treatment completed:
 - If < 5 years old: if ≥ 6kg give 12.5mg (½ of 25mg tablet) daily. If < 6kg give 6.25mg (¼ of 25mg tablet) daily.
 - If ≥ 5 years old: give 25mg daily (1 tablet).

R - rifampicin

H - isoniazid

Z - pyrazinamide

E - ethambutol

Eto - ethionamide

Manage the child with DS-TB and HIV

- Give **co-trimoxazole** and HIV routine care throughout TB treatment ↗ 97. If not on ART, check when to start ART ↗ 100 or restart ART ↗ 101.
- If on ART and not on dolutegravir (DTG), assess eligibility for switch to DTG ↗ 102. If baby ≤ 4 weeks old or ≤ 3kg, discuss with HIV/TB expert ↗ 170.
- At TB diagnosis, do CD4 count and viral load.
- Manage further according to ART regimen:

Child on or starting DTG

Give **DTG** dose 12 hourly according to weight until 2 weeks after TB treatment completed ↗ 168.

Child on lopinavir/ritonavir (LPVr) or atazanavir/ritonavir (ATVr)

- If unable to switch to DTG:
 - Doctor to adjust **LPVr** dose ↗ 168.
 - Avoid ATVr while on TB treatment. If on ATVr, doctor to change to **LPVr** and adjust dose ↗ 168.

Assess the child for shortened TB treatment regimen

Does the child have any of the following:

- Requires urgent attention ➔ 108.
 - Previous TB
- Extrapulmonary TB other than cervical TB lymphadenitis/simple pleural effusion
 - DR-TB (including INH mono-resistant TB)
- Asymmetric or persistent wheeze
 - Respiratory sample that is AFB smear positive
- Severe acute malnutrition (SAM)¹

No to all

Yes to any

Is CXR available?

Yes

No

Is the child's age 3 months to < 16 years² at the start of TB treatment?

No

Yes

Is child living with HIV?

Yes

No

Is VL < 1000 in last 3 months *and* child on ART for > 3 months?

No

Yes

Do CXR:
Does child have features of severe pulmonary TB on CXR ➔ 110?

Yes

No

Not eligible for treatment shortening

Child could be eligible for treatment shortening

On follow up, does the child have **all** of the following:

- Adherent to treatment
- Month 1: TB symptoms and signs improved with weight gain
- Month 4: TB symptoms and signs resolved and appropriate/improving weight trend
- Month 4 (if cervical TB): nodes smaller

No

Yes

- Extend to standard 6 month regimen.
- Check for the following and discuss:
 - Drug resistant TB
 - Non-TB disease (malignancy/HIV)
 - Poor treatment adherence

Complete shortened
4 month regimen

Not eligible for treatment shortening

¹Any of: swelling of both feet, weight-for-length/height below -3 line, BMI-for-age below -3 line or MUAC < 11.5cm. ²For children aged 13-15 refer to the NDOH guideline: MANAGEMENT OF TUBERCULOSIS IN CHILDREN AND ADOLESCENTS (September 2024).

Look for and manage TB treatment side effects

Side effect	Likely cause	Management
Nausea, vomiting, abdominal pain (may have jaundice)	Most TB medications	• If jaundice or tender/enlarged liver, refer to hospital same day. Check fingerprick glucose ↗ 32. • If vomiting, doctor to check for other cause (e.g. drug-induced liver injury, TB meningitis, urinary tract infection, upper respiratory tract infection). Discuss/refer if unsure. • If nausea/vomiting only after taking TB treatment <i>without</i> jaundice or tender/enlarged liver: advise to take treatment after eating or at night. If persists after 1 week, discuss/refer.
Skin rash/itch	Most TB medications	Assess and manage ↗ 75.
Seizures/psychosis	Isoniazid	Manage seizure ↗ 28. Refer to hospital same day.
Change in vision	Ethambutol	Stop ethambutol and refer to eye specialist same day.
Joint pain	Pyrazinamide	• Assess joint symptoms ↗ 74. • Give ■paracetamol 15mg/kg 6 hourly as needed up to 5 days ↗ 163:37. If pain not responding to paracetamol, give ■ibuprofen 5-10mg/kg 8 hourly with food ↗ 161:28 for up to 3 days. Avoid ibuprofen if asthma, heart failure or kidney disease. If persistant pain discuss/refer. • Advise to use ice/heat on affected joint.
Dark or orange urine	Rifampicin	Reassure this is normal while taking rifampicin.
Pain/numbness/burning of hands and feet	Isoniazid	Peripheral neuropathy likely. Ensure child is on ♦pyridoxine ↗ 113 and discuss/refer.

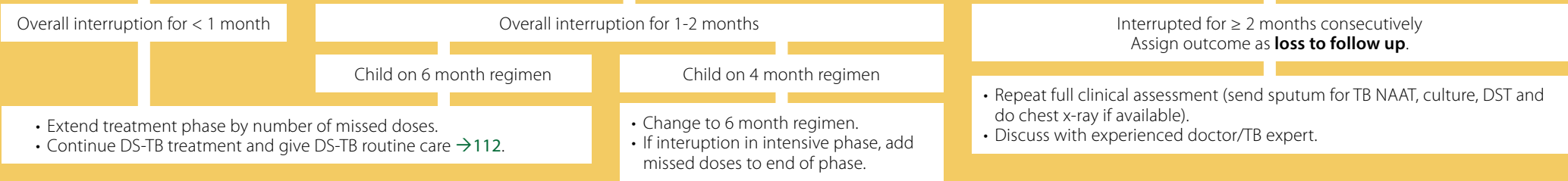
Review the child with DS-TB

- If child is well (TB symptoms are improving and child is gaining weight), review at 2 weeks, 4 weeks, then monthly.
- Advise to return sooner if worsening or side effects develop.
- Check for TB yearly for 2 years after TB treatment completed ↗ 108.
- If child started TB treatment in first 6 weeks of life, repeat BCG after completing treatment.
 - If CLHIV, only give BCG if stable on ART *and* CD4 ≥ 25%¹ (≤ 5 years) or ≥ 200¹ (> 5 years).
 - Give BCG 24 hours after the last DS-TB treatment dose.
 - Avoid BCG if: CLHIV not on ART/unwell/CD4 < 25%/CD4 < 200¹. Start TPT until the child is eligible for BCG ↗ 106. If inborn errors of immunity or other immunodeficiency, discuss with specialist.
- If child presents with recurrent respiratory symptoms after treatment completed, exclude recurrent TB ↗ 108. If TB excluded, consider post-TB lung disease.
- If child has ongoing significant TB exposure² with index client with TB, assess for TB preventive treatment (TPT) after TB treatment completed ↗ 106.

¹Do a CD4 count at TB treatment completion (do not wait for the annual count). ²A significant TB exposure: any child who in the last 12 months shared an enclosed space (school/creche/home/similar) for ≥ 1 night or frequent/extended daytime periods with an adult/adolescent with pulmonary TB ("index client"), during the 3 months before index client started TB treatment.

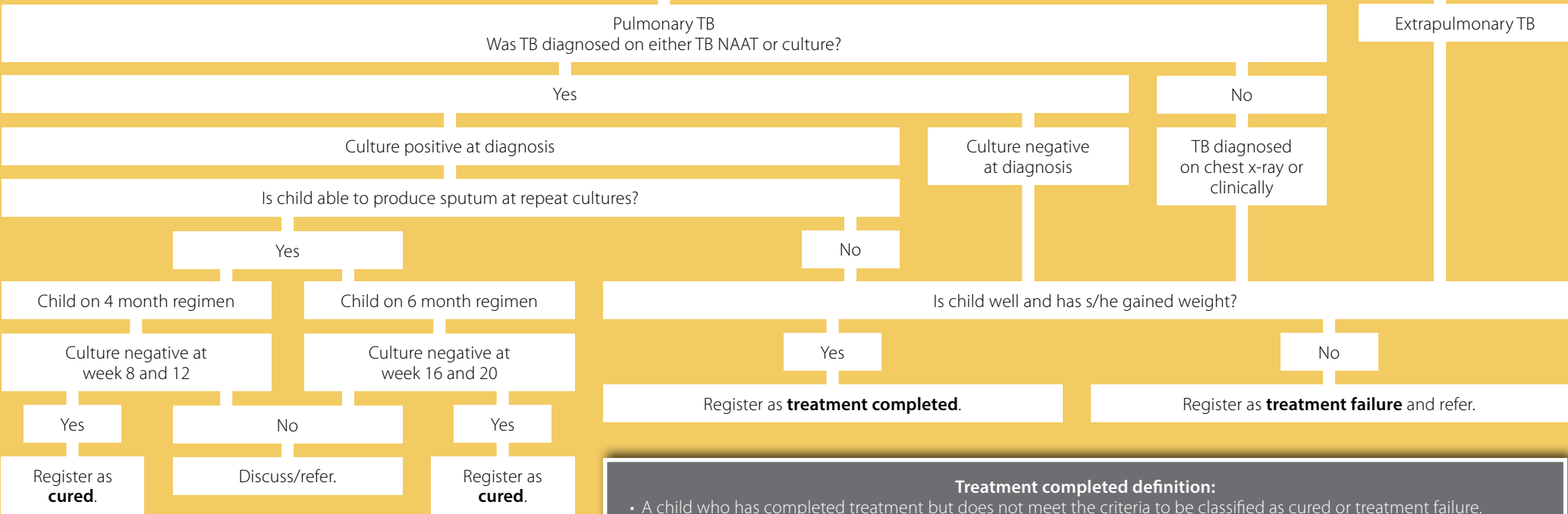
Manage the child who interrupts DS-TB treatment

- Look for reason for treatment interruption and give enhanced adherence support → 152.
- If TB symptoms recur, send specimen for TB NAAT, culture and DST.
- Manage the child who interrupted DS-TB treatment according to duration of interruption/number of missed doses and ability to obtain specimens for TB testing:



Once DS-TB treatment completed, decide on treatment outcome

- Any child who dies during TB treatment: register as **died**.
- Any child who transfers to another district and treatment outcome is unknown: register as **transferred out**.



Treatment completed definition:

- A child who has completed treatment but does not meet the criteria to be classified as cured or treatment failure.
- Treatment response in a child with sputum smear/culture-negative TB or extrapulmonary TB is assessed monthly by monitoring the weight of the child and symptom resolution.

The child with allergy

- Allergy likely if: recurrent sneezing and runny/itchy/blocked nose, itchy/watery/red eyes, itchy rash, eczema or recurrent tight chest.
- If exposed to possible allergen¹ or sudden generalised itch/rash or face/tongue itch/swelling and any of: tight throat/hoarseness, difficulty breathing, wheeze, shock², dizziness/collapse, abdominal pain/nausea/vomiting/diarrhoea, **anaphylaxis** likely. Manage and refer urgently →37.

Assess the child with allergy not needing urgent attention. Record child's condition and care plan in RtHB, print and give child/carer allergy action plan (Allergy foundation South Africa) → 170.

Assess	When to assess	Note
Symptoms	Every visit	• If recurrent cough, wheeze, tight chest or difficulty breathing, exclude asthma → 59. If known asthma, give routine asthma care → 119. • If patches of dry, scaly, itchy skin on wrists, ankles, inside elbows or behind knees (on face in infants), eczema likely → 118. • If itchy, red, raised wheals that appear suddenly and usually disappear within 24 hours, urticaria likely → 75. • If recurrent sneezing or itchy/runny/blocked nose most days for > 1 month (may have line across nose from repeated rubbing), allergic rhinitis likely → 50. • If both eyes watery and itchy, allergic conjunctivitis likely → 48.
Triggers	Every visit	• Ask if using new medication. If yes, stop use. If sudden rash/face swelling following new medication, refer to specialist. • If sudden rash or face swelling following new food, refer urgently to specialist. • Consider other triggers: insect bites, dogs/cats in home, pollen, cigarette smoke, mould, cockroaches. If food trigger suspected, refer/discuss with specialist. • Refer to specialist if unable to identify trigger after 6 months.
Allergy control	Every visit	Allergy uncontrolled if symptoms interfere with sleep, school or sport. Refer to specialist if poor response to maximum treatment.
Adherence	Every visit	If using creams or ointments, check these are being applied correctly. If using an inhaler, check technique → 57.
Routine care	Every visit	Integrate routine care into every visit → 12.

Advise the child with allergy and his/her carer

- If trigger found, advise child/carer to try to avoid it.
- Ensure child/carer understands the need for medication. Advise to bring medication (inhalers, nasal spray, creams /ointment) to every visit to ensure correct use.
- If previous anaphylaxis, arrange MedicAlert® bracelet if not yet done → 170.
- If allergic to medication/food, advise child/carer to always inform health worker and school about allergy.
- If adrenaline prescribed, check child/carer knows when and how to use it:
 - Advise to use if **anaphylaxis** likely (see above).
 - Advise child/carer how to use prescribed adrenaline before calling ambulance:

If using adrenaline auto-injector (like Epipen®)

- Remove from protective case and pull off blue safety release cap.
- Ensure fingers are not over either end and make a fist around the pen to ensure a tight grip.
- Remember “blue to the sky, orange to the thigh.”
- Firmly push orange tip against upper outer thigh so it clicks. Inject through clothes.
- Hold in place, count to 3, then remove.

If using adrenaline vial and needle with syringe

- Attach needle to syringe.
- Ensure no liquid in top end of adrenaline vial (flick top end to move fluid to bottom of vial), then open vial.
- Draw up correct adrenaline dose for weight/age → 157:4.
- Hold syringe with needle facing up. Inject out adrenaline until correct dose remains in syringe, if needed. If air in syringe, flick syringe until air is at the top and inject air out until correct dose remains in syringe.
- Stick needle straight down into upper outer thigh and inject dose into muscle.
- Remove syringe and needle and discard safely.

Treat the child with allergy

- If chronic allergy, check adherence and technique for using medication (inhalers, nasal spray, creams/ointment) before adjusting or adding medication. Review child 3 monthly.
- If mild new onset allergic reaction (generalised itchy rash/itchy nose/sneezing/nausea/abdominal pain/vomiting *without* other symptoms), give ▶cetirizine once daily until symptoms resolve: if 2-6 years old, give 5mg. If ≥ 6 years old, give 10mg. Refer if symptoms of anaphylaxis (see above), no better after 24 hours or rash recurs.

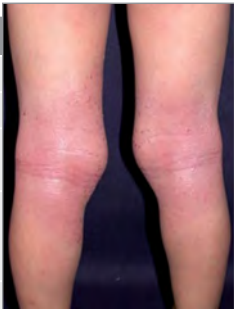
¹Common allergens include medications, new food or insect bite/sting within the last few hours. ²If ≥ 2 of cold hands/feet, weak/fast pulse → 175, CRT ≥3 seconds, decreased level of consciousness → 174, **shock** likely.

Eczema

The child with eczema has itchy, scaly skin that is usually red in the baby and dry in the older child. Affects inner (flexural) surfaces of elbows and knees, cheeks, scalp and neck. Usually not in groin/axilla.

Assess the child with eczema at every visit. Record child's condition and care plan in RthB. Print and give eczema action plan from Allergy foundation of South Africa ↗ 170.

Assess	Note
Control	Eczema is not controlled if any of: 1) skin very itchy, thickened and scaly, 2) increased redness, rawness and oozing, 3) symptoms interfere with sleep, school or sport.
Adherence	Check how often emulsifying ointment is being applied: if less than twice a day, educate on importance of frequent use for eczema control.
Infection	- If skin oozing, crusting and scaling, weeping eczema likely. If yellow pustules that crust, impetigo likely. Treat below. - If crops of ulcers/blisters, herpes simplex (eczema herpeticum) likely. Manage as extensive herpes ↗ 51 and refer.
Other allergy	- If purple rings around eyes, runny/blocked nose, mouth breathing, line across nose from repeatedly rubbing nose, allergic rhinitis likely ↗ 50. - If ≥ 2 years old and recurrent dry cough/whoeze/tight chest/difficulty breathing, consider asthma ↗ 59. - If child develops hives after eating certain foods, consider food allergy. Refer urgently to specialist for allergy testing.
Triggers	Ask about triggers and advise to avoid/limit irritants: cigarette smoke exposure, soaps and detergents. If allergic trigger ¹ identified, refer to specialist for testing.
Routine care	Integrate routine care into every visit ↗ 12.



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Advise the child with eczema and his/her carer

- Advise keeping the skin moisturised at all times is the key to improving eczema.
- Avoid soaps/fabric softener/harsh washing powder/scented creams/perfumes.
- Avoid direct skin contact with woollen or rough clothes and overheating by blankets at night. Keep nails short and clean.
- When applying corticosteroid cream, apply a thin layer. See picture for amount to apply. Carer to avoid rubbing onto own palms.
- Wash daily to remove crusts and prevent infection.



One fingertip unit is the amount of cream/ointment squeezed from the tip of the index finger to the first crease of adult-sized finger.

Treat the child with eczema

- Avoid soap, use **aqueous cream** instead.
- Use **emulsifying ointment** as a moisturiser as often as possible, at least twice a day.
- If **not controlled** and adherent, give **hydrocortisone 1% cream** twice a day for 7 days then taper over next 7 days. Use adjacent table to explain amounts to use. Apply only to eczema patches, use sparingly on face and avoid eye area. If not improving after 7 days advise to return.
- If not improving or severe eczema discuss/refer to specialist and give instead **betamethasone 0.1% cream** once a day for 7 days then taper over next 7 days (avoid face, neck and joint skin folds).
- If itching, give **chlorpheniramine** 0.1 mg/kg at night ↗ 159:15. If no better after 2 weeks, and ≥ 2 years old, change to **cetirizine** once daily: if 2-6 years old, give 5mg; if ≥ 6 years old, give 10mg.
- If **weeping eczema** likely, wet wrap daily until better: apply **hydrocortisone 1% cream** then a thick layer of **emulsifying ointment** to affected area. Soak gauze in luke-warm water and cover this, then cover again with dry dressing.
- If **impetigo** likely, clean infected areas with **povidone iodine scrub**, then wrap with **povidone iodine** soaked gauze and cover with dry dressing. Repeat twice a day for a week. Also give **cephalexin** ↗ 159:13 or **flucloxacillin** ↗ 161:23 for 5 days. Review in 7 days.
- If no improvement in 2 weeks, eczema herpeticum or severe/extensive eczema, refer.
- Review 3 monthly.

	Face and neck	Arm and hand	Leg and foot	Front	Back and buttocks
Age	Number of fingertip units: according to area affected and age of child				
3 months -1 year old	1	1	1.5	1	1.5
1-3 years old	1.5	1.5	2	2	3
3-6 years old	1.5	2	3	3	3.5
6-10 years old	2	2.5	4.5	3.5	5

Adapted from Long C, C. and Finlay A. Y. The finger-tip unit - a new practical measure. Clinical and Experimental Dermatology, 1991; 16: 444-447.

¹Dust mites, animals in the home, pollen, cockroaches, mould, foods. ²If severe penicillin allergy (history of anaphylaxis, urticaria or angioedema), give instead **azithromycin** 10mg/kg once daily for 3 days ↗ 158:10.

Asthma

- Doctor to confirm asthma diagnosis within 1 month. Ideally, arrange spirometry to accurately diagnose and assess asthma. If spirometry unavailable, use peak expiratory flow rate (PEFR).
- Calculate peak expiratory flow rate (PEFR) at every visit if ≥ 6 years old ➔ 58. If < 6 years old and responding to treatment below, continue. If not responding to treatment or diagnosis uncertain, refer.

Assess the child with asthma at every visit. Record child's condition and care plan in RtHB, print and give child/carer asthma action plan (Allergy foundation South Africa) ➔ 170.	
Assess	Note
Asthma symptoms to determine asthma severity and control	• If wheeze/tight chest or difficulty breathing not responding to inhaler/s manage acute exacerbation ➔ 56. • Asthma is poorly controlled if hospitalisation with acute exacerbation in past year or if in past month any of: 1) daytime cough, wheeze or difficulty breathing more than twice a week, 2) runs/ plays less or tires easily due to asthma and child requires salbutamol to relieve symptoms 3) inhaled salbutamol needed more than twice a week, 4) night waking/night coughing due to asthma 5) PEFR < 80% (or a drop of > 20%) ➔ 172. • If none of above, asthma is well controlled. Record child's best PEFR reading in RtHB and folder to compare follow up readings.
Symptoms	Manage other symptoms as on symptom pages. If child using inhaled corticosteroid and white patches on cheeks/gums/tongue/palate, oral candida likely ➔ 51.
Allergy	Ask about other symptoms of allergy: if recurrent sneezing or itchy/running/blocked nose, watery/itchy eyes, itchy/dry skin or itchy, red, raised wheals ➔ 117.
Adherence	Check that child and/or carer can use inhaler and spacer correctly ➔ 57. Ensure child is adherent to treatment before adjusting or adding treatment.
Routine care	Integrate routine care into every visit ➔ 12.

- Advise the child with asthma and his/her carer**
- Advise to identify and avoid triggers¹ that may worsen asthma. If smoking in house, alert to risks and encourage smoker to quit ➔ PACK Adult.
 - Demonstrate inhaler technique ➔ 57. Ensure child/carer understands medication: short acting beta-agonist (e.g. salbutamol) only relieves symptoms and does not control asthma. Inhaled corticosteroid (e.g. budesonide) prevents and controls symptoms.
 - Recognise and manage acute exacerbation: if wheeze/tight chest or difficulty breathing not responding to inhaler/s, go to nearest emergency unit urgently.

Treat the child with asthma

- Advise yearly influenza vaccination during influenza vaccine campaign.
- If not on treatment: start treatment at any step below, according to severity of symptoms listed in steps, and review in 1 month.

Step 1 If ≥ 6 years old and symptoms < twice a month and no exacerbation in past year: Give inhaled ➔salbutamol 200mcg as needed and ➔budesonide ² 200mcg whenever salbutamol is taken.	Step 2 If symptoms ≥ twice a month or exacerbation in past year: • If < 6 years with wheeze ≥ 3 times a year, give inhaled ➔budesonide² 100 mcg 12 hourly plus ➔salbutamol 100-200mcg (1-2 puffs) 6-8 hourly. • If ≥ 6 years, give inhaled ➔budesonide² 200mcg 12 hourly plus ➔salbutamol 200mcg 6 hourly as needed.	Step 3 If symptoms most days/waking up from asthma symptoms ≥ once a week • If < 6 years give inhaled ➔budesonide² 200mcg 12 hourly plus ➔salbutamol 200mcg 6 hourly as needed. Arrange doctor review as soon as possible. • If ≥ 6 years, continue inhaled ➔budesonide² 200mcg 12 hourly plus ➔salbutamol 200mcg 6 hourly as needed and arrange doctor review as soon as possible.
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- If already on treatment:
 - If well controlled, continue treatment and review 3 monthly.
 - If poorly controlled: 1) ensure no ongoing exposure to triggers¹, adherent and can use inhaler and spacer correctly ➔ 57, 2) step up treatment as above and review monthly, 3) if poorly controlled on step 2 or 3, arrange posterior-anterior (PA) and lateral chest x-ray and doctor review. If still not controlled after 3 months, refer to specialist.
 - If recent exacerbation, review in 1 week. If child received prednisone/hydrocortisone for acute exacerbation, continue prednisone once daily for 7 days ➔ 164:41. If not responding within 2 days, refer.
 - If day-time and night-time symptoms are controlled and no acute exacerbations for > 6 months:
 - Gradually decrease inhaled budesonide/beclomethasone. If on 200mcg 12 hourly decrease to daily dose. If on daily dose, decrease to as needed together with salbutamol.
 - If symptoms recur, increase budesonide/beclomethasone to previously controlled dose.

¹Irritant triggers: cigarette smoke and chemicals. Allergic triggers: animals, dust, mould, pollen. Avoid grass if confirmed to be an allergen. ²If on lopinavir/ritonavir or atazanavir/ritonavir, avoid budesonide. Check eligibility to change to DTG-based ART ➔ 102, or use beclomethasone instead..

Breastfeeding

Give urgent attention to the breastfeeding baby with any of:

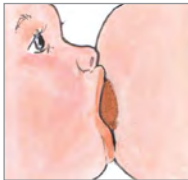
- Unable to feed
 - Vomits everything
 - No attachment/suckling
- Chokes/coughs when feeding
 - Difficulty swallowing (milk pools in mouth)
- Difficulty breathing, blue lips or sweats during feeds
 - Cleft palate

Manage and refer urgently:

- If baby < 2 months old and unable to feed: give ◆ **ceftriaxone**¹ 80mg/kg IM ↪ 159:11. Avoid injecting ≥ 1g IM at one injection site. Keep warm.
- Check glucose: If ≤ 28 days old and < 2.6mmol/L ↪ 9. If > 28 days and < 3mmol/L (< 4 if diabetes) ↪ 32. Otherwise, prevent hypoglycaemia: give 3mL/kg of expressed breastmilk every hour via NGT.

Assess breastfeeding mother and baby to identify feeding problem

Assess	When to assess	Note
Routine care	Every visit	Integrate routine care of the child into every visit : if < 2 months old ↪ 11, if ≥ 2 months old ↪ 12. If any other symptoms, manage as on symptom pages.
Feeding frequency	Every visit	• If not feeding on demand (as often and as long as baby wants day and night), there is a feeding problem ↪ 121. • If baby < 6 weeks old not feeding at least 8 times in 24 hours, there is a feeding problem ↪ 121.
Solids	Every visit	• If baby < 6 months old getting other foods/fluids, there is a feeding problem ↪ 121. • If solids have not been started after 6 months old, there is a feeding problem ↪ 121.
Mother	Every visit	• If mother has a body mass index (BMI)² < 18.5 or mid-upper arm circumference (MUAC) < 23cm, refer to nutritional therapeutic programme (NTP). • Screen for increased psychosocial risk and mental health problem. If baby < 2 months ↪ 11, if baby ≥ 2 months old ↪ 12. • Check HIV status, contraceptive needs and TB symptoms ↪ PACK Adult. • If breast problem (painful breast, breast lump or cracked/sore nipples) ↪ PACK Adult.
Growth	Every visit	Measure and record in RtHB: weight-for-age, length-for-age, weight-for-length ↪ 15.
Baby's mouth	Every visit	If white patches in mouth (inside of cheeks/lips and on tongue), oral thrush likely ↪ 51. If teeth present when baby is born, refer to dentist for possible removal.
Breastfeeding position	At first visit and if feeding problem	Baby's body should be close to mother, facing mother's breast, with nose opposite mother's nipple and with baby's whole body supported in a straight line. If baby is not positioned like this, there is a feeding problem ↪ 121.
Attachment and suckling	At first visit and if feeding problem	• If blocked nose, clear with ◆ sodium chloride 0.9% 1 drop into each nostril and suction nose. • If possible, observe mother breastfeeding for 4 minutes and assess attachment and suckling:



Good attachment/suckling likely if:

- Mouth wide open
- Lower lip turned outwards
- More areola (dark part of the breast) visible above than below baby's mouth
- Chin touching breast
- Slow, deep sucks and swallowing sounds



Poor attachment/suckling likely if:

- Baby sucking on nipple, not areola (dark part of the breast)
- Rapid shallow sucks
- Smacking or clicking sounds
- Cheeks drawn in
- Chin not touching breast
- If poor attachment or suckling, there is a feeding problem ↪ 121.

HIV risk	Every visit	• If mother HIV unknown/negative, do HIV test in mother 3 monthly while breastfeeding ↪ PACK Adult. • If mother tests HIV positive, do HIV PCR in baby same day ↪ 95, start post exposure prophylaxis (PEP) in baby ↪ 92 and ART in mother ↪ PACK Adult. • If mother known HIV positive, check HIV PCR test done on baby at birth (or at first presentation) and follow up result. Ensure PEP given ↪ 92. • If baby has HIV, ensure baby on ART and give routine HIV care ↪ 97. Continue breastfeeding until 2 years old.
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Advise the breastfeeding mother ↪ 121.

¹Avoid giving Ringer's lactate together with IV/IM ceftriaxone. If severe penicillin allergy (previous anaphylaxis, urticaria or angioedema), discuss with doctor. ²BMI is weight (kg) ÷ height (m) ÷ height (m).

Advise the breastfeeding mother and correct feeding problems

If feeding problem, refer to breastfeeding counsellor (or lactation consultant if available) or support group and give feeding advice:

How to feed	What to feed	How often to feed (feeding frequency)
Position baby: • Seat mother comfortably. • Baby faces mother's breast (baby's head should not be turned) with nose opposite nipple and body close to mother's body. • Mother supports baby's whole body, not just neck and shoulders. Help baby attach: • Touch baby's lips with mother's nipple. • Express a few milk drops onto baby's lip. • Wait until baby's mouth opens widely, then move baby quickly onto mother's breast. • Aim baby's lower lip well below nipple.	0-6 months old • Encourage exclusive breastfeeding for 6 months: baby gets only breastmilk and prescribed medicines if needed (no formula, water, cereal). This decreases risk of diarrhoea, pneumonia and allergies. Other foods may damage gut and allow infections in (including HIV). If mother HIV positive, ensure viral load < 50 → PACK Adult. • If concerns about milk supply: - Reassure that mother naturally produces enough milk for child's needs. - Increase frequency and length of feeds and feed day and night. - Advise to rest, drink plenty of fluids and encourage partner support. Advise about expressed breastmilk: • If mother away from baby, explain how to express and store breastmilk (see below). Give expressed breastmilk with cup: 1) hold baby upright with arms wrapped and head supported 2) rest half full cup on lower lip 3) avoid pouring; baby will sip and spill some milk 4) rest between sips. • The exclusively breastfed baby needs about 750mL/day between 1-6 months old.	≥ 6 months old • From 6 months old, introduce solids → 123. • If mother HIV positive, ensure viral load < 50 → PACK Adult. • Continue to breastfeed until at least 2 years old. How often to feed (feeding frequency) • Breastfeed on demand as often as baby wants, day and night for as long as baby wants per feed. A hungry baby turns head to find breast (rooting), puts hands in mouth or makes suckling noises. • A baby < 6 weeks old should feed at least 8 times in 24 hours. • If poor growth, advise mother to wake baby to feed after 3 hours if baby has not woken by him/herself (during day and night).

If mother wants to formula feed instead, ask the following questions. If the answer is no to/mother is doubtful about ≥ 1, formula feeding is *not* recommended. If yes to all → **122**:

- 1 Is there piped water and a flush toilet in house?

2 Is there money to buy formula (12 months supply), feeding equipment, cleaning materials and to cover costs of fuel/travel, for extra clinic/hospital visits?

3 Is mother able to prepare formula hygienically on demand day and night?
- 4 Is mother sure she will not breastfeed as well as formula feed during the first 6 months?

5 Has mother disclosed HIV status to partner/someone in household?

6 Is nearest health care facility easily accessible?

Review baby with newly diagnosed feeding problem in 2 days, thereafter review every 5 days until feeding problem corrected. If young baby has lost weight or problem unlikely to be corrected, refer.

How to express breastmilk

To express, first wash hands. Then stimulate the milk reflex by massaging, stroking or gently shaking breasts. When milk/colostrum appears, express the breast until empty (at least 10 minutes per breast):



- 1 Wash hands. Position thumb just behind the edge of areola (dark part of the breast) and rest of fingers to form the letter "C". Avoid cupping the breast.



- 2 Push straight into chest wall. Avoid spreading fingers apart. For large breasts, first lift and then push back.



- 3 Roll thumb and fingers forward at the same time bringing the milk from the "back to the front".

- 4 Repeat rhythmically to completely drain reservoirs: position, push, roll... position, push, roll... position, push, roll...
- 5 Rotate thumb and fingers to milk other reservoirs, moving all around the areola. Avoid squeezing breast, sliding hands over the breast or pulling nipple.

How to store breastmilk

- Use hard plastic (or glass) container with large opening and tight lid to store the breastmilk.
- Boil container and lid for 10 minutes before use.
- Write time and date that milk expressed on container.
- Store in fridge for up to 6 days or in cool place for 8 hours.
- When ready to use milk: warm by placing upright in container of clean warm water (do not microwave). Gently swirl.
- Before feeding, drop small amount milk on inside of wrist to check milk is not too hot for baby.
- Check person feeding baby knows how to cupfeed.

Formula feeding

Give urgent attention to the formula feeding baby with any of:

- Unable to feed
 - Vomits everything
- Chokes/coughs when feeding
 - Difficulty swallowing (milk pools in mouth)
- Cleft palate
 - Difficulty breathing, blue lips or sweats during feeds
- Baby lost ≥ 10% body weight

Manage and refer urgently:

- If baby < 2 months old and unable to feed: give ◆ **ceftriaxone**¹ 80mg/kg IM ↪ **159:11**. Avoid injecting ≥ 1g IM at one injection site. Keep warm.
- Check glucose: If ≤ 28 days old and < 2.6mmol/L ↪ **9**. If > 28 days old and < 3mmol/L (< 4 if diabetes) ↪ **32**. Otherwise, prevent hypoglycaemia: give 3mL/kg of expressed breastmilk/formula feed every hour via NGT.

Assess the formula feeding carer and baby to identify feeding problem at every visit

Assess	Note
Routine care	Integrate routine care of the child into every visit: if < 2 months old ↪ 11 , if ≥ 2 months old ↪ 12 . If any other symptoms, manage as on symptom pages.
Type of formula	If child getting formula not appropriate for his/her age, there is a feeding problem.
Preparation	Ask carer if s/he: washes hands, boils water for 3 minutes, measures water and formula according to tin instructions with scoop supplied, makes enough for only one feed at a time. If no to any, there is a feeding problem.
Feeding frequency	< 6 weeks old: 8 feeds/24 hours, 6 weeks - 6 months old: 6 feeds/24 hours, 6-12 months old: 4 feeds/24 hours. If baby feeds less frequently than this, there is a feeding problem.
Solids	If solids have been started too early (< 6 months old) or have not been started after 6 months old, there is a feeding problem.
Cleaning	Ask carer if s/he: washes all containers with hot soapy water and rinses, sterilises cup at least once/day (or if using bottles, after each use), sterilises containers by boiling in pot of water for at least 10 minutes and keeps pot covered until containers needed. If no to any, there is a feeding problem.
Social	• If formula feeding not accepted at home, no access to clean water or carer cannot afford formula each month for 12 months, there is a feeding problem. • If mother absent/has died, unable to care for baby due to illness, poses threat to baby or is on medications contraindicated in breastfeeding/on PI-based or 3rd line ART with viral load > 1000, refer to NTP² for formula and discuss post-exposure (PEP) options for baby with HIV expert/hotline ↪ 170. Ensure mother receives HIV routine care ↪ PACK Adult.
Growth	Measure and record in RtHB: weight-for-age, length/height-for-age, weight-for-length/height, mid-upper arm circumference (MUAC) ↪ 15 .
Cup feeding	Check that carer knows how to use a cup to feed as it is safer than bottle feeding: 1) hold baby upright with arms wrapped and head supported 2) rest half full cup on lower lip 3) avoid pouring; baby will sip and spill some milk 4) rest between sips.

Advise the carer about formula feeding and address feeding problems

What to feed

- Do not give other foods/fluids before 6 months old.
- Give formula appropriate for age as indicated on tin.
- From 6 months old, introduce solids ↪ **123**.
- Continue with formula until 12 months old, then give pasteurised full cream milk.

How much and how often to feed

Weight (kg)	Number of feeds	Amount per feed
0-3.9	8	50mL
4-4.9	7	75mL
5-6.4	6	125mL
6.5-6.9	6	150mL
7-7.9	6	175mL
8-8.9	6	200mL
≥ 9	4	250mL

How to prepare feeds

- Wash hands. Boil water for 3 minutes.
- Measure water first, then add formula carefully, using tin instructions and scoop supplied.
- Mix formula while water still hot, use clean spoon to stir.
- Allow to cool to body temperature.
- Use cup to feed: safer (cleaner) than bottle feeding.
- Only make enough formula for one feed at a time.
- Discard leftover formula milk within two hours.

How to clean containers

- Wash with hot soapy water and rinse.
- Sterilise cup at least once/day (if using bottles, sterilise after each use).
- Cover containers with water in pot and boil for at least 10 minutes.
- Keep pot covered until containers needed.

Review in 2 days, then every 5 days until feeding problem corrected. If < 3 months old, check if breastfeeding can be re-established. If young baby has lost weight or problem unlikely to be corrected, refer.

¹Avoid giving Ringer's lactate together with IV/IM ceftriaxone. If severe penicillin allergy (previous anaphylaxis, urticaria or angioedema), discuss with doctor. ²Nutrition Therapeutic Programme.

Eating

Assess eating to identify if feeding problem. Ask carer to recall what child has eaten in the last 24 hours.

Assess	When to assess	Note
Routine care	Every visit	Integrate routine care of the child into every visit ↗ 12. If any other symptoms, manage as on symptom page.
Solids	If 6-12 months old	If solids have been started too early (< 6 months old) or have not been started after 6 months old, there is a feeding problem.
Variety of food	Every visit	• If 6-12 months old and not introducing baby to a variety of foods, there is a feeding problem: introduce porridge, vegetables, fruit, and protein-rich foods¹. • If ≥ 12 months old and not giving the child protein¹ at least once a day and fresh fruit or vegetables twice a day, there is a feeding problem. • If child not eating healthy meals because of unhealthy snacking (sweets/chips/chocolates), there is a feeding problem.
Quantity of food	Every visit	If not getting at least half a cup (125mL per meal) by 12 months old, there is a feeding problem.
Frequency of food	Every visit	• If 6-8 months old and getting fewer than 2 meals/day or not getting breast/formula milk as well as meals, there is a feeding problem. • If 8-12 months old and not getting increasing number of meals so that by 12 months old they are getting 5 meals/day, there is a feeding problem. • If ≥ 12 months old and getting < 5 small meals/day (3 family meals/day and 2 nutritious snacks like bread with peanut butter, fruit, yoghurt), there is a feeding problem.
Fluids	Every visit	If drinking lots of juice, tea or sugary drinks, there is a feeding problem.
Social	Every visit	Ask who looks after child and feeds child most of time. If concerns about poor parental care, refer to social worker/community health worker.
Mouth/teeth	Every visit	• Assess for problems making eating difficult: if mouth symptoms ↗ 51. If throat/swallowing problem ↗ 52. • If dental caries ↗ 54.
Growth	Every visit	Measure and record weight-for-age, length/height-for-age, weight-for-length/height (or BMI), mid-upper arm circumference (MUAC) ↗ 15.

Advise the carer on eating according to child’s age

6-12 months old

When to start solids:

- Introduce solids at 6 months old.
- Continue breastfeeding/formula feeding (offer baby breastmilk/formula first, then offer soft foods).

What to feed:

- Introduce new food every 2-3 days in order: soft porridge/ cereal, mashed/pureed vegetables, fruit, protein¹.
- From 9 months, give foods rich in iron².
- Give clean safe water regularly. Avoid juice/tea/sugary drinks.

How much and how often:

- Gradually increase amount and frequency of feeds:
- **6-9 months old:** give 2 meals/day plus breast/formula milk. Start with 2-3 tablespoons per meal and slowly increase to half a cup. Give milk first, then give food.
- **9-12 months old:** increase to 3 meals/day. Give half a cup per meal. Also give 2 nutritious snacks (fruit, yoghurt) between meals. Give food first, then give milk.

1-2 years old

- Breastfeed as often as child wants.
- If no longer breastfeeding, give 2 cups full cream milk or maas every day (avoid giving more as this may reduce appetite for food).

How much and how often:

Give 3 meals/day: ¾ to full cup per meal. Also give 2 nutritious snacks (bread with peanut butter, fruit, yoghurt).

What to feed:

- Give protein-rich foods¹ at least once a day.
- Give fresh fruit/vegetables twice a day.
- Give foods rich in iron², vitamin A³ and vitamin C⁴.
- Avoid adding salt or sugar to food.
- Avoid sugary/fizzy drinks/coffee, give water instead.

How to feed:

Actively feed child and encourage him/her to eat on their own.

2-5 years old

- Give child his/her own serving (1 cup) of family foods 3 times/day.
- Also give 2 nutritious snacks (bread with peanut butter, fruit, yoghurt).

≥ 5 years old

What to eat:

- Eat a variety of food.
- Eat plenty of fruit and vegetables every day.
- Make starch part of most meals.
- Eat protein¹ regularly.
- Have milk, maas or yoghurt every day.
- Use fats (butter, margarine), sugar and salt sparingly.
- Avoid sweetened drinks and coffee.

How much:

- Eat 3 meals/day and 2 nutritious snacks (fruit, yoghurt).
- Stop eating when full.

How often:

- Do not skip meals, especially breakfast.

¹Protein-rich foods: meat, chicken, fish, cooked eggs, beans, dahl, soya, peanut butter. ²Iron-rich foods: liver, kidney, dark green leafy vegetables (like spinach), cooked egg, beans, peas, lentils, fortified cereals. ³Vitamin A-rich foods: vegetable oil, liver, yellow sweet potatoes, dark green leafy vegetables like spinach (imifino), mango, pawpaw, full cream milk. ⁴Vitamin C-rich foods: oranges, naartjies, melons, tomatoes.

If feeding/weight problem, advise about correct eating habits

Help child and carer to correct eating habits and weight problems:

If not growing well



- Add a teaspoon of margarine, vegetable oil or peanut butter (or Imnut®) to porridge.
- Increase frequency of feeds to at least 5 meals per day.

- Offer meals when child is alert and happy. Give more food if child shows interest.
- Use correct spoon size, put food within reach of child, actively feed child, try sitting child on lap while eating.
- If blocked nose, clear with ►sodium chloride 0.9% 1 drop into each nostril.
- Encourage child to drink a glass of water with every meal once eating solids. If < 1 year old, give 50-100mL of water and increase as s/he gets older. Advise carer that this is the healthiest fluid option and prevents constipation.
- Use varied, favourite foods.
- Give foods of suitable consistency.
- Offer small frequent feeds.

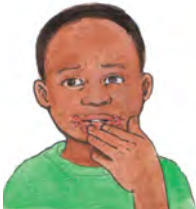


If poor appetite or fussy eater

- Avoid unhealthy snacks (chips, sweets, chocolates) in between meals.
- Avoid juice/tea/sugary drinks - these make the child feel full.
- Avoid giving too much formula/milk as this may also reduce appetite.



If child has mouth ulcers/sores, offer soft foods that don't burn mouth like eggs, mashed potatoes, pumpkin or avocado.



If overweight or obese

Help to maintain weight and increase growth appropriately: invite child/carers to address 1 lifestyle risk factor or dietary change at a time and build on these:

- Plan how to fit change into child's day. Explore what might hinder or support this. Together, set reasonable target/s for next visit.
- Emphasise that support from the carers is very important for success. Encourage carers to improve their own lifestyle choices, diet and weight.
- Emphasise that weight management is a lifelong process and not a brief period of change.

Eat a healthy balanced diet:



Reduce portion sizes - eat less.

- Avoid snacking on chips, sweets, chocolates, fizzy drinks in between meals.
- If hungry between meals, snack on fruit, nuts or unsweetened yoghurt.
- Eat more fruit and vegetables.



- Avoid sugar and fried food.
- Avoid fast foods.



Eat meals together as a family. Avoid watching TV when eating.

Get active:



- Limit screen time to < 1 hour per day: this includes TV, movies, video games, phones, internet and social media.
- Encourage child to go outside and play or join a team sport.
- Take the stairs instead of lifts.
- If safe, walk or ride to school instead of taking transport.



Encourage whole family to do moderate intensity activity (brisk walking, dancing, housework, gardening) for at least 30 minutes/day (adult) and 60 minutes/day (child). This can be accumulated in 10 minute sessions.

If feeding/weight problem, review 1-2 monthly until eating habits have been corrected.

Poor growth in the baby < 2 months old

The baby with any of the following has poor growth:

- > 10% loss of birth weight¹ in first week of life
- Weight below birth weight after 10 days of age
- Weight gain unsatisfactory (growth curve flattening or crossing z-score lines)
- Any weight loss if birth weight < 2.5kg

Give urgent attention to the baby with poor growth and any of:

- Unable to feed
- Apnoea (episodes of no breathing > 10 seconds)
- Tires/sweats during feeds
- Lethargy or decreased level of consciousness ↗ 174
- Vomits everything
- Temperature < 35.5°C or ≥ 38°C
- Bulging fontanelle
- Reduced movements
- Difficulty breathing: respiratory rate > 60, grunting, nasal flaring or chest indrawing
- Diarrhoea (watery stools and/or > 3 stools/day)
- If ≤ 28 days old and glucose < 2.6mmol/L
- If > 28 days old and glucose < 3mmol/L (< 4 if diabetes)

Manage and refer urgently:

- If difficulty breathing, give ♦oxygen at 1L/minute via nasal prongs.
- Assess and manage child's fluid needs ↗ 26.
- Check glucose, if ≤ 28 days old and < 2.6mmol/L ↗ 9. If > 28 days old and < 3mmol/L (< 4 if diabetes) ↗ 32. Otherwise, prevent hypoglycaemia: if alert, encourage breastfeeding or give ♦formula/♦F75®/♦sugar water² 3mL/kg/hour orally (use NGT if baby refusing or unable to feed). Feed at least 2 hourly until transfer.
- Treat for infection: give ♦ceftriaxone³ 80mg/kg IM ↗ 159:11. Avoid injecting ≥ 1g IM at one injection site.
- Keep baby warm: place baby skin to skin with mother or dress warmly including head and feet and cover with blanket.

Approach to the baby with poor growth not needing urgent attention:

- If baby < 2.5kg, discuss/refer to dietitian same day.
- Check for feeding problem: if breastfeeding ↗ 120, if formula feeding ↗ 122. Screen for psychosocial risk, mother-baby bonding and/or mental health problem/s ↗ 11.

Manage further according to age and presence of feeding problem:

Baby < 2 weeks old *or* feeding problem

Review in 2 days

- If baby has lost weight since last visit:
 - If < 2 weeks old, refer.
 - If feeding problem, refer to dietitian same day. If dietitian unavailable same day, discuss/refer.
- If baby gaining weight and feeding problem corrected, review again in 2 weeks.
- If baby not gaining weight or feeding problem persists, give feeding advice again and review again in 5 days. If still not gaining weight, refer to next level of care.

Baby ≥ 2 weeks old *and* no feeding problem

Review in 7 days

- If baby gaining weight, review again at next immunisation visit.
- If baby not gaining weight, review again in 2 weeks.
- If baby has lost weight since last visit:
 - If dietitian available same day, refer to dietitian.
 - If dietitian unavailable same day, discuss/refer.

If weight gain unsatisfactory or feeding problem persists on follow up, refer.

¹Birth weight (kg) ÷ 10 = 10% of birth weight. ²Dissolve 4 teaspoons of sugar (20g) into 200mL water. ³Avoid giving Ringer's lactate together with IV/IM ceftriaxone. If severe penicillin allergy (previous anaphylaxis, urticaria or angioedema), discuss with doctor.

Not growing well: diagnosis

- If not already done, assess growth ↗ 15. A child not growing well may have any of:
 - Weight-for-age, length/height-for-age, weight-for-length/height or BMI-for-age below -2 line
 - Mid-upper arm circumference (MUAC) < 12.5cm
 - Abnormal growth curve pattern on growth chart (flattening, falling or crossing z-score lines on 2 consecutive visits)
 - Unintentional weight loss ≥ 5% of body weight (weight lost ÷ weight at last visit x 100 = % weight loss)

Are there any of: swelling of both feet, weight-for-length/height below -3 line, BMI-for-age below -3 line or MUAC < 11.5cm?

Yes

No

Severe acute malnutrition (SAM) likely

- Does child have any of:
- < 6 months old
 - Weight ≤ 4kg
 - Unable to drink/feed
 - Tires/sweats during feeds
 - Vomiting everything
 - Diarrhoea (watery stools and/or > 3 stools/day)
 - Lethargy or decreased consciousness ↗ 174
 - Jaundice
 - ≥ 2 of: cold hands/feet, weak/fast pulse ↗ 175, 3) CRT¹ ≥ 3 seconds, decreased level of consciousness ↗ 174: **shock** likely
 - Infected skin lesions (red/warm/painful)
 - Hazy/cloudy eye/s
 - Swelling of legs/face
 - Increased respiratory rate ↗ 175
 - Temperature < 36°C or ≥ 38°C
 - Hb < 7g/dL or spontaneous bleeding
 - Glucose < 3mmol/L
 - Leucocytes/nitrites on urine dipstick
 - Seizure/fit ↗ 28

Yes

No

Is child 6 months - 5 years old?

Yes

Do appetite test ↗ 128

Fails appetite test

Severe acute malnutrition (SAM) with medical complications

- If oxygen saturation < 90%, increased respiratory rate ↗ 175 or decreased level of consciousness ↗ 174, give ♦oxygen at 2L/min via nasal prongs.
- Assess and manage child's fluid needs ↗ 26.
- If glucose < 3mmol/L ↗ 32.
- Give ♦ceftriaxone² 80mg/kg IM ↗ 159:11. Avoid injecting ≥ 1g IM at one injection site.
- Give additional dose of ♦vitamin A ↗ 164:44.
- Feed at least 2 hourly until transfer. If refusing/not breastfeeding, give 50mL ♦formula/♦F75[®]/♦sugar water³ orally. If refusing orally, give via NGT.
- Keep warm: place child skin to skin with mother/carer and/or cover with blanket.
- Refer urgently to level 2 (regional/secondary) hospital. If < 6 months old, refer urgently to level 3 (tertiary) hospital.

Passes appetite test

Severe acute malnutrition (SAM) without medical complications likely

Refer same day to level 1 (district) hospital.

- Are there any of:
- Weight-for-length/height below -2 line or
 - BMI-for-age below -2 line or
 - MUAC < 12.5cm

Yes

No

Moderate acute malnutrition (MAM) likely

Is outpatient care available and carer able to care for child at home?

No

Yes

Chronic malnutrition (stunting) likely

Growth faltering likely

Give routine care ↗ 127.

¹Capillary refill time (CRT): hold hand/foot higher than level of heart. Press soft pad of finger/toe until it turns pale, then release pressure and note time taken for colour to return. ²Avoid giving Ringer's lactate together with IV/IM ceftriaxone. If severe penicillin allergy (previous anaphylaxis, urticaria or angioedema), discuss with doctor. ³Dissolve 4 teaspoons of sugar (20g) into 200mL water.

Not growing well: routine care

Give routine care to the child not growing well with any of:

- Acute malnutrition
 - Severe acute malnutrition (SAM) without medical complications after discharge from hospital *or*
 - Moderate acute malnutrition (MAM)
- Chronic malnutrition (stunting)
- Growth faltering

Assess the child not growing well. Record child's condition and care plan in RtHB

Assess	When to assess	Note
Symptoms	Every visit	• If SAM and any of: unable to drink/feed, tires/sweats during feeds, vomits everything, profuse diarrhoea, lethargy or decreased consciousness → 174, swelling of both feet, jaundice, infected skin lesions (red/warm/painful), increased respiratory rate → 175, severe acute malnutrition (SAM) with medical complications likely, manage and refer urgently → 126. • Manage other symptoms as on symptoms pages. If abnormal facial features or suspected congenital problem, refer to doctor.
Feeding	Every visit	Check for feeding problem: if breastfeeding (or mixed feeding) → 120, if formula feeding → 122, if eating solids → 123.
TB	Every visit	• Exclude TB at diagnosis and at any time if any of: significant TB exposure¹, cough, unexplained fever, drenching night sweats, poor weight gain, weight loss, failure to thrive, lethargy, decreased playfulness, visible neck mass, blood-stained sputum, headache, persistent vomiting (without diarrhoea) → 108. • If newly diagnosed TB, strong suspicion of TB or on TB treatment, discuss/refer same day.
Mother/carer	Every visit	• Ask about carer's general and mental health, HIV status, contraceptive needs and TB symptoms. If any problem → PACK Adult. • If breastfeeding mother has BMI² < 18.5 and/or MUAC³ < 23cm, refer to nutritional therapeutic programme (NTP).
Social	Every visit	• If carer struggling with/feeling overwhelmed by parenting <i>and</i> would like help → 145. • Ask who looks after child most of the time. If concerns about neglect → 144. Screen for psychosocial risk → 12. • If acute malnutrition and carer not able to care for child at home, refer same day.
Mental health	If ≥ 10 years old: at diagnosis	Screen for depression/anxiety → 139 and eating disorder/substance abuse → PACK Adult.
Routine care	Every visit	Integrate routine care into every visit → 12. If other long-term health conditions, ensure these are adequately treated.
Weight-for-age	Every visit	• If unintentional weight loss ≥ 5% of body weight (weight lost ÷ weight at last visit x 100), refer. • If weight loss on 2 consecutive visits, refer. • If no weight gain on 3 consecutive visits, refer to dietitian if not yet done.
Weight-for-length/height or BMI-for-age	Monthly	• If < 5 years old, assess weight-for-length/height. If ≥ 5 years old, assess BMI-for-age. • If chronic malnutrition (stunting) or growth faltering: if drops below -2 line, acute malnutrition likely, assess severity and manage → 126.
MUAC ³	If 6 months to 5 years old: monthly	If chronic malnutrition (stunting) or growth faltering: if drops below 12.5cm, acute malnutrition likely, assess severity and manage → 126.
Mouth/teeth	At diagnosis	If white patches in mouth (inside of cheeks/lips and on tongue), oral thrush/candida likely → 51. If dental caries → 54.
Oedema	Every visit	If swelling of both feet, severe acute malnutrition (SAM) with medical complications likely: manage and refer urgently → 126.
Hb	At diagnosis	• If Hb < 10g/dL in child < 5 years old or Hb < 11g/dL in child ≥ 5 years old, anaemia likely → 46. If Hb < 7g/dL, refer. • If child getting Ready-to-use Therapeutic Food (RUTF), avoid giving additional iron.
HIV	At diagnosis	If CLHIV, discuss/refer same day. If HIV status unknown, test for HIV → 95.
Glucose	At diagnosis	Assess and manage glucose → 32. If diabetes, refer.
Thyroid function	If > 10 years old: at diagnosis	If increased pulse rate → 175, palpitations, tremor, unable to tolerate hot weather or thyroid enlargement, check TSH. If abnormal, refer.

Continue to advise and treat the child not growing well → 128.

¹Any child who in the last 12 months shared an enclosed space (school/creche/home/similar) for ≥ 1 night or frequent/extended daytime periods with an adult/adolescent with pulmonary TB ("index client"), during the 3 months before index client started TB treatment. ²Body mass index: weight (kg) ÷ height (m) ÷ height (m). ³Mid-upper arm circumference.

Advise the child not growing well and his/her carer

- Educate that good nutrition is vital for normal development of the brain and body. Advise carer that first 1 000 days of child's life are vital to development.
- Give feeding advice: if breastfeeding (or mixed feeding) ➔ 120, if formula feeding ➔ 122, if ≥ 6 months old ➔ 123.
- Give hygiene advice: wash hands with soap and water regularly, especially before handling food/after using toilet/changing nappies. Wash fruit/vegetables. Use boiled water if no access to clean water.
- Encourage carer to interact and play with child as much as possible. Encourage sensory stimulation by singing songs, tellings stories and playing with different colours, shapes, textures and sounds.
- If child < 18 years needs child support grant, advise carer to take child's birth certificate and carer's ID to South African social security agency (SASSA) to apply.

Treat the child not growing well

- Check that routine vitamin A, deworming and immunisations are up to date ➔ 12.
- Refer to nutritional therapeutic programme (NTP) for a total of 6 months and complete NTP register. Ensure a monthly supply of correct product:
 - < 6 months old: infant formula
 - 6-12 months old: infant formula *plus* infant cereal
 - ≥ 12 months old: instant/enriched porridge *plus* energy drink *plus* Ready-to-use Therapeutic/Supplementary Food (RUTF/RUSF). Avoid RUTF/RUSF if nut allergy.
- Refer to dietitian if any of:
 - baby not breastfeeding and < 2.5kg
 - growth faltering
 - severe or moderate acute malnutrition
 - poor response to NTP after 3 months
 - child needs NTP for > 6 months
- Refer to community health worker, social worker and link with local NGOs (like Philani) ➔ 170.
- If available, refer to physiotherapist and occupational therapist for rehabilitation, physical and emotional stimulation.
- If **chronic malnutrition (stunting)** (length/height-for-age below -2 line), give NTP and refer to dietitian immediately.

Review the child not growing well

- If feeding problem, review in 5 days. If no better, discuss/refer.
- If no feeding problem or feeding problem better, review every 2 weeks until growing well¹, then monthly until NTP completed (6 months total) , then discharge from NTP.

How to do an appetite test

- The child must be ≥ 6 months old: give Ready-to-use-Therapeutic-Food (RUTF/F75®/10% dextrose) according to weight (see table).
- Test may take up to one hour. Do not force child to eat. Offer child plenty of water to drink.
- If child finishes minimum amount of feed, s/he passes the appetite test.
- If child does not finish minimum amount of feed: s/he fails the appetite test.

Minimum amount to be given to child			
Body weight (kg)	RUTF Imunut® Sachet (92g)	F75®	10% dextrose
< 4	12g	45mL	50mL
4-6.9	23g	70mL	80mL
7-9.9	30g	100mL	150mL
10-14.9	45g	150mL	175mL
15 -29.9	70g	200mL	200mL
≥ 30	92g	250mL	250mL

Advise to return immediately if worsens: unable to drink/feed, tires/sweats during feeds, vomiting everything, profuse diarrhoea, lethargy or decreased consciousness, swelling of both feet, fever.

¹Growing well: weight-for-age and weight-for-length/height or BMI-for-age on or above -2 line, MUAC ≥ 12.5cm and upward growth curve for 2 consecutive visits.

Overweight/obesity: routine care

- The **overweight** child has a weight-for-length/height (WFL/H) between the +2 line and +3 lines; or if ≥ 5 years: a BMI between the +1 and +2 lines.
- The **obese** child has a WFL/H on or above +3 line; or if ≥ 5 years: a BMI on or above +2 line. The **very obese** child who is ≥ 5 years has a BMI on or above +3 line.
- Refer to dietitian for initial assessment and healthy meal plan. Give routine care in meantime:

Assess the overweight child. Record child's condition and care plan in RtHB

Assess	When to assess	Note
Routine care	Every visit	Integrate routine care into every visit ↪ 12. If unusual facial features or syndrome suspected, refer to doctor.
Symptoms	Every visit	Manage symptoms as on symptom pages. Ask about hip/leg pain ↪ 72 and back pain ↪ 70.
Medications	At diagnosis	If on long-term medications (steroids, antiseizure medication, antidepressants, antipsychotics), discuss with doctor.
Diet	Every visit	Ask about diet and eating habits ↪ 124.
Activity	Every visit	If < 1 hour/day moderate intensity activity (brisk walking, dancing, housework, gardening) or > 1 hour/day screen time (TV, video games, phone) ↪ 124.
Sleep	At diagnosis	If poor sleep with persistent snoring or breathing problems at night, like gasping or apnoea (episodes of no breathing for > 10 seconds) ↪ 50.
Mental health	Every visit	If over past few months, child has been miserable, stressed or angry ↪ 139.
School problems	At diagnosis	If poor attendance, bullying, learning problems, difficulty interacting with other children ↪ 140.
Carer	At diagnosis	Check carer's Body Mass Index (BMI): weight (kg) ÷ height (m) ÷ height (m). If > 25, assess and manage overweight/obesity ↪ PACK Adult.
Weight-for-age	Every visit	• If ≥ 7 years old or complications (glucose, liver or musculoskeletal problems), aim for a weight loss of 0.5kg/month. • If < 7 years old and no complications (glucose, liver or musculoskeletal problems), aim to keep weight same as child grows until no longer overweight.
Length/height-for-age	Every 6 months	If length/height-for age (L/HFA) below -2 line, refer to paediatrician same month.
Weight-for-length/height	If < 5 years old: 3 monthly	Aim to keep weight same as child grows until WFL/H eventually below +2 line. If < 2 years old and WFL/H on or above +3 line, refer to paediatrician.
BMI	If ≥ 5 years old: 3 monthly	BMI is weight (kg) ÷ height (m) ÷ height (m). Aim to keep weight same as child grows until BMI eventually below +1 line.
Blood pressure	At diagnosis, then yearly	If raised ↪ 175, discuss with doctor.
Teeth	At diagnosis	If dental caries ↪ 54.
Total cholesterol	At diagnosis, then 2 yearly	If ≥ 5.2 mmol/L, refer for fasting lipid profile: if ≥ 4.4 mmol/L, repeat total cholesterol in 6 months after supportive measures to lose weight and improve diet.
Glucose	If obese: at diagnosis, then yearly	If ≥ 10 years old and ≥ 1 of the following, check fasting glucose after 8 hour overnight fast: 1) family member with type 2 diabetes 2) mother had diabetes during pregnancy 3) darkening of skin folds and creases 4) hypertension/dyslipidaemia: - If fasting glucose < 5.6mmol/L, reassure. If 5.6-6.9mmol/L, impaired fasting glucose likely, discuss further tests with doctor. If ≥ 7mmol/L, diabetes likely; refer.
ALT	If obese: at diagnosis, 2 yearly	If ALT > 100, refer.
Thyroid function	At diagnosis	Check TSH if any of: dry skin, brittle hair, constipation, puffy face, intolerant to cold or thyroid enlargement. Refer to doctor if result abnormal.
Hb	If pallor ¹	If Hb < 10g/dL in child < 5 years old or Hb < 11g/dL in child ≥ 5 years old, anaemia likely ↪ 46. If Hb < 7g/dL, refer.

Advise the overweight child and/or carer

- Alert child and carer to increased risks associated with obesity:
 - Joint (hip, knee and back) and medical problems (hypertension, type 2 diabetes, cholesterol problems, heart attack, stroke, liver disease, heartburn, breathing problems) including asthma and snoring.
 - School and social problems (bullying, teasing, anxiety, depression, poor self-esteem, isolation, relationship problems).
- Encourage a balanced healthy diet and daily exercise ↪ 124. Encourage parents/carers and siblings to change to healthy lifestyle as well.

Review every 3 months.

¹Look for pale palms of the hands and conjunctival pallor: paleness of the lower inner eyelid.

Epilepsy

- If child fitting now →28. If child not known with epilepsy and has had a recent fit, assess further →29.
- A doctor (usually specialist) should decide if long-term treatment should be started in a child with ≥ 2 seizures more than 24 hours apart with no identifiable cause.

Assess the child with epilepsy: record epilepsy diagnosis and care plan in RtHB

Assess	When to assess	Note
Long term health conditions	Every visit	Ensure other long term health conditions are adequately treated. If child has cerebral palsy → 146.
Adherence and side effects	Every visit	Ask if child takes medication every day. If not, explore reasons for poor adherence → 152. Ask about side effects of treatment (below).
Other medication	Every visit	If child started antibiotics, TB or HIV treatment, refer to doctor to assess for drug interactions.
Seizure frequency	Every visit	Review seizure diary. If still fitting after 2 months <i>and</i> adherent to treatment (correct dose) with no obvious triggers ¹ or medication interactions, refer to specialist.
Mental health	Every visit	If over past few months, child has been miserable, stressed or angry → 139 or if problematic change in behaviour → 136.
School problems	Every visit	If failing grades, not coping with school work or bullying/violence at school → 140.
Family planning	Girl ≥ 10 years old	If on valproate and menarche, ensure girl on reliable contraception → PACK Adult and discuss change of antiseizure medication with specialist.
Routine care	Every visit	Integrate routine care into every visit → 12. Manage symptoms as on symptom page.
Development	6 monthly	• Check milestones → 12: if not talking properly → 89, if not moving properly → 90, if hearing problem → 49, if vision problem → 48. • If loss of milestones, discuss/refer to specialist.

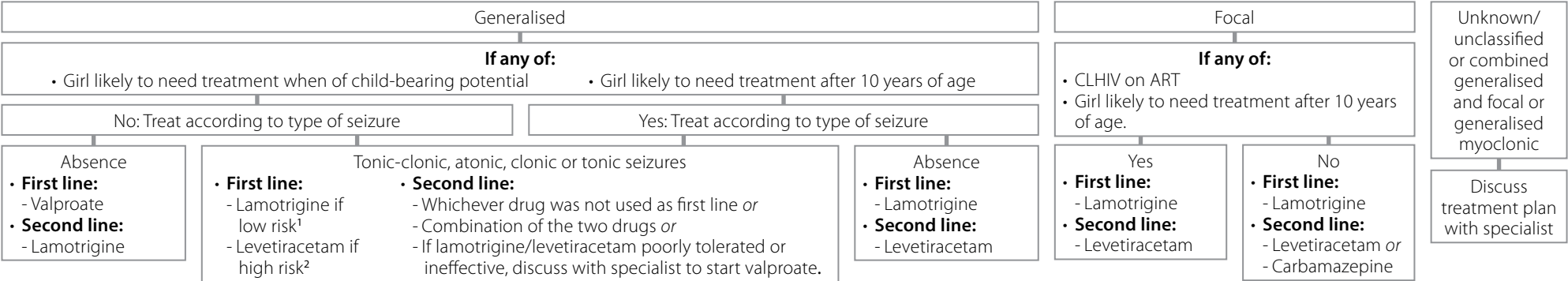
Advise the child with epilepsy and his/her carer

- Explain what to do if child fits at home → 29. Avoid possible triggers¹.
- Educate about epilepsy and need for adherence to be fit free.
- Advise to keep a home record/seizure diary to record frequency and length of fit, possible triggers¹ and changes in medication. Encourage carer to take a video of event.
- Help carer to get MedicAlert® bracelet → 170. Refer for support (Epilepsy SA) → 170. Carer to inform teachers, explain what to do if child fits and what activities child should avoid.
- Reduce chance of injury: supervise swimming/bathing/crossing roads, shield fireplaces/cookers, avoid contact sports (rugby), advise not to lock doors (bed/bathroom).

Treat the child with epilepsy

- Treatment is usually initiated by a specialist, starting with a single medication at low dose and increased slowly every 2 weeks until fits stop or side-effects intolerable.
- **If already on treatment:**
 - If girl on valproate, discuss risks² and carer to sign acknowledgement of risk form³ yearly.
 - If ≥ 10 years old, explain need to change antiseizure medication and discuss with specialist. If unable to change (drug resistant epilepsy or other antiseizure medication ineffective or intolerable), ensure girl is on reliable contraception → PACK Adult.
 - If no further fits, continue antiseizure medication at same dose.
 - If still having fits:
 - If poor adherence: support adherence, continue same dose and review in 2 weeks.
 - If medication interactions: adjust medication as needed and review in 2 weeks.
 - Confirm that diagnosis and type of epilepsy is correct. If unsure, discuss/refer to specialist.
 - If none of above: increase antiseizure medication → 131. If already on maximum dose for 4 weeks, switch antiseizure medication once. If already on second antiseizure medication, discuss/refer to specialist.
 - If fits worsen/persist despite maximum treatment, discuss/refer to specialist.
- **If switching medication** due to poor control, intolerable side effects or girl ≥ 10 years old on valproate:
 - Discuss switching medication with specialist. Continue old antiseizure medication and add new medication, increasing dose 2 weekly until no further fits. Then slowly reduce old antiseizure medication dose 2 weekly, until stopped.
- If fit free, review 6 monthly. If no fits for 2 years: refer to specialist to assess whether to stop treatment. If fits recur, refer/discuss with specialist.

Treat the child with epilepsy according to type of fits



Start with a single antiseizure medication at low dose and increase slowly every 2 weeks until fits stop or side-effects intolerable:

Medication	Dose	Maximum dose	Notes	Side effects
■Lamotrigine (to be initiated by designated prescriber)	• Start dose: 0.2mg/kg daily for 2 weeks • Increase by: 0.2mg/kg/day in divided doses 12 hourly every 2 weeks • Maintenance: 1-5mg/kg/dose 12 hourly	10mg/kg/day in divided doses	• Use if low risk¹ focal, tonic-clonic or absence seizures. • Avoid if liver/kidney disease. • May interact with other antiseizure medication and rifampicin/rifapentine.	• Urgent: skin rash ➡ 75. • Self-limiting: nausea, vomiting, blurred/double vision, dizziness, drowsiness, insomnia, fatigue.
■Levetiracetam (to be initiated by designated specialist)	• 1-6 months: - Start dose: 7mg/kg/dose 12 hourly - Increase by: 7mg/kg/dose 12 hourly - Maintenance: 21mg/kg/dose 12 hourly • ≥ 6 months - 4 years: - Start dose: 10mg/kg/dose 12 hourly - Increase by: 10mg/kg/dose 12 hourly - Maintenance: 25mg/kg/dose 12 hourly • ≥ 4-12 years: - Start dose: 10mg/kg/dose 12 hourly - Increase by: 10mg/kg/dose 12 hourly - Maintenance: 30mg/kg/dose 12 hourly	• 1-6 months: 42mg/kg/day • ≥ 6 months - 4 years: 50mg/kg/day • ≥ 4-12 years: 60mg/kg/day	• If high risk² tonic-clonic or absence seizures. • Avoid if previous allergic reaction.	• Urgent: - Aggression: manage ➡ 136 and discuss/refer same day. - Psychosis: discuss/refer same day. • Self-limiting: dizziness, fatigue, drowsiness, weakness, headache.
■Phenobarbital	Start and maintain: 3.5-5mg/kg/dose as a single dose at night.	5mg/kg/day	• Choose if < 6 months old (specialist review). • Avoid if absence seizures, > 2 years old or CLHIV on ART.	Drowsiness, behaviour problems, hyperactivity.
■Carbamazepine ³	• Start dose: 5mg/kg/day in divided doses 8-12 hourly • Increase by: 2mg/kg/day in divided doses 8-12 hourly • Maintenance: 10-20mg/kg/day in divided doses	20mg/kg/day in divided doses (maximum 1g/day)	• Use if focal seizures. • Avoid if absence/myoclonic seizures, CLHIV on ART or girl ≥ 10 years old. • May interact with other antiseizure medication, isoniazid and rifampicin/rifapentine.	• Urgent: skin rash ➡ 75. • Self-limiting: drowsiness, dry mouth, dizziness, ataxia, nausea, loss of appetite, constipation, abdominal pain. If drowsiness affects school performance, refer to specialist.
■Valproate ^{3,4}	• Start dose: 5mg/kg/dose 8-12 hourly • Increase by: 5mg/kg weekly up to 15-20mg/kg/day in divided doses 8-12 hourly. • Maintenance dose: 20-30mg/kg/day in divided doses 8-12 hourly	40mg/kg/day in divided doses	• Use if absence, myoclonic or generalised tonic/clonic seizures. • Avoid if liver disease or girl ≥ 10 years old. • If girl: carer to sign acknowledgement of risk form yearly. • May interact with other antiseizure medication and AZT.	• Urgent: jaundice, vomiting, abdominal pain: stop medications and refer urgently. If < 2 years old, lethargy or loss of appetite and nausea, check ALT. • Self-limiting: nausea, diarrhoea, constipation.

¹Low-risk: infrequent seizures, no previous hospitalisation for seizures, no history of status epilepticus. ²High-risk: frequent seizures, previous history of hospitalisation for seizures, previous history of status epilepticus. ³Give syrup formulation 8 hourly and tablet formulation 12 hourly. ⁴If able to swallow tablet, give controlled release (CR) formulation 12 hourly. If unable to swallow tablet, give crushable formulation (100mg tablets) 8 hourly.

Bronchiectasis

Bronchiectasis is a chronic lung condition where the small airway walls are thickened and widened from inflammation and infection. A specialist must confirm the diagnosis.

Assess the child with bronchiectasis. Record child's condition and care plan in RthB.

Assess	When to assess	Note
Symptoms	Every visit	• Manage other symptoms as on symptom pages. If coughing up blood, refer urgently. • If signs of respiratory distress (lower chest wall indrawing, nasal flaring, accessory muscle use¹, difficulty feeding/talking, sats < 92%) ➔ 55. • If worsening cough, increased sputum, change in sputum colour, acute exacerbation likely. Manage below.
TB	Every visit	At diagnosis, check for TB even if no symptoms ² or significant TB exposure ³ ➔ 108. At follow-up visits screen for TB symptoms ➔ 108.
Inhaler technique	Every visit if using inhaler	Check that child and/or carer can use inhaler and spacer correctly ➔ 57.
Lung clearance	Every visit	Check if child/carers performing routine chest physiotherapy/lung clearance techniques at home. Refer to physiotherapy.
Palliative care	If disease extensive/deteriorating	If bronchiectasis severe enough to be life-limiting, also give palliative care ➔ 150.
Routine care	Every visit	Integrate routine care into every visit ➔ 12.
Growth	At diagnosis	• Measure and record weight-for-age, length/height-for-age, weight-for-length/height (or BMI), and mid-upper arm circumference (MUAC) ➔ 15. • Refer to dietitian for nutritional support regardless of measurements.
HIV	At diagnosis	If CLHIV, give routine HIV care ➔ 97. If HIV negative or unknown, test for HIV ➔ 95.
Hb	At diagnosis and if coughing up blood	If Hb < 10g/dL in child < 5 years or Hb < 11g/dL in child ≥ 5 years old, anaemia likely ➔ 46. If Hb < 7g/dL, refer.
Chest x-ray	At diagnosis and yearly	• If disease localised to one area of lung, refer for specialist review. • If chest X-ray progressively worsening, refer to specialist.
Sputum	If ≥ 3 exacerbations in 1 year	Send early morning sputum for microscopy, culture and sensitivity and refer to specialist.
Bronchodilator response	If wheeze and not already done	Give ■salbutamol 400mcg (4 puffs) via spacer and assess response after 15 minutes: if wheeze better, child is bronchodilator responsive.

Advise the child with bronchiectasis and his/her carer

- If smoking in the house or child smokes, alert to risks and encourage smoker to quit ➔ PACK Adult.
- Help child/carers to recognise and manage acute exacerbation. If worsening cough, increased sputum, change in sputum colour, advise to return to clinic same day.
- If starting/using salbutamol, demonstrate inhaler technique ➔ 57.
- Educate about need for chest physiotherapy and lung clearance techniques at home. Clearing lungs can help prevent acute exacerbations and further lung damage:
 - Encourage child to blow a piece of rolled up tissue/paper across the table or blow bubbles into air/through a straw into soapy water.
 - Ask child to take 10 deep breaths daily (as big as possible) and give deep cough.
 - Encourage child to do age-appropriate daily physical exercise, as guided by specialist.

Treat the child with bronchiectasis

- If **acute exacerbation**: give ■amoxicillin/clavulanic acid⁴ 45mg/kg 12 hourly ➔ 158:8 for 10 days and review. Continue up to 14 days if needed.
- If bronchodilator responsive, give ■salbutamol via spacer 100-200mcg (1-2 puffs) as needed, up to 4 times a day.
- Advise influenza vaccination during influenza vaccine campaign. If not already given, also give 2 additional doses of ■pneumococcal vaccine (conjugated) 8 weeks apart.

- Review 3 monthly. Doctor to review at least once a year. Advise to return immediately if symptoms worsen.
- Refer to specialist if acute exacerbation not responding to antibiotics within 2 days or > 1 acute exacerbation in 4 months.

¹Any of: subcostal recession, intercostal recession, tracheal tug, use of neck muscles. ²TB symptoms: cough, unexplained fever, drenching night sweats, poor weight gain, weight loss, failure to thrive, lethargy, decreased playfulness, visible neck mass, blood-stained sputum, headache, persistent vomiting (without diarrhoea). ³Any child who in the last 12 months shared an enclosed space (school/creche/home/similar) for ≥ 1 night or frequent/extended daytime periods with an adult/adolescent with pulmonary TB ("index client"), during the 3 months before index client started TB treatment. ⁴If severe penicillin allergy (history of anaphylaxis/urticaria/angioedema), give instead ■azithromycin 10mg/kg once a day for 10 days ➔ 158:10.

Known heart problem

Give urgent attention to the child with known heart problem and any of:

- < 1 year old and difficulty feeding or sweats during feeds: **heart failure** likely
- Any child with known heart problem and difficulty breathing/tires easily/new onset increased pulse rate ↗ 175, **heart failure** likely
- Fainting
- Chest pain
- Sudden weakness of one side of body
- Blue skin/lips
- Temperature ≥ 38°C or fever in last 3 days

Manage and refer urgently:

- If **heart failure** likely, give **furosemide** 1mg/kg IV over 5 minutes ↗ 161:25. Avoid IV fluids.
- Assess and manage child's fluid needs ↗ 26.
- If difficulty breathing/increased respiratory rate ↗ 175 and temperature ≥ 38°C/fever in last few days: give **ceftriaxone**¹ 80mg/kg IM ↗ 159:11. Avoid injecting ≥ 1g IM at one injection site.
- Give **oxygen** at 2L/minute via nasal prongs and raise head of bed to 45 degrees.

Assess the child with known heart problem not needing urgent attention: Record child's heart condition and care plan in RtHB.

Assess	When to assess	Note
Symptoms	Every visit	If sore/red throat ↗ 52, if cough ↗ 55. If difficulty breathing only on exercise, refer for specialist assessment.
Long-term health conditions	Every visit	If bronchiectasis ↗ 132, if CLHIV ↗ 97, if cerebral palsy ↗ 146, if Down syndrome ↗ 149.
Growth	Every visit	Assess growth ↗ 15.
Routine care	Every visit	Integrate routine care into every visit ↗ 12.
Teeth	Every visit	If dental caries ↗ 54. If needing dental extraction, give prophylaxis (see below). Arrange yearly dentist review.

Advise the child with known heart problem and his/her carer

- Ensure child brushes teeth twice a day and encourage good feeding and eating ↗ 123.
- Advise child/carers to go to the clinic the same day if: sore throat, coughing or fever.
- Ensure child attends regular specialist appointments. Encourage family to join a support group/s ↗ 170.
- Encourage child to do daily physical exercise, as guided by specialist.
- If previous rheumatic fever or known rheumatic heart disease, explain the importance of treatment adherence and the risk of damage to heart valves.

Treat the child with previous rheumatic fever or known rheumatic heart disease:

- Advise influenza vaccination during influenza vaccine campaign.
- If previous rheumatic fever *and/or* known rheumatic heart disease: give prophylaxis.
 - Give single dose **benzathine benzylpenicillin**^{2,3} 4 weekly according to weight: if < 30kg, give 600 000 units IM. If ≥ 30kg, give 1.2 million units IM.
 - If child on warfarin, avoid IM injections, give instead **phenoxymethylpenicillin**³ 125mg 12 hourly or **amoxicillin**³ daily according to weight: if < 30kg, give 125mg. If ≥ 30kg, give 250mg.
 - Decide when to stop prophylaxis:
 - If rheumatic valvular disease: give lifelong prophylaxis.
 - If no rheumatic valvular disease:
 - If first episode of rheumatic fever when child < 11 years old, give until age 21.
 - If first episode of rheumatic fever when child ≥ 11 years old, give for 10 years.
- If known with a heart valve problem, also give antibiotic prophylaxis if:
 - Requiring dental extraction, give **amoxicillin** 50mg/kg (up to 2g) 1 hour before procedure and repeat dose **amoxicillin** 6 hours after the procedure. If penicillin allergy, refer.
 - Child needs surgery, refer to specialist.

¹Avoid giving Ringer's lactate together with IV/IM ceftriaxone. If severe penicillin allergy (previous anaphylaxis, urticaria or angioedema), discuss with doctor. ²Dissolve benzathine benzylpenicillin 1.2 million units in 3.2mL **lidocaine 1% without adrenaline**. ³If severe penicillin allergy (history of anaphylaxis, urticaria or angioedema), give instead **azithromycin**. If < 11 years, give 10mg/kg three times per a week; if ≥ 11 years, give 250mg daily.

Chronic arthritis

- The child with chronic arthritis has had painful and/or swollen joints for ≥ 6 weeks that limits daily activities.
- Ensure arthritis is diagnosed and managed by a specialist. Ensure regular dates for specialist review in place; these are usually 3-4 monthly.

Assess the child with chronic arthritis. Record child's condition and care plan in RtHB.

Assess	When to assess	Note
Symptoms	Every visit	If joint pain, swelling, fever or worsening function, treat as below.
Eye test	Every 4 months	Arrange eye tests at eye OPD for first five years from diagnosis to screen for uveitis.
Routine care	Every visit	Integrate routine care into every visit ➔ 12.
Mental health	Every visit	Screen for depression and anxiety at every visit ➔ 139.
Growth	Every visit	Plot growth ➔ 15. If weight loss ➔ 126.
Teeth	Yearly	Arrange yearly dentist review. If gum or tooth problem ➔ 54.
Joints	Every visit	• If swollen/tender joint/s with limited movement and early morning stiffness, active arthritis likely. Treat as below. • Arrange regular physiotherapy or occupational therapy to give child exercises to keep joint mobile and maintain muscle strength. • Arrange splints if worried about permanent joint tightening/deformity. Refer to occupational therapist for hand, knee and elbow splints. • Check if shoes fit well and that child can walk easily. If problem, refer to orthopaedic nurse/orthotist to arrange shoe inserts.
HIV	At diagnosis	If not yet done, test for HIV ➔ 95.
FBC, creatinine, ALT	If on methotrexate: 2-4 monthly	If results abnormal, arrange specialist referral. Specialist will determine frequency of blood tests.

Advise the carer of child with chronic arthritis

- Educate carer that arthritis may take months to years to improve. Advise that early treatment prevents joint damage and lessens length of illness.
- Stress importance of continuing the treatment and attending the clinic and specialist dates.
- If child well and no joint symptoms at visit, do not stop prescribed specialist medication as joint symptoms may flare up if medication is stopped.
- Encourage physical exercise on a daily basis. Swimming and cycling are helpful.
- Encourage a healthy and balanced diet. See advice box in general assessment ➔ 12.
- Advise that child can live a full and happy life as long as joint symptoms are managed.
- Provide carer and child with social support services ➔ 170.

Treat the child with chronic arthritis

- Child may be on steroids (methylprednisolone) and/or methotrexate.
- If on methotrexate, give **■folic acid** 5mg weekly, on the day after methotrexate.
- Give **■ibuprofen** 5-10mg/kg 8 hourly with food ➔ 161:28 for up to 5 days. Avoid ibuprofen if asthma, heart failure or kidney disease. If no response to treatment after 2 days or fever without other identified cause, refer to specialist same day.

Arrange 3-4 monthly specialist review.

Suicide

Give urgent attention to the child who has attempted or considered suicide

Has child just attempted suicide?

Yes

- First assess and manage airway, breathing, circulation and level of consciousness ↗ 24.
- If overdose or exposure to harmful substance ↗ 36.
- If exposure to carbon monoxide (exhaust fumes), give 100% face mask ♦ oxygen.

No

Does child have current thoughts or plans to commit suicide?

Yes

Explore how often, how serious intent to die is and if a plan exists.

No

Has child had thoughts or plans of suicide in past month or performed act of self-harm or suicide in past year?

Yes

Is child agitated, violent, distressed, uncommunicative or does child have hallucinations or delusions?

Yes

Imminent risk of suicide

No

No imminent risk of suicide

Manage as below.

- Avoid leaving child alone. Ensure parent/carer is accompanying child at all times. Remove any possible means of suicide (like shoelaces, pills).
- If aggressive or violent, ensure safety: assess child with other staff, use security personnel or police if needed. If sedation needed, discuss with paediatrician or psychiatrist.
- Refer urgently: while awaiting transport, monitor closely. If child or parent/carer refuses admission, discuss involuntary admission under the Children's Act urgently with paediatrician or psychiatrist.

Assess the child with no imminent risk of suicide

Assess	Note
Mental health	• If irritable, sad, unable to enjoy anything, crying a lot or feeling lonely for most of day for ≥ 2 weeks, depression likely, refer/discuss with mental health team. • If angry, withdrawn or change in mood/behaviour/feelings/sleep/appetite and not coping at school/work/home ↗ 139.
Substance use	If concerns about use of alcohol or drugs, link to psychosocial services (counsellor/social worker/support group/helpline) ↗ 170.
Family/home situation	Refer all cases to social worker to assess family/home situation.
Chronic condition	Check child is receiving appropriate routine care and is adherent to long-term medication. If HIV ↗ 97, if life-limiting illness ↗ 150. If pregnant, discuss with specialist.

Advise the child with no imminent risk of suicide and his/her parent/carer

- Reassure parent/carer that talking about suicide does not increase the risk of suicide. Discuss with child reasons to stay alive. Together with parent/carer and child, work out a safety plan:
 - Spend time working out what behaviours/actions are considered warning signs of a crisis for this child. Encourage carer to closely monitor child. Discuss coping strategies:
 - Self help: encourage physical exercise, relaxation techniques.
 - Distraction activities: together identify people/social settings that may distract. List names/numbers.
 - Ways to make environment safe: remove firearms from house, keep medications/toxic substances locked away.
 - Provide hotlines/agencies they can contact during a crisis ↗ 170.
 - Together identify trusted people to ask for help when it gets too much.
- Where available, refer child to psychologist or counsellor and link child and parent/carer with support group/s ↗ 170.

Review child at least weekly for 2 months. If still no imminent risk of suicide, follow up monthly.

Behaviour problems

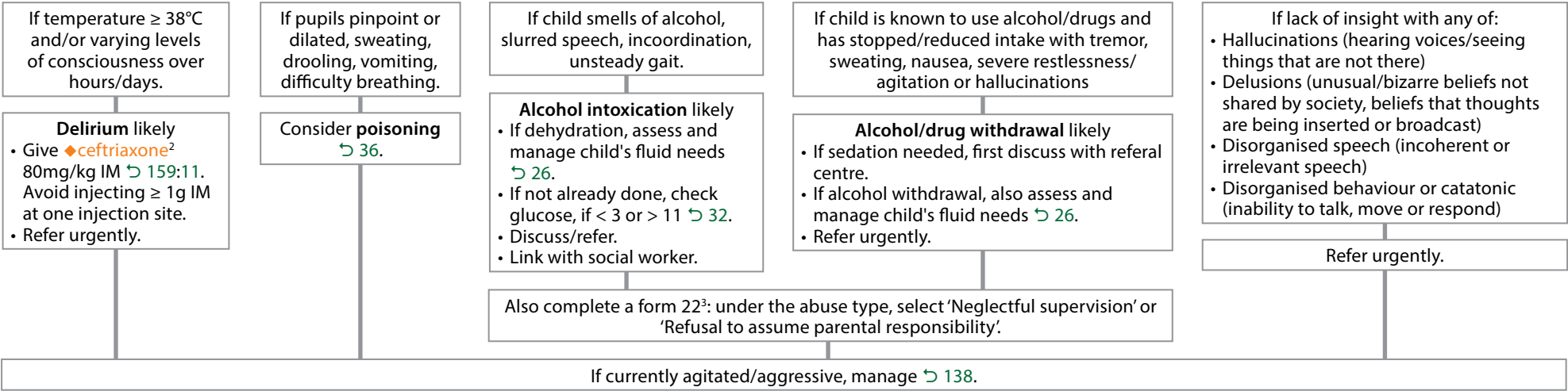
If reported behaviour problem (in classroom or at home) →137. If current behaviour problem, assess and manage below.

Give urgent attention to the child with behaviour problems that include any of:

- At risk of harming self or others → 135
 - Just had a fit → 28
 - Recent head injury → 33
 - Sudden onset of abnormal thoughts/behaviour
 - Hearing/seeing things that are not there
 - Intoxicated/withdrawal needing restraint or sedation
 - Confused → 31
 - Varying level of consciousness → 174
- Temperature ≥ 38°C
 - Agitated with any of:
 - Challenging, insulting or provocative behaviour
 - Frequently changing body position, pacing
 - Tense posturing like gripping tightly or clenching fists
 - Aggressive acts like pounding walls, throwing objects, hitting
 - Loud, aggressive speech or angry behaviour

Manage and refer urgently:

- Check breathing, respiratory rate, BP, pulse, CRT¹, glucose, temperature and pupil response:
 - If difficulty breathing, increased respiratory rate → 175, oxygen saturation < 90% or oxygen saturation machine not available, give face mask ♦oxygen.
 - If ≥ 2 of: cold hands/feet, weak/fast pulse → 175, CRT¹ > 2 seconds, decreased level of consciousness → 174, **shock** likely → 26.
 - If fingerprick glucose < 3 or > 11 → 32.
- Look for delirium, poisoning, intoxication, withdrawal or psychosis and manage before referring urgently:



If behaviour problem not needing urgent attention →137.

¹Capillary refill time (CRT): hold hand/foot higher than level of heart. Press soft pad of finger/toe until it turns pale, then release pressure and note time taken for colour to return. ²Avoid giving Ringer's lactate together with IV/IM ceftriaxone. If severe penicillin allergy (previous anaphylaxis, urticaria or angioedema), discuss with doctor. ³Form 22 prompts further detailed investigation into a case of suspected child abuse/neglect. Any adult working with a child in a professional capacity may complete the form.

Approach to the child with behaviour problems not needing urgent attention

First check for physical causes:

Check for pain, problems with vision, difficulty hearing or communication problem, sleep problem and developmental delay:

Look for pain

- Ask about other symptoms and manage as per symptom page.
- Check ears for foreign body ↗ 49, teeth for caries ↗ 54 and mouth for painful blisters/ulcers ↗ 51.

Ask about vision, hearing and communication problem

- If vision problem ↗ 48.
- If speech/language problem ↗ 89.
- If hearing problem ↗ 49.

If sleep problems ↗ 88.

Check development

- Assess milestones if < 6 years old ↗ 134.
- If > 6 years old: if delay in reading and writing or delay in self-care (such as dressing, bathing, brushing teeth) ↗ 90.

Ask about emotional distress:

If sad, withdrawn, irritable, worry, stress, anxiety or emotional outbursts, assess and manage further ↗ 139.

Screen for social risk/stressors:

Screen for school problems, parenting difficulty, child abuse, depression and substance abuse in the carer:

- Ask if behaviour worse in a particular setting (school or home) or with a particular person, try to explore and address problem.
- If behaviour problem occurs only at school or if school refusal/ bullying/poor school grades ↗ 140.
- Ask about multiple carers with different rules/parenting style. Ask if father figure present and if family member currently serving prison sentence. Provide additional parenting support ↗ 145.

Ask carer if aware of any abuse of child, carer or siblings. Ask child if anyone hurts or upsets him/her. If yes to either, **child abuse** likely ↗ 144.

- Screen for depression and substance abuse in carer:
- In the past month, has carer felt down/depressed/hopeless or little interest/ pleasure in doing things? If yes to either ↗ **PACK Adult**.
 - Does carer drink ≥ 6 drinks/session every month, drink every day, use illegal drugs or misuse prescription or over-the-counter medication ↗ **PACK Adult**.

If other household members with mental health illness, recommend that they attend the clinic to ensure control.

Asses behaviour:

Assess for ADHD, substance use in child and autism spectrum:

If overactive, unable to stay still for long, easily distracted, not finishing tasks, restless and symptoms affecting school and home life, consider **attention deficit hyperactivity disorder** ↗ 141.

If concerns about use of alcohol or drugs, consider substance abuse.

Link to psychosocial services (counsellor/social worker/ support group ↗ 170) and refer/discuss with mental health team.

- Consider **autism spectrum** and refer to paediatrician if child:
- Prefers solitary play or has difficulty in new social contexts
 - Avoids eye contact
 - Displays repetitive behaviour, craves routine, becomes distressed with changes
 - Displays an unusual sensory profile (taste, textures, sounds)
 - Uses language unusually - copies sounds, uses unknown words, speaks about favourite subjects a lot

- Reassure parent that a child will do well if they can. Behavioural problems usually indicate a stress behaviour and not a misbehaviour. If carer struggling with parenting ↗ 145.
- Review in 1 month. If no better and unable to find cause, discuss/refer to paediatrician or mental health practitioner.

Manage the agitated/aggressive child

If not yet done, assess the child with behaviour problems [↗ 136](#).

Manage the agitated/aggressive child:

- **Try to first calm the agitated/aggressive child without restraints or sedation:**
 - Ensure the safety of yourself, the child and those around you: ensure security personnel present, call police if needed. They should disarm child if s/he has a weapon.
 - Move to a quieter area/room: reduce noise, dim lights, close curtains, minimise visitors.
 - Introduce yourself and try to verbally calm the child:
 - Avoid direct eye contact, sudden movements or approaching child from behind. Stand at least two arm's lengths away. Ensure exit is not blocked.
 - Use an honest, non-threatening manner. Use simple language. Avoid talking down, arguing or commanding child to calm down. Explain each step and what is to come next.
 - Discuss restraint and offer reward for calmer behaviour.
 - Ask about hunger, thirst, pain or discomfort and if needed, address these.
 - Listen to child, identify his/her feelings and desires and offer choices. Take all threats seriously.
- **Only if verbal attempts to calm the child fail *and* child is a danger to self or others, consider restraints or sedation:**
 - Before restraint/s or sedation:
 - If not yet done, assess and manage possible causes of abnormal thoughts or behaviour [↗ 136](#).
 - If child < 12 years old, avoid restraints: discuss urgently with family physician, paediatrician or psychiatrist.
 - If restraints used, document type, duration and site and check restraint sites every 15 minutes. Counsel the family/caregiver regarding reasons for restraints.
 - If sedation needed, discuss with paediatrician or psychiatrist.
 - Refer urgently to district psychiatrist at District hospital. Document history and time and dose of each medication given.

Emotional distress

Give urgent attention to the child with emotional distress and any of:

- Suicidal thoughts/attempt or at risk of harming self or others ➔ 135.

Approach to the child with emotional distress not needing urgent attention

First check for physical causes:

Ask about and manage symptoms as on symptom page:

- If abdominal pain ➔ 60
- If weak or tired ➔ 45
- If headache ➔ 44
- If bedwetting and > 5 years old ➔ 69

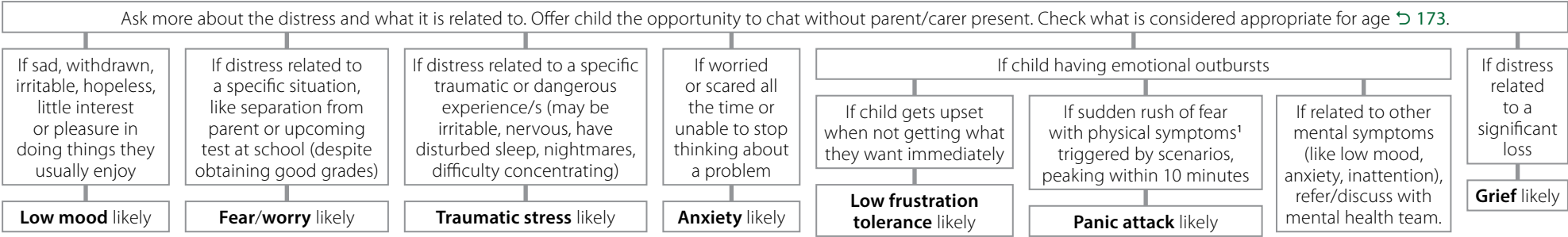
Assess and manage long-term health conditions:

- If asthma ➔ 119.
- If HIV ➔ 97.
- If eczema ➔ 118.
- If epilepsy ➔ 130.
- If TB ➔ 112.
- If allergies ➔ 117.

Review medication:

Prednisone, efavirenz, metoclopramide, contraceptives can cause mood changes. Discuss with doctor.

Ask child and carer more about the emotional distress:



Address contributing factors where possible:

- Ask about routine and quality of **sleep**: does child have regular times for going to bed and waking; is child waking at night and unable to go back to sleep. If sleep problem ➔ 88.
- If distress due to specific experiences at school either academically or socially, address **school problems** ➔ 140.
- If distress related to a family member, consider **abuse** – verbal (persistent criticising, bullying), neglect, physical. Ask child if anyone hurts/upsets him/her. If yes ➔ 144.
- If distress related to **parents who fight**, explain how this may negatively impact child. If needed, discuss/refer to social worker.
- If concerns about use of **alcohol or drugs**, link to psychosocial services (counsellor/social worker/support group/helpline ➔ 170).
- Screen for **depression in carer**: in the past month, has carer felt down/depressed/hopeless or little interest/pleasure in doing things? If yes to either ➔ **PACK Adult**.

Advise carer and child:

- Encourage parent to be involved: acknowledge feelings, encourage listening, talking, understanding and respect. Offer parenting support ➔ 145 and link with support ➔ 170.
- Encourage routines: ensure healthy sleep and nutrition; encourage creative, fun activities that child enjoys; encourage safe outdoor play; spending time with family/friends.

Follow up in 1 month or sooner if worsening. If distress persistent and interfering with activities of daily living (school, home, social life), refer/discuss with mental health team.

¹Trembling, shaking, breathlessness, rapid/pounding heart beat, chest pain/discomfort, nausea, numbness/tingling, sweating, feeling of choking, dizziness/faintness, chills/hot flushes. Fear may be linked with fear of dying, losing control or going crazy and feelings of unreality.

School problems/bullying

Assess the child with school problems

Assess	Note
Symptoms	Ask about other symptoms and manage as on symptom page. If hearing problems ↗ 49. If communicating/talking problems ↗ 89. If vision problems ↗ 48.
Long term health conditions	If HIV ↗ 97, if epilepsy ↗ 130, if asthma ↗ 119, if allergic rhinitis ↗ 50 or eczema ↗ 118.
School work	- Ask about repeated grades (failed grade or years that child was pushed through) and if coping with school work. Ask which subjects child struggles with. Ask if parents help with school work. If not coping with school work and no vision/hearing problem, refer to school-based support team. Write a brief letter to the team, indicating that medical/physical causes for poor school performance have been excluded and request further assessment for possible learning challenges (like dyslexia¹ or intellectual disability). Review in 1 month to check if assessment has occurred. If not, refer to district-based support team. - If learner referred by school-based support for medical assessment, complete a Form DBE 126: Health and disability assessment form².
School refusal	- Establish how many days child was absent from school in last few terms. - Try to establish reason for not attending school, explore stressors by asking child/carer if: - There is something at school that causes feelings of stress, anxiety or low mood. - There are unpleasant social situations that make child not want to go to school (like bullying at school, unsafe school premises). Ask if there are drugs at school. Reassure child that the conversation is confidential in case they are afraid of gang involvement. - There are evaluations or tests that make child not want to go to school. - There are things that child does instead of attending school that makes them feel better about themselves or results in increased attention from family member/s. - If issue/s identified needing urgent assessment and intervention, refer/discuss with mental health team.
Behaviour	If behaviour problem ↗ 137.
Stressors	- Ask about violence at school, bullying, carer alcohol/drug use or if any other family crisis. See below. - If not enough food at home or no lunch at school, assess growth ↗ 15 and refer to Nutritional Support Programme if needed. - If appropriate, ask carer if aware of any abuse of child, carer or siblings. Ask child if anyone hurts/upsets him/her. If yes ↗ 144.
Sleep	If difficulty sleeping ↗ 88.
Mental health	- If angry, withdrawn or change in mood/behaviour/feelings and not coping ↗ 139. - If previous traumatic event/accident and disturbed sleep, nightmares, irritability or difficulty concentrating for ≥ 1 month, post-traumatic stress disorder likely, refer.
Substance use	If child using alcohol or drugs, refer to social worker and link carer/child to support group ↗ 170.
Parenting	- Ask who is responsible for the child most of the time. If harm or neglect suspected ↗ 144. - If carer struggling with parenting ↗ 145.

Advise the child with school problems

- Establish regular daily routine for sleeping, eating and free time. Limit all screen time to < 2 hours/day (TV/gaming/phones/ipads/tablets).
- Set aside specific time for homework daily. Encourage parents to try and support homework.
 - Encourage good nutrition: ensure breakfast before school and lunch at school. If not enough food at home or no lunch at school, refer to Nutritional Support Programme. Avoid fast foods, sugar and caffeine.
- Explore ways to address any stressors identified above. Encourage communication with teacher/school.
- Encourage safe after-school care and encourage activities with peers like sport and music to help reduce risk of gang/drug involvement.
- Address violence and drugs at school and unsafe school premises: involve social worker and school principal.
- If bullying: advise parents to talk to the school. If cyber-bullying, advise carer to monitor activity on social media and take action to stop it.

If school problem persists despite above measures, refer/discuss with mental health team

¹Dyslexia is a specific learning disability characterised by difficulties in word reading and/or spelling. ²Access Form DBE 126 using the following weblink (go to page 71, Annexure D): <https://www.education.gov.za/LinkClick.aspx?fileticket=2b87EaySbcw%3D&tabid=617&portalid=0&mid=2371>

Attention deficit/hyperactivity disorder (ADHD): diagnosis

Assess for ADHD in the child ≥ 6 years old who has symptoms of inattention, hyperactivity, or impulsivity. If child < 6 years old, refer/discuss with mental health team or paediatrician.

Does child have 6 or more of the following inattention symptoms (check carefully in girls - inattention symptoms more common than hyperactive symptoms):

- Difficulty staying focused on tasks that aren't highly stimulating or need sustained mental effort
- Often does not seem to listen when spoken to directly
- Often appears to be daydreaming
- Does not complete tasks
- Is easily distracted
- Lacks attention to detail
- Makes careless mistakes in school
- Loses things often
- Is forgetful in daily activities
- Has difficulty remembering to complete upcoming daily tasks/activities
- Difficulty planning/managing/organising schoolwork, tasks and activities

No

Does child have 6 or more of the following hyperactivity/impulsivity symptoms:

- Excessive activity
- Often runs about
- Talks too much
- Blurts out answers in school
- Leaves seat when expected to sit still
- Difficulty engaging in activities quietly
- Difficulty sitting still without fidgeting (younger children)
- Difficulty waiting turn in conversation, games, or activities
- Feelings of physical restlessness, a sense of discomfort with being quiet or sitting still (adolescents)
- Interrupts or intrudes on others' conversations or games
- Behaves, acts or makes decisions without thinking (risky/reckless)

Yes

No

Yes. Have these symptoms been present for ≥ 6 months?

No

Yes. Are symptoms severe enough to have a negative effect on school, home or social functioning?

No

Yes. Check for underlying causes and if needed, assess and manage further before diagnosing as likely ADHD.

Check vision and hearing

If ear problems ↪ 49

Check for pallor

If pallor, do Hb: if Hb < 11 ↪ 46

Check for thyroid disease

If clinically suspected¹, check TSH.

Check for diabetes

If increased thirst, frequent urination, check glucose ↪ 32.

Check for sleep problems

If sleeping < 9 hours/night ↪ 88.

Screen for substance abuse

If concerns about use of alcohol/drugs, consider drug levels.

Check for medication side effects

If on treatment (like bronchodilators or thyroid replacement), discuss.

- Follow up 2 monthly to assess if symptoms improving.
- Advise parent/carer to:
 - Keep a structured daily routine.
 - Keep things simple: make only one request at a time.
 - Praise any improvements in behaviour. Avoid punishment.
 - Avoid overstimulation.
 - Encourage outdoor play.
 - Limit screen time (TV/phone/ipad/tablet) to < 2 hours per day.
 - Avoid fast foods, sugar and caffeine, fizzy drinks and foods with additives.
- If problems persist, discuss with/mental health team/paediatrician.

Is child known/suspected of having any other mental health illnesses (including tics)? Screen for anxiety and post traumatic stress disorder (PTSD). Ask:

- If previous or ongoing life experience/s result in disturbed sleep, nightmares, fear, irritability and difficulty concentrating?
- If child is feeling worried or scared all the time or is unable to stop thinking about a problem?

Yes

Refer/discuss with mental health team.

No

Attention deficit/hyperactivity disorder likely

- Refer to family physician or trained doctor to confirm diagnosis.
- Arrange for following documents to be completed and sent with child for assessment (if needed, help parent/carer):
 - Parent/carer to complete ADHD parent questionnaire and SNAP questionnaire² and for copies of school reports.
 - Request that child's school teacher also completes a SNAP questionnaire².

¹Hyperthyroidism - thyroid enlargement, increased pulse rate, weight loss/failure to gain weight, increased sweating, heat intolerance, bulging eyes, lid lag, tremor, poor sleep, exaggerated changes in mood. Hypothyroidism - thyroid enlargement, dry skin, brittle hair, constipation, puffy face, intolerant to cold. ²The SNAP scale is a questionnaire which helps to objectively track the frequency of symptoms of ADHD and can be completed by parents and teachers. Access SNAP questionnaire http://www.shared-care.ca/files/Scoring_for_SNAP_IV_Guide_26-item.pdf. Or calculate electronically: https://qxmd.com/calculate/calculator_147/snap-iv-26-teacher-parent-rating-scale. If capacity for more thorough assessment exists, use SNAP-IV 90-item assessment.

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Attention deficit/hyperactivity disorder (ADHD): routine care

A family physician or trained doctor or mental health nurse must confirm the diagnosis using careful history, examination and observation reports and questionnaires.

Assess the child known with ADHD

Assess	When to assess	Note
Symptoms	Every visit	Ask about ADHD symptoms. Ask how child and family are coping at home.
SNAP ¹ scale	Every visit	Ask parent/carer <i>and</i> child's teacher to each complete a SNAP ¹ form. Record scores and compare to assess trend of ADHD symptoms: - for inattention symptoms (questions 1-9): aim for a score less than 13/27. - for hyperactivity/impulsivity symptoms (questions 10-18): aim for a score less than 13/27.
School progress	Every visit	• Ask about school attendance. Ask to see school report 6 monthly to track school performance. If school problems/bullying, assess further → 140. • Check if child has to access school-based support team, remedial teacher, facilitator, psychologist, occupational therapist and link with resources, if available. If child has not been linked to resources, contact Head of specialised learner and educator support².
Adherence and side effects	If on methylphenidate	Ask about adherence to methylphenidate (correct dose and time given) and side effects → 143.
General health/other conditions	First visit	• Integrate routine care into every visit → 12. Ensure any other long-term health conditions are well controlled. • If epilepsy on anticonvulsant, blood clotting disorders or known with another mental health illness, discuss if on methylphenidate.
Mental health	Every visit	• If life experience/s result in disturbed sleep, nightmares, fear, irritability and difficulty concentrating, consider post traumatic stress disorder. • If excessive fear, worry or child is unable to stop thinking about a problem, consider anxiety. • If irritable, sad, unable to enjoy anything, crying a lot or feeling lonely for most of day for 2 weeks, consider depression. • If concern of alcohol or drug use, link to psychosocial services (counsellor/social worker/support group, helpline) → 170.
Parent/carer	Every visit	• At diagnosis, ask about family history of ADHD: if similar symptoms in sibling/parent, consider ADHD in parent, discuss/refer to mental health team. • Screen for depression: in past month, has parent/carer: felt down/depressed/hopeless or little interest/pleasure in doing things? If yes to either → PACK Adult. • Screen for substance abuse: does carer drink ≥ 6 drinks/session every month, drink every day, use illegal drugs or misuse prescription or over-the-counter medication → PACK Adult.
Weight	Every visit	• Methylphenidate may lead to changes in appetite: measure weight at every visit and track weight gain and weight loss. • If losing or not gaining weight, encourage good breakfast before taking treatment and a good supper, with healthy 'power snacks' (nuts, raisins, fruit) at school, ask for teachers assistance. Suggest a snack before bed. If needed, skip weekend doses of methylphenidate. If problem persists, discuss with paediatrician.
Height	6 monthly	Calculate BMI ³ , plot and interpret growth → 15.
Blood pressure	Every visit	Measure blood pressure using correct size cuff and interpret result → 175. If increased BP, refer/discuss with paediatrician.
Heart rate	Every visit	Check heart rate and interpret result → 175. If heart rate above normal limit, refer/discuss with paediatrician.
Risk of heart disease	At diagnosis	Ask about heart problems in child or family: ask about dizziness/fainting/blackouts, especially related to exercise, palpitations, family history of sudden death in a young family member. Doctor to examine cardiac system. If any concerns, do ECG and refer to paediatrician and/or cardiologist.

Continue to advise and treat the child known with ADHD →143.

¹The SNAP scale is a questionnaire which helps to objectively track the frequency of symptoms of ADHD and can be completed by parents and teachers. Access SNAP-IV form: http://www.shared-care.ca/files/Scoring_for_SNAP_IV_Guide_26-item.pdf. Afrikaans and Xhosa also available. Or calculate electronically: https://qxmd.com/calculate/calculator_147/snap-iv-26-teacher-parent-rating-scale. If capacity for more thorough assessment exists, use SNAP-IV 90-item assessment. ²<https://wcedonline.westerncape.gov.za/contact/districts>. ³Body Mass Index: weight(kg) ÷ height (m) ÷ height (m).

Advise the child with ADHD and his/her parent/carer

- Try to keep to a structured routine with child. Try to do things in the same order every day. Set aside specific time for homework daily. Warn child about any changes of plan ahead of time.
- Keep things simple: make only one request at a time. Follow up on the instruction, ensure child complies. Praise/reward child when s/he obeys.
- Give praise when there is any improvement in child’s behaviour. Avoid punishment – the child is not to blame for being overactive.
- Avoid overstimulation: play with one friend at a time, take part in one activity at a time. Avoid background television and radio. Avoid crowded places where possible.
- Allow plenty of outdoor play in garden/park (or street , if safe) to ‘blow off steam’. Limit screen time (TV/phone/ipad/tablet) to < 2 hours per day.
- Avoid fizzy drinks and foods with additives (e.g. heavily coloured sweets, fish fingers) if they seem to make things worse. Avoid fast foods, sugar and caffeine.
- Advise that appetite may be poor, so encourage bigger breakfast before taking treatment and a good dinner, as well as healthy ‘power snacks’ (nuts, raisins, fruit) at school and before bed.
- Advise that thirst may also be affected: child needs to be reminded to drink water during the day. Ask teacher to assist with this. Dehydration may cause headaches, which may be confused as a medication side effect.

Treat the child known with ADHD

- A specialist or doctor authorised to prescribe, needs to start and titrate dose of methylphenidate (Ritalin®) and renew prescription 6 monthly.
 - Start methylphenidate only if: child ≥ 6 year old, trial of classroom/home interventions have been unsuccessful, medical problems have been excluded and ADHD symptoms are severe.
 - Avoid methylphenidate if known hypersensitivity, hyperthyroidism, glaucoma, pregnant/breastfeeding, heart disease, hypertension. Refer to tertiary hospital if any high risk factors¹.
- Once stable, clinic doctor to re-prescribe each month.

Medication	Starting dose	Titrate	Usual maintenance dose	Maximum dose
Methylphenidate SA (short acting)	• Give 5-10mg half an hour before school starts. • If needed, give another 5-10mg 3-4 hours after initial dose. • If needed, give further 5-10mg 3-4 hours after second dose, no later than 3pm.	If needed, increase dose by 0.1mg/kg/ dose (or 5-10mg/day), 2-4 weekly.	0.5-1mg/kg/day every weekday (no weekend/school vacation holidays)	1mg/kg/day or 60mg/day.

Look for and manage methylphenidate side effects

Chest pain, shortness of breath, palpitations	Stop methylphenidate, refer/discuss with paediatrician.
Raised BP or heart rate	
Convulsions	
Psychosis (disordered thoughts, loses touch with reality, hearing or seeing things that are not there)	Stop methylphenidate and refer to psychiatrist same day.
Nausea, vomiting, abdominal pain, drowsiness, headache	• Reassure that these side effects are temporary and only last a short time. • If symptoms persist > 5 days, stop methylphenidate and refer/discuss with paediatrician/psychiatrist.
Poor appetite or weight loss	If losing or not gaining weight, encourage good breakfast and supper, with healthy ‘power snacks’ (nuts, raisins, fruit) at school, ask for teachers assistance. If needed, skip weekend doses of methylphenidate. If problem persists, discuss with paediatrician.
Insomnia, nightmares	Advise to give last dose before 3pm. If severe insomnia persists, refer/discuss with paediatrician.
Irritability/mood swings (unhappy, crying a lot)	Reassure that these are usually temporary and resolve. If persists, stop methylphenidate and refer/discuss with paediatrician/psychiatrist.
Tics, nervous movements	Decrease dose. Reassure that these are usually temporary and resolve. If persists, refer/discuss with psychiatrist.

Review the child with ADHD

- Specialist/doctor to review 2-4 weekly until stable. Give next appointment date, letter needed for school. Once stable, clinic doctor to review 3-6 monthly. Parent/carer to collect medication monthly.
- If symptoms and/or functioning worsen or no better after 6 months, refer/discuss with paediatrician/psychiatrist.
- If on treatment more than 2-3 years or sustained reduction in ADHD symptoms, consider trial off treatment: refer/discuss with paediatrician/psychiatrist.
- If mental health problem (like **depression, anxiety, post traumatic stress disorder**) develops, refer/discuss with paediatrician/psychiatrist.

¹Cardiovascular disease, congenital heart disease, vasculitis, stroke, history of syncope, family history of heart disease/long QT syndrome, tics, family history of Tourette’s syndrome; psychosis, bleeding disorders, anorexia, bipolar disorder.

Suspected child abuse/neglect

Child abuse/neglect likely if: carer or child discloses history of abuse, inconsistent history and examination, delay in presentation of injury, abuse clear on examination or interaction between carer and child seems odd.

Give urgent attention to the child where abuse/neglect is likely with any of:

- Definite history of rape/sexual assault
- Injuries needing urgent attention ➔ 33
- At risk of being harmed and in need of shelter

Management:

- If rape/sexual assault: arrange urgent referral to closest designated facility for management of rape and sexual assault, like a Thuthuzela Care Centre.
 - Prevent HIV and hepatitis B: give PEP ➔ 93 as soon as possible, if within 72 hours of rape.
 - Do not bath or wash child. Make sure child is comfortable. Be kind. No need to examine or ask further questions.
- If urgent referral not available also consider giving emergency contraception and STI prevention, discuss with specialist or Thuthuzela Care Centre.

Approach to the child with suspected abuse/neglect not needing urgent attention:

Look for warning signs that make abuse more likely and assess for other types of abuse simultaneously (do this in an area which is quiet):

• History of physical assault. • If eardrum perforated, refer to ENT. • Old and new scars, grasp marks on arms/chest/face, bruises, bruises of different ages, burns/cigarette burns, unusual or patterned wounds specifically on skin, ears, eyes or in/around mouth.	If any of the following with no other obvious cause: • Vaginal or urethral discharge/bleeding or ano-genital warts/ulcers ➔ 68. • Persistent urinary frequency/burning urine ➔ 69. • Knowledge/interest in sexual acts inappropriate for age, or seductive behaviour. • Sudden emotional/behavioural changes. Child is not him/herself. • If sexually active and any of: sex not consensual, child < 12 years old or child ≥ 12 years old with a partner not in peer group¹.	• Poor growth with no obvious cause. • Clothes ill-fitting, dirty or inappropriate clothes. • Unbathed, matted hair, body odour. • Untreated illnesses or physical injuries. • Frequently left unsupervised or unsafe. • Often late or missing from school.	• Child excessively withdrawn, fearful or anxious. • Child frightened of being bullied or exploited.
Physical abuse likely Manage injuries ➔ 33.	Sexual abuse a possibility Doctor to obtain consent and examine with chaperone: do not perform internal examination. If tears in ano-genital area or unsure, discuss/refer to closest designated facility, like a Thuthuzela Care Centre.	Neglect likely	Emotional abuse likely

Assess general health of the child ➔ 12.

Manage the abused/neglected child

- Complete form 22² and report to social services agency ➔ 170 or hand over to social worker to send to local Department of Social Development.
- If neglect and inadequate food, refer to nutritional support programme/NGO (like Philani) ➔ 170.
- Assess home environment - ask questions like: how often are there parties at home that get out of hand; or does child ever see parents getting physical with one another.
 - Ask about stressors in household like: loss of a breadwinner from death/imprisonment/separation/divorce; or a substance user who may be stealing food/money or showing aggression; or a carer misusing alcohol/substances; or a carer/siblings being mistreated. If any concerns, refer to social worker.
- If bullying at school, contact teacher to work with carer to stop the abuse. If bullying at home, refer carer/s and child to psychologist for family therapy.
- If **physical abuse or sexual abuse**: clearly record child and carer's story in their own words - include identity of perpetrator and child's name and date and sketch all injuries and scars.
 - Inform carer/s of all relevant investigations being done and referrals being made.
 - Notify police: if police not present yet, phone SAPS FCS (Family Violence, Child Protection and Sexual Offences) detective ➔ 170 to begin investigation. Fill in J88 if requested.
 - Involve social worker to arrange place of safety for child. If social worker unavailable, contact FCS detective ➔ 170. If unable to respond same day, refer to hospital until suitable placement arranged.
 - Help carer identify sources of support for child. Refer to available trauma counsellor, mental health nurse, psychologist or helpline ➔ 170. Also refer to community health worker to do home visit.

¹Partner should also be 12-16 years old with < 2 year age gap. ²Form 22 prompts further detailed investigation into a case of suspected child abuse/neglect. Any adult working with a child in a professional capacity may complete the form.

Parenting support

Approach the family with parenting difficulties to identify a cause, give general parenting advice and help access support.

Assess the carer and child with parenting difficulty

Assess	When to assess	Note
General health of carer	Every visit	• If carer unwell, assess and manage ↪ PACK Adult. • If delivery in past 6 weeks, give postnatal care ↪ PACK Adult.
Mental health of carer	Every visit	Screen for depression/anxiety in carer: in the past month, has carer felt down/depressed/hopeless or little interest/pleasure in doing things? If yes to either ↪ PACK Adult .
Substance abuse	Every visit	Does carer drink ≥ 6 drinks/session every month, drink every day, use illegal drugs or misuse prescription or over-the-counter medication? If yes, ↪ PACK Adult .
Special needs	If needed	If child known with special needs, give routine care. If cerebral palsy ↪ 146 . If Down syndrome ↪ 149 .
Psychosocial risk	Every visit	• If family or relationship problems, violence at home, difficult life event in last year, family member serving time in prison, lack of partner/family support, carer < 20 years old, financial difficulty, refugee status or bereavement, help access support below. • Ask carer who looks after child most of time. If concerns about abuse or neglect ↪ 144.
Parent/carers and child relationship	Every visit	• If baby < 1 year old: if carer not interacting with baby or not responding appropriately to baby (avoids eye contact with baby or does not comfort/feed baby when crying), screen for depression in the carer if not yet done and if needed, refer to mental health nurse/psychologist/doctor. • Ask if persistent conflict, tension or dysfunction between parent/carers and child: ask if poor communication (misunderstandings, frustration, resentment), control issues (carer overly authoritarian, demanding, rebelliousness in child), trust problems, emotional or physical absence (child feels unsupported or unloved).
Behaviour/sleep of child	If needed	If problem that persists despite parenting strategies below, assess thoroughly: behaviour problem ↪ 136 , sleep problems ↪ 88 .
School	If needed	If school problem persists despite parenting strategies below ↪ 140 .

Advise the carer with parenting difficulties

- If child has two parents, encourage both to be actively involved in parenting. If multiple carers (extended family, nanny), encourage discussion to ensure rules and parenting styles are consistent.
- Encourage carer to discuss concerns with crèche, preschool and school staff.
- Help the carer to consider strategies to cope with parenting and address issues impacting on parent/carers and child relationship:

Encourage a healthy bond

Encourage carer to be sensitive, reassuring, and consistent, especially during the 1st year. Avoid leaving baby < 2 years old to cry alone.



Establish routine

- Encourage routines for sleeping, meals, playing, homework and chores.
- Encourage supervision after school.



Provide consistent discipline

- Actively listen to child, respect his/her wishes and feelings and encourage him/her to express his/her opinions.
- Be firm, kind, reasonable and consistent.
- Set clear boundaries. Explain reasons for rules.
- Encourage praise/reward for good behaviour.
- Talk about issues when everyone is calm. Avoid shouting, shaming, smacking or other forms of emotional or physical punishment.



Encourage carer to look after him/herself



Spend time with supportive friends or family
Find a creative or fun activity to do.



Get active



Get enough sleep

Access support
Link client with helpline (like The Parent Centre) ↪ [170](#).



Offer to review the carer in 1 month.

Cerebral palsy: routine care

Assess the child with cerebral palsy: record child's condition and care plan in RtHB.

Assess	When to assess	Note
Symptoms	Every visit	• Manage symptoms on symptom pages: If coughing ↗ 55, if constipation ↗ 66, if snoring on most nights ↗ 50, if change in sleeping pattern ↗ 88. • If significant TB exposure¹ ↗ 106. If TB symptoms², test for TB ↗ 108.
Seizures/fits	Every visit	If known epilepsy ↗ 130 or fitting/uncontrolled fits ↗ 28.
Vision, talking and hearing problems	Every visit	• If squint, cataracts or other problems with vision ↗ 48. • If talking or hearing problem, ensure assessment by speech therapist or audiologist. If no previous assessment ↗ 89.
Feeding	Every visit	If difficulty swallowing, coughing/choking with feeds or unable to chew, refer to speech therapist. If specialised feed required, also refer to dietitian.
Social risk	Every visit	• If in need of full-time care, apply for child dependency grant. Advise to take child's birth certificate and carer's ID to SASSA (South African Social Security Agency) to apply. • If concerns about abuse or neglect, refer to social worker.
Rehabilitation, home needs and equipment	Every visit	• Refer to physiotherapist and occupational therapist to assist with improving and maintaining muscle strength, balance, motor skills, and to prevent contractures. • If adaptation of home environment needed in terms of feeding, toileting and mobility or problem with equipment, refer to occupational therapist.
Behaviour problems	Every visit	If problem or a noticeable change in behaviour ↗ 136. Also exclude abdominal pain ↗ 60, joint pain ↗ 74 or teeth pain ↗ 54 as a cause.
Schooling	Every visit	Check if in school or appropriate alternative placement (like recreational therapy) ↗ 170 and if coping. If not, refer to school-based support team or occupational therapy.
Carer	Every visit	Assess carer's mental health: in the past month, has carer felt down/depressed/hopeless or little interest/pleasure in doing things? If yes to either ↗ PACK Adult. If carer is pregnant, advise to book early.
Palliative care	At diagnosis and if deteriorating	If degree of cerebral palsy is life-limiting, also give palliative care ↗ 150.
Routine care	Every visit	Integrate routine care into every visit ↗ 12.
Growth	Every visit	Weigh, plot and look at trend. If child using walking aids to be mobile, plot growth on cerebral palsy chart ↗ 147, if child confined to wheelchair at all times ↗ 148. If not growing well ↗ 126. If overweight ↗ 129.
Back/limbs	Every visit	• Check back: look for curved spine (kyphosis and scoliosis). If found for first time, refer to physiotherapist. • Check limbs: if marked stiffness in limb/difficulty moving joints, pain on moving limb/dressing child or if walking pattern has changed, refer to paediatrician. • If child was previously walking and now stopped, refer urgently. • Check hips: if child in wheelchair and pain or dislocation on movement, refer to paediatrician.
Teeth	Every visit	If dental caries ↗ 54. Ensure that carer brushes child's teeth twice daily.
Skin	Every visit	Check skin over pressure areas, if pressure sore found ↗ 83. If surrounding skin red/warm or pus/offensive smell, infected bed sore likely ↗ 83.

Advise the child with cerebral palsy and his/her carer

- Cerebral palsy can range from mild (one hand stiff) to severe (quadriplegic). The child with cerebral palsy can have normal intelligence. Ensure a formal assessment is done by a specialist.
- Ensure life-long physiotherapy or occupational therapy is in place, especially if problems with spine/limbs or walking.
- Cerebral palsy can be difficult to deal with so allow carer time to express feelings and to ask questions. Encourage family to get involved in social support networks ↗ 170.

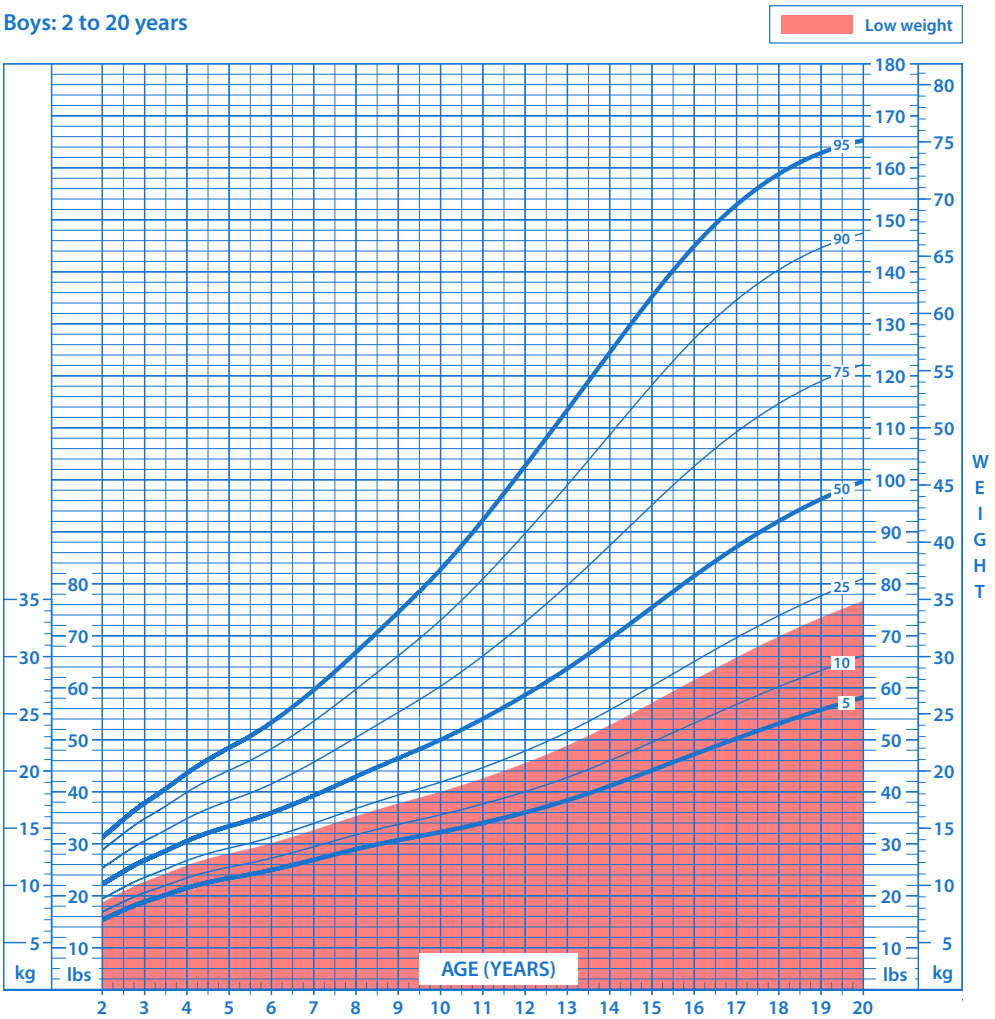
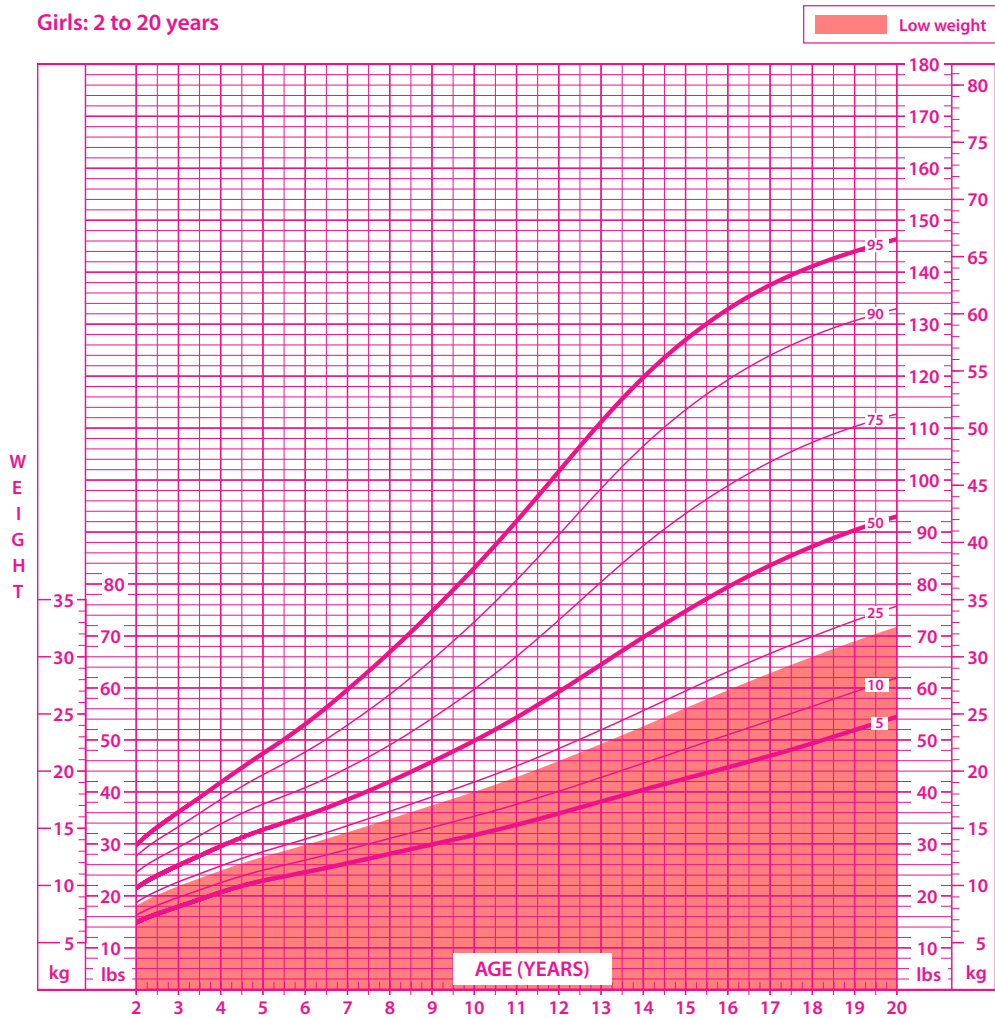
Treat the child with cerebral palsy

Continue treatment prescribed by specialist.

Ensure child attends 6-12 monthly paediatrician check-ups.

¹Any child who in the last 12 months shared an enclosed space (school/creche/home/similar) for ≥ 1 night or frequent/extended daytime periods with an adult/adolescent with pulmonary TB ("index client"), during the 3 months before index client started TB treatment. ²Cough, unexplained fever, drenching night sweats, poor weight gain, weight loss, failure to thrive, lethargy, decreased playfulness, visible neck mass, blood-stained sputum, headache, persistent vomiting (without diarrhoea).

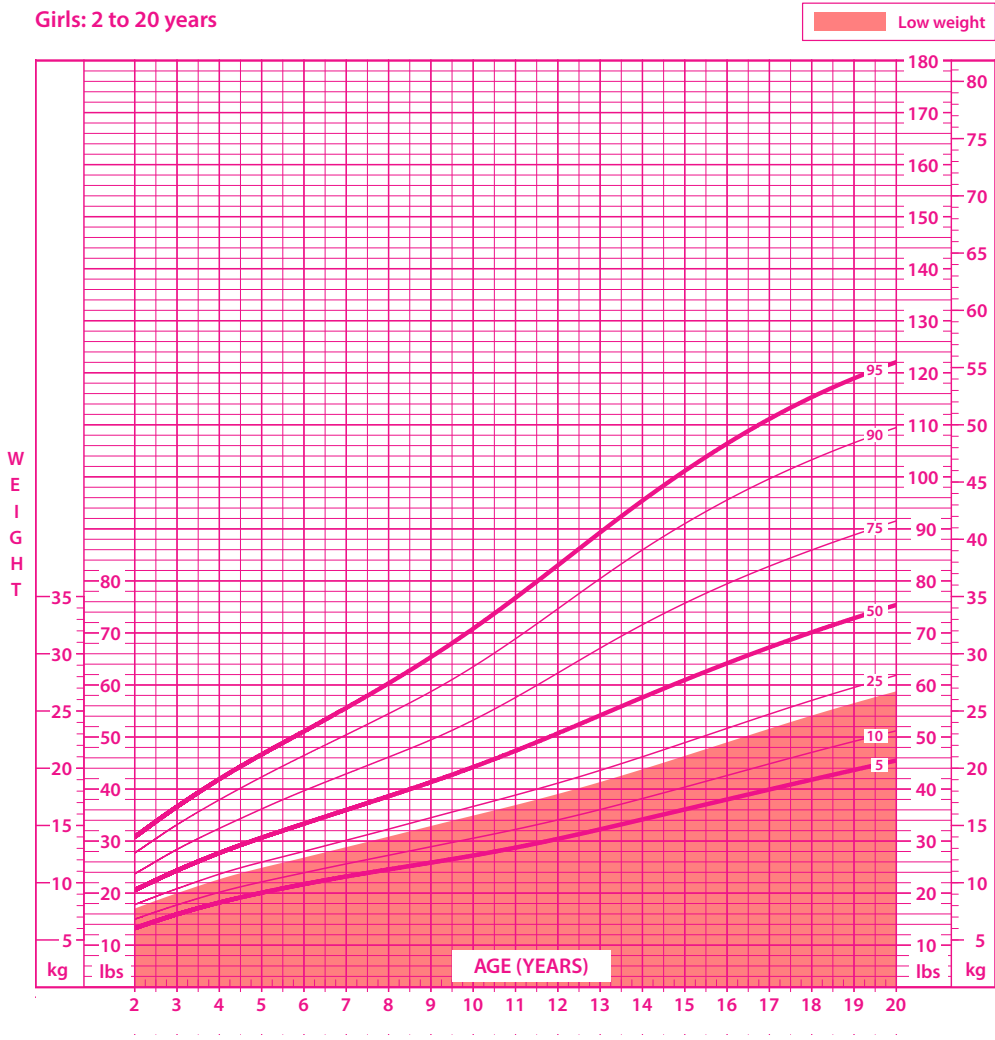
Weight-for-age chart: Cerebral palsy (GMFCS IV)



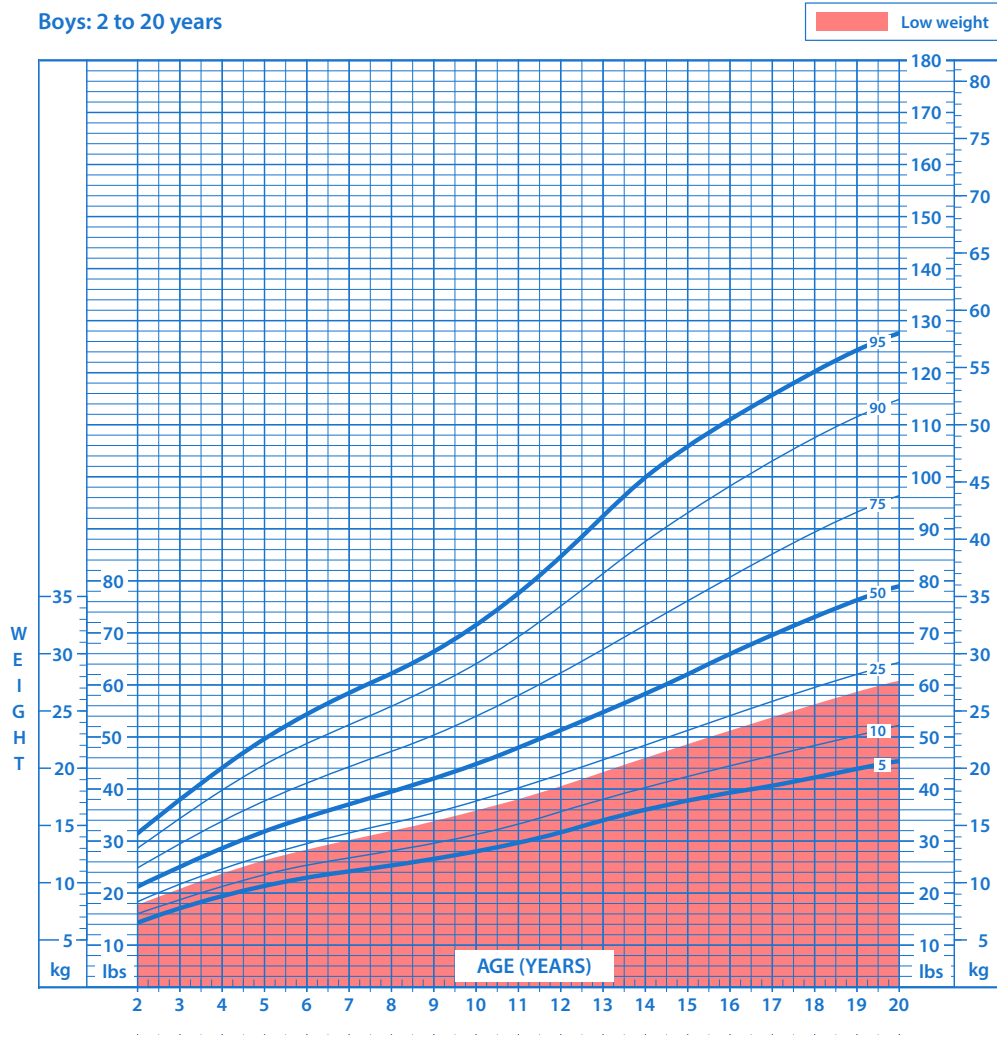
SOURCE: Life Expectancy Project (2011). Based on data from the California Department of Developmental Services and California Bureau of Vital Statistics. <http://www.LifeExpectancy.org/Articles/NewGrowthCharts.shtml>

Weight-for-age chart: Cerebral palsy (GMFCS V)

Girls: 2 to 20 years



Boys: 2 to 20 years



SOURCE: Life Expectancy Project (2011). Based on data from the California Department of Developmental Services and California Bureau of Vital Statistics. <http://www.LifeExpectancy.org/Articles/NewGrowthCharts.shtml>

Down syndrome: routine care

Assess the child with Down syndrome: record child's condition and care plan in RthB.

Assess	When to assess	Note
Symptoms	Every visit	- Manage symptoms on symptom pages. If urinating large volumes, very thirsty, losing weight and very tired: check glucose ↗ 32. If constipation ↗ 66. - If child snores continuously, obstructive sleep apnoea likely ↗ 50. - Check ear and eardrum. If ear pain/discharge or red/bulging eardrum ↗ 49.
Feeding	Every visit	If struggles with feeding, refer to speech therapist for feeding assistance.
Heart disease	At birth	If neonate, refer to cardiology OPD for assessment. If known heart problem, give routine care ↗ 133.
Carer	Every visit	- Assess carer's mental health: in the past month, has carer felt down/depressed/hopeless or little interest/pleasure in doing things? If yes to either ↗ PACK Adult. - Check family planning in place ↗ PACK Adult.
Vision	Every visit	- Refer to ophthalmologist for formal eye and vision screen between 1-3 years old. - School nurse to do eye test at school entry and refer if problem found. - If squint, cloudy cornea or absent red reflex ↗ 48.
Talking, hearing problems	Screen before 3 months old, at 6 months, 12 months, then yearly	- Refer to audiologist for hearing test before 3 months old and between 1-3 years old. - If talking problem, refer to speech therapist.
Behaviour	Every visit	If problematic behaviour or concerned about child's behaviour ↗ 136.
Schooling	Every visit	- At 5 years old, child must be assessed by a paediatrician to decide school placement. - Check child is attending and coping at school. Refer to occupational therapist if problems. - If learning problem, refer to remedial teacher/occupational therapist/school based support team.
Family planning	If started period	If girl has started period, refer to doctor to discuss contraception method.
Social risk	Every visit	- If in need of full-time care, apply for child dependency grant. Advise to take child's birth certificate and carer's ID to SASSA¹ to apply. - If concerns about abuse or neglect, refer to social worker.
Routine care	Every visit	Integrate routine care into every visit ↗ 12.
Growth	Every visit	- Measure and record weight-for-age, length/height-for-age, weight-for-length/height (or BMI) ↗ 15. - Refer to dietitian for weight and nutritional assessment.
Teeth	Every visit	If dental caries ↗ 54. Ensure child (with carer help) brushes teeth twice daily.
TSH	At 6 months old, then yearly	If TSH abnormal, refer.
Haemoglobin	Yearly	If Hb < 10g/dL in < 5 years old or Hb < 11g/dL in child ≥ 5 years old, anaemia likely ↗ 46.

Advise the carer of child with Down syndrome

- Make carer aware child may always have weaker muscles than other children. Ensure regular occupational therapy is in place until walking and running well.
- Encourage family to join a support group/s ↗ 170.
- Encourage carer to play and interact with child: provide contact/hold child regularly, sing/dance with child, read books, encourage outside activities, kick/throw a ball.

- Advise influenza vaccination during influenza vaccine campaign.
- Doctor to review child 6 monthly until 1 year old, yearly from 1-3 years old and 2 yearly thereafter unless otherwise instructed.

¹South Africa Social Security Agency.

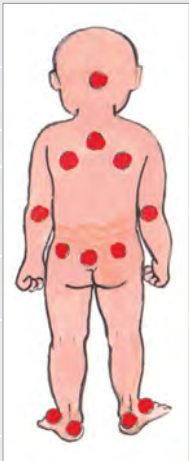
Life-limiting illness: routine palliative care

A child with a life-limiting illness can be given curative and palliative care at the same time. A doctor should confirm if the child needs palliative care:

- Child with life-threatening illness where cure is possible but could fail *and/or*
- Child with disease where cure not possible but can be managed (like HIV on ART) *and/or*
- Child with disease where cure not possible and no option for active management (like inoperable congenital heart disease, Trisomy 13, Trisomy 18) *and/or*
- Child with irreversible yet non-progressive conditions (like cerebral palsy).

Assess the child needing palliative care

Assess	Note
Symptoms	• If constipation, diarrhoea, nausea/vomiting, abdominal cramps, itchiness, anxiety, cough/difficulty breathing or dry mouth, manage ➔ 151. • Manage other symptoms as on symptom pages.
Pain	• Assess if child has cancer pain (constant and progressive) or non-cancer pain (pain > 3 months, nerve pain/tissue damage/any other pain child suffering with). • Assess the severity of the pain (mild, moderate or severe) using the R-FLACC pain scale¹ ➔ 173. This will help decide which pain medication to start/increase ➔ 151. • If new or sudden pain, temperature ≥ 38°C, tender swelling, redness or pus, treat on symptom page. If no better or uncertain of cause, discuss.
Side effects	• Ask about side effects from pain medication ➔ 151. • If on morphine, advise that nausea, confusion and sleepiness usually resolve after a few days. Check that child is using regular laxative.
Sleep	If child has difficulty sleeping ➔ 88.
Mental health	• Ask about how the child and carer are feeling. Do they have sadness or worry? Refer child and family/carers to counsellor to help identify their support network (family, schools, churches, mosques, community support groups). • Ask if child has suicidal thoughts or plans ➔ 135. If low mood, stress, anxiety or anger ➔ 139 or if problematic change in behaviour ➔ 136.
Carer health	• Screen for depression/anxiety in carer: in the past month, has carer felt down/depressed/hopeless or little interest/pleasure in doing things? If yes to either ➔ PACK Adult. • Ask how the carer and family are coping and what support they need now and in the future. Refer to social worker.
Chronic care	• Assess how much child and family understand about the condition and ask what further information the child and carer need. • Check adherence to treatment. Refer to doctor if child struggling with medications. • Assess ongoing need for chronic care in discussion with child, carer and health care team/specialist. Consider which medication could be discontinued. • If known kidney failure with eGFR < 15 or any unmanageable symptoms, refer to palliative care specialist.
Social risk	• If in need of full-time care, child dependency grant, hospice application or community healthcare workers refer to social worker. • If child abuse or neglect suspected ➔ 144.
Mouth	Check oral hygiene. If ulcers or oral candida ➔ 51. If gum or tooth problem ➔ 54. If difficulty swallowing, discuss/refer.
Pressure ulcers	If bedridden or in wheel chair, check common areas for damaged skin (change of colour) and pressure ulcers (see picture). If pressure ulcer ➔ 83.



Advise the child needing palliative care and his/her carer

- Educate about the condition and prognosis. Explaining what is happening relieves fear and anxiety. Involve child as much as possible in self care and decision making.
- Aim to provide care in child's home. If unable, ensure care is close to home. Emphasize importance of school and "normal" activities and encourage carer to play with, comfort and massage child daily.
- Refer child and carer to community health worker, social worker, physiotherapist, counsellor, spiritual counsellor and/or support group. Deal with bereavement issues ➔ 139.
- Prevent mouth disease: teach carer to wash out child's mouth after meals. Rinse mouth with ½ teaspoon of salt in 1 cup of water. Brush teeth and tongue regularly using toothpaste/dilute bicarbonate of soda.
- Prevent pressure ulcers: wash and dry skin daily. Keep linen dry. Move (lift, avoid dragging) child every 2-4 hours, if unable to shift own weight. Look daily for skin colour changes (see picture).
- If bedridden, prevent contractures: gently bend and straighten joints as far as they go (at least twice a day). Avoid causing pain. Massage muscles.
- The child's appetite will get less as s/he gets sicker. Offer small meals frequently, allow the child to choose what s/he wants to eat and encourage fluid intake.
- Discuss the plan for caring for the child. Advise whom to contact if pain or other symptoms get severe. Ensure child has advanced care plan in Road to Health Book. If not, refer to doctor to write one.
- Educate the carer to recognise signs of deterioration and impending death: s/he may be less responsive, become cold, sleep a lot, have irregular breathing, and will lose interest in eating.

¹The Revised-FLACC pain scale: Face, Legs, Activity, Cry, and Consolability scale to assess pain in children.

Treat the child needing palliative care

• If pain, aim to have child pain free at rest, able to sleep and as alert as possible to manage daily tasks. Start pain medication based on type and severity of pain:

Non-cancer pain

- If **mild** (1-3) pain, start at step 1.
- If **moderate** (4-7) or **severe** (8-10) pain, refer.

Cancer pain

- If **mild** (1-3) pain, start at step 1.
- If **moderate** (4-7) or **severe** (8-10) pain start at step 2.

If unsure, start at lower step and increase pain medication if needed.

- If pain controlled, continue same dose. Once controlled for 1 month, consider reducing dose/stepping down depending on condition. If pain worsens, then increase dose/step up again.
- If cancer pain not controlled within 2 days, move to step 2.
- If non-cancer pain uncontrolled on step 1, refer. If cancer pain uncontrolled on step 2, discuss/refer.

Step	Pain medication	Start dose	Maximum dose	Note
Step 1 Start one or both of:	▶ Paracetamol	15mg/kg 6 hourly ↗ 163:37	60mg/kg daily	If starting, give ▶ paracetamol 15mg/kg in clinic and reassess pain after 4 hours. If no better, add ■ ibuprofen .
	■ ibuprofen	5-10mg/kg 8 hourly ↗ 161:28	40mg/kg daily	• If starting, give ■ ibuprofen 5-10mg/kg in clinic and reassess pain after 4 hours. If no better, add ▶ paracetamol . • Give with food. Avoid if asthma, heart failure or kidney disease.
Step 2 (only if cancer pain) Continue paracetamol/ibuprofen ¹ , or both, and add:	■ Morphine hydrochloride (short-acting, solution)	• ≥ 2 months - 1 year old: 0.1mg/kg 4 hourly • > 1 year old: 0.2mg/kg (up to 10mg) 4 hourly	• No maximum dose. Titrate against pain. • If sedated/confused or respiratory rate low ↗ 175, stop. Discuss with doctor/specialist.	• Also give ▶ lactulose 0.5mL/kg daily to prevent constipation ↗ 162:29. Avoid if diarrhoea. • If constipation, nausea/vomiting or itchiness, manage as below. • If on morphine hydrochloride and breakthrough pain (pain that occurs before next scheduled dose): - Give one extra dose morphine, then continue regular dose at scheduled times for the rest of that day. - If pain persists, increase morphine doses the next day. Calculate new dose: add up total amount of extra morphine given in last 24 hours. Divide this amount by 6 and add this to each regular 4 hourly dose ² .

• If side effects from pain medication or other symptoms, manage:

Constipation

- Mobilise if possible.
- Give abdominal massage³ if bed bound.
- Advise high fibre diet (vegetables, fruit, wholemeal cereals, bran and cooked dried prunes), and adequate fluid intake.
- Give ▶**lactulose** 2.5-10mL daily. If still constipated give twice a day.
- If no better, refer.

Diarrhoea

Manage as on symptom page ↗ 63.

Nausea/vomiting

- Give ▶**metoclopramide** 0.1mg/kg 8 hourly as needed ↗ 162:32.
- Allow the child to choose what s/he wants to eat:
 - Advise bland/non-spicy food and avoid very sweet/fatty foods.
 - Encourage frequent small sips of fluids like water, tea, juice or ginger drinks.
 - Offer small meals frequently.
- Advise caregiver to avoid cooking nearby.
- If no better, refer.

Abdominal cramps

Discuss the use of hyoscine butylbromide with a specialist.

Generalised itchiness

- Advise to:
 - Avoid hot baths, itchy fabrics and scratching as these worsen itch.
 - Wash with ◆**aqueous cream** instead of soap.
 - Moisturise with ▶**emulsifying ointment** twice a day.
 - Keep nails short.
- Give ▶**chlorpheniramine** 0.1mg/kg 6-8 hourly ↗ 159:15 until itch controlled, up to 2 weeks.
- If yellow skin/eyes ↗ 61.
- If no better with treatment or itch due to burn wounds, discuss/refer.

Anxiety

- Discuss the use of diazepam with a specialist.
- If low mood, anger, stress or anxiety ↗ 139.

Cough or difficulty breathing

- If thick sputum, give steam inhalations⁴. Refer to physiotherapy if available.
- If excess thin sputum or persistent dry cough, discuss with palliative care specialist.
- Refer/discuss with specialist if:
 - Severe discomfort (needing morphine)
 - Low saturation (needing home oxygen).

Dry mouth

- Place pieces of ice in child's mouth and apply ▶**petroleum jelly** to lips.
- If child able to swallow, change to soft diet until improvement.
- Give ▶**chlorhexidine 0.2% mouthwash** twice daily.
- If severe or preventing good nutritional intake and no response to above, refer.

- Emphasize the importance of taking pain medication regularly (not as needed), and if using morphine to use a laxative daily to prevent constipation.
- Review 2 days after starting or changing pain medication. If pain persists/worsens despite maximum treatment, side effects intolerable, or unsure about dose: discuss/refer.

¹Stop ibuprofen if no response after 3 days. If good response, continue ibuprofen. ²Example: child on morphine 5mg 4 hourly has 3 episodes of breakthrough pain: 5mg x 3 = 15mg (total extra morphine); 15mg ÷ 6 = 2.5mg. Add 2.5mg to each 5mg regular dose: new dose is 7.5mg 4 hourly. ³Abdominal massage: massage in small clockwise circles in periumbilical area and in U-shaped pattern following direction of stool in colon. ⁴Steam inhalations: breathing in steam from heated water, to loosen sputum.

Support the child taking long-term medication

Assess the child taking long-term medication

Assess	Note
A Adherence	- Ask older child open ended questions like “What makes it difficult for you to take your treatment? Do you sometimes find it difficult to remember to take your medication? How many doses have you missed this week?”. Encourage child to open up with statements like “We all miss doses now and then”. - Ask child/carer about factors that may influence adherence: - Is the cost and time of clinic visits a problem (like transport, loss of income for the day, paying another person to take on responsibilities at home). - Are medications causing any side effects or is there difficulty taking medications (like horrible taste, difficulty swallowing, taken on an empty stomach)? If child on ART stopped taking ART because of these, check if ART can be switched ↗ 102. - Is there a problem with understanding: check child/carer knows the diagnosis, understands the condition and what it means to be well controlled. If CLHIV, check where child/carer is in their journey. Support this communication ↗ 96.
B Recent illness (Bugs)	- Ask about and document recent infection/s or illness. Manage other symptoms as on symptoms pages. - If TB symptoms¹ or significant TB exposure², test for TB ↗ 108. Check if HIV test needed ↗ 95.
C Correct doses	- Ask child/carer to show you medications, tell you the dose and how often s/he should take it. - Check child/carer knows how medication works and why it is important to take it as advised. - Check that dose is correct for age and weight. If on ART ↗ 168. If on TB treatment ↗ 166.
D Drug interactions	- Review other medications and check for known interactions, especially ART, TB treatment or TPT, and epilepsy treatment. Ask if child taking traditional/herbal medications. - Consult the South African Medicines Formulary (SAMF), EMGuidance app, use web-based drug interaction checker (see QR code) or MIC helpline (021) 406 6829.
E Drug resistance	If child on TB or HIV treatment, and not improving, consider drug resistance if other causes have been excluded and child is adherent. Discuss with HIV expert/hotline ↗ 170.
Daily routine	Ask about child/carer's daily routine and if it causes difficulty with adherence. Identify opportunities that can be used as reminders to take medication.
Support	Ask if child/carer receives support from family, friends or others in the community. If carer has a long-term condition (like HIV, TB, diabetes or hypertension), ensure that condition is well-controlled and assess whether carer needs extra support ↗ PACK Adult.
Mental health	If child miserable, stressed or angry ↗ 139. If poor school attendance, problematic change in behaviour, bullying, learning problems, difficulty socialising at school ↗ 140.



Check for HIV medication interactions

Advise the child taking long-term medication

- Be supportive and non-judgemental. If newly diagnosed or poor understanding, spend extra time educating and counselling the child/carer. Explain condition and benefits of medication.
- If difficulty with adherence, avoid blaming child/carer. Rather explore reasons for poor adherence and come up with ideas together to improve.
- Discuss ways to help child/carer to remember to take medication, like star charts, games and rewards, diaries, alarms or pill boxes. Use reminders that form part of daily routine.
- Explain that good adherence is taking medication at the correct dose and time every day. If dose is skipped, advise to take dose as soon as s/he remembers. Medication will still work even if it is taken a bit late. This will improve control and reduce risk of long-term complications.
- Encourage child/carer to involve trusted partner or family member in his/her treatment.

Treat the child taking long-term medication

- Ensure patient adherence plan in child's folder and completed for sessions 1 - 4 of fast track initiation counselling (FTIC)³ and enhanced adherence counselling (EAC)⁴ if needed.
- Link with community health worker.
- Try to keep medication regimen simple with as few syrups/capsules/tablets and doses as possible. Use fixed dose combination formulations if available. Involve child in his/her treatment plan and adapt treatment schedule to daily routine as much as possible. Schedule appointments to align with routine visits or parent/carer appointments if possible. If child/carer misses appointment ≥ 7 days after scheduled date, establish contact and initiate tracing with community health worker.

¹TB symptoms: current cough/fever, drenching night sweats, poor weight gain/failure to thrive, weight loss, decreased playfulness, visible neck mass, blood-stained sputum, persistent headache/vomiting (without diarrhoea). ²Any child who in the last 12 months shared an enclosed space (school/creche/home/similar) for ≥ 1 night or frequent/extended daytime periods with an adult/adolescent with pulmonary TB (“index client”), during the 3 months before index client started TB treatment. ³Child qualifies for FTIC if: medication started within 7 days of diagnosis; if HIV, child > 12 years old and fully disclosed; carer of child < 12 years consents to FTIC and able to attend sessions alone if child has HIV; child has TB and needs ART. ⁴Child qualifies for EAC if: child with HIV has VL ≥ 50 on ART; child with TB has positive TB smear at 2 months; child with other long-term health condition has poor adherence to treatment.

Protect yourself from occupational stress

Experiencing pressure and demands at work is normal. However if these demands exceed knowledge and skills and challenge your ability to cope, occupational stress can occur.

Arrange urgent assessment for the health worker with occupational stress and any of:

- Alcohol or drug intoxication at work
- Aggressive or violent behaviour at work
- Inappropriate behaviour at work
- Suicidal thoughts or behaviour

Adopt measures to reduce your risk of occupational stress

Protect yourself

Look after your health:

- Get enough sleep.
- Exercise regularly, eat sensibly, minimise alcohol and avoid smoking.
- Address your general health and get screened for long-term health conditions ➔ [PACK Adult](#).

Look after your long-term health condition if you have one:

- Adhere to treatment and appointments.
- Avoid diagnosing and treating yourself.
- If you can, confide in a trusted colleague/manager.

Manage stress:

- Delegate tasks as appropriate, and develop coping strategies.
- Talk to someone (friend, psychologist, mentor) or access helpline ➔ [170](#).
- Take time to do a relaxing breathing exercise each day.
- Find a fun or creative activity to do.
- Spend time with supportive family or friends.

Have healthy work habits:

- Manage your time sensibly.
- Take scheduled breaks.
- Remind yourself of your purpose as a clinician.
- Be sure you are clear about your role and responsibilities.

Protect your team

Decide on an approved way of behaving at work:

- Communicate effectively with clients and colleagues ➔ [155](#).
- Treat clients and colleagues with respect.
- Support each other. Consider setting up a staff support group.
- Instead of complaining, focus on finding solutions to problems.

Cope with stressful events:

- Develop or access policies or procedures to deal with events like complaints, harassment/bullying, accidents/mistakes, violence, or death of client or staff member.

Look at how to make the job less stressful:

- Examine the team's workload to see if it can be better streamlined.
- Identify what needs to change to make the job easier and frustrations fewer: equipment, drug supply, training, space, décor in work environment.
- Discuss each team member's role. Ensure each one has a say in how s/he does his/her work.
- Support each other to develop skills to better perform your role.

Celebrate:

- Acknowledge the achievements of individuals and the team.
- Organise or participate in staff social events.

Identify occupational stress in yourself and your colleagues

Possible alcohol or drug problem

- In the past year, have you/has your colleague: drunk ≥ 4 drinks¹/session, used illegal drugs or misused prescription or over-the-counter medications?
- Smells of alcohol

Possible depression

- Indifferent, tense, irritable or angry
- In the past month, have you/has your colleague felt down/depressed/hopeless or little interest/pleasure in doing things?

Recent distressing event

- Diagnosis of long-term health condition
- Bereavement
- Needlestick injury
- Traumatic event

Poor attendance at work

- Frequent absenteeism
- Frequent lateness
- Often takes sick leave

Marked decline in work performance

- Reduced concentration
- Fatigue

The health worker with any of the above may have substance misuse, stress, depression/anxiety or burnout and would benefit from referral for assessment and follow-up.

¹One drink is 1 tot of spirits, or 1 small glass (125mL) of wine or 1 can/bottle (330mL) of beer.

Protect yourself from occupational infection

- Give urgent attention to the health care worker who has had a sharps injury or splash to eye, mouth, nose or broken skin with exposure to any of:**
 - Blood
 - Blood-stained fluid/tissue
- Pleural/pericardial/peritoneal/amniotic/synovial/cerebrospinal fluid
 - Vaginal secretions
- Semen
 - Breast milk

Management:

- If broken skin, clean area immediately with soap and water.
- If splash to eye, mouth or nose, immediately rinse mouth/nose or irrigate eye thoroughly with ►sodium chloride 0.9% or sterile water.
- Assess the need for HIV and hepatitis B post-exposure prophylaxis ➤ **PACK Adult.**

Adopt measures to diminish your risk of occupational infection

Protect yourself

Adopt standard precautions with every child:

- Wash hands with soap/water or use alcohol-based cleaner after contact with child or body fluids.
- Dispose of sharps correctly in sharps bins.

Wear protective gear:

- Wear gloves when handling blood, body fluids, secretions or non-intact skin.
- Wear surgical mask if in contact with child with suspected respiratory viral illness (N95 respirator if performing aerosol-generating procedure or child has suspected or confirmed infectious TB).
- Wear surgical mask with a visor or glasses if at risk of splashes.

Get vaccinated:

- Get vaccinated against hepatitis B (if not yet done) and yearly against influenza.
- Ensure pertussis vaccinations are up to date.

Know your HIV status:

- Test for HIV ➤ **PACK Adult.** ART and TB preventive treatment (TPT) can decrease the risk of TB.
- If living with HIV, you are entitled to work in an area of the facility where exposure to TB is limited.

Protect your facility

Clean the facility:

- Clean frequently touched surfaces (door handles, telephones, keyboards) daily with soap and water.
- Disinfect surfaces contaminated with blood/secretions with 70% alcohol or chlorine-based disinfectant.

Ensure adequate ventilation:

- Leave windows and doors open when possible and use fans to increase air exchange.

Organise waiting areas:

- Prevent overcrowding in waiting areas.
- Fast track children with suspected respiratory viral infections, TB or acute gastroenteritis.

Manage sharps safely:

- Ensure sharps bins are easily accessible and regularly replaced.

Manage infection control in the facility:

- Appoint an infection control officer to coordinate and monitor infection control policies.

Manage possible occupational exposure promptly

Reduce TB risk

Identify children with possible TB promptly:

- The child with TB symptoms¹ and/or significant TB exposure² may have TB.
- Separate patients with possible TB from others in the facility.
- Educate about cough hygiene and give face mask/tissues to cover mouth/nose to protect others.

Diagnose TB rapidly:

- Fast track TB workup and start treatment as soon as diagnosed.

Protect yourself from TB:

- Wear N95 respirator (not a surgical mask) if in contact with a child with suspected or confirmed infectious TB.

Screen and test yourself for TB every 6 - 12 months:

- Screen and test for TB according to your facility policy. If TB test negative, discuss TB preventive treatment (TPT) with your occupational health practitioner.

Reduce risk of respiratory infections (including pertussis, influenza and COVID-19)

- Before managing a child with suspected or confirmed respiratory infection, wear appropriate personal protective equipment (PPE).
- Wash hands with soap and water.
- Wear a surgical mask over mouth and nose during procedures.
- Encourage child to cover mouth/nose with a tissue when coughing/sneezing, to dispose of used tissues correctly and to wash hands regularly with soap and water.
- Advise child/carer to avoid close contact with others while sick.
- If exposed to pertussis in the last month, discuss need for post-exposure prophylaxis against pertussis with your occupational health practitioner.

¹TB symptoms: cough, fever, drenching night sweats, poor weight gain/failure to thrive, weight loss, decreased playfulness, visible neck mass, coughing up blood. ²Any child who in the last 12 months shared an enclosed space (school/creche/home/similar) for ≥ 1 night or frequent/extended daytime periods with an adult/adolescent with pulmonary TB ("index client"), during the 3 months before index client started TB treatment.

Communicate effectively

Communicating effectively with a child/carer during a consultation need not take much time or specialised skills. Try to use straightforward language and take into account child/carer's culture and belief system.

Integrate these four communication principles into every consultation:

Listen

Listening effectively helps to build an open and trusting relationship with the child/carer.

Do

- Give all your attention
- Recognise non-verbal behaviour
- Be honest, open and warm
- Avoid distractions e.g. phones

The child/carer might feel:

- 'I can trust this person'
- 'I feel respected and valued'
- 'I feel hopeful'
- 'I feel heard'

Don't

- Talk too much
- Rush the consultation
- Give personal advice
- Interrupt

The child/carer might feel:

- 'I am not being listened to'
- 'I feel disempowered'
- 'I am not valued'
- 'I cannot trust this person'

Discuss

Discussing a problem and its solution can help the overwhelmed child/carer to develop a manageable plan.

Do

- Use open ended questions
- Offer information
- Encourage child/carer to find solutions
- Respect the child/carer's right to choose

The child/carer might feel:

- 'I choose what I want to deal with'
- 'I can help myself'
- 'I feel supported in my choice'
- 'I can cope with my problems'

Don't

- Force your ideas onto the child/carer
- Be a 'fix-it' specialist
- Let the child/carer take on too many problems at once

The child/carer might feel:

- 'I am not respected'
- 'I am unable to make my own decisions'
- 'I am expected to change too fast'

Empathise

Empathy is the ability to imagine and share the child/carer's situation and feelings.

Do

- Listen for, and identify feelings e.g. 'you sound very upset'
- Allow the child/carer to express emotion
- Be supportive

The child/carer might feel:

- 'I can get through this'
- 'I can deal with my situation'
- 'My health care worker understands me'
- 'I feel supported'

Don't

- Judge, criticise or blame
- Disagree or argue
- Be uncomfortable with high levels of emotions and burden of the problems

The child/carer might feel:

- 'I am being judged'
- 'I am too much to deal with'
- 'I can't cope'
- 'My health care worker is unfeeling'

Summarise

Summarising what has been discussed helps to check the child/carer's understanding and to agree on a plan for a solution.

Do

- Get the child/carer to summarise
- Agree on a plan
- Offer to write a list of options
- Offer a follow-up appointment

The child/carer might feel:

- 'I can make changes in my life'
- 'I have something to work on'
- 'I feel supported'
- 'I can come back when I need to'

Don't

- Direct the decisions
- Be abrupt
- Force a decision

The child/carer might feel:

- 'My health care worker disapproves of my decisions'
- 'I feel resentful'
- 'I feel misunderstood'

Prescribe rationally

Assess the child needing a prescription

Assess	Note
Diagnosis	Confirm child's diagnosis, medication is necessary and benefits outweigh the risks.
Other conditions	If necessary, adjust dose (e.g. co-trimoxazole in kidney disease) or change medication (e.g. avoid ibuprofen if asthma, heart failure or kidney disease).
Other medications	Check all medication (prescribed, over-the-counter, herbal) is necessary and for possible interactions especially if on hormonal contraceptive or treatment for TB, HIV, epilepsy.
Allergies	If known allergy or previous bad reaction, record in child's notes and RtHB and discuss alternative with doctor.
Dose	Ensure dose is calculated according to child's weight. Increase dose as child grows. Show carer how to give medicines at home.
Response to treatment	• If child's condition does not improve, first exclude poor adherence, then consider changing treatment or an alternative diagnosis. • Check for side effects and report medication/vaccine reactions. Use Adverse reporting form¹ and email to pha-mic@uct.ac.za or complete online on Sinjani (http://sinjani.westerncape.gov.za).

Advise the child and carer needing a prescription

- Explain when and how to take medication and what to do if side effects occur. Ask child/carers to repeat your explanation to ensure s/he understands how to take medication.
- Ensure child/carers know generic name of all medication and advise to ask prescriber/pharmacist if s/he does not understand a change to regular medication.
- Educate on importance of adherence and that not adhering to medication may lead to relapse or worsening of condition and in some instances, resistance to medication.
- Over-the counter medications and herbal treatments may interfere with prescribed medication. Encourage child/carers to discuss with prescriber before using them.

Treat the child needing a prescription

- Ensure appropriate prescriber writes prescription: ♦Orange-highlighted medications may be prescribed by a doctor or an authorised prescriber (clinical nurse practitioner/primary care nurse specialist or professional nurse) in accordance with his/her scope of practice within a specified field. ►Blue-highlighted medications may be prescribed by a doctor or clinical nurse practitioner/primary care nurse specialist who is an authorised prescriber. ■Green-highlighted medications may be prescribed by a doctor only. ●Pink-highlighted medications may be initiated by a doctor only, but may be continued by a clinical nurse practitioner/primary care nurse specialist who is an authorised prescriber.
 - Shape icons placed in front of colour-coded medications support users with colour vision deficiency.
- Consult the South African Medicines Formulary (SAMF) or MIC hotline (021 406 6829) if unsure about medicine choice and dosing, side-effects or drug interactions.
- Ensure the prescription contains all the detail it needs - see sample prescription below. Write legibly.

Facility details

Diagnosis and ICD10 code

Generic name of each medicine with dose, strength, frequency, route and quantity (if prescribed as needed) of medicine

Prescriber name, qualification, practitioner number, signature, contact number

Child's full name, ID/passport number, address, age, gender, weight

Child's allergies

If applicable, number of repeats of prescription (maximum for 6 months)

Date of issue

¹Adverse drug reaction report forms available from clinic pharmacy or via: <https://mic.uct.ac.za/educational-material/posters>.

Medication dosing tables

- If child's age and weight not in same row, choose dose according to weight.
- Show carer how to give medication at home.

How to interpret age and weight ranges:

- '10-12kg' means 'from and including 10kg, up to but not including 12kg'. The next range starts with 12kg.
- '< 2 months old' refers to a baby less than 2 months old and not including 2 months.
- '≥ 2 months old' indicates that the baby is 2 months old and older (this includes the whole month that the baby is 2 months old).
- '< 5 years old' refers to a child who is younger than 5 years, not including being 5 years old.
- '≥ 5 years old' refers to a child who is 5 years old and older, and starts on the day the child turns 5.

A

Aciclovir for herpes zoster (chickenpox)

Oral, 20mg/kg/dose (up to 700mg) 6 hourly for 7 days.

1

Weight	Dose	Use one of the following:			Age
		Suspension	Tablet		
		200mg/5mL	200mg	400mg	
>3.5-5kg	100mg	2.5mL	-	-	1-3 months
5-7kg	140mg	3.5mL	-	-	3-6 months
7-9kg	160mg	4mL	-	-	6-12 months
9-11kg	200mg	5mL	1 tablet	½ tablet	12-18 months
11-14kg	240mg	6mL	-	-	18 months-3 years
14-25kg	400mg	10mL	2 tablets	1 tablet	3-7 years
25-35kg	600mg	15mL	3 tablets	1½ tablet	7-11 years
35-55kg	700mg	-	3½ tablets	-	11-15 years

Aciclovir for herpes simplex

Oral, 250mg/m²/dose (up to 400mg) 8 hourly for 7 days.

2

Weight	Dose	Use one of the following:			Age
		Suspension	Tablet		
		200mg/5mL	200mg	400mg	
>3.5-5kg	50mg	1.25mL	-	-	1-3 months
5-7kg	80mg	2mL	-	-	3-6 months
7-11kg	100mg	2.5mL	½ tablet	-	6-18 months
11-14kg	120mg	3mL	-	-	18 months-3 years
14-25kg	160mg	4mL	-	-	3-7 years
25-35kg	200mg	5mL	1 tablet	½ tablet	7-11 years
35-55kg	300mg	7.5mL	1½ tablets	-	11-15 years
>55kg	400mg	-	-	1 tablet	≥ 15 years

Activated charcoal

Oral, 1g/kg (up to 100g) mixed as a slurry with water.

3

Weight	Dose	Age
3.5-7kg	5g	1-6 months
7-11kg	10g	6-18 months
11-17.5kg	15g	18 months-5 years
17.5-35kg	25g	5-11 years
35-55kg	50g	11-15 years
≥ 55kg	100g	≥ 15 years

Adrenaline

IM/IV, 1mg/mL (1:1 000), 0.01mL/kg as a single dose

4

Weight	Dose	Injection 1mg/mL (1:1 000)	Age
< 9kg	0.05mg	0.05mL	< 1 year
9-12kg	0.1mg	0.1mL	1-2 years
12-17.5kg	0.2mg	0.2mL	2-5 years
17.5-40kg	0.3mg	0.3mL	5-12 years
≥ 40kg	0.5mg	0.5mL	≥ 12 years

Albendazole

- Treatment of tapeworm infestation: oral, once daily for 3 days.
- Treatment of worm infestation, other than tapeworm: oral, as single dose.

5

Age	Tablet 200mg	Tablet 400mg
12-24 months	1 tablet	-
≥ 2 years	-	1 tablet

PACK CHILD

Amoxicillin

Oral, 45mg/kg/dose (up to 1500mg) 12 hourly for 5 days

- Single dose: complicated severe acute malnutrition (SAM), measles with pneumonia.
- 5 days treatment: pneumonia, acute otitis media, sinusitis if < 3 years old.

6

Weight	Dose	Use one of the following:				Age
		Suspension		Capsule		
		125mg/5mL	250mg/5mL	250mg	500mg	
2-2.5kg	100mg	4mL	2mL	-	-	Birth-1 month
2.5-3.5kg	125mg	5mL	2.5mL			
3.5-5kg	175mg	7mL	3.5mL	-	-	1-3 months
5-7kg	250mg	10mL	5mL	-	-	3-6 months
7-11kg	375mg	15mL	7.5mL	-	-	6-18 months
11-14kg	500mg	-	10mL	2 capsules	1 capsule	18 months-3 years
14-17.5kg	750mg	-	15mL	3 capsules	-	3-5 years
17.5-25kg	1000mg	-	20mL	4 capsules	2 capsules	5-7 years
25-30kg	1250mg	-	25mL	5 capsules	-	7-10 years
≥30kg	1500mg	-	-	6 capsules	3 capsules	≥ 10 years

Amoxicillin for bacterial tonsillopharyngitis

Oral, 50mg/kg/dose (up to 2g) daily for 10 days.

7

Weight	Dose	Use one of the following:				Age
		Suspension		Tablet		
		125mg/5mL	250mg/5mL	250mg	400mg	
2-2.5kg	100mg	4mL	2mL	-	-	Birth-1 month
2.5-3.5kg	150mg	6mL	3mL	-	-	
3.5-5kg	200mg	8mL	4mL	-	-	1-3 months
5-7kg	275mg	11mL	5.5mL	-	-	3-6 months
7-11kg	400mg	-	8mL	-	-	6-18 months
11-17.5kg	575mg	-	11.5mL	-	-	18 months-5 years
17.5-25kg	750mg	-	15mL	3	-	5-7 years
25-35kg	1000mg	-	20mL	4	2	7-11 years
≥ 35kg	2000mg	-	-	-	4	≥ 11 years

Amoxicillin/clavulanic acid (600/42.9mg per 5ml)

Oral, 45mg/kg/dose (of amoxicillin component) 12 hourly.

8

Weight	Dose (amoxicillin component)	Use one of the following:		Age
		Suspension	Tablet	
		600/42.9mg/5mL	875/125mg	
3.5-5kg	180mg	1.5mL	-	1-3 months
5-7kg	240mg	2mL	-	3-6 months
7-11kg	360mg	3mL	-	6-18 months
11-14kg	480mg	4mL	-	18 months-3 years
14-17.5kg	720mg	6mL	1 tablet	3-5 years
17.5-25kg	900mg	7.5mL	1 tablet	5-7 years
≥ 25kg	1200mg	10mL	1 tablet	≥ 7 years

Atropine

IV, 0.05mg/kg/dose.

9

Weight	Dose	Use one of the following injections		Age
		0.5mg/mL	1mg/mL	
3.5-5kg	0.2mg	0.4mL	0.2mL	1-3 months
5-7kg	0.3mg	0.6mL	0.3mL	3-6 months
7-9kg	0.4mg	0.8mL	0.4mL	6-12 months
9-11kg	0.5mg	1mL	0.5mL	12-18 months
11-14kg	0.6mg	1.2mL	0.6mL	18 months-3 years
14-17.5kg	0.8mg	1.6mL	0.8mL	3-5 years
≥ 17.5kg	1mg	2mL	1mL	≥ 5 years

Azithromycin

Oral, 10mg/kg/dose (up to 500mg) daily for 3 days.

10

Weight	Dose	Use one of the following:			Age
		Suspension	Tablet		
		200mg/5mL	250mg	500mg	
3.5-5kg	40mg	1mL	-	-	1-3 months
5-7kg	60mg	1.5mL	-	-	3-6 months
7-9kg	80mg	2mL	-	-	6-12 months
9-11kg	100mg	2.5mL	-	-	12-18 months
11-14kg	120mg	3mL	-	-	18 months-3 years
14-18kg	160mg	4mL	-	-	3-5 years
18-25kg	200mg	5mL	-	-	5-7 years
25-35kg	250mg	-	1 tablet	-	7-11 years
≥ 35kg	500mg	-	-	1 tablet	≥ 11 years

PACK CHILD

C

Ceftriaxone

- IM, 80mg/kg/dose (up to 1500mg) immediately as a single dose.
- Give injection into upper thigh, not buttocks.
- If giving ≥ 1g IM, split the dose and give half into each thigh.

11

Weight	Dose	Use one of the following injections mixed with water for injection (WFI):			Age
		250mg/2mL (250mg diluted in 2mL WFI)	500mg/2mL (500mg diluted in 2mL WFI)	1000mg/3.5mL (1000mg diluted in 3.5mL WFI)	
2-2.5kg	190mg	1.5mL	0.75mL	-	Birth-1 month
2.5-3.5kg	225mg	1.8mL	0.9mL	-	
3.5-5.5kg	310mg	-	1.25mL	-	1-3 months
5.5-7kg	440mg	-	1.75mL	-	3-6 months
7-9kg	625mg	-	2.5mL	-	6-12 months
9-11kg	750mg	-	3mL	-	12-18 months
11-14kg	810mg	-	3.25mL	-	18 months-3 years
If giving ≥ 1g IM, split the dose and give half into each upper thigh					
14-17.5kg	1000mg	-	4mL	3.5mL	3-5 years
≥ 17.5kg	1500mg	-	-	5.5mL	≥ 5 years

Ceftriaxone

- IV/IM, 100mg/kg/dose (up to 1875mg) immediately as a single dose.
- If giving IM injection, give injection into upper thigh, not buttocks.
- If giving ≥ 1g IM, split the dose and give half into each thigh.

12

Weight	Dose	Use one of the following injections mixed with water for injection (WFI):			Age
		250mg/2mL (250mg diluted in 2mL WFI)	500mg/2mL (500mg diluted in 2mL WFI)	1000mg/3.5mL (1000mg diluted in 3.5mL WFI)	
2-2.5kg	238mg	1.9mL	0.9mL	-	Birth-1 month
2.5-3.5kg	280mg	-	1.1mL	-	
3.5-5.5kg	388mg	-	1.6mL	-	1-3 months
5.5-7kg	550mg	-	2.2mL	-	3-6 months
7-9kg	782mg	-	3.1mL	-	6-12 months
9-11kg	938mg	-	3.75mL	-	12-18 months
If giving ≥ 1g IM, split the dose and give half into each upper thigh					
11-14kg	1013mg	-	4.0mL	3.5mL	18 months-3 years
14-17.5kg	1250mg	-	-	4.6mL	3-5 years
≥ 17.5kg	1875mg	-	-	6.9mL	≥ 5 years

Cephalexin

- If > 7 years old: oral, 500mg 6 hourly for 5 days.
- If ≤ 7 years old: oral, 25mg/kg/dose 12 hourly for 5 days (use table below).

13

Weight	Dose	Syrup		Capsule	Age
		125mg/5mL	250mg/5mL	250mg	
2.5-3.5kg	75mg	3mL	1.5ml	-	Birth-3 months
3.5-5.1kg	100mg	4mL	2mL	-	3-6 months
5.1-7.5kg	150mg	6mL	3mL	-	6-12 months
7.5-10.1kg	200mg	8ml	4ml	-	12-18 months
10.1-14.1kg	250mg	10ml	5ml	1 capsule	18 months-3 years
14.1-18.1kg	350mg	-	7ml	-	3-5 years
18.1-25kg	500mg	-	10ml	2 capsules	5-7 years

Cetirizine

Oral, once daily.

14

Weight	Dose	Use one of the following:		Age
		Syrup	Tablet	
		1mg/mL	10mg	
12-21kg	5mg	5mL	-	2-6 years
≥ 21kg	10mg	10mL	1 tablet	≥ 6 years

Chlorpheniramine

- Oral, 0.1mg/kg/dose (up to 4 mg) 6-8 hourly for up to 2 weeks
- If < 2 years old, give 12 hourly.
- In mild cases, only give at night.
- Chlorpheniramine is sedating.

15

Weight	Dose	Use one of the following:		Age
		Suspension	Tablet	
		2mg/5mL	4mg	
3.5-5kg	0.4mg	1ml	-	1-3 months
5-7kg	0.6mg	1.5ml	-	3-6 months
7-9kg	0.8mg	2ml	-	6-12 month
9-12kg	1mg	2.5ml	-	12 months-2 years
12-14kg	1.2mg	3mL	-	2-3 years
14-17.5kg	1.6mg	4mL	-	3-5 years
17.5-25kg	2mg	5mL	-	5-7 years
25-35kg	3mg	7.5mL	-	7-11 years
≥ 35kg	4mg	-	1 tablet	≥ 11 years

PACK CHILD

Ciprofloxacin

Oral, 15mg/kg/dose (up to 500mg) 12 hourly for 3 days.

16

Weight	Dose	Use one of the following:			Age
		Suspension	Tablet		
		250mg/5mL	250mg	500mg	
9-11kg	150mg	3mL	-	-	12-18 months
11-14kg	200mg	4mL	-	-	18 months-3 years
14-17.5kg	250mg	5mL	1 tablet	-	3-5 years
17.5-25kg	300mg	6mL	-	-	5-7 years
≥ 25kg	500mg	10mL	2 tablets	1 tablet	≥ 7 years

Ciprofloxacin

Oral, 20mg/kg/dose (up to 520mg) as a single dose for cholera.

17

Weight	Dose	Use one of the following:			Age
		Suspension	Tablet		
		250mg/5mL	250mg	500mg	
9-11kg	180mg	3.6mL	-	-	12-18 months
11-14kg	220mg	4.4mL	-	-	18 months-3 years
14-17.5kg	280mg	5.6mL	1¼ tablets	-	3-5 years
17.5-25kg	350mg	7mL	-	-	5-7 years
≥ 25kg	520mg	10.4mL	2 tablets	1 tablet	≥ 7 years

Co-trimoxazole

Oral, daily dose.

18

Weight	Dose	Use one of the following:		
		Suspension 200/40mg/5mL	Single strength tablet 400/80mg	Double strength tablet 800/160mg
3-6kg	100/20mg	2.5mL	¼ tablet	-
6-14kg	200/40mg	5mL	½ tablet	-
14-25kg	400/80mg	10mL	1 tablet	½ tablet
≥ 25kg	800/160mg	-	2 tablets	1 tablet

D

Diazepam

- Rectal, 0.5mg/kg/dose (up to 10mg) for fits as a single dose.
- If second dose needed, give IV diazepam ➔ 160:20

19

Weight	Dose	Ampoule 10mg/2mL	Age
3-6kg	2mg	0.4mL	< 6 months
6-10kg	2.5mg	0.5mL	6 months-1 year
10-18kg	5mg	1mL	1-5 years
18-25kg	7.5mg	1.5mL	5-8 years
≥ 25kg	10mg	2mL	≥ 8 years

Diazepam for fits

IV, 0.25mg/kg/dose

- If < 5 years old: up to 5mg.
- If ≥ 5 years old: up to 10mg.

20

Weight	Ampoule 10mg/2mL (5mg/mL)
4-5kg	0.2mL
5-6kg	0.25mL
6-7kg	0.3mL
7-8kg	0.35mL
8-9kg	0.4mL
9-10kg	0.45mL
10-11kg	0.5mL
11-12kg	0.55mL
12-13kg	0.6mL
13-14kg	0.65mL
14-15kg	0.7mL
15-16kg	0.75mL
16-17kg	0.8mL
17-18kg	0.85mL
18-19kg	0.9mL
19-20kg	0.95mL
20-25kg	1.0mL
≥ 25kg	0.25 x weight ÷ 5 = number of mL to give

Adapted from Pocketbook of hospital care for children, Management of common childhood illnesses, National Department of Health (South Africa), 2016.

F

Ferrous gluconate

Oral, 1-2mg/kg/dose (elemental iron) 8 hourly with food.

21

Weight	Dose	Syrup 250mg/5mL 30mg elemental iron per 5mL	Age
3-6kg	10mg	1.7mL	Birth-3 months
6-10kg	20mg	3.3mL	3-12 months
10-25kg	40mg	6.7mL	1-5 years

Ferrous sulphate

Oral, 1-2mg/kg/dose (elemental iron) 8 hourly with food.

22

Weight	Tablet 170mg ± 55mg elemental iron per 5mL	Age
10-25kg	½ tablet	1-5 years
≥ 25kg	1 tablet	≥ 5 years

PACK CHILD

Flucloxacillin
Oral, 12-25mg/kg/dose 6 hourly for 5 days.

23

Weight	Dose	Use one of the following:		Age
		Syrup	Capsule	
		125mg/5mL	250mg	
2.5-5kg	62.5mg	2.5mL	-	Birth-3 months
5-11kg	125mg	5mL	-	3-18 months
11-25kg	250mg	10mL	1 capsule	18 months-7 years
≥ 25kg	500mg	-	2 capsules	> 7 years

Fluconazole
Oral, 6mg/kg/dose once daily.

24

Weight	Dose	Use one of the following:			Age
		Suspension	Capsule		
		50mg/5mL	50mg	200mg	
3.5-5kg	25mg	2.5mL	-	-	1-3 months
5-7kg	30mg	3mL	-	-	3-6 months
7-9kg	50mg	5mL	1 capsule	-	6-12 months
9-11kg	60mg	6mL	-	-	12-18 months
11-14kg	70mg	7mL	-	-	18 months-3 years
14-17.5kg	100mg	10mL	2 capsules	-	3-5 years
17.5-25kg	125mg	12.5mL	-	-	5-7 years
25-35kg	150mg	15mL	3 capsules	-	7-11 years
≥ 35kg	200mg	-	-	1 capsule	≥ 11 years

Furosemide
IV, 1mg/kg (up to 40mg), over 5 minutes.

25

Weight	Dose	Injection 10mg/mL	Age
3.5-5kg	4mg	0.4mL	1-3 months
5-7kg	6mg	0.6mL	3-6 months
7-9kg	8mg	0.8mL	6-12 months
9-11kg	10mg	1mL	12-18 months
11-14kg	12mg	1.2mL	18 months-3 years
14-17.5kg	15mg	1.5mL	3-5 years
17.5-25kg	20mg	2mL	5-7 years
25-35kg	30mg	3mL	7-11 years
≥ 35kg	40mg	4mL	≥ 11 years

H

Hydrocortisone for anaphylaxis
IM/slow IV, 5mg/kg (up to 100mg) immediately.

26

Weight	Dose	Injection 100mg/2mL	Age
< 11kg	25mg	0.5mL	1 month-2 years
11-14kg	50mg	1mL	2-3 years
14-17.5kg	75mg	1.5mL	3-5 years
≥ 17.5kg	100mg	2mL	≥ 5 years

Hydrocortisone for acute asthma
IM/slow IV, 4mg/kg (up to 100mg) immediately.

27

Weight	Dose	Injection 100mg/2mL	Age
< 11kg	25mg	0.5mL	1 month-18 months
11-14kg	40mg	0.8mL	18 months-3 years
14-17.5kg	50mg	1mL	3-5 years
17.5-25kg	60mg	1.2mL	5-7 years
25-40kg	80mg	1.6mL	7-12 years
>40kg	100mg	2mL	≥ 12 years

Ibuprofen
Oral, 5-10mg/kg/dose (up to 400mg) 8 hourly with food.

28

Weight	Dose	Use one of the following:		Age
		Syrup	Tablet	
		100mg/5mL	200mg	
9-11kg	80mg	4mL	-	12-18 months
11-14kg	100mg	5mL	-	18 months-3 years
14-17.5kg	120mg	6mL	-	3-5 years
17.5-25kg	150mg	7.5mL	-	5-7 years
25-40kg	200mg	10mL	1 tablet	7-12 years
≥ 40kg	400mg	-	2 tablets	≥ 12 years

L

Lactulose
Oral, 0.5mL/kg/dose (up to 15mL) once daily. If poor response, increase to 12 hourly.

29

Weight	Syrup 3.3g/5mL	Age
5-7kg	3mL	3-6 months
7-9kg	4mL	6-12 months
9-11kg	5mL	12-18 months
11-14kg	6mL	18 months-3 years
14-17.5kg	7.5mL	3-5 years
17.5-35kg	10mL	5-11 years
≥ 35kg	15mL	≥ 11 years

Lidocaine without adrenaline

- Infiltrate around wound as local anaesthetic.
- Maximum dose: 3mg/kg.

30

Weight	Maximum dose	1% vial (10mg/mL)	2% vial (20mg/mL)	Age
2.5-3.5kg	7mg	0.7mL	0.35mL	< 1month
3.5-5kg	10mg	1mL	0.5mL	1-3 months
5-7kg	15mg	1.5mL	0.75mL	3-6 months
7-9kg	20mg	2mL	1mL	6-12 months
9-11kg	25mg	2.5mL	1.25mL	12-18 months
11-14kg	30mg	3mL	1.5mL	18 months-3 years
14-17.5kg	40mg	4mL	2mL	3-5 years
17.5-35kg	50mg	5mL	2.5mL	5-11 years
35-55kg	100mg	10mL	5mL	≥ 11 years

M

Mebendazole

- Routine 6 monthly deworming: oral, from 12 months–5 years.
- Treatment of worms: from 12 months old.

31

Age	Use one of the following:		
	Suspension 100mg/5mL	Tablet	
		100mg	500mg
12-24 months	5mL 12 hourly for 3 days	1 tablet 12 hourly for 3 days	-
≥ 2 years	25mL as a single dose	5 tablets as a single dose	1 tablet as a single dose

Metoclopramide
Oral, 0.1 mg/kg/dose (up to 4.5mg) 8 hourly as needed for up to 5 days.

32

Weight	Dose	Suspension 5mg/5mL	Age
9-11kg	1mg	1mL	12-18 months
11-14kg	1.2mg	1.2mL	18 months-3 years
14-17.5kg	1.6mg	1.6mL	3-5 years
17.5-25kg	2mg	2mL	5-7 years
25-35kg	3mg	3mL	7-11 years
≥ 35kg	4.5mg	4.5mL	≥ 11 years

Metronidazole
Oral, 7.5mg/kg/dose (up to 400mg) 8 hourly for 5 days.

33

Weight	Dose	Use one of the following:			Age	
		Suspension 200mg/5mL	Tablet			
			200mg	400mg		
9-11kg	80mg	2mL	-	-	12-18 months	
11-14kg	100mg	2.5mL	½ tablet	-	18 months-3 years	
14-17.5kg	120mg	3mL	-	-	3-5 years	
17.5-25kg	160mg	4mL	-	-	5-7 years	
25-35kg	200mg	5mL	1 tablet	½ tablet	7-11 years	
35-55kg	300mg	7.5mL	1½ tablets	-	11-15 years	
≥ 55kg	400mg	-	-	1 tablet	≥ 15 years	

Midazolam

- Buccal (between the cheek and gum), 0.5mg/kg/dose (up to 10mg).
- Check formulation: use 5mg/mL formulation for buccal administration.

34

Weight	Dose	Buccal 5mg/mL	Age
7-9kg	4mg	0.8mL	6-12 months
9-11kg	5mg	1mL	12-18 months
11-14kg	6mg	1.2mL	18 months-3 years
14-17.5kg	7.5mg	1.5mL	3-5 years
≥ 17.5kg	10mg	2mL	≥ 5 years

Midazolam
IV, 0.25mg/kg (up to 9mg).

35

Weight	Midazolam 5mg/5mL ampoule (1mg/mL)	Midazolam 15mg/3mL ampoule (5mg/mL)
< 4kg	-	0.2mL
4-5kg	1.0mL	
5-6kg	1.3mL	0.3mL
6-7kg	1.5mL	
7-8kg	1.8mL	0.4mL
8-9kg	2.0mL	
9-10kg	2.3mL	0.5mL
10-11kg	2.5mL	
11-12kg	2.8mL	0.6mL
12-13kg	3.0mL	
13-14kg	3.3mL	0.7mL
14-15kg	3.5mL	
15-16kg	3.8mL	0.8mL
16-17kg	4.0mL	
17-18kg	4.3mL	0.9mL
18-19kg	4.5mL	
19-20kg	4.8mL	1.0mL
20-25kg	5.0mL	
25-35kg	7.5mL	1.5mL
≥ 35kg	9mL	1.8mL

Adapted from Pocketbook of hospital care for children, Management of common childhood illnesses, National Department of Health (South Africa), 2016.

Morphine, oral

- 0.05mg/kg/dose 6 hourly if < 1 month.
- 0.1mg/kg/dose 4-6 hourly if 1-12 months.
- 0.2-0.4mg/kg/dose 4-6 hourly if ≥12 months.

36

Weight	Dose	Use one of the following:		Age
		Syrup	Tablet	
		1mg/mL	10mg	
<4.4kg	0.2mg	0.2mL	-	0-1 month
4.4-9kg	0.7mg	0.7mL	-	1-12 months
9-11kg	2.5mg	2.5mL	-	12-18 months
11-14kg	4mg	4mL	-	18 months-3 years
14-17.5kg	5mg	5mL	-	3-5 years
17.5-25kg	6mg	6mL	-	5-7 years
≥ 25kg	10mg	10mL	1 tablet	≥ 7 years

Paracetamol

Oral, 10-15mg/kg/dose (up to 1g) 6 hourly as needed for up to 5 days

- If < 1 month, maximum daily dose 60mg/kg/day.
- If ≥ 1 month, maximum daily dose 90mg/kg/day (up to 4g daily).

37

Weight	Dose	Use one of the following:		Age
		Syrup	Tablet	
		120mg/5mL	500mg	
3.5-5kg	48mg	2mL	-	1-3 months
5-7kg	72mg	3mL	-	3-6 months
7-9kg	96mg	4mL	-	6-12 months
9-11kg	120mg	5mL	-	12-18 months
11-14kg	144mg	6mL	-	18 months-3 years
14-17.5kg	180mg	7.5mL	-	3-5 years
17.5-25kg	240mg	10mL	½ tablet	5-7 years
25-35kg	360mg	15mL	-	7-11 years
35-55kg	500mg	-	1 tablet	11-15 years
≥ 55kg	1 000mg	-	2 tablets	≥ 15 years

PACK CHILD

Phenobarbital
IM, 20mg/kg (up to 1g) slow injection.

38

Weight	Injection 200mg/mL	
	20mg/kg IM Volume 0.1mg/kg	
4kg	0.4mL	
5kg	0.5mL	
6kg	0.6mL	
7kg	0.7mL	
8kg	0.8mL	
9kg	0.9mL	
10kg	1.0mL	
11kg	1.1mL	
12kg	1.2mL	
13kg	1.3mL	
14kg	1.4mL	
15kg	1.5mL	
16kg	1.6mL	
17kg	1.7mL	
18kg	1.8mL	
19kg	1.9mL	
≥ 20kg	2.0mL	

Adapted from Pocketbook of hospital care for children, Management of common childhood illnesses, National Department of Health (South Africa), 2016.

Phenobarbital
Oral, crushed. Give via nasogastric tube (NGT), 20mg/kg (up to 240mg) as a single dose.

39

Weight	Dose	Tablet 30mg	Age
2.5-3.5kg	60mg	2 tablets	Birth-1 month
3.5-5kg	75mg	2½ tablets	1-3 months
5-7kg	120mg	4 tablets	3-6 months
7-11kg	180mg	6 tablets	6-18 months
11-14kg	210mg	7 tablets	18 months-3 years
≥ 14kg	240mg	8 tablets	≥ 3 years

Praziquantel
Oral, 40mg/kg (up to 1.8g) as a single dose.

40

Weight	Dose	Tablet 600mg	Age
12-17.5kg	600mg	1 tablet	2-5 years
17.5-25kg	900mg	1½ tablets	5-7 years
25-35kg	1200mg	2 tablets	7-11 years
≥ 35kg	1800mg	3 tablets	≥ 11 years

Prednisone
Oral, 1-2mg/kg/dose (up to 40mg) daily for 7 days.

41

Weight	Dose	Tablet 5mg	Age
11-14kg	20mg	4 tablets	2-3 years
14-17.5kg	30mg	6 tablets	3-5 years
≥17.5kg	40mg	8 tablets	≥ 5 years

Promethazine
IM/slow IV. > 2 years: 0.25mg/kg.

42

Weight	Dose	Use one of the following injections		Age
		25mg/mL	50mg/2mL	
12-17.5kg	2.5mg	0.1mL	0.1mL	2-5 years
17.5-25kg	5mg	0.2mL	0.2mL	5-7 years
25-35kg	7.5mg	0.3mL	0.3mL	7-11 years
35-55kg	15mg	0.6mL	0.6mL	≥ 11years

V

Vitamin A routine preventive
Routine treatment: oral, single dose 6 monthly (from 6 months-5 years old).

43

Age	Dose units	Use one of the following:		
		Capsule, snip off narrow end 50 000IU	Capsule, snip off narrow end 100 000IU	Capsule, snip off narrow end 200 000IU
6-12 months	100 000IU	2 capsules	1 capsule	-
12 months-5 years	200 000IU	-	2 capsules	1 capsule

Vitamin A additional dose/s

- If severe acute malnutrition, diarrhoea > 14 days, or symptoms of vitamin A deficiency: give single additional dose. Wait at least 1 month after last dose of vitamin A.
- If giving as part of measles treatment: give two additional doses (one dose now and one dose the following day).

44

Age	Dose units	Use one of the following:		
		Capsule, snip off narrow end 50 000IU	Capsule, snip off narrow end 100 000IU	Capsule, snip off narrow end 200 000IU
< 6 months	50 000IU	1 capsule	½ capsule	-
6-12 months	100 000IU	2 capsules	1 capsule	-
12 months-5 years	200 000IU	-	2 capsules	1 capsule

TB preventive treatment (TPT) dosing tables

- Choose one of the tables below according to chosen regimen and dose according to weight.
- If unable to swallow whole tablet, encourage child to swallow split tablets if possible, or dissolve/crush and mix with small amount of soft food such as yoghurt/honey/jam or give in water (see below).

Check for possible medication interactions:

- If on dolutegravir and 4R, give dolutegravir dose 12 hourly according to weight ➔ 168.
- If on lopinavir/atazanavir/ritonavir, avoid 3HP and 4R. Give instead ♦6H (if < 25kg or previously treated with TPT) or ♦12H (if ≥ 25kg and *not* previously treated with TPT). If on 4R, discuss/refer.
- If older child on oral contraception/subdermal implant and 3HP, advise to use condoms during and for 2 weeks after 3HP stopped. Injectable contraceptives may be used if on 3HP ➔ **PACK Adult**.
- If on antiseizure medication or on diabetes medication, discuss.

Isoniazid (6H and 12H):

- 6H is *daily* isoniazid for 6 months.
- 12H is *daily* isoniazid for 12 months.
- Give before eating (on an empty stomach).

Weight (kg) <i>If < 2kg, discuss with specialist</i>	Isoniazid (daily)		
	100mg tablet	OR 100mg tablet/s in water	OR 300mg tablet
2-3.5	¼ tablet	2mL ¹	-
3.5-5	½ tablet	4mL ¹	-
5-7.5	¾ tablet	6mL ¹	-
7.5-10	1 tablet	Crush and mix 1 tablet in small amount of water	-
10-15	1½ tablets	Crush and mix 1½ tablets in small amount of water	-
15-20	2 tablets	Crush and mix 2 tablets in small amount of water	-
≥ 20	3 tablets	Crush and mix 3 tablets in small amount of water	1 tablet

Isoniazid and rifapentine (3HP):

- 3HP is *weekly* isoniazid and rifapentine for 3 months.
- Give with food or immediately after eating.

Weight (kg)	Choose only one option (weekly)		
	Isoniazid 300mg tablet	+ Rifapentine 150mg tablet	OR Isoniazid/rifapentine 300/300mg tablet
25-30	2 tablets	4 tablets	2 tablets
≥ 30	3 tablets	6 tablets	3 tablets

Rifampicin and Isoniazid (3RH):

- 3RH is *daily* rifampicin and isoniazid for 3 months.
- Give before eating (on an empty stomach).

Weight (kg) <i>If < 2kg, discuss with specialist</i>	RH (daily)			
	75/50mg tablet ²	OR 75/50mg tablet/s in water ²	OR 150/75mg tablet	OR 300/150mg tablet
2-3	½ tablet	Mix 1 tablet in 4mL water. Give 2mL of mixture ³ .	-	-
3-4	¾ tablet	Mix 1 tablet in 4mL water. Give 3mL of mixture ³	-	-
4-6	1 tablet	Mix 1 tablet in 4mL water. Give ALL of mixture.	-	-
6-8	1½ tablets	Mix 1½ tablets in 5-10mL water. Give ALL of mixture.	-	-
8-12	2 tablets	Mix 2 tablets in 5-10mL water. Give ALL of mixture.	-	-
12-16	3 tablets	Mix 3 tablets in 5-10mL water. Give ALL of mixture.	-	-
16-25	4 tablets	Mix 4 tablets in 5-10mL water. Give ALL of mixture.	-	-
25-38	-	-	2 tablets	-
38-55	-	-	3 tablets	-
≥ 55	-	-	-	2 tablets

Rifampicin (4R):

- 4R is *daily* rifampicin for 4 months.
- 150mg capsules: open into 10ml water. Calculate volume of mixture based on dose/kg.

< 10 years old	15mg/kg daily
≥ 10 years old	10mg/kg daily

¹Crush one ♦**isoniazid** 100mg tablet in 8mL water, give required dose and discard unused solution. ²If RH 75/50mg formulation unavailable, discuss giving RH 150/75mg or RH 60/60mg with pharmacist instead. ³Discard unused solution.

Drug-sensitive TB treatment dosing tables

Children should be taught and encouraged to swallow whole tablets or, if required, fractions of tablets to avoid large volumes of liquid medication.



TB medication for children: Easy dosing guide

Confirmed or clinically diagnosed drug-susceptible/presumed drug-susceptible non-severe TB, severe pulmonary TB and extrapulmonary TB (EPTB)
excluding TB meningitis/central nervous system (CNS) TB/miliary TB

Treatment phase	Intensive phase Once daily, 7 days a week		Continuation phase Once daily, 7 days a week			Treatment phase
Duration	2 months		Non-severe TB 2 months	Severe PTB and most EPTB 4 months	Bone and joint TB 10 months	Duration
Target dose (range)	Isoniazid (H): 10(7-15)mg/kg, Rifampicin (R): 15(10-20)mg/kg, Pyrazinamide (Z): 35(30-40)mg/kg, Ethambutol (E): 20(15-25)mg/kg					Target dose (range)
Formulation	HRZ 50/75/150mg dispersible tablet (scored)	</				

¹To make oral suspension: for weight band 3–3.9kg, for each dose, disperse one HRZ 50/75/150mg tablet (2 months intensive phase) or one HR 50/75mg tablet (continuation phase) in 4mL water, administer 3mL, discard unused suspension. For other weight bands: disperse required number of tablets and fractions of tablets in a small amount of water (5-10mL) and administer all of the suspension to the child orally or via nasogastric tube.

²If oral suspension required, for each dose, crush one ethambutol 400mg tablet into a fine powder, disperse in 8mL water to a concentration of 400mg/8mL (50mg/mL), administer required dose as indicated in above chart, discard unused suspension.

HRZ – isoniazid + rifampicin + pyrazinamide; **HR** – isoniazid + rifampicin; **HRZE** – isoniazid + rifampicin + pyrazinamide + ethambutol; **E** – ethambutol

Adapted from 2024 NDoH Management of Tuberculosis in Children and Adolescents: A Clinical Guideline For The Diagnosis and Treatment of Drug -susceptible TB in Children and Adolescents in South Africa, September 2024 and World Health Organisation Operational Handbook on Tuberculosis Module 5: Management of tuberculosis in children and adolescents (2022).

Children should be taught and encouraged to swallow whole tablets or, if required, fractions of tablets to avoid large volumes of liquid medication.

Confirmed/presumed drug-susceptible TB meningitis or central nervous system (CNS) TB or miliary TB

Single phase treatment: 6-9 months. Once daily, 7 days a week

Target dose range and maximum doses	H: 15-20mg/kg, max dose 450mg R: 22.5-30mg/kg, max dose 900mg	Z: 35-45mg/kg, max dose 2g	Eto: 17.5-22.5mg/kg, max dose 1g	Target dose range and maximum doses	
Formulation	HR 50/75mg dispersible tablet (scored) OR	Z 500mg tablet (scored) OR	Eto 250mg tablet (not scored) OR	Formulation	
Weight (kg)	50/75mg/4mL suspension ¹	500mg/8mL suspension ²	250mg/8mL suspension ³	Weight (kg)	
< 2	Obtain expert advice			< 2	
2-2.9	¾ tablet (3mL) ¹	1mL	1.5mL	2-2.9	
3-3.9	1½ tablets	2mL	2mL	3-3.9	
4-4.9	< 3 months: 1½ tablets ≥ 3 months: 2 tablets	2.5mL	2.5mL	4-4.9	
5-5.9	2½ tablets	3mL	3mL	5-5.9	
6-7.9	3 tablets	½ tablet or 4mL	½ tablet or 4mL	6-7.9	
8-8.9	3½ tablets			8-8.9	
9-9.9			¾ tablet or 6mL	¾ tablet or 6mL	9-9.9
10-11.9	4 tablets	1 tablet or 8mL	1 tablet or 8mL	10-11.9	
12-12.9					12-12.9
13-14.9	4½ tablets			1¼ tablets or 10mL	13-14.9
15-15.9	5 tablets				15-15.9
16-16.9	6 tablets	1¼ tablet or 10mL	1¼ tablets or 10mL	16-16.9	
17-17.9					17-17.9
18-19.9			1½ tablets or 12mL	1½ tablets or 12mL	18-19.9
20-24.9	7 tablets	1½ tablet			20-24.9
25-29.9	HR 150/300mg tablet 3 tablets	2 tablets	2 tablets or 16mL	25-29.9	
30-34.9		2½ tablets	2½ tablets or 20mL	30-34.9	
35-39.9		3 tablets	3 tablets or 24mL	35-39.9	
40-49.9		3½ tablets	3½ tablets or 28mL	40-49.9	
≥ 50		4 tablets	4 tablets or 32mL	≥ 50	

¹To make an oral suspension for weight band 2-2.9kg: for each dose, disperse one HR 50/75mg tablet in 4mL of water, administer 3mL, discard unused suspension. For other weight bands, an oral suspension can be made by dispersing required number of tablets and fractions of tablets in a small amount of water (5-10mL) and administering all of the suspension to the child orally or via nasogastric tube.
²To make an oral suspension, crush one 500mg pyrazinamide tablet into a fine powder, disperse in 8mL water to prepare a concentration of 500mg/8mL (62.5mg/mL), administer required dose as indicated in above chart, discard unused suspension.
³To make an oral suspension, crush one 250mg ethionamide tablet into a fine powder, disperse in 8mL water to prepare a concentration of 250mg/8mL (31.3mg/mL), administer required dose as indicated in above chart, discard unused suspension.

HR – isoniazid + rifampicin; H – isoniazid; Z – pyrazinamide; Eto – ethionamide

Adapted from 2024 NDoH Management of Tuberculosis in Children and Adolescents: A Clinical Guideline For The Diagnosis and Treatment of Drug -susceptible TB in Children and Adolescents in South Africa, September 2024 and World Health Organisation Operational Handbook on Tuberculosis Module 5: Management of tuberculosis in children and adolescents (2022).

Antiretroviral and co-trimoxazole/multivitamin dosing table

ABC – abacavir; **ALD** – abacavir+lamivudine+dolutegravir; **am** – in the morning; **ATVr** – atazanavir+ritonavir; **AZT** – zidovudine; **bd** – twice daily dosing; **cap/s** – capsule/s; **DT** – dispersible tablet; **DTG** – dolutegravir; **FC** – film coated; **FDC** – fixed dose combination; **LPVr** – lopinavir+ritonavir; **od** – once daily dosing; **pm** – at night; **RTV** – ritonavir; **sol** – solution; **tab/s** – tablet/s; **TLD** – tenofovir+lamivudine+dolutegravir; **3TC** – lamivudine

Medication	ABC + 3TC	ABC	3TC	AZT	DTG	DTG ¹ while on rifampicin	LPVr (FDC: LPV + RTV)	LPVr ¹ while on rifampicin		ATVr (ATV + RTV)	Medication													
	• DT FDC: ABC/3TC 120/60mg (scored) • Tabs FDC: ABC/3TC 600/300mg, ALD 600/300/50mg	• Sol: 20mg/mL • Tabs: 60mg (scored, dispersible), 300mg (not scored)	• Sol: 10mg/mL • Tabs: 150mg (scored)	• Sol: 10mg/mL • Tabs: 100, 300mg (not scored) • FDC: AZT/3TC 300/150mg	• DTG DT: 10mg • DTG FC: 50mg • FDC: TLD 300/300/50mg, ALD 600/300/50mg • FDC DT: pALD 60/30/5mg	3 x DTG DT 10mg tablets = 1 x DTG FC 50mg tab (i.e. they are not bioequivalent)	• Sol: 80/20mg/mL² • Tabs: 200/50mg, 100/25mg	Choose only one option: Super-boost with RTV powder 100mg/packet Double-dose 200/50mg tabs OR 100/25mg tabs OR		FDC: ATVr 300/100														
2-2.9kg	0.5 x 120/60mg tab od	-	-	-	0.5 x 10mg tab od OR 1 x pALD 60/30/5mg od	0.5 x 10mg tab bd OR 1 x pALD 60/30/5mg od + 0.5 x 10mg tab 12 hours later	-	-	-	-	2-2.9kg													
3-5.9kg		3mL bd OR 1 x 60mg tab bd	3mL bd	6mL bd			1mL bd	LPVr bd (see purple columns) + RTV powder 100mg (1 packet) bd	Avoid	No preparation available	3-5.9kg													
6-9.9kg	1.5 x 120/60mg tabs od	4mL bd OR 1.5 x 60mg tabs bd	4mL bd	9mL bd	1.5 x 10mg tabs od OR 3 x pALD 60/30/5mg od	1.5 x 10mg tab bd OR 3 x pALD 60/30/5mg od + 1.5 x 10mg tabs 12 hours later	1.5mL bd				6-9.9kg													
10-13.9kg	2 x 120/60mg tabs od	od: 4 x 60mg tabs od OR 12mL od	od: 12mL od	12mL bd OR 1 x 100mg tab bd	2 x 10mg tabs od OR 4 x pALD 60/30/5mg od	2 x 10mg tabs bd OR 4 x pALD 60/30/5mg od + 2 x 10mg tabs 12 hours later	2mL bd OR 2 x 100/25mg tabs am + 1 x 100/25mg tab pm		3 x 100/25mg tabs bd		10-13.9kg													
14-19.9kg	2.5 x 120/60mg tabs od	5 x 60mg tabs od OR 1 x 300mg tab od	1 x 150mg tab od	2 x 100mg tabs am + 1 x 100mg tab pm OR 15mL bd	2.5 x 10mg tabs od OR 5 x pALD 60/30/5mg od	2.5 x 10mg tabs bd OR 5 x pALD 60/30/5mg od + 2.5 x 10mg tabs 12 hours later	2.5mL bd OR 2 x 100/25mg tabs bd OR 1 x 200/50mg tab bd	LPVr bd (see purple columns) + RTV powder 200mg (2 packets) bd	4 x 100/25mg tabs bd OR 2 x 200/50mg tabs bd		14-19.9kg													
20-24.9kg	3 x 120/60mg tabs od	1 x 300mg tab + 1 x 60mg tab od OR 6 x 60mg tabs od		2 x 100mg tabs bd OR 20mL bd	3 x 10mg tabs od OR 1 x 50mg tab od OR 6 x pALD 60/30/5mg od	3 x 10mg tabs bd OR 1 x 50mg tab bd OR 6 x pALD 60/30/5mg od + 3 x 10mg tabs 12 hours later OR 6 x pALD 60/30/5mg od + 50mg DTG tab 12 hours later	3mL bd OR 2 x 100/25mg tabs bd OR 1 x 200/50mg tab bd				20-24.9kg													
25-29.9kg	1 x 600/300mg tab od OR ALD 600/300/50mg od	2 x 300mg tabs od	2 x 150mg tabs od	1 x 300mg tabs bd OR 1 x AZT/3TC 300/150mg tab bd	1 x 50mg tab od OR ALD 600/300/50mg od	1 x 50mg tab bd OR ALD 600/300/50mg od + 50mg DTG tab 12 hours later	3.5mL bd OR 3 x 100/25mg tabs bd OR 1 x 200/50mg tab bd + 1x 100/25mg tab bd	LPVr bd (see purple columns) + RTV powder 300 mg (3 packets) bd	6 x 100/25mg tabs bd OR 3 x 200/50 mg tabs bd	1 x ATVr 300/100mg od	25-29.9kg													
30-34.9kg					1 x 50mg tab od OR ALD 600/300/50mg od OR TLD 300/300/50mg od	1 x 50mg tab bd OR ALD 600/300/50mg od + 50mg DTG tab 12 hours later OR TLD 300/300/50mg od + 50mg DTG tab 12 hours later	5mL bd OR 4 x 100/25mg tabs bd OR 2 x 200/50mg tabs bd		8 x 100/25mg tabs bd OR 4 x 200/50mg tabs bd		30-34.9kg													
≥ 35kg	<table><tr><td>Weight (kg)</td><td>3- 5.9kg</td><td>6 - 13.9kg</td><td>14 - 24.9kg</td><td>≥ 25kg</td></tr><tr><td>Co-trimoxazole</td><td>2.5mL od</td><td>5mL OR ½ tab</td><td>10mL OR 1 tab od</td><td>2 tabs od</td></tr><tr><td>Multivitamin</td><td>2.5mL od</td><td>2.5mL od</td><td>5mL od</td><td>10mL od</td></tr></table>	Weight (kg)	3- 5.9kg	6 - 13.9kg	14 - 24.9kg	≥ 25kg	Co-trimoxazole	2.5mL od	5mL OR ½ tab	10mL OR 1 tab od	2 tabs od	Multivitamin	2.5mL od	2.5mL od	5mL od	10mL od								≥ 35kg
Weight (kg)	3- 5.9kg	6 - 13.9kg	14 - 24.9kg	≥ 25kg																				
Co-trimoxazole	2.5mL od	5mL OR ½ tab	10mL OR 1 tab od	2 tabs od																				
Multivitamin	2.5mL od	2.5mL od	5mL od	10mL od																				

¹Adapted from antiretroviral drug dosing chart for children 2025, by the Child and Adolescent Committee of the SA HIV Clinicians Society in collaboration with the Department of Health.

¹Continue for 2 weeks after stopping rifampicin. ²Improve taste by coating mouth with peanut butter or numbing taste buds with ice before giving LPVr solution and/or feeding child soft food after giving LPVr solution.

Practical tips for giving ART correctly

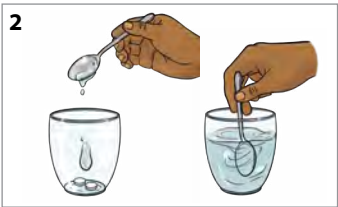
Support carer to give ART correctly

If child is taking abacavir + lamivudine (ABC + 3TC) and dolutegravir (DTG) dispersible tablets:

- Explain that ABC/3TC and DTG dispersible tablets can be dissolved whole, split or crushed in a small amount of water. Child to avoid chewing tablet.
- Use steps 1-3 below to explain how to prepare and give tablets:



- Add tablets to clean, empty glass/cup.
- If giving 1½ or 2½ tablets, split tablet down the middle on solid line. Store leftover ½ tablet in bottle/packet tablet came in as it contains a drying agent to help preserve tablets correctly until they are needed.



Add 4 teaspoons of fluid/food¹ to glass/cup and stir until tablets fully dissolve. If tablets dissolve incompletely or lump together, stir in 2 more teaspoons of fluid/food¹ until tablets fully dissolve.

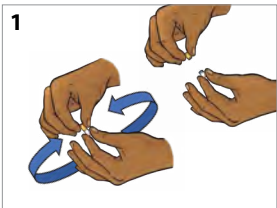


- Give within 30 minutes with spoon or allow child to drink mixture directly from glass/cup.
- If mixture left over, add 2 further teaspoons of fluid/food¹, gently swirl or stir and give to child to drink/eat. Repeat until all medication swallowed.

- If child not taking both ABC/3TC and DTG (taking only 1 of the 2), follow the same steps above, but adjust dosing:
 - If dispersing ½ - 1½ tablets, mix with 1 teaspoon of fluid/food¹. If dispersing ≥ 2 tablets, mix with 2 teaspoons of fluid/food¹.
- If child vomits within 30 minutes of taking dolutegravir, repeat dose once.

If child is taking lopinavir + ritonavir (LPVr) pellets in capsules:

- Explain that yellow and white capsules contain white pellets.
- Advise to avoid swallowing capsules whole or stirring/dissolving/crushing/chewing pellets. Pellets still need to be seen before child swallows these.
- Use steps 1-3 below to explain how to prepare and give pellets. Follow the same steps for every capsule:



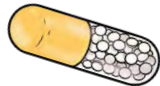
- Hold capsule with yellow side up, then tap top of capsule to release pellets into lower half of capsule.
- Twist to loosen and lift yellow half of capsule. Pour all pellets onto a teaspoon.



Pour pellets onto food that child enjoys eating (like yoghurt, mashed potato, porridge). If child refuses to swallow food while pellets are visible, cover pellets with food.

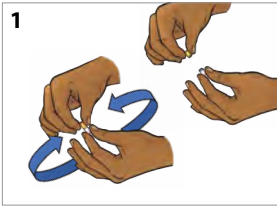


- Feed child immediately to prevent bad taste. Offer child a drink of fluid if some of the mixture remains in mouth.
- Encourage child to finish every spoon. Discard any food with uneaten pellets.

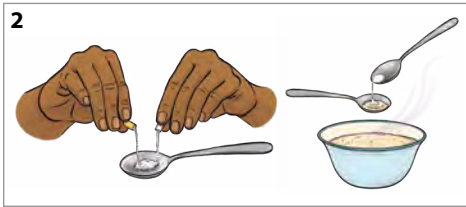


If child is taking '4-in-1' abacavir + lamivudine + lopinavir + ritonavir (ABC + 3TC + LPVr) granules in capsules:

- Explain that brown and white capsules contain white granules.
- Advise to avoid swallowing capsules whole.
- Use steps 1-3 below to explain how to prepare and give granules. Follow the same steps for every capsule:



- Hold capsule with brown side up, then tap top of capsule to release granules into lower half of capsule.
- Twist to loosen and lift brown half of capsule. Pour all granules onto a teaspoon.



Add fluid/food¹ to cover the granules on spoon or add 4 teaspoons of fluid/food¹ to glass/cup and stir until granules fully dissolve.



- Feed child immediately. Offer child a drink of fluid if some of the mixture remains in mouth.
- Encourage child to finish every spoon. Discard any fluid/food with uneaten granules.



¹Clean water, breast milk, milk, juice, yoghurt, porridge, mashed potato. If food used, crush tablet to help with dissolving tablet in food.

Helpline numbers

Helpline	Services provided	Contact details
General counselling		
ACVV	Help nurture and protect those in need.	• 021 461 7437 • www.acvv.org.za
Cape Mental Health	Provides mental health services in the Western Cape.	021 447 9040
Childline South Africa	For children who are in crises, abuse or at risk of abuse and violence.	• 116 (24 hour toll-free) • www.childlinesa.org.za
Family Services Directory	Area specific directory of family services available with contact details provided.	https://pmhp.za.org/wp-content/uploads/Western-Cape-Family-Services-Directory.pdf
The South African Depression and Anxiety group	Telephonic counselling and assistance with mental health.	• 0800 567 567 • www.sadag.org
Safe Schools	For learners who are in crises. Help schools to mobilise community support for safe schools.	0800 45 46 47
Suicide Crisis Helpline	Assist with counseling and support where needed.	0800 554 433
The Parent Centre	Parenting information, support and training.	• 021 762 0116 (8am-4pm) • www.theparentcentre.org.za
Western Cape Education Department	Schooling information or child not enrolled in school.	0861 819 919
Health care worker		
Poison Information Helpline of the Western Cape	Advice on management of exposure to or ingestion of poisonous substances.	0861 555 777 (24 hour line)
Notifiable Medical Conditions: Provincial CDC	Notification and information on outbreaks.	• provincialcdc@westerncape.gov.za • Coordinator: 021 830 3727/072 356 5146; Charlene.Lawrence@westerncape.gov.za • Administrative Clerk: 021 815 8660/1; Felencia.Daniels@westerncape.gov.za • NMC Surveillance Manager: 064 742 4005; Washiefa.Isaacs@westerncape.gov.za
Medicines Information Centre (MIC)	Medicine related queries (drug interactions, side effects, dosage, treatment failure) or to report an adverse drug reaction (ADR).	• 021 406 6829 (8.30am-4.30pm) • www.mic.uct.ac.za
Provincial office: Communicable Disease Control - expanded programme on immunisation (EPI)	For reporting of investigation of adverse event following immunisations (AEFI).	• 021 830 3727 • 021 815 8810/8660/8740
SASSA information line	Assessment and arranging of grants.	• 0800 601 011 toll-free (7.30am-4pm) • www.sassa.gov.za
National HIV & TB Health Care Worker Hotline	For HIV and TB related clinical queries.	• 0800 212 506 toll-free (8.30am-4.30pm) • 071 840 1572 • pha-mic@uct.ac.za
National Institute for Communicable Diseases Hotline (NICD)	Guidance on communicable diseases and outbreaks, including rabies.	0800 212 552
Philani Maternal, Child Health and Nutritional Project	Helps prevent and support child malnutrition with food aid and support.	• 021 387 5124 (8.30am-4.30pm) • www.philani.org.za
WCDHW: Employee Health and Wellness programme	Confidential counselling for emotional and financial issues.	0800 611 093 (24 hours)

Helpline	Services provided	Contact details
Abuse		
Crime stop: Provincial Family Violence, Child Protection and Sexual Offences	To notify police of physical or sexual abuse.	10111 or 112
Department of Social Development	To notify a case of suspected child abuse or neglect.	0800 220 250 (7.00am-7.00pm)
Rape Crisis Cape Town Trust	Advice and support for cases of sexual abuse. Information on closest crisis or Thuthuzela care centre.	• 021 447 9762 (English); 021 361 9085 (isiXhosa); 021 633 9229 (Afrikaans) • WhatsApp counselling: 083 222 5164
Safeline	Abuse counselling, court preparation, anti-abuse awareness campaigns and group therapy.	0800 035 553 or 021 705 2147 (8am-3pm Monday to Friday)
Western Cape Directory of Services for Victims of Crime and Violence	Contact details of individuals, associations and organisations contributing to service delivery within the field of Victim Empowerment in the Western Cape.	https://www.westerncape.gov.za/social-development/files/wcg-blob-files?file=2025-11/western-cape-directory-of-services-for-victims-of-crime-and-violence-november-2025.pdf&type=file
Long-term condition		
Allergy Foundation of South Africa	Advice and support for children with allergies/asthma/eczema and their carers.	• 081 405 8442 • info@allergyfoundation.co.za • www.allergyfoundation.co.za
Down Syndrome Association	Advice and support groups for children with Down Syndrome and their families.	• 072 652 2377 (8.30am-4pm Monday to Thursday, 8.30am-1pm Friday) • www.downsyndrome.org.za
Western Cape Cerebral Palsy	Advice and support for parents and guardians of children with cerebral palsy.	• 021 685 4150 (8.30am-3pm) • www.wccpa.org.za
Hi Hopes	Deaf children hearing support.	• Ms De Jager 083 552 2110 (8am-4pm) • hihopeswc@gmail.com, www.hihopes.co.za
Paedspal	Care and support for children living with life-limiting illness.	• 021 200 5873 (8.30am-4.30pm) • www.paedspal.org.za
Epilepsy SA	Education, counselling and support groups for children with epilepsy and their families.	• 0860 374 537 • www.epilepsy.org.za
National HIV/AIDS helpline	Counselling and information for children living with HIV.	0800 012 322 (24 hour)
Childhood Cancer Foundation of South Africa	Support for families affected by cancer.	• 086 111 3500 • bdu@choc.org.za
Paediatric Cardiac Society of SA	Advice, counselling and information for children with heart disease.	www.pcssa.org
Administration		
MedicAlert®	Assistance with application for MedicAlert® disc or bracelet.	• 086 111 2979; 021 461 000 (24 hour emergency line) • www.medicalert.co.za
Legal Aid Advice Line	Information and guidance on any legal matter.	0800 110 110 (7am-7pm Monday to Friday)
Telephone directory/additional resources		

Predicted peak expiratory flow rate (PEFR) for height

- Compare child’s PEFR to below table according to height: aim for PEFR between 80-100%.
- A normal PEFR does not exclude poorly controlled asthma. If other symptoms, step up treatment ➔ 119 or discuss/refer.

Height (cm)	PEFR (L/min) zones		
	60%	80%	100%
95cm	51	68	85
96cm	54	72	90
97cm	57	76	95
98cm	60	80	100
99cm	63	84	105
100cm	66	88	110
101cm	69	92	115
102cm	72	96	120
103cm	75	100	125
104cm	78	104	130
105cm	81	108	135
106cm	84	112	140
107cm	87	116	145
108cm	90	120	150
109cm	93	124	155
110cm	96	128	160
111cm	99	132	165
112cm	102	136	170
113cm	105	140	175
114cm	108	144	180
115cm	111	148	185
116cm	114	152	190
117cm	117	156	195
118cm	120	160	200
119cm	123	164	205
120cm	126	168	210
121cm	129	172	215
122cm	132	176	220
123cm	135	180	225
124cm	138	184	230
125cm	141	188	235

Height (cm)	PEFR (L/min) zones		
	60%	80%	100%
126cm	144	192	240
127cm	147	196	245
128cm	150	200	250
129cm	153	204	255
130cm	156	208	260
131cm	159	212	265
132cm	162	216	270
133cm	165	220	275
134cm	168	224	280
135cm	171	228	285
136cm	174	232	290
137cm	177	236	295
138cm	180	240	300
139cm	183	244	305
140cm	186	248	310
141cm	189	252	315
142cm	192	256	320
143cm	195	260	325
144cm	198	264	330
145cm	201	268	335
146cm	204	272	340
147cm	207	276	345
148cm	210	280	350
149cm	210	284	355
150cm	216	288	360
151cm	219	292	365
152cm	222	296	370
153cm	225	300	375
154cm	228	304	380
155cm	231	308	385
156cm	234	312	390

Height (cm)	PEFR (L/min) zones		
	60%	80%	100%
157cm	237	316	395
158cm	240	320	400
159cm	243	324	405
160cm	246	328	410
161cm	249	332	415
162cm	252	336	420
163cm	255	340	425
164cm	258	344	430
165cm	261	348	435
166cm	264	352	440
167cm	267	356	445
168cm	270	360	450
169cm	273	364	455
170cm	276	368	460
171cm	279	372	465
172cm	282	376	470
173cm	285	380	475
174cm	288	384	480
175cm	291	388	485
176cm	294	392	490
177cm	297	395	495
178cm	300	400	500
179cm	303	404	505
180cm	306	408	510

Based on Polgar and Promadht: Pulmonary function testing in children: techniques and standards, 1979. Adapted from www.allergyfoundation.co.za.

Age-appropriate behaviour

Age-appropriate fears and anxieties in children ¹	
Babies & Toddlers (9 months – 2 years old)	Loud noises, separation anxiety, fear of the dark, fear of strangers.
Young Children (2-5 years old)	Fear of storms, fire, water, darkness, nightmares, and animals.
Middle Childhood (6-12 years old)	• Fear of monsters, ghosts, germs, natural disasters, physical illness, and being badly injured. • Anxiety about school or about performing in front of others.

Age-appropriate disruptive or challenging behaviour in children ¹	
Toddlers and young children (18 months – 5 years old)	• Refusing to do what they are told, breaking rules, arguing, whining, exaggerating, saying things that aren't true, denying they did anything wrong, being physically aggressive and blaming others for their misbehaviour. • Brief tantrums (emotional outbursts with crying, screaming, hitting, etc.), usually lasting less than 5 minutes and not longer than 25 minutes, typically occur less than 3 times per week. Developmentally typical tantrums should not result in self-injury or frequent physical aggression toward others, and the child can typically calm themselves down afterward.
Middle Childhood (6-12 years old)	Avoidance of or delay in following instructions, complaining or arguing with adults or other children, occasionally losing their temper.

Assess pain in the child

Over 5 minutes, look at child's facial expression, position of legs, activity level, type of cry and ability to be consoled. Then use the table below to score each of these. Add scores to get a total and use this to rate the pain². Use pain rating to inform choice of pain relief medications.

- If score is 0: rate as **no pain**.
- If score is 1-3: rate as **mild pain**.
- If score is 4-6: rate as **moderate pain**.
- If score is 7-10: rate as **severe pain**.

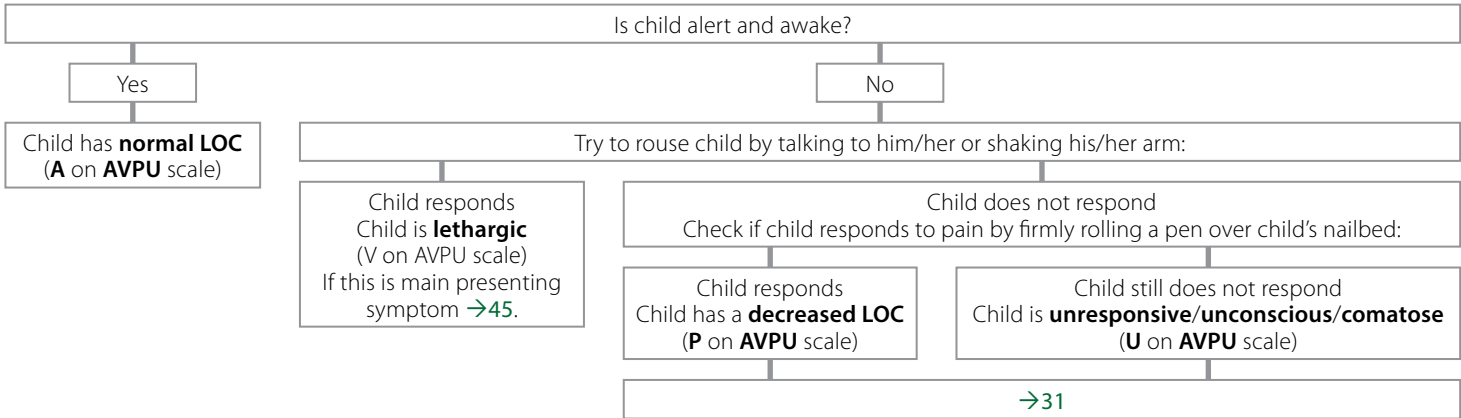
Face		Legs		Activity		Cry		Consolability	
0	No particular expression or smile	0	Normal position or relaxed	0	Normal position, lying quietly, moves easily	0	No cry (awake or asleep)	0	Content, relaxed
1	Occasional grimace or frown, withdrawn, uninterested, appears sad or worried	1	Uneasy, restless, tense, occasional tremors	1	Squirming, shifting, back and forth, tense, mildly agitated, shallow/splinting respirations, intermittent sighs	1	Moans or whimpers, occasional complaints/verbal outbursts/ grunting	1	Reassured by occasional touching, hugging or being talked to, distractible
2	Constant grimace/frown, quivering chin, clenched jaw, looks distressed, expression of fright or panic	2	Kicking or legs drawn up, spasticity, constant tremors, jerking	2	Arched, rigid or jerking, severe agitation, breath-holding, gasping, sharp intake of breath, severe splinting	2	Crying steadily, screams or sobs, frequent complaints/outbursts, constant grunting	2	Difficult to console or comfort; pushing away caregiver or comfort measures

¹Adapted from WHO mhGAP Intervention Guide for mental, neurological and substance use disorders in non-specialized health setting. Mental Health Gap Action Programme. Version 2.0. World Health Organization. Geneva, 2016. ISBN: 978 92 4 1549790.
²Revised FLACC observational pain tool (Face, Legs, Activity, Cry, Consolability) was developed by Sandra Merkel, Terri Voepel-Lewis, and Shobha Malviya at C. S. Mott Children's Hospital, University of Michigan Health System, Ann Arbor, USA.

Assess level of consciousness (LOC)

- If rapid, simple assessment of LOC needed, use **AVPU** scale.
- If more detailed assessment of LOC needed (like head injury or considering intubation), use **Glasgow Coma Scale (GCS)**.

Assess level of consciousness (LOC) with the AVPU scale:



Assess level of consciousness with AVPU

A	Alert
V	responds to Voice
P	responds to Pain
U	Unresponsive/Unconscious

Assess level of consciousness (LOC) using the Glasgow Coma Scale (GCS)

Add scores to give a single score out of 15: if GCS ≤ 9, intubate child.

Best motor response	Best verbal response	Eye opening
6 Spontaneous movements (obeys commands)	5 Age-appropriate vocalisation/smile/response to sound, interacts (coos, babbles), follows objects	4 Spontaneous
5 Withdraws to touch (localizes pain)	4 Cries, irritable	3 To voice
4 Withdraws from pain	3 Cries to pain	2 To pain
3 Abnormal flexion to pain	2 Moans to pain	1 None
2 Extends to pain	1 None	
1 None		

Quick reference chart

If emergency situation, use a Broselow® paediatric emergency tape, if available, to estimate weight, endotracheal tube size and emergency medicine doses.

Decide if pulse rate is normal for age		
Age	Pulse rate (beats/minute)	
	Pulse rate decreased if:	Pulse rate increased if:
< 1 year	< 110	≥ 160
1-2 years	<100	≥ 150
2-5 years	< 95	≥ 140
5-12 years	< 80	≥ 120
≥ 12 years	< 60	≥ 100

Decide if respiratory rate is normal for age		
Age	Respiratory rate (breaths/minute)	
	Respiratory rate decreased if:	Respiratory rate increased if:
< 2 months	< 30	≥ 60
2-12 months	< 30	≥ 50
1-5 years	< 25	≥ 40
5-12 years	< 20	≥ 25
≥ 12 years	< 15	≥ 20

Decide if blood pressure is normal for age		
Age	Systolic blood pressure (mmHg)	
	Blood pressure decreased if:	Blood pressure increased if:
< 1 year	< 80	> 90
1-2 years	< 85	> 95
2-5 years	< 85	> 100
5-12 years	< 90	> 110
≥ 12 years	< 100	> 120

Estimate weight according to age	
< 12 months	Weight (kg) = (0.5 x age in months) + 4
1-5 years	Weight (kg) = (2 x age in years) + 8
5-12 years	Weight (kg) = (3 x age in years) + 7

Decide on endotracheal tube (ETT) size (mm) and depth (cm) Use cuffed tube. If only uncuffed available, use one size bigger.				
Age	Weight (kg)	ETT size (mm) (internal diameter)	ETT depth (cm)	
			Oral (measurement at lips)	Nasal (measurement at nostril)
Preterm	1	2.5	7	8.5
Preterm	2	2.5 - 3	8	9.5
Term	3	3 – 3.5	9.5	11.5
2 months	4.5	3.5	11	12.5
1 year	10	4	12	14
18 months	12	4.5	13	15
2 years	15	5	14	16
4 years	17	5.5	15	17
6 years	21	6	16	19
8 years	25	6.5	17	20
10 years	31	7	18	21

Decide on maintenance fluid rate	
Weight	24 hour fluid need
< 10kg	100mL/kg
10-20kg	1000mL + (50mL for every kg body weight over 10kg) e.g.: if 14kg: 1000mL + (50 x 4) = 1200mL/24 hours
≥ 20kg	1500mL + (20mL for every kg body weight over 20kg) Up to 2000mL in girls and 2500mL in boys e.g.: if 23kg: 1500mL + (20 x 3) = 1560mL/24 hours



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PACK
Practical Approach to Care Kit

Primary Care Guide for the Child 2026· Western Cape Edition

The Practical Approach to Care Kit (PACK) Child guide is a comprehensive guide for the primary care of the child up to 13 years old. It uses simple algorithms to evaluate and treat the child with common symptoms and a standardised checklist format to care for the child with a long-term health condition. It supports the clinician to integrate routine care of the child into every visit. It is designed for use in Western Cape public sector primary care consultations.

PACK Primary Care Guide for the Child complies with and integrates Western Cape Provincial and South African National policies. Prescribing provisions are displayed clearly for each medicine, its dose and indication, to capacitate staff to manage children.

The development and revision of the guide was a collaborative process with substantial input from managers, clinicians and academics, as well as feedback from end-users of previous editions (see acknowledgements page). A more thorough explanation of the development process and role of contributors can be found at www.knowledgetranslation.co.za.

This **PACK Primary Care Guide for the Child** was compiled by the Knowledge Translation Unit (KTU), a unit of the Knowledge Translation Foundation (KTF), University of Cape Town. The Knowledge Translation Unit declares it has no competing interests in pharmaceutical companies whose products or services are related to the guide topics.

This **PACK Primary Care Guide for the Child** forms part of a suite of clinical tools designed for use in primary care. In various stages of development and pilot are **PACK Adult** for the adult ≥ 18 years old and **PACK Youth** for the 10-24 year old. These clinical tools are supported by Training Manuals and an Implementation Toolkit to ensure the programme is embedded in the health system to streamline the delivery of primary care.

PACK training can be accessed on the PDC online school:
wcgh-pdc-online-school.thinkific.com

Practical Approach to Care Kit

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